STATE UNIVERSITIES AND COLLEGES – ASSOCIATION OF COLLEGES OF AGRICULTURE IN THE PHILIPPINES (SUC-ACAP)

REVIEW MATERIALS FOR
THE LICENSURE EXAMINATION FOR AGRICULTURISTS

Crop Science

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SUC-ACAP-LEA

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PREFACE

Republic Act 8435 or the Agriculture and Fisheries Modernization Act of 1997, simply known as AFMA, was enacted on December 22, 1997. It is a policy instrument defining measures to modernize Philippine agriculture for the country to compete in the global market. A basic human resource development framework in modernizing agriculture under the Act is to enhance the professional competency of agriculturists by upgrading the standards of the agriculture and fisheries education that will ensure to similarly upgrade the quality and sustainability of, and promote global competitiveness, at all levels.

One strategy for professionalizing the agriculture sector is the Licensure Examination for Agriculturists (LEA). First implemented in 2003, LEA serves as a standardized evaluation tool to identify graduates of Agriculture and Agriculture-related degree programs who are academically and technically fit to be admitted in the agricultural profession. The three-day examination consists of multiple-choice type questions in the following component subjects: Crop Science, Soil Science, Crop Protection, Animal Science, Agricultural Economics and Marketing, and Agricultural Extension and Communication.

The COVID-19 pandemic has been a major concern of all higher education institutions. When review sessions for licensure board exams are already supposed to be grinding, SUCs are confronted by the reality that face-to-face review is undeniably not possible. Under the circumstance, universities face new pressures to provide an engaging, motivating, and effective teaching and learning experiences for both reviewees and reviewers.

Following this, the SUC-ACAP, with its goal of advancing the agriculture professional competency among their member-institutions, and the College of Agriculture and Food Science, UP Los Baños embarked on a joint initiative to enhance the capability of the SUC faculty particularly those engaged in the conduct of review classes in preparation for the Licensure Exam for Agriculture (LEA). To achieve this, the joint initiative conducted three online lectures, namely, 1) Innovative Teaching Methods and Tools Designed for Online Review Classes by no less than the Chancellor of the UP Open University, Dr. Melinda dP. Bandelaria. This online course aimed at introducing the faculty participants to the particularities of conducting review classes online with the use of innovative methods and tools; 2) Test Construction: the Multiple-Choice Type Test from a seasoned UP faculty, Dr. Blanda R. Sumayao, which gave the faculty-participants the opportunity to review the principles and rules for constructing multiple-choice type tests; and 3) Design and Development of Review Materials for LEA, again with Dr. Bandelaria with the objective of providing the faculty-participants the necessary knowledge and skills in designing and developing review materials and planning for the delivery of online review sessions.

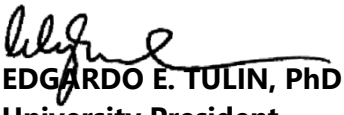
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The Project initiative also gave the faculty-participants the opportunity to review their course syllabus viz-a-viz CMO #14 S. 2008 or the CHED-prescribed outline which enabled them to design and develop this harmonized syllabus for each of the six core subject matter area. The review materials are thus a product of many minds and talents, that is, of faculty-participants from the different SUCs. I would like to acknowledge all the writers/contributors who willingly presented themselves to the initiative and for strictly following deadlines in submitting drafts; to all University Presidents and the Deans of the Colleges of Agriculture who have been highly committed to the objective of this Project initiative and for the support provided to the faculty-participants; to all Cluster and Sub-Cluster Coordinators for the constant reminders and follow-ups; and to the SUC-ACAP Secretariat for the untiring administrative support.

And lastly, we are grateful to the College of Agriculture and Food Science, UP Los Baños for being true to its commitment of being a public service University, providing scholarly and technical assistance to its sister and brother SUCs. Specifically, we thank the CAFS faculty who served as Subject Area Coordinators/Specialists who unselfishly guided our faculty participants throughout their journey in the project, as well as the Technical Assistants and Administrative Staff who have lend full support to attaining the objectives of the joint initiative.

As I have always reiterated in many occasions, let us be reminded that student preparation for the licensure is just one of the factors that would determine performance. It is still incumbent upon us SUCs or higher educational institutions to ensure that we will be able to provide the necessary prerequisite services and institutional support in order for students to ultimately pass successfully and help advance the students' professional careers post licensure exam.

The preparation for publication of these review materials was supported by the SUC-ACAP.



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FOREWORD

Unlike in *Lord of the Rings* and *Harry Potter* films, there are no wizards, Balrogs, Voldemort, precious rings, or elderly wands, on the journey to becoming a professional agriculturist. But there are narrow bridges and mazes that you must travel and cross over. One of these challenges is the Professional Licensure Exam.

In the current climate where globalization, changing natural conditions, production, and social composition; migratory movements, among others, have significantly impacted most agricultural landscapes in the developing world, more than ever, the Licensure Exam for Agriculture has become all the more relevant as it prepares graduates of Agriculture to embrace the tasks ahead of them with utmost professionalism by way of recognizing their expert knowledge and practice. The Licensure Examination for Agriculture was established to recognize the expert knowledge and practice of aspiring professional agriculturists.

This Review Material produced by the faculty participants of the joint initiative of the UPLB College of Agriculture and Food Science (UPLB-CAFS)- State Universities and Colleges-Association of Colleges of Agriculture in the Philippines (SUC-ACAP) is a comprehensive resource that covers all relevant subject areas in the field of agriculture which hopes to serve as a valuable review tool to students of SUC-ACAP member institutions.

I wish you the best in your journey and beginning your career as a professional agriculturist.



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CROP SCIENCE 1

I. INTRODUCTION

A. AGRICULTURAL DEVELOPMENT

Overview of the Philippine Agriculture

The Philippines is an agricultural country consisting of mountainous terrains, dense forests, plains, coastal areas and rich in biodiversity and high percentage of flora and fauna. The leading crops are rice, corn, sugarcane, coconut, banana, mango, pineapple, cassava, coffee, sweet potato and eggplant. In terms of harvest area, the most extensively grown crops are rice, coconut, corn, sugarcane, banana, cassava, coffee, mango, sweet potato and abaca. The share of the Philippine agriculture in the gross domestic product (GDP) from 2009 to 2019 was 8.82%, industry contributed approximately 30.17% and the services sector contributed about 61%. According to International Labor Organization (ILO, 2020) of the 58% total employment services of the country, 23% are employed in agricultural sectors. This made the sector as the key driver of the economy not only in the rural areas but in the urban as well.

Learning Objectives

At the end of the topic, the participant should be able to:
1. trace the agricultural systems and practices in the Philippines,
2. identify new farming technologies in the Philippines, and
3. list down the contributions of new technologies in crop production systems.

Major Agricultural Systems in the Philippines

1. **Lowland Irrigated Farming System –** It implies complete water control and the area is flooded. Around 1300−1500 mm is a typical amount of water needed for irrigated rice in Asia. Irrigated rice receives an estimated 34−43% of the total world's irrigation water, or about 24−30% of the entire world's developed fresh water resources (IRRI). A survey of 2019-2020 crop year shows that China produced 146.7 million metric tons of milled rice, a higher volume than any other country. The Philippines ranks number 8 with 11.93 tons from among the ten leading countries producing milled rice in the world (2019-2020, Statista).

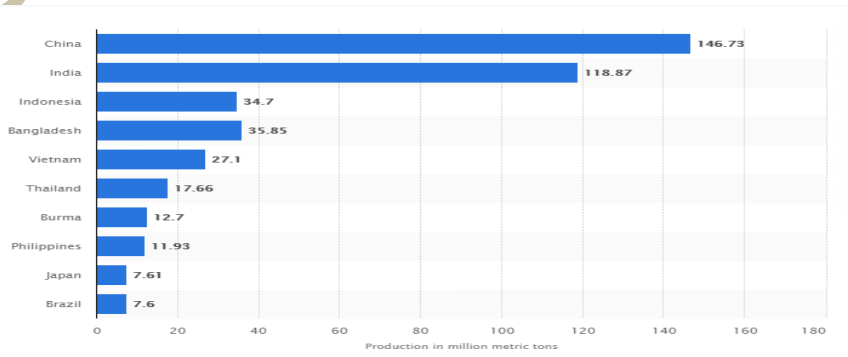


Figure 1. Graph showing the world's leading rice producing countries (2019-2020, Statista).

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2. **Rainfed Farming** – It is a type of farming that relies on rainfall and water is applied from other sources, such as freshwater from streams, rivers and lakes or groundwater.

3. **Upland Farming** – crops grown in dry fields that are not flooded and the area is flooded only for short periods.

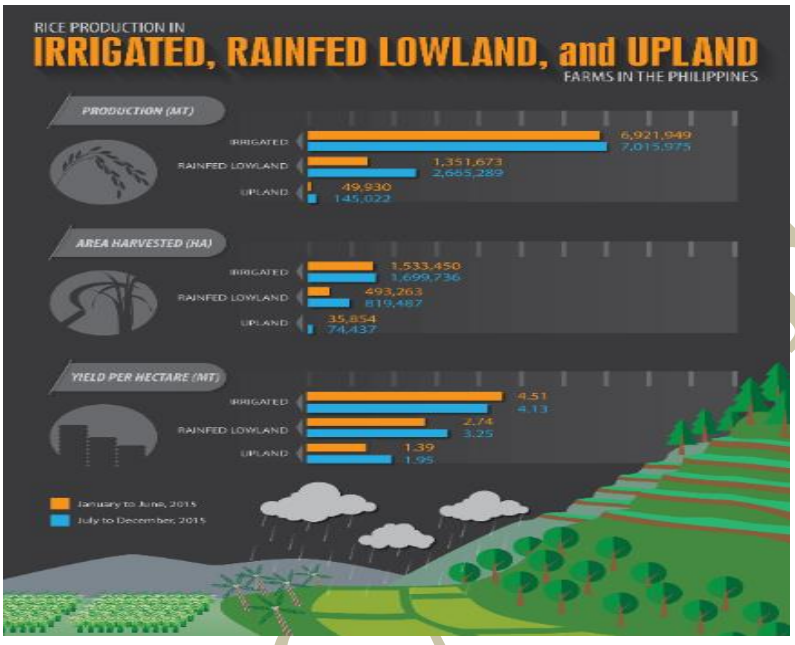


Figure 2. Rice production systems in the Philippines (IRRI, 1990).

Good Agricultural Practices

The Philippine Department of Agriculture's vision is to modernize small-holder agriculture and fisheries to a diversified rural economy that is dynamic, technologically advanced and internationally competitive. Good agricultural practices is being promoted in order to ensure food safety, assure quality of agricultural products while keeping high regard for environmental protection and that of workers health, safety and welfare, and facilitate access of Philippine agricultural crops to neighboring ASEAN and other foreign markets.



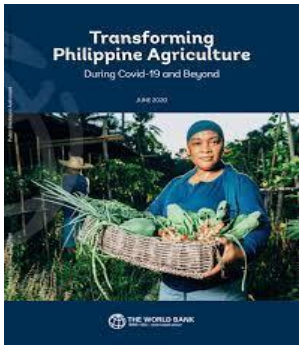


Figure 3. Good agricultural practices.

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4. **Integrated Farming Systems –** integration of natural resources in a regulated mechanism into farming activities to achieve maximum replacement of off-farm inputs and sustain farm income.

Table 1.1 Terminology for rice-growing environments (IRRI 1984c).

Ecosystem	Subecosystem
Irrigated	Irrigated, with favorable temperature Irrigated, low-temperature, tropical zone Irrigated, low-temperature, temperate zone
Rainfed lowland	Rainfed shallow, favorable Rainfed shallow, drought-prone Rainfed shallow, drought- and submergence-prone Rainfed shallow, submergence-prone Rainfed medium-deep, waterlogged
Deepwater	Deep water Very deep water
Upland	Favorable upland with long growing season (LF) Favorable upland with short growing season (SF) Unfavorable upland with long growing season (LU) Unfavorable upland with short growing season (SU)
Tidal wetlands	Tidal wetlands with perennially fresh water Tidal wetlands with seasonally or perennially saline water Tidal wetlands with acid sulfate soils Tidal wetlands with peat soils

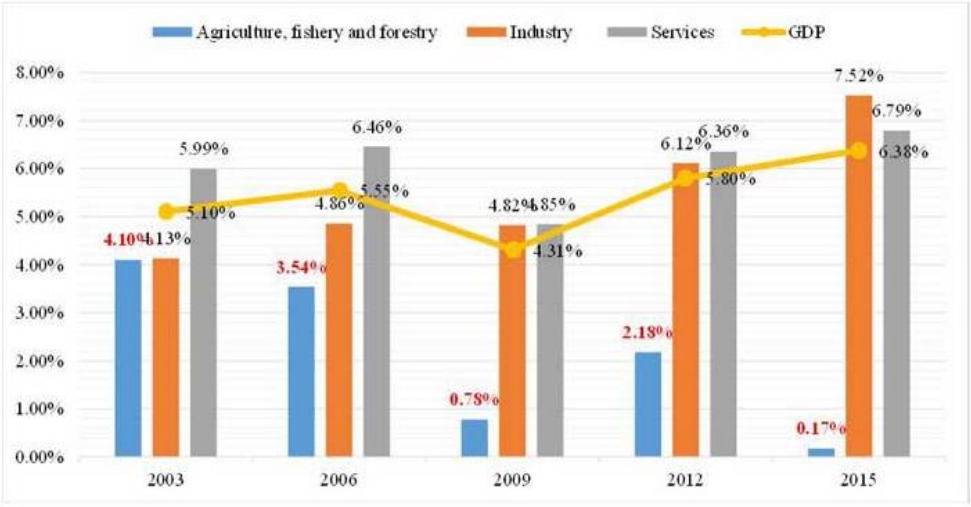


Figure 4. Gross annual growth rates by sector, Philippines 2003-2015 (Source: PSA, 2018)

Economic Contribution

In terms of gross value, agricultural production rose by more than 4% year on-year in the fourth quarter of 2018 to reach P521.2bn ($9.7bn). Agriculture, hunting, forestry and fishing contributed 8.9% to GDP in the same quarter. According to the data from the Philippine Statistics Authority, crop production accounted for 50.4% of total agricultural output, which is equivalent to P295.4bn ($5.5bn) in value terms during the final quarter of 2018.

Problems in Agriculture in the Philippines

Long standing challenges that hamper productivity include limited access to credit and agricultural insurance, low farm mechanization and inadequate postharvest facilities, inadequate irrigation, scant support for research and development (R&D), weak extension service and incomplete agrarian reform program implementation

Farming Technology Generated and Adopted by the Filipino Farmers

To address self-sufficiency in important food staples, new farming technologies empowering Filipino farmers are made by both the government and the private sector to improve farming methodologies. Among the generated and adopted technologies are listed below.

1. F_1 hybrid seeds in rice, corn, vegetable production and other crops to ensure sustainable production.
2. Biotechnology research and development
3. Intelligent farming equipment to increase field productivity
4. The completion of solar-powered irrigation system
5. Japanese-made drones to spray fertilizer on vegetable farms located on mountain slopes.
6. Farmers Guide Map to assist in identifying suitable crops to be planted on their lands when provided with data on climate and soil adaptability
7. Rice Crop Manager app, a web-based platform, that provides recommendations to farmers in crop management based on their own farm conditions.
8. AgriDOC app that enables farmers to record their activities and monitor farm inputs, improve their farm tasking and guide them in farm management
9. KROPS app that lets farmers market their own products at their own prices and help potential buyers find sources of farm products within their own area.
10. Smart Plant Production in Controlled Environments (SPICE) program, designed to develop and promote urban farming and high-tech plant conservation.

Ten Features of a Philippine Farm of the Future (Ranada, 2015)

1. Multiple crops
2. Water harvesting
3. Renewable energy
4. High-tech sensing equipment
5. Own weather station
6. Connected to apps
7. Farming machines
8. Right varieties
9. Makes the most of extreme conditions
10. Natural pesticide and fertilizer (NFIs)

Technological innovations such as the use of global positioning system (GPS), geographic information system (GIS), and precision farming equipment, have greatly shaped agriculture from the old farming practices and make farming more efficient. Although these innovative equipment are not fully adapted, several methods contributed much in crop and animal production:

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1. **Vertical Farming** or high-rise farming is an indoor and urban farming technology involving large-scale agricultural production in multi-story buildings.
2. **Livestock Farming Technology** adoption of “ready to use technology” to increase production and improve productivity to help and ensure the availability, accessibility and affordability of livestock products. The growth in livestock production is due to adoption of new technologies such as: imported breeds, veterinary medicines and feed ingredients.
3. **Hydroponics.** Crops grown in soil-less condition which uses mineral-enriched water solutions.



Figure 5. Photo showing the hydroponics system.

4. **Aquaponics.** Bringing together fish and plant farming in one recirculating system.
5. **Greenhouse.** A regulated conditions to prevent fluctuations in external environments and also optimum conditions for growth and photosynthesis. e.g. controlled temperatures, controlled humidity, higher CO_2 content.
6. **Nursery** is a highly controlled conditions to protect young plantlets and prevent exposure from harsh conditions. e.g. high humidity, warm temperatures, low light intensities.
7. **Urban Agriculture (Bayanihan Act of 2020)** an establishment of a resilient food system through urban farming that would ensure food security especially during the time of pandemic.

Importance of Agricultural Technology

1. Higher crop productivity
2. Decreased use of water, fertilizer and pesticides
3. Reduced impact on natural ecosystem
4. Less runoff of chemicals into rivers and groundwater
5. Increase worker safety

Summary

- This module presented the topic how agriculture plays a dominant role in the Philippine economy. The role of agriculture and its contribution to national income and employment growth and the agricultural sector (comprising of crops, livestock and poultry, fishery, and forestry) which performed remarkably well.

- The status of Philippine agriculture as well as the challenges and strategies for sustainable development that are addressed and supported by the government as well as the non-government organizations who participated and adopted the modern techniques in farming.

- The adapted technological innovations that contributed have greatly that makes farming more efficiently.

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B. THE PHILIPPINE AGRICULTURE

Overview

The Philippines is an agricultural country with a land area of 30 million hectares wherein 47% of which is agricultural land. With the richness in the land, natural resources, hardworking farmers, and agri-research institutions, agriculture is the backbone of the economy (Atilano, 2017).

This module is composed of five sub-topics about agriculture under the Philippine setting. It traces the development of agriculture in the Philippines leading to the different production systems. Participants are expected to develop an interest in the farming systems of Filipino farmers.

Learning Objectives

At the end of the topic, the participant should be able to:
1. trace the historical development of Philippine agriculture from pre-colonial to the present time,
2. describe an ecosystem where agriculture is a component,
3. enumerate the existing natural resources of the Philippines and learn to conserve them through agriculture,
4. differentiate the agricultural production systems of the Philippines,
5. describe the smallholding farming system of Filipino farmers, and
6. discuss the importance of balanced agro-individual in agricultural production.

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Historical Development of Philippine Agriculture

1. Pre-Colonial Agriculture – Most Filipino farmers were engaged in hunting, fishing, slash and burn agriculture, and still, others developed the rice terraces.
   - The main crops being produced were rice, gabi, yam, banana, corn, coconuts, and citrus.
2. Spanish Colonialism – The colonization of the Philippines by the Spaniards started with Ferdinand Magellan on March 16, 1521. The Spanish rule started the formation of a two-class structure in Philippine agriculture: the landlords or *hacenderos* and the peasants or *sacadas*. They developed plantations mainly growing sugar and hemp intended for export.
3. American Colonialism – Under the U.S rule, the *hacenderos* flourished into powerful agrarian elites brought about by tariff barriers imposed by the Americans and opened the Philippine market to US manufacturers.
   - The Philippines was under the control of the Japanese for 300 years after which it was liberated by General Douglas Mc Arthur. The granting of Philippine independence by the US brought close economic ties and special treatment for US investors.
   - Introduced plants during the Philippine colonization were the following: mulberry, cocoa, wheat, cucumber, cantaloupe, watermelon, coffee, cereals, peas, and vegetables.
4. The Green Revolution – Rice is the staple food of the Filipinos. In the early 1960s, Western countries began to develop and introduce new rice-growing technologies to increase rice production which is known as “the green revolution.”
   - 1968 – IR8, or “miracle rice,” was formed. IR8, “miracle rice,” produced ten times the amount of rice produced by traditional varieties. As a result of the switch to farming IR8, annual rice production in the Philippines increased from 3.7 to 7.7 million tons in two decades. The large increase in rice production allowed the Philippines to become an exporter of rice for the first time in the 20th century.
   - The Green Revolution was a period when the productivity of global agriculture increased drastically as a result of new advances. During this period, new chemical fertilizers and synthetic herbicides, and pesticides were created. The chemical fertilizers made it possible to supply crops with extra nutrients and, therefore, increased yield.
   - The newly developed synthetic herbicides and pesticide-controlled weeds, deterred or killed insects, and prevented diseases, which also resulted in higher productivity. In addition to the chemical advances utilized during this time, high-yield crops were also developed and introduced. High-yielding varieties (HYVs) are crops that are specifically designed to produce more overall yield (Cunningham, 2020).

Environmental and agricultural problems of green revolution:

- Chemical fertilizers used in conjunction with miracle rice eroded the soil.
- Increased rice production led to increased water consumption.
- Pesticides and fertilizers used in rice farming polluted water and caused siltation.
- Declining water quality posed a threat to future rice production, due to the high amount of clean water required to grow rice.
- The Philippines did not have sufficient funding to improve irrigation systems and fell behind neighboring countries.

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- Vulnerability to recurring natural disasters posed a large threat to the agriculture-based economy.

Export-Oriented Agriculture

With the development of Philippine agriculture, the country started exporting large amounts of agricultural products to different countries around the world in 2008 (Table 1). These include coconut, sugarcane, banana, pineapple, mango, and coffee (PSA 2020).

Table 1. Philippine exports by major type of goods January 2020 to 2019, FOB in thousand USD).

Major Type of Goods	2020		2019		Growth Rate %
	January [p]	% Share	January [r]	% Share	
	(1)	(2)	(3)	(4)	(5)
Total Exports	**5,788,774**	**100.00**	**5,278,985**	**100.00**	**9.7**
Total Agro-Based Products	**429,698**	**7.42**	**375,411**	**7.11**	**14.5**
Agro-Based Products	**360,598**	**6.23**	**298,670**	**5.66**	**20.7**
Coconut Products	**111,648**	**1.93**	**96,709**	**1.83**	**15.4**
Copra		-		-	-
Coconut Oil	89,688	1.55	69,260	1.31	29.5
Desiccated Coconut	16,878	0.29	18,811	0.36	(10.3)
Copra Meal/Cake	4,271	0.07	5,495	0.10	(22.3)
Others	811	0.01	3,143	0.06	(74.2)
Sugar and Products	**490**	**0.01**	**463**	**0.01**	**5.7**
Centrifugal and Refined	47	0.00		-	-
Molasses		-		-	-
Others	442	0.01	463	0.01	(4.5)
Fruits and Vegetables	**248,461**	**4.29**	**201,498**	**3.82**	**23.3**
Canned Pineapple	21,068	0.36	8,026	0.15	162.5
Pineapple Juice	152	0.00	7	0.00	2,221.9
Pineapple Concentrates	10,572	0.18	6,129	0.12	72.5
Bananas	159,454	2.75	160,572	3.04	(0.7)
Mangoes	1,614	0.03	1,336	0.03	20.8
Others	55,601	0.96	25,428	0.48	118.7
Other Agro-Based Products	**69,100**	**1.19**	**76,741**	**1.45**	**(10.0)**
Fish, Fresh or Preserved Of Which: Shrimps and Prawns	28,032	0.48	30,669	0.58	(8.6)
Coffee, Raw, not Roasted	2	0.00		-	-
Abaca Fibers	1,763	0.03	1,713	0.03	2.9
Tobacco Unmanufactured	8,353	0.14	14,697	0.28	(43.2)
Natural Rubber	9,491	0.16	7,257	0.14	30.8
Ramie Fibers, Raw or Roasted	6	0.00	3	0.00	149.0
Seaweeds, Dried	757	0.01	1,915	0.04	(60.5)
Rice		-	22	0.00	(100.0)
Others	20,696	0.36	20,467	0.39	1.1
Forest Products	**27,971**	**0.48**	**24,485**	**0.46**	**14.2**
Logs		-		-	-
Lumber	16,071	0.28	12,141	0.23	32.4
Plywood	6,998	0.12	9,406	0.18	(25.6)
Veneer Sheets/Corestocks		-	13	0.00	(100.0)
Others	4,902	0.08	2,925	0.06	67.6
Mineral Products	**482,172**	**8.33**	**361,222**	**6.84**	**33.5**
Copper Concentrates	13,351	0.23	17,956	0.34	(25.6)
Copper Metal	151,219	2.61	137,407	2.60	10.1
Gold	181,386	3.13	76,842	1.46	136.0
Iron Ore Agglomerates	4,575	0.08	2,158	0.04	112.1
Chromium Ore		-	1,288	0.02	(100.0)
Nickel		-		-	-
Others	131,641	2.27	125,571	2.38	4.8
Petroleum Products	**38,025**	**0.66**	**46,146**	**0.87**	**(17.6)**
Manufactured Goods	**4,703,504**	**81.25**	**4,374,506**	**82.87**	**7.5**
Electronic Products	3,233,806	55.86	2,792,214	52.89	15.8
Components/Devices (Semiconductors)	2,481,546	42.87	2,036,211	38.57	21.9
Electronic Data Processing	448,337	7.74	480,103	9.09	(6.6)
Office Equipment	50,245	0.87	28,526	0.54	76.1
Consumer Electronics	76,549	1.32	56,600	1.07	35.2
Telecommunication	37,254	0.64	37,711	0.71	(1.2)
Communication/Radar	55,574	0.96	73,576	1.39	(24.5)
Control and Instrumentation	51,080	0.88	64,592	1.22	(20.9)
Medical/Industrial Instrumentation	12,216	0.21	3,521	0.07	246.9
Automotive Electronics	21,006	0.36	11,374	0.22	84.7
Other Electronics	262,618	4.54	284,096	5.38	(7.6)
Garments	77,964	1.35	73,586	1.39	5.9
Textile Yarns/Fabrics	19,643	0.34	16,357	0.31	20.1
Wood Manufactures	20,323	0.35	20,320	0.38	0.0
Furniture and Fixtures	31,956	0.55	34,313	0.65	(6.9)
Chemicals	106,288	1.84	103,961	1.97	2.2
Non-Metallic Mineral Manufactures	23,049	0.40	18,656	0.35	23.5
Machinery and Transport Equipment	315,151	5.44	443,710	8.41	(29.0)
Processed food and Beverages	96,523	1.67	101,551	1.92	(5.0)
Iron and Steel	2,080	0.04	11,770	0.22	(82.3)
Baby Carr., Toys, Games and sporting goods	20,238	0.35	19,802	0.38	2.2
Basketwork, Wickerwork and Other Articles of Plaiting Materials	2,791	0.05	2,895	0.05	(3.6)
Misc.Manufactured Articles, n.e.s.	73,666	1.27	76,659	1.45	(3.9)
Others	348,955	6.03	325,897	6.17	7.1
Special Transactions	**107,404**	**1.86**	**97,215**	**1.84**	**10.5**
Re-Export	**13,445**	**0.23**	**11,193**	**0.21**	**20.1**

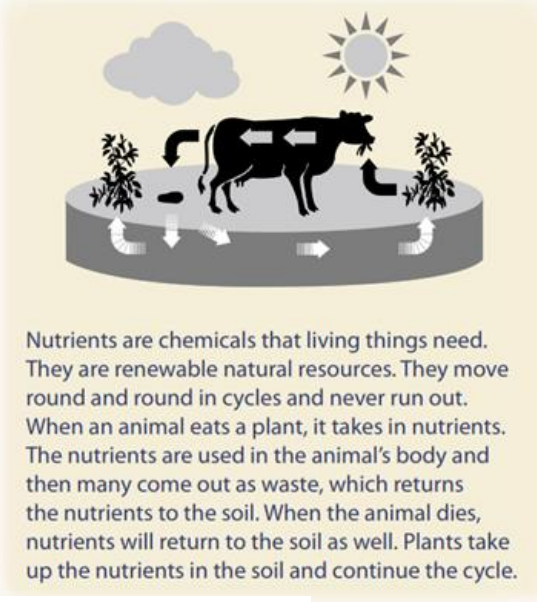
Notes
Details may not add up to total due to rounding
a - no export data
b - less than $1,000
- - no growth rate/no percent share
p - preliminary
r - revised

Resources and its Preservation

Natural resources are materials or things that people use from the earth.

Two types of natural resources:

1. ***Renewable resources*** – They are called renewable because they can grow again or never run out.
   - They are the ones that can grow again. Trees are a good example. If cut down, they can regrow from seeds and sprouts. Animals are another example. Baby animals are born and grow up. They replace older animals that die.
   - Air and water are renewable natural resources, too. They don't regrow like trees or have babies like animals. But they are always being renewed. They move in cycles. They go from one place to another, and often back where they started, again and again. Finally, renewable resources can regrow or be replaced within a person's lifespan.
2. ***Non-renewable resources*** – These are things that can run out or be used up. They usually come from the ground.
   - There are fixed amounts of these resources. They are non-living things, and they are sometimes hard to find. They don't regrow and they are not replaced or renewed. They include the fossil fuels that are burned for energy (natural gas, coal, and oil). Minerals, used for making metals, are also nonrenewable natural resources.
   - Nonrenewable natural resources are things that take longer than a person's lifespan to be replaced. They can take millions of years to form.



Nutrients are chemicals that living things need. They are renewable natural resources. They move round and round in cycles and never run out. When an animal eats a plant, it takes in nutrients. The nutrients are used in the animal's body and then many come out as waste, which returns the nutrients to the soil. When the animal dies, nutrients will return to the soil as well. Plants take up the nutrients in the soil and continue the cycle.

Source: www.cas.psu.edu

All-natural resources should be used wisely. We must conserve natural resources. Conserve means to not use up, spoil, or waste things. This is especially true for nonrenewable resources.

The Philippines is rich in natural resources. It has fertile, arable lands, diverse flora and fauna, extensive coastlines, and rich mineral deposits. Its fertile land produces a variety of crops such as coconut, banana, pineapple, sugarcane, and rubber. Some of these agricultural crops are raised for export. Rice, which is the staple food of Filipinos, is produced, largely, in the Central Plains of Luzon (Region III).

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Natural resource means something that is a component of the physical environment, either biotic (living) or abiotic (non-living) which can satisfy certain human wants or needs, and provide goods and services. This includes land, air, water, sunshine, minerals, soil, fish, etc.

The Philippines holds the distinction as one of the seventeen countries whose combined plant and animal species account for 70-80% of the world's biological diversity. It has approximately 52,177 species of flora and fauna. From among these living things, an estimated 14,000 plant species and 1,130 animals with backbones (terrestrial vertebrates) are found in forest areas (*https://philippines-atbp.jimdofree.com/geography-resources/).*



Narra tree (*Pterocarpus indicus)* – **National Tree of the Philippines** (*https://rmdludovice17.wordpress.com/2014/03/04/philippine-indigenous-tree-narra-tree/)*

1. Land

- Almost half of the Philippines' total land area (around 15 million hectares) is classified as timberland. Philippine forests produce timber for local consumption and export. Hardwood products are known throughout the world for their distinct appearance and high quality, appropriate for home furnishings, such as narra. Most Philippine forests are of the tropical rainforest type. Besides extensive reserves of tropical evergreen hardwoods, the country also has considerable areas of pine in the mountainous regions of Northern Luzon.

2. Water

- The Philippines boasts of many different kinds of natural water forms, such as bays, rivers, lakes, falls, gulfs, straits, and swamps. Because it is made up of islands, the country's coastline, if laid end-to-end, would measure around 17.5 thousand kilometers.
- This includes marine life such as fishes, turtles, shells, pearls, and species found in freshwater bodies.

3. Flora and Fauna

- The Philippines is rich in flora and fauna with an estimated 2M species of plants and animals, many of which are unique to the islands. Its tropical rainforests are among the most species-rich ecosystems on earth and it still has many virgin forests.
- The Philippine soil is very fertile due to its volcanic nature. Abundant rain and sunshine, as well as the wide range of habitats and elevations, account for an incredible variety of plant life in every category, from mosses and lichens (including 1,000 species of fern) to giant trees (about 3,000 species). About 60% of the 10,000plant species grow only in the archipelago.
- Among the important species in the Philippines are bamboo (with species), mangroves, coconut and nipa palms, mahogany, narra, sampaguita, orchids, pili nut, abaca, durian, and mango.

4. Fauna-Land Creatures (Among these are eagles, reptiles, insects, and mammals)

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The Philippine eagle (*Pithecophaga jefferyi*) – the world's longest eagle species, it has been declared the **Philippine National Bird**
(https://en.wikipedia.org/ wiki/Philippine_eagle)

- The Philippine eagle, formerly called the "monkey-eating eagle" (because it eats monkeys), is now known as haribon (from the words haring ibon and is the best-known Philippine member of the bird family.
- About 25,000 species of insects are found in the Philippines. Among them are butterflies that have vivid colors that are of great interest to collectors.
- Carabao, a native water buffalo, is known as the beast of burden in the Philippines. It is commonly utilized in farming, from land preparation to transporting farm produce to the market. It is also becoming a major source of milk.

4. Mineral resources

- The Philippines is known for its vast mineral resources. Its metal deposits are estimated at 21.5 billion metric tons, while non-metal deposits are about 19.3 billion metric tons.

<u>Metal deposits:</u>

1. Nickel ranks first in terms of deposits and size. It is found in Surigao del Norte, Davao, Palawan, Romblon, and Samar.
2. Iron abounds in Ilocos Norte, Nueva Ecija, Camarines Norte and Cotabato.
3. Copper is found in Zambales, Batangas, Mindoro, Panay and Negros.
4. Gold – the major gold-producing areas in the Philippines are districts of Baguio and Paracale located on the Island of Luzon and Marasa, Nueva Vizcaya, and the locations in Mindanao such as Surigao and Masbate.



Oceana gold mine, Didipio, Kasibu, Nueva Vizcaya

<u>Non-metal deposits</u> include cement, lime, marble, asbestos, clay, guano, asphalt, feldspar, sulfur, talc, silicon, phosphate, and marble.

Preservation of Resources

Concerned about its resources, the Philippines devised strategies to protect its natural heritage: education and proper legislation on environmental issues. The country established national parks, recreation areas, and wildlife sanctuaries.

- REPUBLIC ACT NO. 9147 – this Act shall be known as the "Wildlife Resources Conservation and Protection Act." This Act promotes the conservation and protection of wildlife species and their habitats to promote ecological balance and enhance biological diversity and regulate the collection and trade of wildlife.
- REPUBLIC ACT NO. 7586 – this Act shall be enforceable for all wildlife species found in all areas of the country, including protected areas, otherwise known as the National Integrated Protected Areas System (NIPAS) Act, and critical habitats. This Act shall also apply to exotic

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species that are subject to trade, are cultured, maintained, and/or bred in captivity or propagated in the country.

- ✓ *Conservation programs exist for the Philippine eagle, eastern sarus crane, tamaraw, Philippine crocodile, dugong, marine turtles, and some rare deer species.*
- ✓ *There is a growing consciousness among people of the need to conserve forests and marine life.*
- ✓ *A law banning log exportation has been passed. In 1992 Palawan was cited as a national treasure because its forests and sea life are relatively well-preserved that a total ban on logging is now being.*

Production/Farming Systems

- Farming systems are ecosystems that are modified and managed by man to produce outputs that are useful to him.
- Agriculture in the country remains to be one of the major contributors to the economic growth of many countries in the Asia-Pacific region. More than 70% of the country's population is dependent on agriculture.
- Two-thirds (2/3) of the population is still living in a rural area where agriculture is the main source of livelihood. Twenty-four percent (24%) (or 71,900km2) of the total land area of the Philippines is allotted to agriculture (Figure 1).

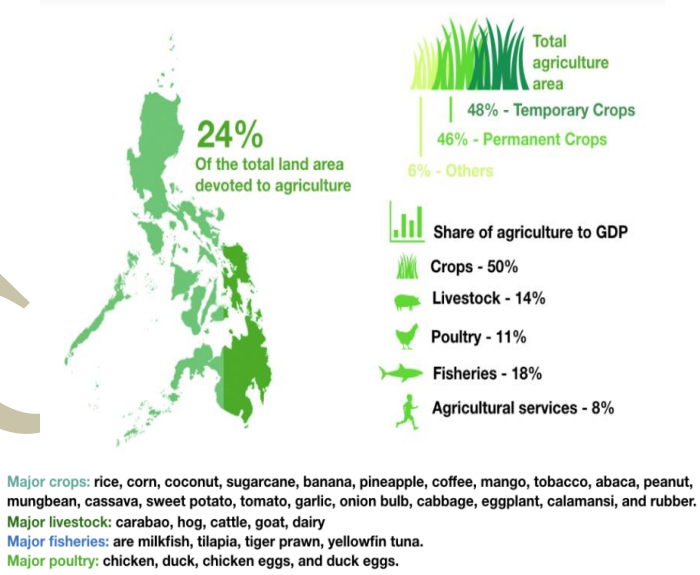


Figure 1. A quick overview of the Philippine agriculture (*PSA.gov.ph - 2019*).

- The major crops in the Philippines are rice, corn, coconut, sugarcane, banana, pineapple, coffee, mango, tobacco, abaca, peanut, mungbean, cassava, sweet potato, tomato, garlic, onion bulb, cabbage, eggplant, calamansi, and rubber.
- Major livestock is carabao, hog, cattle, goat, dairy and the major fisheries are milkfish, tilapia, tiger prawn, yellow-fin tuna. Poultry products in the country are chicken, duck, chicken eggs, and duck eggs.

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- The production of rice, known locally as palay, is largely concentrated in Central Luzon, Cagayan Valley, and Ilocos, which account for 19%, 13%, and 10%, respectively, of total national production.
- Cagayan Valley is the largest producer of corn, generating 23% of national production, followed by Northern Mindanao (17%) and SOCCSKSARGEN (16%).

- Davao City grows 15% coconuts and 38% bananas of the country's total production for each crop. Western Visayas grows 59% of the country's total production of sugarcane.
- The production of livestock breaks down into four main categories: pigs, cattle, carabao (water buffalo), and goats. Central Luzon and CALABARZON are the largest pig-farming regions in terms of production; Northern Mindanao, Ilocos, and CALABARZON are the main producers of cattle; the top-three carabao-breeding regions are Western Visayas, Davao, and Eastern Visayas; and the biggest producers of goats are Ilocos, Central Visayas, and Central Luzon. Central Luzon is the main source of chickens, ducks, and duck eggs, while CALABARZON is the main producer of chicken eggs.
- Mindanao is an important contributor to the fisheries segment. The Bangsamoro Autonomous Region in Muslim Mindanao is the province with the largest volume of fish production, totaling 856,152 tons in 2017. Also, the Zamboanga Peninsula region accounts for almost 70% of the country's overall sardine production.
- Central Luzon also plays a leading role in the segment, with the region being the source of 40% of tilapia production and 50% of tiger prawn production.

Types of Crop-based Farming System

In the Philippines, approximately 50% of lowland rice fields are rainfed, and 50% with adequate irrigation facilities.

Meanwhile, the uplands (where upland farming is practiced) are zones where both agriculture and forestry are practiced on rolling to steep land. The problem of deforestation resulted from the movement of the population to sloping lands.

a) Shifting cultivation – the traditional system (kaingin) in most forest areas throughout the tropics.
b) Lowland rice-based system – age-old, very successful system in Asia.
c) Upland cereal-based system – a predominant system in seasonally humid and semi-arid.
d) Smallholder scale mixed farming – complete integration of animal husbandry and crop production components. In this system, animals provide manure for crops, and crop residues are used to feed animals.
e) Irrigated smallholder farming – an age-old system in the Middle East and spreading to other dry areas.
f) Small-holder farming with plantation (Perennial) crops – usually with one dominant cash crop.
g) Agroforestry – omnipresent, age-old system.



Subsistence rice farming (Buyasyas, Sta. Fe, N. Vizcaya)

Food Production Systems

a) Subsistence

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- Agricultural types and systems depend heavily on the character of production, i.e. whether crops are produced in a subsistence (the farmer has to produce to live) or a market economy (commercial).
- Farmer usually resists changes in production methods since if the changes turned out unproductive, the livelihood and survival are at a risk (Beets, 1990).

b) Commercial (Agricultural trade)

- Economic growth, urbanization, and withdrawal of labor from the agricultural sector led to the increasing commercialization of the agricultural system.
- This leads to the greater market orientation of farm production, progressive substitution out of non-traded inputs in place of purchased inputs, and the gradual decline of integrated farming systems and their replacement by specialized enterprises for the crop, livestock, poultry, and aquamarine products.
- In 2018, the Philippines is a net importer of agricultural products. Cereals continue to dominate demand (Oxford Business Group, 2020). Filipino importers purchased almost a billion dollar-worth of wheat (US$982.1 million) from foreign suppliers during 2015. That metric far surpasses the combined total for all other imported cereals. In second place, Philippine imports of rice amounted to $464.7 million followed by imported corn at $177.6 million (Santiago, 2017).
- Coconut oil is the Philippines' biggest agricultural export (Pines, 2020). Cavendish banana is the second top agricultural export product of the country and the second leading supplier of bananas globally behind Ecuador (FAO, 2020).

Small-holding Farming and Agribusiness

- Smallholder farmers are family units that exist in a family-labor intensive system with low capital input. These farmers generally have limited opportunities for livelihood improvement because of limited farm size, poor knowledge in production and marketing, and difficulties accessing finance (Oakeshott, 2016).
- Small farmers are an extremely diverse group, often resource-poor, and are responsible for growing the majority of the world's food (Ong'wen and Wright, 2007). They produce various agricultural commodities, from food to fiber, wood, rubber, and cotton; they also raise livestock.
- A small farm is generally managed and worked by one household except in peak labor seasons (Obschatko, 2006). The size of farms varies with the ecological, social, and economic conditions within which farmers work.
- In general, the land size of the small farm holder is sufficient to provide a significant portion of the family's needs, and in most cases, the farm income is supplemented with income from other sources.
- The Philippine agricultural sector has 4.5 million farms directly supporting 25 million Filipino farmers and families (PSA, 2015).
  Most of these farmers are poor smallholders in rural communities where some poverty rates are as high as 70%. The smallholder farmers not only contend with community poverty but



A small-hold type rice farm in the Philippines

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are also disconnected from the markets and have few opportunities for livelihood improvement. Many rely on traders to organize the harvest and transport their produce to the markets (Oakeshott, 2016).

- The majority of the farms in the country are all small farms averaging about 2 hectares. These are simple farms that are owned and managed by single families ranging from subsistence to commercial production. A typical farming system consists of major crops, with rice, corn, and coconut as common base crops, and a few heads of livestock and poultry.

Table 2. Comparison between small-scale farming and industrial agriculture

SMALL-SCALE FARMING	INDUSTRIAL AGRICULTURE
✓ Benefits can be multi-functional	✓ Dominated by the profit motive
✓ Diversity in cultures, cropping systems, and on-farm ecology	✓ Uniformity and monoculture
✓ Strong relations to land due to reliance on ecological health for livelihood generation	✓ Exploitative relations to land
✓ Knowledge is diverse, placed-based, more likely to be communal, and based on a notion of stewardship	✓ Techno-scientific knowledge covered by patents and other forms of intellectual property
✓ Secondary income sources sometimes needed to meet subsistence	✓ Maybe part of conglomerates with various sources of income
✓ Labor intensive with labor primarily provided by the family	✓ Low labor inputs using non-family labor
✓ General low levels of capitalization	✓ Highly capitalized
✓ Diverse ownership pattern and forms of exchange	✓ Diverse ownership patterns in the context of capitalist market-based exchange
✓ Place within dense networks and embedded in rural communities	✓ Absentee ownership and export orientation means network more strongly tied to metropolitan centers and global production networks than local communities
✓ Production for subsistence and local exchanged prioritized	✓ Production networks span the globe and supported by direct and indirect subsidies
✓ Potential to be environmentally sustainable	✓ Environmentally exploitative

Source: Ong'wen and Wright, 2007

Factors (micro and macro levels) preventing farmers from harnessing their potentials

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1. Distorted and inefficient markets – the markets for Philippine agricultural products have not been perfectly competitive due to several distorting factors, among which are:
   a. Lack of market information and the ability to analyze the same – Farmers in the country are not aware of how agricultural markets are doing in other parts of the country or where opportunities might be. This lack of knowledge on prices elsewhere makes farmers price-takers, leaving them at the mercy of traders.
   b. Poor transport facilities – Rural infrastructure in the country, however, is in deplorable condition. As the rural population continues to grow, an increasing percentage of the farm do not have access to farm-to-market roads.
   c. Inadequate post-harvest facilities – The inadequacy of storage, handling, processing, and other post-harvest equipment and facilities has limited farmers' options in marketing their produce.
   d. Excessive government intervention/regulation – Price controls, government monopolies, subsidies, and other forms of government interventions in the past have distorted agricultural markets, thereby constraining production growth, increasing imports of food and feed grains, and weakening the country's food security situation. Government subsidies made farmers more dependent on the government and likewise made them vulnerable to political exploitation.

2. Inadequate investments
   a. Lack of financing and credit facilities – Most agricultural producers are small farmers with inadequate financing. Their financial requirements are usually provided by traders at high interest rates. Small farmers, therefore, borrow money from informal sources because these are accessible, require no collateral and voluminous documents like those required by formal lending institutions.
   b. Uneven tariff protection – The low tariff protection received by the agriculture sector compared to the non-agriculture sector is one of the factors which have contributed to the unattractiveness of agriculture to investments.
   c. Lack of organized farmers – Many farmers in the country are unorganized and this limits their ability to market their products more effectively. Being organized into cooperatives or producers' associations will give them stronger bargaining power to command better prices for their produce.
   d. Lack of entrepreneurial skills among farmers – The farmers' lack of entrepreneurial skills has prevented them from properly utilizing marketing investment information for longer-term production and marketing planning and from taking advantage of economic opportunities. The possession of these skills would make their operations more efficient and profitable, thereby enabling them to earn more.

Opportunities for Government Interventions

The main concern of the government is to make the country's agriculture more productive and competitive for the benefit of farmers. Because despite all the structural changes in the country's economy, it continues to remain largely an agricultural economy. Among the more important concerns which the government have to attend to are:

1. The modernization of agriculture will be realized if farmers are converted from mere tillers of the soil to full-pledged/successful agribusiness entrepreneurs. There is a need to establish a cadre of extension workers properly honed on the rigors of marketing principles and preferably, with sufficient entrepreneurial skills.

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2. Revitalize and maximize the full extension of access to sustained rural credit in favor of the farmers through the establishment of a network of branches of financial institutions in the countryside, of which funds generated therefrom shall be used exclusively for agricultural loans and credit as well as for the benefit of agrarian reform beneficiaries.
3. Provision to our farmers of timely, accurate, and responsive business information and efficient trading services which will link them to profitable markets for their products; activities to be undertaken under this component will include market-matching activities, market assistance services, and exploration of export market potential.
4. A master plan on the infrastructure support for dynamic agriculture should immediately be evolved to include the requirements for the construction of all farm-to-market roads, good provincial and national roads, post-harvest facilities, adequate transportation and storage facilities, and communication systems in both the urban and rural areas.
5. Strengthen access to land and tenure security.

Smallhold farmers are important drivers of development in the countryside but are commonly marginalized and mostly vulnerable to the rapidly changing social, political, and environmental conditions – *Aquino, Lim, and Ani (2013).*

Role and Contribution of Small Farmers

1. Food security
   Small farmers greatly contribute to food security. Well-supported small-scale systems have the potential to meet humanity's food needs equitably and sustainably. Farmers' contribution to both household food security and farm outputs is thus disproportionately high (Singh et al., 2002).



2. Productivity and efficiency
   Small farmers, produce food efficiently and effectively, and extensive evidence shows that small-scale farming systems are extremely productive, generating a higher output per unit of land than extensive systems. Comparisons reveal that small farms have a higher output than large-scale farms, producing considerably more per unit of production area (Rosset, 1999).
3. Economic development
   Small-scale farming systems are generally more integrated into local economies than industrial farming in several ways. A rural economy dominated by a small-scale farming system will have more farmers per unit of area than industrial farming. These farmers are more likely to sell produce locally, purchase goods at local stores, and support local services.
   Small-scale farms tend to use more labor and contribute to regional employment. There are flow-on effects here, too, as employed laborers feedback their earnings into the local economy stimulating local businesses. This leads to a generation of wealth in rural areas and a more diverse and functioning rural community.
4. Employment
   Small-scale agriculture is not small in aggregate terms, accounting for much more employment and staple food production than larger commercial behemoths.
5. Sustainable human development
   Sustainable human development relies upon diverse and resilient communities within which local populations can access services and meet their needs for human

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development. Within small-scale systems, the decentralized ownership and management structures mean a greater proportion of the population can make decisions affecting their lives.

6. Ecology and maintenance of biodiversity
   It was small farmers who developed and nurtured not only the staggering genetic diversity of food products but also developed the vast range of complex agricultural techniques and methods that helped agriculture to adapt to ecosystems throughout the world.
   Small farmers have acted and continue to act as stewards for agrodiversity through the stewardship and development of locally adapted and genetically diverse food sources, animal management, and vegetation associations.
7. Safe and healthy food
   Small farmers contribute to the provision of healthy and nourishing food. They tend to produce a more diverse range of products that can facilitate access to a healthy diet including a mix of protein, carbohydrates, and fat as well as a range of vitamins and minerals.

Promoting the Development of Agribusiness Enterprises

Agribusiness is defined as the totality of all operations involved in the manufacturing of farm supplies, production, operations on the farm, as well as processing and distribution of farm commodities and items made from them". Agribusiness is viewed as a vital sector contributing to economic prosperity. In most Asian countries, agribusiness remains crucial in terms of labor absorption and export earnings, and therefore government support in promoting agribusiness enterprises is a crucial requirement.

Agribusiness is an important sector in the Philippine economy. It contributes substantially to gross national product (GNP) and it is a source of employment for many of the population. Furthermore, since most agribusiness activities are in rural areas, their development can be an effective instrument in alleviating poverty in any country. Agribusiness is a source of foreign exchange earnings as well as food and fiber, which are considered to be strategically important for national security.

<u>Among the common problems faced by agribusiness are:</u>

- international trade barriers;
- limited access to credit, high-interest rates, and fluctuating exchange rates;
- poor trade policy;
- export and import taxes on raw materials;
- bureaucratic practices;
- high transportation costs;
- raw materials of poor quality and unreliable supply;
- the lack of support facilities; and
- the lack of professional managers

<u>Marketing promotion for agribusiness:</u>

Marketing is defined as "the performance of all business activities involved in the flow of agricultural products and services from the point of initial agricultural production until they are in the hands of consumers". Hence, marketing is a productive activity that creates utilities, i.e., form utility, place utility, time utility, and possession utility, in the process of bringing the products from the farm to the consumer.

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Agribusiness sector:

- Agriculture, forestry, and fisheries form the core of Philippine agribusiness. The sector also includes those enterprises in the manufacturing sector that are engaged in agro-based industry, as well as the portion of the service sector that deals with agricultural/agribusiness activities. The majority of Philippine agribusinesses are small- and medium-sized enterprises (SMEs).
- Figure 2 shows agricultural exports of the same countries from 1990-2009. By comparison, Philippine agricultural exports declined from comprising about the same 20% share in 1970 of ASEAN-6 total agricultural exports as Thailand and Indonesia – while Malaysia held a larger 30% share – to be the lowest among the ASEAN-6 in 2009, around 3% of the ASEAN-6 total. During these nearly two decades (1990-2009) total, ASEAN-6 agricultural exports grew strongly, increasing about 20-fold, while the Philippines fell behind Vietnam and Singapore.

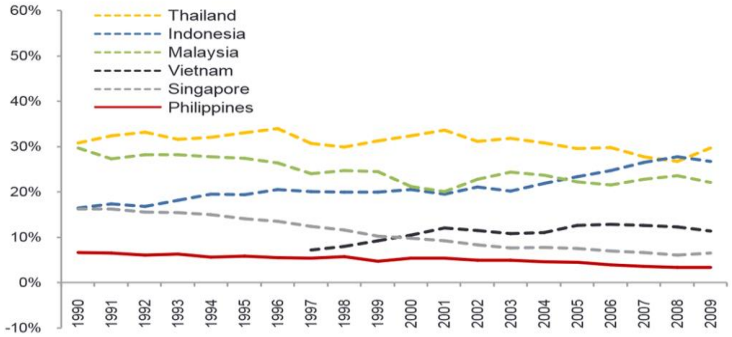


Figure 2. Agricultural goods exports, ASEAN-6, % of total, 1990-2009 (*Joint Foreign Chambers, 2010).*

- Filipino farmers are capable farmers. However, they face high domestic transport, labor, and other production costs. The cost of Philippine labor is many times Cambodia's. Insecticide costs twice as much in the Philippines as elsewhere in the region. Because the country is archipelagic, domestic transportation costs are high. Bureaucratic "overkill" is also a problem.

Productivity and Income in Farming Systems

- Productivity can be simply defined as the increment in valued product per unit of time and is usually measured as yield or income per unit of land. In previous years, 63% of crop productivity growth in the tropics was derived from higher yields.
- Yields increased considerably and FAO projects that this trend will continue. Although it is not known to what extent the past increases have been at the expense of sustainability and their effect on the environment.
- High yield per unit area and therefore high productivity does not necessarily mean higher income for the farmers as high yields can only be achieved through high levels of inputs, and the farmers' income is directly related to the costs of inputs and outputs.
- Agricultural productivity is measured as the ratio of agricultural outputs to agricultural inputs. This productivity is an important component of food security. Increasing agricultural productivity, especially amongst smallholder farms, is an important way to

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decreasing the amount of land needed for farming and slow environmental degradation through processes like deforestation.

Importance of agricultural productivity

- Aside from providing more food, increasing the productivity of farms affects the region's prospects for growth and competitiveness in the agricultural market, income distribution and savings, and labor migration.
- More efficient distribution of scarce resources. As farmers adopt new techniques and differences, the more productive farmers benefit from an increase in their welfare while farmers who are not productive enough will exit the market to seek success elsewhere.
- The comparative advantage in agricultural products increases, which means that it can produce these products at a lower opportunity cost than can other regions.
- Increases in agricultural productivity lead also to agricultural growth and can help to alleviate poverty in poor and developing countries, where agriculture often employs the greatest portion of the population.

The role of sustainable agriculture in productivity

- Sustainable agriculture means to maintain or allow increases in agricultural productivity for the long term, by making certain the resources used are as renewable as possible. Sustainable agriculture focuses on meeting the agricultural needs of the present generation while maintaining the ability of future generations to meet their own needs by conserving the environment. It should ensure the welfare of the ecosystems and still uphold its ability to provide food secure nations.



Summary

- The module presented a topic about the development of Philippine agriculture starting with the pre-colonial period, under the Spanish, Japanese and American rules, and up to the present system. It was learned how agriculture in the country was developed from primitive to the modern system.
- The natural resources of the Philippines were also discussed, their importance and contribution to the Philippine economy. Its contribution to the economy provides more production systems or strategies developed and adapted to the Philippine condition which eventually improved agricultural production. Promoting/advocating sustainable agricultural production was also explained which also takes care of the remaining natural resources and serves not only the present population but for the future generation.
- Small-hold farmers as presented, have the potential to create a model of human development that is socially, economically, and environmentally sustainable. Small-scale farming systems are not merely economic units to be manipulated, instead, they need to be accorded respect, and their contributions to the world food supply, to the economies of nations, and the world's diversity of culture and environment need to be acknowledged and supported by the government and private sectors.

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C. MAJOR CROP STATISTICS OF THE PHILIPPINES

Overview

This unit covers the knowledge, skills and attitudes required for the identification of major crops in the Philippines including their geographical location, economic importance and its scientific name. Agriculture dominates Philippine economy. It furnishes employment to about 3 million persons or about 60% of the gainfully employed workers. Agriculture operates provide 40% to 45% of the total national income and about 75% to 80% of the country's export (Cutshall, N.D.).

Learning Objectives:

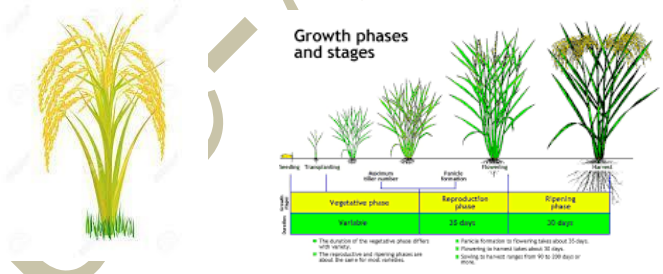
At the end of the topic, the participant should be able to:

1. identify the major crops in the Philippines and their geographical location, and
2. discuss their economic importance.

What are the major crops in the Philippines?

The following are the major crops in the Philippines including their geographical location.

1. **Rice** *(Oryza sativa)* Rice is the staple food of over 90% of the Philippine population. Rice farming is the source of income and employment to 12 million farmers and family members. In 2017 the total paddy rice output met 93% of the country's annual requirement. The population consumed 11.7 million tons of rice. The major rice producing parts of the country are Central Luzon (18.7%), western Visayas (11.3%) and Cagayan Valley (11%).The diversity of ecosystems, cultural management practices, preferences and use contribute to the diversity of Philippine rice. The major rice-producing parts of the country are Central Luzon (18.7%), western Visayas (11.3%), Cagayan Valley (11%), Ilocos region (9.8%), SOCCSKSARGEN (7.5%), and Bicol region (6.8%).



Source: https://en.m.wikipedia.org

2. **Maize** *(Zea mays)* Maize is the second most important agricultural crop in the Philippines. A total of 2.49 million hectares are planted to the crop, the third most extensive after rice and coconut. In 2019, the production volume of corn in the Philippines was almost eight million metric tons, higher than the produced quantity of eight million metric tons in 2018. Aside from rice, corn is considered as another staple crop in the Philippines. The City of Ilagan, today acknowledge as the corn capital of the country lives its economic boom while playing a vital contributory role for the province of Isabela as its capital since its creation as a province

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Source: https://en.m.wikipedia.org

3. **Coconut** *(Cocos nucifera)* Davao Region was the top coconut producer with 540.82 thousand metric tons' output or 13.3 percent of the total coconut production. This was followed by Northern Mindanao and Zamboanga Peninsula with 12.0 percent and 10.9 percent shares, respectively.



Source: https://en.m.wikipedia.org

4. **Sugarcane** *(Saccharum officinarum)* Western Visayas remained as the top producer of sugarcane with 5.61 million metric tons' output or 85.1% share to the total sugarcane production. This was followed by Central Visayas and Northern Mindanao with respective shares of 6.1% and 5.6%.



Source: https://en.m.wikipedia.org

5. **Banana** *(Musa acuminata, M. balbisiana and M. x paradisiaca)* Banana is the most important fruit crop of the Philippines in terms of area and volume of production. Davao Region remained as the top banana producer with 846.23 thousand metric tons or 35.3%

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share to the total production this quarter. This was followed by Northern Mindanao and SOCCSKSARGEN with respective shares of 21.0% and 14.5%.



Source: https://en.m.wikipedia.org

6. **Manila Hemp** *(Musa textilis)* Catanduanes is the biggest abaca producing province contributing 35% of the total production, followed by Davao Oriental with about 8.5%. Due to frequent typhoons in Bicol and Leyte, there are moves by some abaca factories and investors to look for alternative areas to plant abaca.




Source: https://en.m.wikipedia.org

7. **Mango** *(Mangifera indica)* Carabao mango was still the most produced variety with 22.54 thousand metric tons, representing 81.9% of the total mango production. During the period, CARAGA was the top producer of mango with 6.84 thousand metric tons' output, a 24.8% share to the total production




Source: https://en.m.wikipedia.org

8. **Pineapple** *(Ananas comosus)* Pineapple is the third most important fruit crop in the Philippines (after banana and mango) in terms of agricultural crop area and value of production. Northern Mindanao remained as the top pineapple producer which contributed

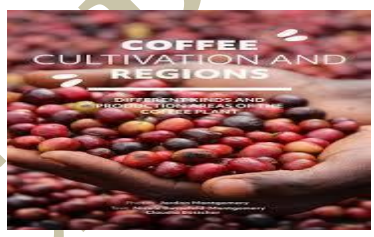
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462.32 thousand metric tons or 64.9% to the total production this quarter. This was followed by SOCCSKSARGEN and Bicol Region with respective shares of 29.8% and 2.1%.




Source: https://en.m.wikipedia.org

9. **Coffee** *(Coffea spp.)* Coffee is an important crop commodity in the country, and most of the local production is consumed domestically. The top coffee producer was SOCCSKSARGEN with 12.70 thousand metric tons' output or 39.9% share to the total production for this quarter. This was followed by Bangsamoro Autonomous Region in Muslim Mindanao (BARMM) and Davao Region with respective shares of 21.0% and 15.9%.

Source: https://en.m.wikipedia.org

10. **Tobacco** *(Nicotiana tabacum)* Tobacco is an important crop commodity in the country. Historically, tobacco was an important export commodity. However, in recent years, all of the local production is consumed domestically. Virginia tobacco is the most dominant tobacco type and constitutes 58% of the tobacco area, equivalent to 23.347 hectares, 52% of the total tobacco farmers or 32,352, and is commercially grown in the four Ilocos provinces of Abra, Ilocos Norte, Ilocos Sur (60%) and La Union (17%)



Source: https://en.m.wikipedia.org

11. **Rubber** *(Hevea brasiliensis)* Rubber is an important industrial crop in the southern part of the country. The top rubber-producing region for the quarter was SOCCSKSARGEN with 70.70 thousand metric tons' output or 45.6% share to the total rubber production. This was followed by Zamboanga Peninsula with 39.6% share.



Source: https://en.m.wikipedia.org

12. **Cassava** *(Manihot esculenta)* Cassava is most important root crop in the Philippines in terms of agricultural crop area and value of production. BARMM registered the highest production of 489.46 thousand metric tons, representing 62.9% of the total output this quarter. Northern Mindanao and Zamboanga Peninsula followed with 17.3% and 3.7% shares, respectively. The major producers of cassava came from the islands of Luzon and Mindanao. The case studies were conducted in Naguillan, Isabela in Luzon and Tampakan, South Cotabato in Mindanao with KU 50 and golden yellow varieties were harvested, respectively.





Source: https://en.m.wikipedia.org

13. **Peanut** *(Arachis hypogaea)* Peanut is the second most important food legume in the Philippines. Among the provinces in the Philippines, the top producers of peanut are Isabela, Pangasinan, La Union, Quirino, Cagayan, Ilocos Norte, Ilocos Sur, Aurora, Albay and Iloilo. However, the Cagayan Valley region produced almost half of the country's total peanut production.



Source: https://en.m.wikipedia.org

14. **Mungbean** *(Vigna radiata)* Mungbean is the most important food legume in the Philippines in terms of agricultural area and value of production. In 2012, area planted to mungbean was 44, 324 hectares and total volume of production was 32,364 metric tons with the province of Isabela as the top mungbean producer contributing 26% to the country's production. However, national average yield per hectare remains quite low at 0.73 metric ton.



Source: https://en.m.wikipedia.org

15. **Onion** *(Allium cepa)* Central Luzon, the top producer of red onion at 19.74 thousand metric tons accounted for 51.9% of the country's total production. This was followed by MIMAROPA with 42.4%, and Ilocos Region, 2.6%.



Source: https://en.m.wikipedia.org

16. **Garlic** *(Allium sativa)* The Department of Agriculture (DA) in Ilocos Region (Region 1) has already crafted a revival program to raise garlic sufficiency. The Ilocos Region is the largest garlic-producing region in the Philippines, contributing more than 65% to the country's total production.



Source: https://en.m.wikipedia.org

17. **Eggplant** *(Solanum melongena)* In the Philippines, eggplant or talong is the number one vegetable in terms of production area, with an average of 21,225 hectares planted every year. The top five eggplant producing areas in the country are Ilocos, Central Luzon, Cagayan Valley, CALABARZON, and the Bicol region Eggplant. During the fourth quarter of 2020, production of eggplant at 23.13 thousand metric tons was -5.9% lower than the 24.57 thousand metric tons output in same quarter of 2019.



18. **Tomato** *(Lycopersicon esculentum)* Tomato is the second most important fruit vegetable in the country in terms of area and volume of production. Production of tomato from October to December 2020 dropped to 26.00 thousand metric tons or by -7.6%, from 28.13 thousand metric tons in the same quarter of 2019. The largest bulk of production came from Northern Mindanao with 14.85 thousand metric tons, accounting for 57.1% of the total output this quarter. This was followed by Ilocos Region and Zamboanga Peninsula with 9.1% and 7.5% shares, respectively.



19. **Cabbage** *(Brassica oleracea)* From October to December 2020, production of cabbage increased to 60.06 thousand metric tons or by 2.2%, from 58.77 thousand metric tons in the same period of 2019.CAR, the major producer of cabbage with 48.84 thousand metric tons shared 81.3% to the total output this quarter.



20. **Cacao -***Theobroma cacao*, also called the cacao tree, is a small evergreen tree in the family Malvaceae, native to the deep tropical regions of Mesoamerica. Its seeds, cocoa beans, are used to make chocolate liquor, cocoa solids, cocoa butter and chocolate. The Philippines is said to be the first country in Asia that planted cacao. However, the country is still an importer of cocoa products such as chocolate, cocoa powder, cocoa beans, and cocoa butter. In 2014, a total of USD 102.3-M worth of import was recorded, while export value was at USD 24.3-M only.



21. **Durian-** *Durio zibethinus L.* Davao City in Mindanao is dubbed as the “Durian Capital of the Philippines.” The fruit is such a big deal in the entire Davao region, aiding the local economy in terms of exporting and tourism, that they even hold a festival in honor of it annually.

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Economic importance of identified major crops in the Philippines

Rice

- Rice remains the agricultural commodity with foremost political and economic significance in the Philippines. Rice contributes 13 percent to the Consumer Price Index (CPI), 16 percent to the Gross Value Added (GVA) of agriculture, and 3.5 percent to the Gross Domestic Product (GDP) (Gonzales, 1999).

Corn

- Corn is one of the most important staple crops in the Philippines. It ranks second to rice in the utilization of agricultural resources. It is used not only for human consumption but also for animal feeds and industrial uses, hence it is a vital crop in the development of the livestock and manufacturing industries.

Rubber

- With its National Greening Program (NGP), the DENR considers rubber as one of the high value crops because of its potential to help livelihood in the communities, given the huge role of the Philippines in global rubber production (FMB, 2014). It ensures the proper plantation procedures of high value crops in its NGP.

Manila hemp

- Hard fiber, such as Manila hemp (abaca), has become an important commodity in modern times. It is used as material for strategic goods, such as ship rigging and machinery ropes, everyday goods, such as packaging twine and paper, and clothing, such as hats and dresses.

Coffee

- The most typical reason behind the popularity of coffee is its energy-boosting effects. Most Filipinos see it as an instant pick-me-up when they're feeling under the weather. Among its various products, the 3-in-1 mixes drive the sales growth (https://beantomug.com>filipino-cul).

Pineapple

- With their potential to be consumed fresh, juiced, canned, and as a preserve, pineapple's main economic use is culinary. The fruits contribute to over 20% of the world's production of tropical fruits and are the only member of the Bromeliaceous family that produces edible fruits.

Eggplant

- In the Philippines, eggplant production accounts for more than 30.0% of the total volume of production of the most important vegetables in the country Eggplant production provides an important source of cash income, particularly for small, resource-poor farmers
- It can be grilled, stuffed, roasted, served in soups and stews and on kabobs, and used in curries and stir-fries. Eggplant is nutritious, being low in calories, fat, and sodium. It is

high in fiber, and provides additional nutrients such as potassium, magnesium, folic acid, vitamin B6 and A.

Banana

- The Philippine banana industry contributes significantly to the agriculture sector and the economy in general. In 2000, the banana sector contributed about 7% to the total value of production in agriculture. Banana is also one of the country's top export earners.
- Out of the country's 2.2 million farms, only 75% are now considered productive. And yet, in some cases, banana farming is more lucrative than rice cultivation. In ideal conditions, lakatan is highly profitable compared to rice, according to research made by the Isabela State University-Santiago campus.

Garlic

- Garlic is an important seasoning for Filipinos. 'Ilocos White' is the most common variety of garlic planted for commercial production in the Philippines. It is moderately resistant to insect pests and diseases

Onion

- Onions are one of the most important vegetable crops in the Philippines, with pests representing a major production constraint. Overall, the paper finds that reduced pesticide use as a result of the onion IPM project resulted in health-related, environmental, and economic benefits for the farmers.

Cabbage

- Cabbage is an excellent source of vitamins and dietary fiber. It also contains significant amounts of glutamine, an amino acid that has anti-inflammatory properties. Cabbage can also be included in dieting programs, as it is a low calorie food.

Durian

- Durians are sold whole, or cut open and divided into segments, which are wrapped in clear plastic. The flesh is mostly eaten fresh, often out-of-hand. It is best after being well chilled in a refrigerator. Sometimes it is simply boiled with sugar or cooked in coconut water, and it is a popular flavoring for ice cream.
- Durian flesh is canned in syrup for export. It is also dried for local use and export. Blocks of durian paste are sold in the markets. The unripe fruit is boiled whole and eaten as a vegetable.
- The seeds are eaten after boiling, drying, and frying or roasting. In Java, the seeds may be sliced thin and cooked with sugar as a confection; or dried and fried in coconut oil with spices for serving as a side-dish.
- Young leaves and shoots are occasionally cooked as greens. Sometimes the ash of the burned rind is added to special cakes.

Sugarcane

- Sugarcane molasses

Annual molasses production output has averaged about 916 000 metric tons. Of this, about 67 percent is exported, 17% is used by the distilleries, and 16 percent is for animal feed, etc. (National Economic and Development Authority, 1985).

- Sugarcane bagasse
  Bagasse production output averaged 6.45 million metric tons from 1980 to 1985. For many years, bagasse has been used mainly as fuel by sugar mills during milling seasons, but there is plenty of bagasse available that cannot be used as fuel. A small amount is used as feed, the surplus bagasse being left just to rot around the sugar mills (Rojas, 1985).

- Sugarcane tops
  During the past five years, estimated yearly cane top production output has been 4.7 million metric tons, with an average production of 11.5 tons per hectare (PSA, 2020). Around sugar-producing areas, small farmers collect cane tops during milling seasons as feed for their cattle, carabaos, horses, goats and sheep. Among medium and large livestock raisers, the cane tops are chopped and fed to animals or ensiled for use during off-milling seasons.

Summary

Agricultural economics plays a role in the economics of development, for continuous level of farm surplus is one of the wellsprings of technological and commercial growth in general, one can say that when a large fraction of a country's population depends on agriculture for its livelihood. Average income are low.

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II. PHYSIOLOGICAL PROCESSES AFFECTING CROP PRODUCTION

A. RESPIRATION

Overview

In this lesson students will learn that respiration is a process that occurs in cells. Respiration is the release of energy, not the production of energy. They will learn that energy from glucose can be converted into ATP and that it is ATP that powers reactions in the body**.**

Cellular respiration is a process that occurs in the mitochondria of all organisms. In this process, both plants break down simple sugars into carbon dioxide and water and release energy in the form of adenosine triphosphate (ATP). The ATP is used for all the processes that occur within a cell that need energy. Cellular respiration or aerobic respiration is a series of chemical reactions which begin with the reactants of sugar in the presence of oxygen to produce carbon dioxide and water as waste products. This process occurs in the mitochondria, the powerhouse of the cell *(Starr, 2010).*

Respiration occurs in all living cells, including leaves and roots. Since respiration does not require light energy, it can be conducted at night or during the day. However, respiration does require oxygen which can be problematic for roots which are overwatered or in soils with poor drainage. If roots are inundated for long periods of time they cannot take up oxygen and convert glucose to maintain cell metabolic processes. As a result, waterlogging and excessive irrigation can deprive roots of oxygen, kill root tissue, damage trees, and reduce yield.

What is important to recognize is that ultimately all the food that is respired for life processes comes from photosynthesis. This topic deals with cellular respiration or the mechanism of breakdown of food materials within the cell to release energy, and the trapping of this energy for synthesis of ATP. As you have learned in the previous lesson, photosynthesis, takes place within the chloroplast.

Key Concepts

- The compounds that are oxidized during this process are known as respiratory substrates.
- Carbohydrates are oxidized to release energy
- All the energy contained in respiratory substrates is not released free into the cell, or in a single step during oxidation process.
- These steps are controlled by various enzymes.
- This energy trapped in ATP is utilized in various energy-requiring processes of the organisms, and the carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in the cell.

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Learning Objectives

At the end of the topic, the participant should be able to:

1. identify the components of the respiration process,
2. distinguish the different stages of respiration, and
3. list the factors affecting the respiration rate.

Cellular respiration occurs in the mitochondria, a small organelle similar to the size of chloroplasts. Plants produce the glucose and other metabolites that their cells need. Other compounds produced are stored as starches and complex sugars. Respiration is expressed in the formula as:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 32\ ATP\ (energy)$$

Types of Respiration

There are two types of respiration: the aerobic and anaerobic respiration. Aerobic respiration is the release of energy where oxygen is required while anaerobic respiration is energy release in the absence of oxygen.

1. **Aerobic Respiration-** the final product of glycolysis, pyruvate is transported from the cytoplasm into the mitochondria. The crucial events in aerobic respiration are:

   • the complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO2, and
   • the transfer of the electrons to molecular O2 with simultaneous synthesis of ATP

2. **Anaerobic Respiration-** the conversion of glucose into 3-C pyruvate through glycolysis. 3-C pyruvate is then converted into ethanol + carbon dioxide through fermentation.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

Ethanol is the alcohol found in alcoholic drinks like beer and wine. In bread-making, bubbles of carbon dioxide gas expand the dough and help the bread rise. (https://en.unesco.org/themes/building-knowledge-societies/oer/265196eng.pdf)

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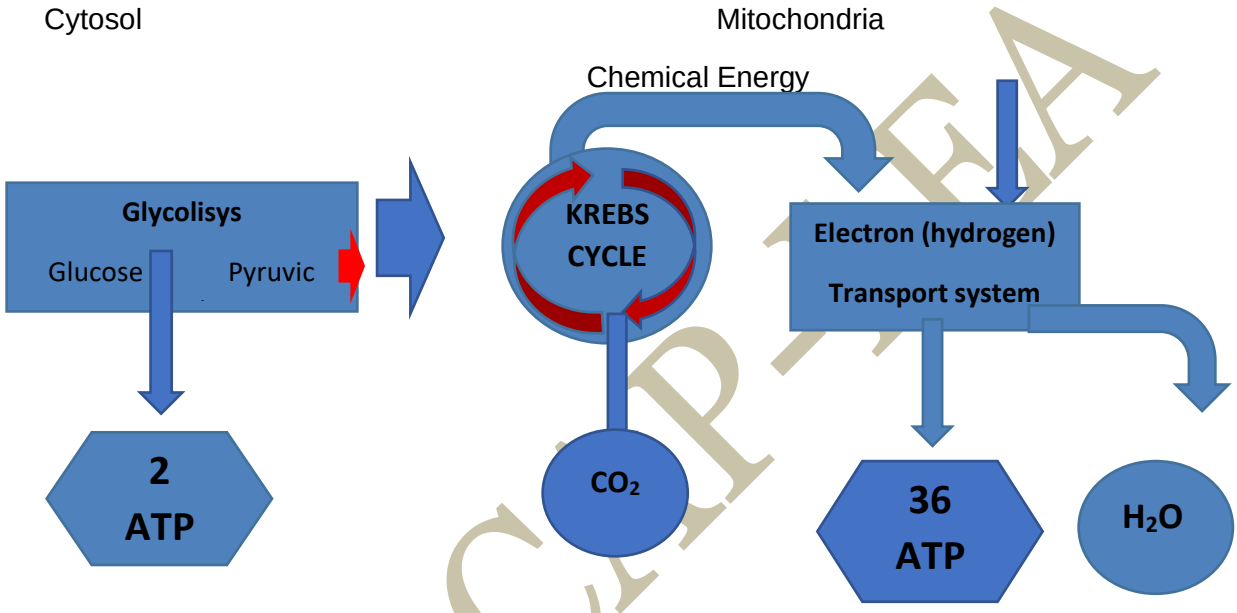


Figure 1. Aerobic respiration

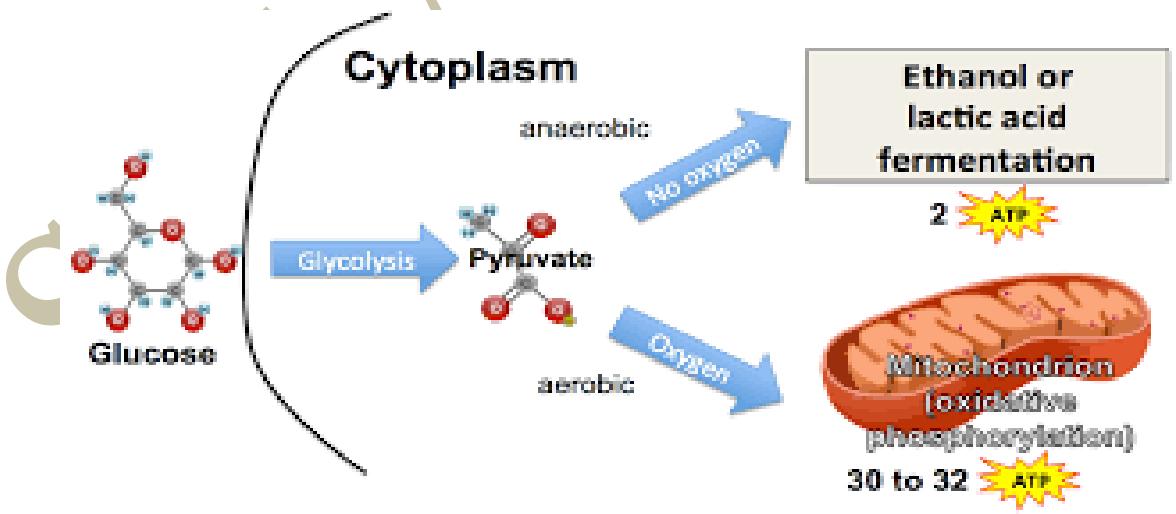


Figure 2. Anaerobic respiration
Source: google

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Summary of differences between aerobic and anaerobic respiration

AEROBIC	ANAEROBIC
Takes place in the presence of oxygen	Takes place in the complete absence of oxygen
Leads to complete oxidation of organic substrate.	Incomplete oxidation of organic substrate takes place
most common in higher organisms (both plants and animals)	Takes place in lower organisms such as bacteria, fungi, and in higher animals under limiting conditions of oxygen (e.g. in muscles when oxygen present is insufficient)
$C_6H_{12}O_6$ ➔ $6CO_2 + 6H_2O + 6O_2$ + 38 ATP	$C_6H_{12}O_6$ ➔ 2 Ethyl alcohol + $2CO_2$ +2 ATP (as in yeast) OR $C_6H_{12}O_6$ ➔ 2 Lactic acid + 2 ATP (as in muscles)
Takes place in the cytoplasm, and mitochondria in eukaryotes and plasma membrane in prokaryotes.	Takes place in the cytoplasm

Stages in respiration

1. **Glycolysis** - the stage where glucose molecule is split into pyruvic acid resulting in the formation of reduced nucleotides, NADH and ATP. This is taken place in the cytosol which yields two molecules of ATP.

    a. Phosphorylation of glucose to fructose 1,6-diphosphate - activation of glucose and use of 2 ATPs
    b. Splitting of glucose into two 3-carbon sugar phosphates, which are interconvertible. Note: this is the origin of the term glycolysis meaning splitting of glucose.
    c. Oxidation by dehydrogenation. Each 3-C sugar phosphate is oxidized by removal of hydrogen, making a reduced NAD i.e. NADH and 2 ATPs.

Enhance your knowledge on glycolysis through this link: https://youtu.be/A1nJRoPGkRs

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2. **Citric Acid Cycle** - the second stage in respiration which takes place in the mitochondria. The pyruvic acid produced in glycolysis will be oxidized to carbon dioxide and reduced from NAD^+ to $NADH_2$ and forms acetyl-CoA. In the presence of oxygen, pyruvic acid enters the mitochondria and is Plants and animals decarboxylated (removal of CO2) and dehydrogenated (removal of H) to acetyl CoA. Acetyl CoA is thus the connecting link between glycolysis and the next series of reactions that yield more energy in the form of ATP. Acetyl CoA can also be generated from fats and proteins.

   Go to this link for further learnings. https://www.youtube.com/watch?v=d9hffYCwfEA

3. **Electron Transport System** - the metabolic pathway through which the electron passes from one carrier to another and it is present in the inner mitochondrial membrane. Electrons from NADH produced in the mitochondrial matrix during citric acid cycle are oxidized by an NADH dehydrogenase (complex I), and electrons are then transferred to ubiquinone located within the inner membrane.

   To further enhance your learning on this watch the video on Youtube channel through this link: https://www.youtube.com/watch?v=d9hffYCwfEA

Factors affecting respiration rate

The rate of respiration can be measured by the amount of CO_2 released per unit time. As expected, it varies in different organs and with age. In general the factors which affect respiration include internal factors such as minerals, structure of respiratory tissue or organ, the activity of the respiratory enzymes and the type of substrate; and external factors such as oxygen, water, and temperature.

1. **Temperature -** The temperature between 30-35°C is most suitable for respiration. This is because the enzymes can work best in this range. Respiration is reduced beyond 50°C and also at very low temperatures (0-10°C).
2. **Oxygen -** the rate of respiration increases with rise in oxygen concentration. As O_2 concentration increases from zero, the rate of respiration increases. However, beyond a limit the rate of increase falls.
3. **Carbon dioxide -** rate of respiration decreases if CO_2 is allowed to accumulate surrounding the respiratory tissue.
4. **Water -** respiration is very slow if the water content of the protoplasm is low as in dry, matured seeds. Dormant seeds show very low rate of respiration. It water is supplied to dry seeds, respiration starts immediately.

Summary

- Respiration happens in the cells of plants, mainly inside mitochondria, which are located in a cell's cytoplasm.
- Glycolysis is the first step in in both aerobic and anaerobic respiration which means, it may proceed even with or without the presence of oxygen.
- The second step in respiration is citric acid cycle which loads electrons and hydrogens.

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- The third step in respiration process is electron transfer which transports proteins and drives ATP formation from ADP.
- In anaerobic respiration, two ATP's are yielded while in aerobic respiration, thirty six ATP's are yielded. Thus, the total energy released in cellular respiration is 38 ATP's.
- Respiration rate is affected by temperature, amount of oxygen, carbon dioxide and water.
- The influence of respiration in plants is manifested in land preparation where flooding and water logging by which the root system is exposed. Upland crops require a well aerated soil and drainage while lowland rice dwells in puddled soil or under continuous flooding.

B. MINERAL NUTRITION

Overview

Basic knowledge and understanding with regards to different soil factors that affects plant nutrition is necessary in the practice of agriculture. Understanding the basic functions of each essential element, knowing its available form, the timing of application and the movement of nutrient within plants are useful for the practitioner of agriculture. Moreover, soil mineralogical, chemical, physical and biological properties are as important as well. These soil properties often dictate the fertility of the soil and the performance of crops is reliant on these factors. However, human intervention such as fertilizer application may solve soil fertility problems but with the proper management practices. These topics will be discussed in detail in this module.

The Essential Elements

Understanding the principles of soil fertility is important to efficiently manage nutrients and also to conserve environmental resources. There are seventeen (17) chemical elements known to be essential for plant survival and productivity and fourteen (14) of these elements are derived from the soil. Each essential plant nutrient differs in the amount required by plants, movement within crops is also varying with one another. Thus, it is important to understand the characteristics and properties of each element as well as its available forms and its function.

Learning Objectives

At the end of the topic, the participant should be able to:

1. identify the different essential elements needed by plants, and
2. understand the different mobility characteristics and movement of each essential element.

The Essential Elements

In order for a particular element to be classified as essential for plant nutrition it must meet at least four (3) basic criteria. These are:

1. A deficiency of the essential element makes it impossible for a plant to complete its life cycle.
2. The deficiency is specific to particular essential element. The deficiency of an essential element can be overcome only by the supply of that element.
3. An essential element is directly involved in the metabolism of the plant.

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As mentioned earlier, out of 17 essential elements, only 14 are obtained mainly from the soil. The other 3 which includes Carbon (C), Hydrogen (H), and Oxygen (O) are taken from air and soil water. Although they constitute approximately 95% of the plant biomass, they are generally given little attention in plant nutrition because they are always in sufficient supply.

Essential Mineral Elements, Functions and Available Forms Taken Up by Plants

The essential mineral elements are categorized into two which is based on the concentration required by the plant to function properly. The elements that are needed at higher concentration (as percentage of the dry weight of the plant) are known as the major elements or macronutrients. The macronutrients include N, P, K, Ca, Mg, and S. In contrast the minor elements or micronutrients are required at lower concentration. B, Cl, Cu, Fe, Mn, Mo, Ni, and Zn are included in this category.

Macronutrients

Nitrogen (N)

Functions in plants:

- Found in both inorganic and organic forms in the plant
- Combines with C, H, O, and sometimes S, to form amino acids, amino enzymes, nucleic acids, chlorophyll, alkaloids, and purine bases
- Organic N predominates as high-molecular-weight proteins in plants
- Inorganic N can accumulate in the plant, primarily in stems and conductive tissue, in the nitrate (NO_3^-) form

Available forms for root absorption:

- Exists in the soil solution as either the NO_3^- anion or the NH_4^+ cation, the uptake of either form influenced by soil pH, temperature, and the presence of other ions in the soil solution.
- The NH_4^+ cation participates in cation exchange in the soil; nitrite (NO_2^-) may be present in the soil solution under anaerobic conditions and is toxic to plants at very low levels (<5 ppm).

Phosphorus (P)

Functions in plants:

- A component of certain enzymes and proteins, adenosine triphosphate (ATP), ribonucleic acids (RNA), deoxyribonucleic acids (DNA), and phytin.
- ATP is involved in various energy transfer reactions, and RNA and DNA are components of genetic information.

Available forms for root absorption:

- Exists in most soils in about equal amounts of organic or inorganic forms.
- Dihydrogen phosphate ($H_2PO_4^-$) and monohydrogen phosphate (HPO_4^{2-}) are the two anion forms of P in the soil solution, which form and their concentration depending on soil water pH.
- Al, Fe, and Ca phosphates are the major inorganic sources of P, the relative amount among these three forms being a function of soil water pH.
- Release of P into the soil solution with the decomposition of crop residues and microorganisms can be a major source of P for plant utilization.

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- Plant availability is influenced by soil water pH.

Potassium (K)

Functions in plants:

- Involved in maintaining the water status of the plant, the turgor pressure of its cells, and the opening and closing of its stomata.
- Required for the accumulation and translocation of newly formed carbohydrates.

Available forms for root absorption:

- As the K^+ cation in the soil solution.
- As exchangeable K^+ adsorbed to soil colloids.
- As fixed K in the lattice of 2:1 clays.
- As a component in K-bearing minerals.
- An equilibrium exists between K in the soil solution, exchangeable K, and fixed K.
- When K fertilizer is applied to the soil, the equilibrium shifts toward exchangeable and fixed K, a shift that is reversed as K is removed from the soil solution by root absorption.
- Plant availability is influenced by soil water pH.

Calcium (Ca)

Functions in plants:

- Plays an important part in maintaining cell integrity and membrane permeability; enhances pollen germination and growth.
- Activates a number of enzymes for cell mitosis, division, and elongation.
- May also be important for protein synthesis and carbohydrate transfer.
- Its presence may serve to detoxify the presence of heavy metals in the plant.

Available forms for root absorption:

- Calcium exists as the Ca^{2+} cation in the soil solution and as exchangeable Ca on soil colloids.
- Usually, the cation of highest concentration in the soil in both soluble and exchangeable forms for soils high in pH (>8.0), soils that may contain sizable quantities of Ca as precipitates of calcium carbonate ($CaCO_3$) and calcium sulfate ($CaSO_4$).
- It is generally assumed that if the soil pH is within the acceptable range in the rooting media, Ca should be of sufficient concentration to ensure plant- Ca sufficiency, assuming that other factors are also within their normal ranges.
- Soil pH seems to have little effect on Ca uptake.

Magnesium (Mg)

Functions in plants:

- A component of the chlorophyll molecule.
- Serves as a cofactor in most enzymes that activate phosphorylation processes as a bridge between pyrophosphate structures of ATP or ADP and the enzyme molecule.
- Stabilizes the ribosome particles in the configuration for protein synthesis.

Available forms for root absorption:

- Exists as the Mg^{2+} cation in the soil solution.

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- As exchangeable Mg on soil colloids, usually the cation being next to the highest in concentration in the soil in both soluble and exchangeable forms when the soil is slightly acid to neutral in pH.
- Availability declines significantly when the soil water pH is less than 5.4.

Sulfur (S)

Functions in plants:

- Involved in protein synthesis.
- Is part of the amino acids cystine and thiamine.
- Is present in peptide glutathione, coenzyme A, and vitamin B1, and in glucosides such as mustard oil and thiols, which contribute the characteristic odor and taste to plants in the Cruciferae and Liliaceae families.
- Reduces the incidence of disease in many plants.

Available forms for root absorption:

- Over 90% of available S exists in the soil organic matter, which has an approximate 10:1 nitrogen: sulfur (N:S) ratio available, depending on organic matter decomposition rates.
- The sulfate (SO_4^{2-}) anion is the primary available form found in the soil solution.
- In general, most of the available SO4 is found in the subsoil as the anion and can be easily leached from the surface horizon.
- Availability may depend on that deposited in rainfall (acid rain) and/or that released from organic matter decomposition.
- At high soil pH (>7.0), S may be precipitated as calcium sulfate ($CaSO_4$), while at lower pH levels (<4.0), the SO_4^{2-} anion may be adsorbed by Al and Fe oxides.
- Plant availability is influenced by soil water pH.

Micronutrients

Boron (B)

Functions in plants:

- Believed to be important in the synthesis of one of the bases for RNA (uracil) formation.
- In cellular activities (e.g., division, differentiation, maturation, respiration, growth, etc.).
- Has long been associated with pollen germination and growth, improving the stability of pollen tubes.
- Relatively immobile in plants.
- Transported primarily in the xylem.

Available forms for root absorption:

- Most of the B in soil exists in organic plant and microorganism residues, released by residue decomposition, thereby becoming the major supply source for crop utilization.
- Exists in the soil solution as the borate anion (BO_3^{3-}).
- Recent findings suggest that B can also exist as undissociated $B(OH)_3$.
- Primary loss of B from soils is by leaching.
- Leaching is also a common technique of removing excess B from the surface soil and rooting zone.
- Total B in the soil ranges between 20 and 200 ppm.
- Amount available for plant absorption ranges from 1 to 5 ppm in the soil solution.

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- Plant availability is influenced by soil water pH.

Chlorine (Cl)

Functions in plants:

- Involved in the evolution of oxygen (O_2) in photosystem II in the photosynthesis process.
- Raises the cell osmotic pressure.
- Affects stomatal regulation.
- Increases the hydration of plant tissue.
- May be related to the suppression of leaf spot disease in wheat and fungus root disease in oat.

Available forms for root absorption:

- Exists in the soil solution as the Cl^- anion.
- Moves in the soil by mass flow.
- Chloride anion (Cl^-) competes with other anions, such as NO^{3-} and SO_4^{2-}, for uptake.

Copper (Cu)

Functions in plants:

- Constituent of the chloroplast protein plastocyanin.
- Serves as a part of the electron transport system linking photosystems I and II in the photosynthesis process.
- Participates in protein and carbohydrate metabolism and nitrogen (N_2) fixation.
- Is part of the enzymes that reduce both atoms of molecular oxygen (O_2) (cytochrome oxidase, ascorbic acid oxidase, and polyphenol oxidase).
- Is involved in the desaturation and hydroxylation of fatty acids.

Available forms for root absorption:

- Exists in the soil primarily in complexed forms as low-molecular-weight organic compounds such as humic and fulvic acids.
- Cupric ion (Cu^{2+}) is present in very small quantities in the soil solution.
- Uptake rates are lower than for most other micronutrients.
- Plant availability is influenced by soil water pH.

Iron (Fe)

Functions in plant:

- Important component in many plant enzyme systems, such as cytochrome oxidase (electron transport) and cytochrome (terminal respiration step).
- Component of protein ferredoxin and is required for NO_3 and SO_4 reduction, nitrogen (N_2) assimilation, and energy (NADP) production; functions as a catalyst or part of an enzyme system associated with chlorophyll formation.
- Thought to be involved in protein synthesis and root-tip meristem growth.
- The plant and soil chemistry of Fe is highly complex, with both aspects still under intensive study.

Available forms for root absorption:

- Exists in the soil as both the ferric (Fe^{3+}) and ferrous (Fe^{2+}) cations.
- The Fe^{2+} form, whose availability is affected by the degree of soil aeration, is thought to be the active form taken up by plants.

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- Iron-sufficient plants are able to acidify the rhizosphere as well as release Fe-complexing substances, such as siderophores, which enhance Fe availability and uptake, while Fe-inefficient plants do not exhibit similar root characteristics.
- Plant availability is influenced by soil water pH.

Manganese (Mn)

Functions in plants:

- Involved in oxidation-reduction processes in the photosynthetic electron transport system.
- Essential in photosystem II for photolysis, acts as a bridge for ATP and enzyme complex phosphokinase and phosphotransferases, and activates IAA oxidases.
- Not known to interfere with the metabolism or uptake of any of the other essential elements.

Available forms for root absorption:

- Exists in the soil solution as either Mn^{2+}, Mn^{3+}, or Mn^{4+} cations and as exchangeable Mn.
- Cation Mn^{2+} is the ionic form taken up by plants.
- Availability is significantly affected by soil pH, decreasing when the pH increases above 6.2 in some soils; while in other soils, the decrease may not occur until the soil water pH reaches 7.5.
- Availability can be reduced significantly by low soil temperatures.
- Soil organic matter can reduce Mn availability, decreasing its availability with an increase in organic matter content.

Molybdenum (Mo)

Functions in plant:

- Is a component of two major enzyme systems, nitrogenase and nitrate reductase, nitrogenase being involved in the conversion of nitrate (NO_3) to ammonium (NH_4).
- The requirement for Mo is reduced greatly if the primary form of nitrogen (N) available to the plant is NH_4.

Available forms for root absorption:

- Primary soluble soil form is the molybdate anion (MoO_4^{2-}), whose availability is increased tenfold for each unit increase in soil pH.
- In the soil, Mo is strongly absorbed by Fe and Al oxides, the formation of which is pH dependent.

Zinc (Zn)

Functions in plants:

- Involved in the same enzymatic functions as Mn and Mg with only carbonic anhydrase activated by Zn.
- The relationship between P and Zn has been intensively studied as research suggests that high P can interfere with Zn metabolism as well as affect the uptake of Zn through the root.
- High Zn can induce a Fe deficiency, particularly those sensitive to Fe.

Available forms for root absorption:

- Exists in the soil solution as the Zn^{2+} cation, as exchangeable Zn, and as organically complexed Zn.
- Availability is affected by soil pH, decreasing with increasing pH.
- Availability can also be reduced when the available soil P level is very high.

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- Plant availability is influenced by soil water pH.

Nickel (Ni)

Functions in plants:

- Component of plant urease.
- Benefits growth of N-fixing plant species.
- Barley seeds will not germinate when deficient in Ni.

Available forms for root absorption:

- Is available and absorbed as the Ni^{2+} cation.

Nutrient Movement and Uptake

Nutrients must come in contact with the root surface. Generally, there are three ways for the nutrient ions in soil to reach the root surface: root interception, with the possibility of contact exchange, diffusion of ions in the soil solution; and movement of ions by mass movement with the soil solution.

Root Interception

As the root system develops and exploits the soil completely, soil solution and soil surfaces retaining adsorbed ions are exposed to the root mass and adsorption of these ions by the contact exchange mechanism is accomplished. The quantity of nutrients that can come in direct contact with the plant roots is the amount in a volume of soil equal to the volume of roots. It can be assumed that roots usually occupy 1% or less of the soil. Because roots grow through soil pores which may have higher than average nutrient content, it is estimated that roots would contact a maximum of 3 % of the available nutrients in the soil.

Mass Flow

Movement of ions in the soil solution to the surfaces of roots is accomplished largely by mass flow and diffusion. Mass flow, a convective process, occurs when plant nutrient ions and other dissolved substances are transported with the flow of water to the root that results from transpirational water uptake by the plant. Some mass flow can also take place in response to evaporation and percolation of soil water.

Amounts of nutrients reaching roots by mass flow are determined by the rate of water flow or the water consumption of plants and the average nutrient concentrations in the soil water. The level of a particular nutrient around the root will fluctuate depending on the balance between the rate at which it reaches this zone by mass flow and the rate of uptake by the root. Mass flow supplies an overabundance of calcium and magnesium in many soils and most of the mobile nutrients, such as nitrogen and sulfur, if concentrations in the soil solution are sufficient.

Diffusion

Diffusion occurs when an ion moves from an area of high concentration to one of low concentration by random thermal motion. As plant roots absorb nutrients from the surrounding soil solution, a diffusion gradient is set up. Plant roots absorbing nutrients in this manner thus create a sink to which nutrients diffuse. A high plant requirement or a high root “absorbing power” results in strong sink or a high diffusion gradient, favoring ion transport.

Back Diffusion

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Under some conditions the concentration of certain ions may build up at the root surface because the root is unable to absorb them at a sufficiently rapid rate. This results in a phenomenon known as “back diffusion.” In which the concentration gradient, and hence the movement of certain ions, will be away from the root surface and back toward the soil solution. Normally such a condition will not occur, but as roots do not absorb all nutrient ions at the same rate, there may on occasion be a buildup of those ions that are less rapidly absorbed, particularly during periods when the plant is absorbing moisture rapidly. It should be noted that elevated levels of one or more nutrients in the rhizosphere can have important effects on the uptake of other nutrients.

Complimentary Ion Effect

The complementary ion effect is defined as the influence of one adsorbed ion on the release of another from the surface of a colloid. When only a portion of the cations held by soil colloids is being exchanged, the release of a given cation from exchangeable form is easier as the retention strength or the strength of bonding of the complementary exchangeable cations increases. For example, ammonium in the soil solution is exchanging with calcium on soil colloids. This exchange will take place more readily when the complementary cation on the exchange complex is aluminum rather than sodium. Ammonium will replace much more sodium than aluminum. The more strongly held trivalent aluminum tends to satisfy a greater part of the cation exchange capacity and permits the exchange of ammonium for calcium to proceed more completely.

C. GROWTH AND DEVELOPMENT

Overview

This module discusses growth and development in plants. Plant growth and development are essential processes of life and propagation of a species (Gardner et al 2017). This serves as an introduction to plant development. It started with a brief discussion of the distinctions between growth, differentiaion and development to measurement of growth and factors affecting growth. The internal growth mechanisms of plants are likewise explained. Towards the end, hormones, its function, effects and agricultural uses are reviewed.

Learning Objectives

At the end of the topic, the participant should be able to:

1. differentiate development, growth, and differentiation,
2. cite two types of internal growth mechanisms in plants and give examples,
3. list down the characteristics responses in plants that are elicited by plant growth hormones, and
4. identify the five main classes of hormones synthesized by plants and their roles on the growth, development and physiological processes of plants.

What is growth and development?

Sometimes in our conversation we interchangeably used growth and development to describe various aspect of changes in life cycle of a plant. However, in botanical sense they refer to

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separate events in the organization of the mature plant body. The three terms used to describe various aspects of the changes that a plant goes through in its life cycle are:

1. Growth
2. Differentiation
3. Development

In order to understand the descriptions of plant development and its regulations, we need to clarify these three terms.

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What is Development?

Development is an umbrella term which refers to the sum of all the changes that organism goes through its life cycle (Hopkins and Huner, 2003). This term is encompassing starting from seed germination through flowering, seed formation maturation, and senescence. Development also applies and manifested in the following changes:

1. pattern of change in cells, tissues and organs.
2. changes in the form of organism or organ such as transition from vegetative stage to flowering stage or leaf primordium to fully expanded leaf
3. changes at subcellular and biochemical levels such as: when chloroplasts appear in leaf cells brought into light and the enzymes of photosynthesis are activated.

Development involves the following events:

1. growth
2. morphogenesis (acquisition of form and structure)
3. differentiation

What is Growth?

Growth is popularly defined as an irreversible increase in size of an organ or its parts or of an individual cell. In a restricted sense, it is cell division and cell enlargement, referring to increase in number (cell division) and increase in size (cell enlargement). Both require protein synthesis which means that growth is accompanied by metabolic processes (Gardner et al., 1985 and NCERT, 2021).

Meristematic Tissues

Growth in plants occurs chiefly at meristems where rapid mitosis provides new cells. As these cells differentiate, they provide new plant tissue. As a review we need to define what meristem s are.

Meristems are tissues in which the cells remain forever young and divide actively throughout the life of the plant. There are various kinds of meristems: the apical meristem and three kinds of lateral – vascular cambium, cork cambium, and intercalary meristem.

Apical meristems

Apical meristems are located in the tips of roots and shoots. Cell divisions and subsequent cellular enlargement in these areas lengthen the above and below ground parts of the plant. The meristems also influence the shapes of the mature plants since the patterns for subsequent growth are laid down in the meristems.

Lateral meristems

1. Vascular cambium - responsible for growth in diameter. Some plants grow in diameter by producing new tissues laterally from the vascular cambium, which extends throughout the length of the plant from the tips of the shoots to the tips of the roots. Tissues produced by cell divisions of the vascular cambium are secondary tissues.

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2. Cork cambium - cork cambia (also called phellogens) are found in the bark of roots and stems of woody plants where they produce cork cells.
3. Intercalary meristems – the intercalary meristems of grasses are located along the stems near the nodes. Cell divisions in this tissue push the stem upward. Grasses and other monocots have no lateral meristems, any lateral increase in size is the result of primary tissue cell enlargement, not cell divisions.

Growth in plants can either be:

1. determinate - when an organ or part or whole organism reaches a certain size and then stops growing
2. indeterminate - when cells continue to divide indefinitely. Plants in general have indeterminate growth.

Measurement of Growth

In agriculture, growth is generally defined by agronomist as increase in dry matter (Gardner et al, 1985). This definition includes the process of differentiation which contributes greatly to dry matter accumulation.

Growth can be evaluated by a variety of quantitative measures depending on the interest of the researchers. But why is dry weight accumulation commonly used as a parameter characterizing growth? This is because it usually has the greatest economic importance. Fresh weight is less useful because it fluctuates depending on the moisture content of the plants. Other related growth parameters include: plant height, volume or size and leaf area.

Growth Analysis

The most widely used in analyzing growth is the classical growth analysis (Clipson, 1994) which includes the following components: relative growth rate (RGR), net assimilation rate (NAR), crop growth rate (CGR), leaf area index (LAI), leaf area ratio (LAR) with its two components: leaf weight ratio (LWR) and specific leaf area (SLA).

RGR – increase of plant dry weight relative to total dry weight at a single given time point
NAR – rate of dry matter accumulation per unit leaf area per unit time.
CGR – rate of dry matter production per unit ground area per unit time.
LAR – assimilatory material per unit of plant material present or ratio of leaf area to total plant dry weight
LWR – ratio of leaf weight to total plant dry weight
SLA – ratio of leaf area to total leaf weight.
LAI – total green leaf area per unit land area.

What is Differentiation?

Differentiation is the process in which generalized cells specialized into the morphologically and physiologically different cells. The tissues or organs has distinct morphology and specialized functions. Normally, it is best described as a series of qualitative rather than quantitative changes (Hopkins and Huner, 2003).

Differentiation arises when a dividing cell gives rise to two daughter cells destined to assumed different anatomical characteristics and function. Examples:

1. Division of the zygote gives rise to cells that will become root or shoot of the plant.

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2. Unspecialized parenchyma cells will differentiate into xylem vessels or phloem sieve tube.

Another term is dedifferentiation. Under certain conditions, mature plant cells can be stimulated to divide and differentiate again. This is the case when tissues are wounded or when branches break or leaves are damaged by insects. The plant repairs itself by dedifferentiating.

Growth and differentiation act in coordination and this implies organization. Although growth and differentiation are concurrent events, there are cases where there is growth but without differentiation and differentiation without growth. Differentiation and organization are aspects of morphogenesis.

What is the difference between morphogenesis and differentiation?

Differentiation refers to how cells become specialized, whereas morphogenesis refers to the development of the forms of living organisms. One example of morphogenesis would be plant growth and how new plants change shape to be either straight or twisting or branching.

Internal growth mechanisms

a. Correlations. Plant acquire its shape or form by correlated growth of component parts (Gardner et al.1985). The regulatory effect exerted by one part of the plant on the growth and development in another part is called correlation effects or growth correlation. Some called this allometry. The basis of this interaction can be traced to the role played by growth regulators such as phytohormones or it may be caused by competition of nutrients.

The common correlation effects are exemplified by the following:

a. Influence of the shoot on vascular differentiation
b. Apical dominance
c. Root/shoot ratio or interaction
d. Abscission
e. Tropism
f. Stolon development

Example: Top growth is favored differentially when N and water are plentiful. On the other hand, root growth is favored when N and water are limited as reflected by shoot-root ratios.

b. Endogenous rhythm. This is a term used to describe biological processes which alter periodically although the external conditions remain constant. In other words, these are recurring events or oscillations with properties not directly reflecting environmental fluctuation. However, in many instances the rhythm may be started by certain environmental factors. Once started it may persist for several cycles even under constant or fluctuation of the environmental parameter that started it.

There are several rhythms known to plants which are classified according to the duration of each cycle or frequency of recurrence. The three most common are:

a. Annual rhythm
b. Lunar rhythm
c. Circadian rhythm (repeated every 24 hours)

The closing and opening of foliage and flowers of certain plants even when transferred to continuous light or darkness are among the familiar circadian rhythm in plants. Vivid examples are:

i. The folding of the foliage of acacia tree before sunset.

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ii. Increase in respiration of certain flowers at specific time of the day even under constant ambient temperature.

Factors affecting growth

Factors affecting growth can be broadly categorized as external (environmental) and internal (genetic).

External factors:

1. Climatic – radiation, temperature, water, daylength, wind and gases.
2. Edaphic – soil texture, structure, organic matter, CEC, pH, base saturation and nutrient availability
3. Biological – weeds, insects, diseases, nematodes, various types of herbivores, and soil microorganisms such as N_2 –fixing bacteria and mycorrhiza (symbiotic fungal association with plant roots)

Internal factors:

1. Resistance to climatic, edaphic and biological stresses
2. Photosynthetic rates
3. Respiration
4. Partitioning of assimilates and N
5. Chlorophyll, carotene and other pigment contents
6. Type and loction of meristems
7. Capacity to store food reserves
8. Enzyme activity
9. Direct gene effects (heterosis, epistasis)
10. Differentiation

Since factors under genetic control that contribute to yield are numerous, the above list is only partial.

Hormones and growth regulation

What are hormones and how do they regulate growth?

Plant growth and development are controlled by extremely low concentration of chemical substances, phytohormones or plant growth regulators (PGR) (Gardner et al., 1985). Plant hormone can be defined as an organic compound that can stimulate physiological responses even at very low concentrations (Davies, 2004). Hormones are synthesized in the stems, buds, root tips and translocated to other parts of plant (Went and Thimann, 1937), synthesized in one part of the plant and translocated to another part wherein it causes a physiological response at very low concentration.

The five main classes of plant hormones are auxins**, gibberellins, cytokinins, abscisic acid** and **ethylene**. New classes of hormones are identified as **brassinosteriods, salicylic acid, methyl jasmonate.**

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1. **Auxin** - Indole Acetic Acid (IAA) is the principal natural auxin. It is synthesized from the amino acid tryptophan. An analog of auxin is 2,4-dichlorophenoxyacetic acid (2,4-D) and 2,4,5-trichlorophenoxyacetic acid which are popular systemic broadleaf herbicides.

   Functions of Phytohormones (Source: Dr. Oosterhuis, Powerpoint Slides)

   Functions of Auxins and their Effects:

   a) Cellular effects
      - Increases in nucleotides (DNA, RNA), protein and enzyme synthesis.
      - Increases in proton exchange, membrane charge and K uptake
      - Influences on the phytochrome reaction with red and FAR-red light
   b) Phototropic responses
   c) Gravitropism
   d) Apical dominance
   e) Promote rooting of cuttings
   f) Delay leaf and fruit abscission
   g) Callus growth
   h) Cell wall extensibility

   Agricultural uses of Auxins:
   - Selective broadleaf herbicide
   - Preventive of fruit drop
   - Fruit set in pineapples
   - Rooting compounds
   - Prevent sprouting in potatoes
   - Prevent biennial bearing

2. **Gibberellins** - was first isolated in the fungus, *Gibberella fujikuroi,* which causes bakanae disease in rice. The fungi produce so much of gibberellic acid (GA) that resulted in increased in height of the infected rice plants. This endogenous compound belongs to the same chemical family (terpenoids) of chlorophyll and carotene; which are synthesized in fruits, seeds, buds, young leaves, root tips and all plant organs at varying concentrations and translocated in xylem and phloem. The three principal metabolites commonly involved in the biosynthesis of GA are mevalonic acid (precursor for formation of isoprene), Isoprene (gives rise to kaurene) and kaurene (formed form isoprene). This hormone is widely commercialized as GA_3.

   Functions of Gibberellins and their Effects:

   a) Stimulate internode elongation in dwarf plants
   b) Overcome seed dormancy, bud sprouting and flowering
   c) Release of amylase and result in start hydrolysis in germinating seeds
   d) Development of parthenocarpy (seedless) fruits

   Agricultural uses of Gibberellins:

   - Promotes coleoptile elongation in rice

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- Induces growth in Thompson's seedless grapes
- Produce sterile male in crop hybridization
- Promote amylase activity and starch hydrolysis in barley for the malting industry
- Development of seedless grapes, watermelons and tomatoes

3. **Cytokinins** - is the growth hormone that usually stimulates cell division (cytokinesis). It is discovered by Skoog in 1950. This hormone is available in plants as zeatin. The site of synthesis in plants is found in young roots, developing fruits and seeds. Kinetin is the first synthetic cytokinin discovered.

Functions of Cytokinins and their Effects:

a) Cell division
b) Shoot initiation
c) Leaf enlargement
d) Delays senescence
e) Stimulates lateral bud growth
f) Promotes germination
g) Adventitious root development

Agricultural uses of Cytokinins:

- Used in plant tissue culture
- Increased fruit setting
- Increased germination
- Used in plant breeding

4. **Abscisic acid (ABA)** - was discovered by Wareing and Addicott in separate experiments. ABA biosynthesis begins in the chloroplast via the mevalonic and isoprene pathway. This growth inhibitor hormone is translocated in the xylem and phloem. Like GA, cytokinin, chlorophyll and carotene, ABA is a terpenoid.

Functions of Abscisic Acid and their Effects:

a) Suppresses growth and development
b) Delays senescence
c) Apical dominance
d) Induces stomatal closure
e) Response to water stress
f) Inhibits activities of enzymes
g) Inhibits proton extrusion and K uptake
h) Induce dormancy by shutting synthesis of RNA and proteins

Agricultural uses of Abscisic acid

- Reduce growth
- Reduce lodging in cereals
- Defoliant
- Prevents water stress

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5. **Ethylene (C_2H_4)** - is a gaseous hormone thus, movement within the plant does not require the vascular transport system. Although large concentration of ethylene is produce by ripe climacteric fruits, ethylene production is also present in various parts of the plants like leaves, stem, roots, flowers and seeds. The metabolic precursor of ethylene is methionine. Ethephon ($C_2H_6ClO_3P$) is a popular ripening compound used in crop production.

   Functions of Ethylene and their Effects:

   a) Enhance ripening of climacteric fruits
   b) Geotropic response
   c) Rapid senescence in diseased plants
   d) Released by plants when under physical stress (drought, flooded, wounded)
   e) Leaf and fruit abscission

   Agricultural uses of Ethylene

   - Cotton defoliation
   - Hasten senescence and dehiscence of walnut
   - Suppress growth of tobacco seedlings for easier transplanting
   - Break dormancy in peanut
   - Increase latex flow in rubber plant
   - Promote abscission in cherries, walnuts, olives, apples, peaches
   - Promote fruit and uniform ripening in tomatoes, grapes, pineapples
   - Promote leaf senescence in tobacco

6. **Other Plant Growth Regulators**

   a. Brassinosteroids- are classified as growth promoting substances that accelerate cell division and cell elongation
   b. Methyl jasmonates- are senescence promoting substances. These compounds are also involved in the plant's defense against water stress, wounding, insect attack and pathogen attack.
   c. Salicylic acid- belongs to a diverse group of plant phenolics. Salicylic acid is also involve in the resistance of plants to pathogen because it induces the production of pathogenesis-related proteins.

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Summary

- Development is the sum of growth and differentiation. Growth is an irreversible increase in size that chiefly occurs in the meristems while differentiation refers to how cells become specialized.
- Plant and development are continuous process in the life cycle of a plant, depending on the availability of meristems, assimilates, hormones and favorable environment.
- Plant's internal growth mechanisms involves: correlation or allometry and endogenous rhythm.
- Factors affecting growth are broadly categorized as external (environmental) and internal (genetic).
- Hormones at micromolar concentrations, control plant growth and development processes. They also serve as chemical messengers in plants to aid in coordination of developmental and environmental responses.
- The main classes of hormones are auxins, gibberellins, cytokinins, abscisic acid and ethylene. The other classes are: brassinosteriods, salicylic acid, and methyl jasmonate.

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D. PLANT MOVEMENT

Overview

This module is all about plant movements, its meaning, types and processes that affect plant growth and development. Plants movement known as tropism are crucial for plants survival from germination to how plant position its flower and seed dispersal. Plants can sense their environment and respond to sensory information through visible movement to optimize their survival, growth and reproduction. It can be manifested through the changes in the spatial orientation or conformation of an organ or any of its parts. Plant movement includes movement towards light, opening and closing of flowers, growth and developing of roots in searching of water and nutrients. Inflorescence, fruits and shoots are all capable of moving by sensing changes in environment thereby adjusting their position, function and behavior accordingly.

Learning Objectives

At the end of the module, the participant should be able to:

1. define two types of plant movements,
2. differentiate types of autonomic and paratonic movement of plants, and
3. explain the meaning and types of tropic, nastic and tactic movements in plants.

Before we start our discussion, I would like to ask the reviewees if the plants move. Though plants do not, even in the restricted sense of locomotion but this is not quite true unlike animals. Most plants are fixed unable to move from one place to another but they are capable of some form of movement.

So, the answer is yes, plant do move though it's not obvious as the animals but plants move in the form of bending, twisting or elongation of certain plant parts or organs toward resources.

How do the plants move? There are two types of plant movement:

1. **Autonomic (Spontaneous movement)** refers to the movement of whole or parts of the plants due to the influence of an internal stimulus. There are three types of Autonomic (Spontaneous movement):

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a. Growth movement - are due to unequal growth on two sides of plant organs like stem, roots, tendrils, buds and so on. Three types of growth movement
   i. Epinasty- shown by leaves and petals if the upper surface shows more growth leading to opening of buds as shown in the petals
   ii. Hyponasty- opposite of epinasty. If the development of the lower surface area shows more growth leading to the closure of the bud as shown by the petals
   iii. Nutation- growing tip of the young stem moves in a zigzag style due to alternate changes in development on opposite side of the apex.

b. Turgor movement - a change in position of plant part due to change in turgor pressure of various cells. These movements are not permanent, plant gain its initial state after some period of time. Example, sleep movement of bean plant and some members of the bean family lower leaves in the evening and raise them in the, morning. These sleeping movement are due to everyday changes in the turgor pressure in the pulvinus. Pulvinus is the place of attachment of leaves with the shoot made of parenchymatous cells. When the turgor pressure of the lower side of pulvinus increases, the leaves rise and end up of being horizontal and if turgor pressure of the lower side of pulvinus reduces, the leaves lower and go to the "sleeping position". Another example is the rapid movement of leaflets in Mimosa when touched.

c. Movement of locomotion- plants moves from one place to another due to internal stimuli. Example; Zoospores and gametes of lower plants move from one place by another by means of cilla or flagella.

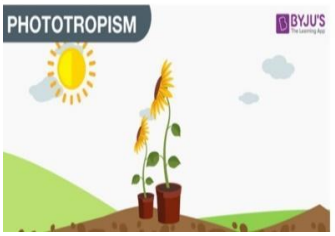
2. **Paratonic (induced movement)** refers to the movement of whole or parts of the plants due to the influence of an external stimulus.
Types of Paratonic (induced movement):

   a. **Tropic movement**. Tropism is a kind of turning movement towards or away from some kind of stimulus. Growth movement occur in the direction of the stimulus known as tropic movement, the response acts on the protoplasm from one side hence plants display positive and negative movement in response to stimulus. Positive tropism when the movement is towards the direction of stimuli while negative tropism when the movement away from the stimuli.

Types of Tropic movements

i. Phototropism: Response to light

Plants grow towards the direction of light. The plants have a chemical called auxin which reacts to light. This causes the cells of the plants to elongate. Growth towards light is positive phototropism while that away from the light is negative phototropism. In general, the stems grow towards the direction of light is called positively phototropic while the roots called negatively phototropic since it grows away from the light source negativephototropism.

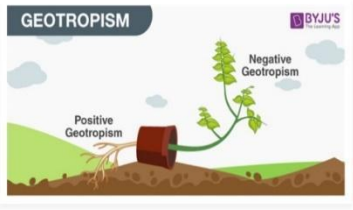


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ii. Gravitropism or Geotropism: Response to gravity

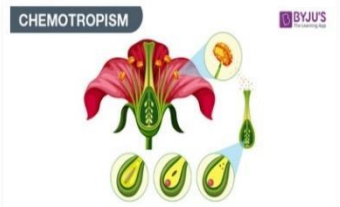
Plants show some growth in response to gravity. Stems are negative to the gravitropism while roots are positive to the gravitropism. This is also referred to as geotropism. Primary roots and certain other portions of the root system show positive geotropism by growing directly towards the center of gravity while the stems are called as negatively geotropic as they grow away from the center of gravity. The leaves are transversely geotropic as they select their position at right angles to the center of gravity.



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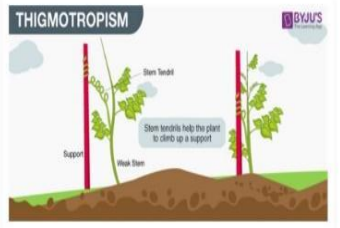
iii. Chemotropism: Response to chemicals

Plants grow in response to chemicals. There are few chemical substances which are actively responsible for bringing a curvature movement in plant organs. Some examples of chemotropic movements are the conversion of a flower into fruit, the growth of pollen tube down the style during fertilization, movement of tentacles in Drosera., etc. Another example is, during fertilization in plants, the stigma produces a chemical in response to which the pollen tube grows towards the ovule.



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iv. Thigmotropism: Response to contact with the solid object

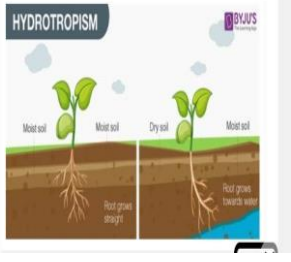


The growing or developing movements made through plants in response to contact with a solid object. Thigmo a Latin word means touch so Thigmotropism is sometimes known as touch tropism. This movement is usually seen in tendrils and twiners. This movement is also referred to as Haptotropism.

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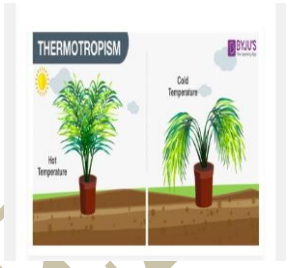
v. Hydrotropism: Response to water

The movement or growth of a plant in relation to the stimulus of water is called hydrotropic movements. In this type of movement, roots show the positive hydrotropic response as they move and grow towards the water. For example, the plant roots grow in the direction of high humidity level. That is why, when we pull a plant, we find its roots growing in all direction.



vi. Thermotropism: Response to atmospheric temperature

It is a form of tropic movement in which a plant or a part of the plant responds to the changing of atmospheric temperature. For example, this type of movement is usually seen in the plants of Rhododendron. The leaves starts twisting or bending in response to cold temperature. Another example, the peduncles of *Anemone nemerosa* grows in response to the heat imparted by the sun.



vii. Sonotropism: Response to sound

Plants grow in response to sound

viii. Electropism : Response to electric field

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b. Nastic Movement. These movements are non-directional responses to stimuli. These movements are independent of the direction of stimulus. These movement can be due to changes in turgor or growth. For example, the response of touch-me-not (*Mimosa pudica*) plant to touch. However, this is not a growth phenomenon. Another example, leaves of Oxalis move in response to intensity of light. They open in daylight and fold when the intensity of light is less. Flowers of sunflower, morning glory and tulips bloom during the day and close in the evening. The closing of venus flytrap respond to the walk of an insect.

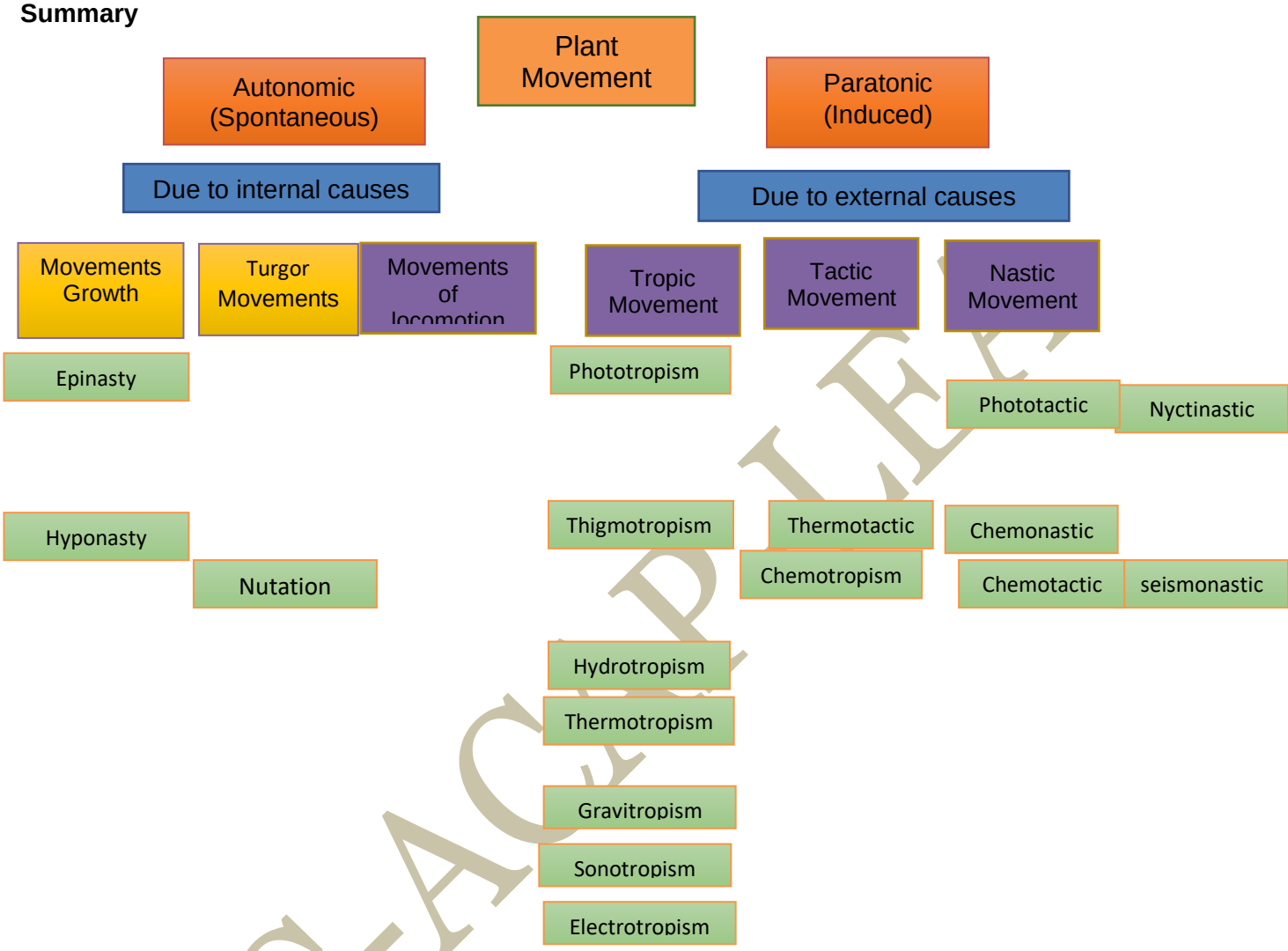
i. Nyctinastic- a nightly process for plants and flowers, close their leaves and petals of flowers at night and open during the day. Example: Prayer plant
ii. Chemonastic- The closing of venus flytrap respond to the walk of an insect.
iii. Seismonastic-

c. Tactic Movement

i. Phototactic- movement of whole organism in response to the stimulus of light.
ii. Chemotactic- movements occur in response to an external stimulus of some "chemicals".
iii. Thermotactic- movement of whole organism in response to external "heat" stimulus.

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Summary



Therefore, we see that the plant movements are the movements of the plant influenced by both internal and external stimulus.

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E. CROP ADAPTATION

Overview

This module discusses crop adaptation in plants. Plants are immobile. They can only occupy and utilize a limited expanse of their environment. Nevertheless, they can effectively compete for essential resources like light, carbon dioxide, water and mineral nutrients. They have mechanisms for moving water, minerals and photo assimilates. This is largely a result of its physiological evolution and its subsequent adaptation to the environment. The principles of plant response to environment, types of plant response to environment, and factors affecting ecological plant distribution will be discussed. Stress physiology and hormonal mediation of stress response are likewise explained.

Learning Objectives

At the end of the lesson, the participant should be able to:

1. define crop adaptation,
2. discuss the principles of plant responses to environment,
3. list down the types of plant response to environment and factors affecting ecological plant distribution, and
4. identify any change in environmental conditions resulting in a plant response that is less optimum (stress) and the hormonal mediation of stress response.

What is crop adaptation?

Crop adaptation essentially refers to the relationships between the principal environmental factors and the growth response of crop plants. The capacity of plants to adapt to their environment enables them to survive under wide range of changing environmental conditions. It involves physiological, morphological as well as genetic and developmental origin.

Principles of Plant response to Environment

The two principles that govern plant response to the environment are saturation and limiting factors. Organisms respond to any environmental parameter according to a common pattern. As the parameter increases, it reaches a threshold above which begins to have an effect, after which the response increases until the system becomes saturated by the parameter. Then, as the parameter's level or concentration continues to increase, the response remains constant or begins to decrease if at such high levels the parameter becomes toxic or inhibitory.

In physiological ecology studies; the distribution of a plant may be limited on one border by the factor present in "least" amount (law of minimum). The growth of a plant is dependent on the limiting factor or the amount of "foodstuff" present in the most minimum quantity. The limiting factor might be one or more of the following: mineral nutrients, water, damage by pests, and competition from weeds, CO_2 concentration or the plant's genes

Climate exerts a strong control over a plants requirement for nutrient, water, and energy, and on the plants resistant to parasites, salinity, flooding and other injuries factors.

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One factor that may compensate for or aggravate deficiencies in another factor (e. g., a humid atmosphere compensates for low rainfall). Plants may become predisposed to one injurious factor by being exposed to another (e.g., rice plants grown in diffuse light become weak and susceptible to rice blast).

The disturbance of one factor can trigger a series of changes connected to the disturbed factor (e.g., fertilization of a forest may increase the growth of the dominant species, increasing shading of the forest floor and resulting in the alteration of the vegetation of the forest floor).

The separate effects of two factors may not be the same when the two factors are combined. There is synergism (opposite response is antagonism) when the two factors interact in such a way that the response to a combination of the two is greater than the sum of the separate response for each factor.

For instance, fertilizing at a certain level of nitrogen may result in a 10% increase in yield, similar to the addition of phosphorus. Combining the N and P fertilizers, however, may give a 100% yield increase instead of 20%. That is a synergistic effect.

Ion competition is common in mineral nutrition. For example, Ca can antagonize the uptake rate of Mg (antagonistic effect) while it can stimulate influx of K^+ and Cl^- (synergistic effect).

Types of Plant Response to Environment

Crop adaptation involves physiological, morphological as well as genetic and developmental origin. Morphological adaptation involves the change in structure of plants that allow them to survive and reproduce under diverse environmental conditions. Morphological adaptations include changes in both the vegetative and floral structures in plants. Vegetative structures like roots, stems and leaves may develop modified structures that function differently such as for storage and other specialized functions. Changes in floral structures include separation of male and female flowers on the same or different plants, temporal separation by differential maturation of male and female flowers or organs, or remodeling of flower structure.

Plants develop aerial and air roots as one of their morphological adaptations to waterlogged conditions. Aerial roots are the ones that develop from aboveground structures. They are modifications of the roots that may function other than for absorption and anchorage. Air roots or also known as respiratory roots are roots that grow upward. These roots grow against gravity, or also known as negative gravitropic extensions, out of the mud to provide adequate aeration through the numerous lenticels and broad cortex composed of aerenchyma tissue. Some roots are also adaptations for food storage. These roots possess abundant storage parenchyma tissues that serve as storage organs such as in sweet potato and other root crops.

Plant responses to the environment can be classified into six categories:

1. Direct (non-delayed) - plant response almost immediately changes as environment changes (e.g.; photosynthesis vs. Irradiance);
2. Triggered (on-off) - plant response begins when an environmental factor crosses a threshold and continues even when the factor has returned to below the threshold level; such response is often delayed (e.g; germination in response to red light)

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3. Modulated (delayed) - plant response is determined by level of environmental factor; amplified, delayed and often interacts with biological clock (e.g., phototropism, photoperiodism);
4. Conditioning effects - plant response gradually changes with continuous exposure to the environmental factor(e.g.; development of drought resistance);
5. Homeostasis - internal conditions are constantly maintained in spite of changes of environment (e.g., internal concentration of growth hormones).
6. Carryover effects - effects of growth conditions carried over two or more generations; contends that the environment does not change the genes (e.g., experiments with inbred pea plants).

Factors Affecting Ecological Plant Distribution

Factors affecting ecological plant distribution are (a) water; (b) temperature; (c) light; (d) soil; (e) atmosphere; and (f) genetics.

Water. A region with an even rainfall distribution generally supports a more luxuriant and diverse vegetation compared to where rainfall is concentrated in the wet season. Evaporation, in the context of other considerations, such as relative humidity of the air, also affects ecological plant distribution.

Temperature. Temperature extremes, high and low, can cause cessation of growth, chlorosis and death. However, plants within all the major vegetation types, have developed adaptations to various temperature regimes. Seasonal variation and temperature periodicity also affect vegetation.

Dormancy is one of the mechanisms through which plants avoid extreme cold temperatures. Another mechanism is the thermo periodic response of stem elongation and flowering in tomatoes. These two growth processes proceed better when day temperature is 26.5^{o}C and night temperature is from 17 to 19°C than when temperature is at a constant 26.5^{o}C.

Light. Ecologically, plants can be classified into sun-loving and shade-loving plants.

Soil. The more important features of soil that affect plants are soil acidity (pH), soil moisture and aeration, soil solutes (including salinity) and soil microorganisms.

Some plants, such as legumes, are calciphytes. They grow best in basic soil that has high amounts of free calcium. They disappear from an area and the soil becomes acidic and free calcium becomes unavailable. On the other hand, acid loving plants (acidophiles) such as oxalis, cannot grow in calcareous soil. Plants that are referred to as halophytes, in contrast to glycopyhtes, thrive best in saline conditions. Some economically important plants like sugar beet, almond and alfalfa are facultative halophytes. They can be grown in both saline and non-saline soils

An area with an imbalance of nutrient elements results in a barren condition. An example of plants thriving under this condition is Rhizophora, a native of mangrove swamps.

Atmosphere. Within defined areas, the flora may be largely altered by the presence of natural or man-made pollution. Forests in areas around large mining and smelting

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operations have been killed or severely modified, rendering them useless for lumber or recreation, by sulfur dioxide (SO_2) fumes. The natural cut flow of hot underground water with high mineral content induces toxicity and /or altered plant water status in the flora

The continuing increase in atmospheric CO2 has resulted in ecosystem responses like reduction in tissue nitrogen concentration due to rubisco and soluble protein reduction. This is accompanied by reduced insect grazing and fungal infection. Increased production of fine roots and increase in methane production have also resulted due to elevated CO2.

Genetics. The distribution of a given species is presumed to be determined by its genetics. This may not be so if there is genetic diversity within species. Plants with similar genetic makeup that exhibit morphological and physiological differences due to varied natural environments are called ecophenes. They are called ecotypes if there are genetic differences in representatives taken from different areas of a species distribution of the population.

Different environments exert different selection pressures that result in different genetic compositions which are directly correlated with geography. Selection also works on the physiological responses to the environment. For instance, the sun ecotypes have several times as much RubisCO as the shade ecotype even when the latter are grown in the sun.

Stress Physiology

An understanding of the factors that limit the unfavorable environment is important. If resistance increases as a result of exposure to prior stress physiology, any change in environmental conditions resulting in a plant response that is less than optimum is considered stressful. Stress results to an aberrant change in physiological processes brought about by one or a combination of environmental (abiotic) and biological (biotic) factors.

Resistance

Resistance is the plant's fitness for the unfavorable environment. If resistance increases as a result of exposure to prior stress, the plant is said to be acclimated or hardened. Acclimation involves physiological responses that improve performance and survival by enhancing growth resource use efficiency and stress tolerance in the modified environment. It could be up or down regulation. On the other hand, adaptation is a genetically determined level of resistance acquired over generations by selection.

Drought stress

Drought stress is a multidimensional disturbance to plant growth that leads to altered functions at morpho-physiological, biochemical and genetic levels in plants. Biotechnological and other genetic engineering techniques have been employed to enhance tolerance to drought. Some of these strategies include the manipulation of the signaling pathways involved in the expression of genes conferring drought stress tolerance. Another plant adaptive mechanism is the decrease in leaf area as an early response to drought stress. When there is water deficit, stomates will close resulting in limiting photosynthetic rate in the chloroplast.

Plant Adaptive Mechanisms to Drought Stress

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1. Drought Avoidance. Drought avoidance is the ability of plants to maintain relatively higher levels of tissue water during soil water deficit periods. Plants have developed specialized features that enable them to overcome water loss such as stomatal characteristics, decreased number of stomata and sunken stomates in the epidermal cells to decrease water loss due to excessive transpiration; cuticular resistance, leaf area and leaf orientation.

2. Drought Tolerance. Drought tolerance is the ability of plants to continue normal physiological activities in very dry conditions for prolonged periods of time. It is mainly controlled by genetic factors. Crop plants with drought tolerance are able to combine high yield potential under optimum conditions with relatively good yield.

3. Drought Acclimation. Acclimation to drought stress is a major protective mechanism in plants. Initially, drought stress causes a build-up of environmental demand on plants causing an initial disturbance in its functions followed by a normalized activity and would result to resistance if the stress is survived. When plants are exposed to repeated drought cycles, enhanced drought resistance is indicated by the high leaf water potential, high turgor pressure, and high relative water content and surviving plants.

Salt Stress and Salinity

Salt injury involves both osmotic/water stress effects and specific ion effects (or nutritional imbalance). A too high concentration of salt lowers (more negative) osmotic potential (and soil water potential) preventing flow of H_2O and minerals to the roots. Overdose of fertilizer is an example wherein the treatment can result in wilted plants. This is also referred to as physiological drought.

Specific ion effects occur when injurious concentrations of Na^+, Cl^- or SO_4^{2-} accumulate in cells which may inactivate enzymes and protein synthesis.

Plants use different strategies to avoid salt injury. One is by exclusion of ions from leaves (e.g.; ions retained in roots) or by compartmentation of ions in vacuoles, an avoidance mechanism. Another is tolerance mechanism via osmoregulation wherein upon exposure to NaCl, the synthesis of new protein (e.g., proline) is induced.

1. Salt Tolerance. High levels of Na+ in the cells cause salt toxicity which impacts enzymes and impairs cellular metabolism (Hasegawa et al. 2000). It can also disturb K^+ absorption and content. Salt tolerant plants are able to maintain cytosolic Na+ and Cl^- ions in a low concentration with adequate K^+ resulting to ion homeostasis. Other adaptive mechanisms pf halophytes include osmotic adjustment (tissue tolerance), vacuolar membrane transport systems, antioxidant defense systems either enzymatic or non-enzymatic, and alterations of the membrane components.

2. Tissue Tolerance. Tissue tolerance is the plant capability to maintain tissue activity through accumulating high Na^+ in older leaves, where leaf senescence occur first, rather than in younger leaves. Moreover, salinity induces osmotic action which affects plant growth due to lowered water potential in the surrounding medium. In order to keep continuous water absorption, the cellular water potential needs to be

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reduced, relative to that of the soil solution, in order to provide water potential gradient between the soil and plant. This water potential gradient drives water entrance to the plant that helps in maintaining the cell turgor and elongation, which eventually promotes growth.

3. Vesicular Transport. Vesicular transport of ions is a widespread mechanism in eukaryotes. It involves conservation of exo- and endocytosis in cells. The transporters or ion channel depends on ion concentration near forming vesicle and electrical charge of vesicular membrane, while transport via intrinsic proteins is determined by its structure. This vesicular traffic of Na^+ and Cl^- is one of the adaptations of halophytes to salt resistance.

4. Antioxidant Defense Systems. High salinity induced the formation of reactive oxygen species (ROS) produced from the electron transport or energy transfer in the mitochondria and chloroplasts that cause oxidative stress in plants. It is therefore important that plant cells must produce antioxidant defense systems to combat the deleterious impacts of ROS. The antioxidant defense systems include either enzymatic or non-enzymatic species, targeted as tissue tolerance mechanism.

5. Salt Includers and Excluders. Transport of ions in halophytes can avoid the injurious effects of salts to cellular metabolism. Halophytes that accumulate large amounts of Na^+ ions, or also known as includes, pump Na^+ into the vacuoles by specific transporters so that accumulation of ions leading to damage is averted thus providing tissue tolerance. Other types of halophytes prevent intracellular Na^+ ion accumulation, they are termed excluders, by maintaining Na^+ in their roots and prevent its transport to the shoot system or transport it from the shoot to the roots via the xylem.

Flooding/ Waterlogging and Low Oxygen Supply

Roots are injured in soils where there is low O_2 (hypoxia) or absence of O_2 (anoxia). In this condition, root respiration is hampered and roots begin to ferment pyruvate into lactate. Roots also fail to absorb and transport nutrients

Acclimation to oxygen deficit involves (a) production of anaerobic stress proteins (e.g., enzymes of glycolysis and fermentation pathway) or (b) morphologically by stomatal closure, leaf shedding or slowing of growth. Some plants can of parenchyma (air spaces in the cortex resulting from the lysis of walls of adjoining parenchyma cell) as in roots of paddy rice.

Heat Stress

Few higher plants (e.g.; CAM plants) are able to tolerate temperatures above 40°C. At such temperatures, photosynthesis is inhibited before respiration. High temperature also impairs the thermal stability of membranes and proteins, and modifies membrane composition and structure causing leakage of ions.

Heat Shock proteins (HSP) induced by high temperature is involved in acclimation. They protect the plants against photo inhibition. HSP could be enzymes associated with the mechanism of thermal tolerance.

High Temperature on Physiological Processes

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1. Growth and Flowering. Heat stress affects plant growth, development, flowering and productivity. Heat stress can vary both in terms of intensity and duration. Effects of heat stress also vary in crop species and stage of growth. High temperatures generally reduce germination percentage and seedling emergence which leads to poor seedling establishment and poor crop stand. Heat stress may also cause reduced productivity by causing physiological injuries to flowers and fruits.

2. Senescence. Oxidative damage of cell membranes caused by reactive oxygen species (ROS) production under high temperatures provides a detrimental effect on chlorophyll and photosystems leading to senescence, a programmed cell death (PCD). In heat stress tolerant crops, there is reduced production of ROS coupled with a better antioxidant defense which plays an important role in stress tolerance.

3. Reproductive Physiology. Temperatures between 30 and 40°C cause significant damage to the key reproductive processes in different crop species. Heat stress affects pollen development, reduced pollen viability and germination thereby resulting to sterility and poor seed set. Significant reproductive damage can occur if maximum temperature is between 30 and 38°C in cereals, 40°C in millets, 36°C in oilseeds and 32–40°C in pulses.

Environmental Pollutants

Like CO_2, gasses such as SO_2, NO_4 and O_3 enter leaves through the stomata, NO_4 dissolves in cells and forms the toxic NO_2. SO_2 closes the stomata. The highly reactive O_3 can bind to plasma membranes and alter cellular metabolism.

Pollutive gases, when dissolved in rainwater, fall as acid rain (pH 3 to 4). Such acidity can release aluminum in soil and cause Al toxicity. Vegetation weakened by air pollution can become susceptible to invasion by the pathogens and pests. Some plants utilize detoxification mechanism to this type of stress.

UV Radiation

Ozone in the stratosphere (1 to 40 km layer above the earth's surface) blocks harmful radiation from reaching the earth's surface. However, chemical compounds released by industries (chlorofluorocarbons, etc.) destroy the ozone layer. If chlorofluorocarbons accumulate at the poles, they create a hole or gap that increases UV-B radiation 9280-320 nm). This radiation is powerful and has biological consequences. It is absorbed by nucleic acids, proteins and chlorophyll resulting to damage.

Too much UV-B radiation may result in: (a) reduced leaf area; (b) chlorosis and reduced photosynthesis (direct effect on PSII); and, (c) decreased reproduction and seed viability. In some ways, plants protect themselves by having thick cuticles or by producing flavonoids which absorb additional UB-V.

Heavy Metals

Mining activities increasingly contaminate the biosphere with heavy metals like Pb, Cd, Zn, Hg, Cu, etc. The most sensitive indicator of heavy metal toxicity in plants is inhibition of root elongation. Other effects of toxicity include impairment of enzyme (like rubisco and carbonic anhydrase) activities, chlorosis of leaves, and membrane damage. Plants can tolerate heavy

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metals through: (a) binding to cell wall; (b) restricted influx through plasma membrane; (c) compartmentation in vacuole; or (d) chelation in cytoplasm, membrane, or cell wall. The latter mechanism involves enhanced synthesis of phytochelatins, proteins that chelate or bind the toxic ions.

Heavy Metals/Metalloids effects on Physiological Processes

1. Root Plasmolysis

   Plants that are exposed to heavy metal toxicity usually exhibit discoloration and plasmolysis after entering the root cells. These plants develop lignified root cells and microtubule dissolution and later results to rapid cell amplification in cell width thereby increasing root diameter.

2. Biomass

   Plants exposed to heavy metal toxicity have reduced biomass accumulation. This is caused by increased permeability of the cell membrane which resulted to leakage of cellular constituents and fundamental nutrients and not being utilized for cellular respiration and plant development. Heavy metal toxicity also cause water loss and decrease water uptake.

3. Cell Division, Elongation, and Expansion

   Plants when exposed to heavy metal toxicity lead to inhibited cell division in the meristematic region of the roots. This reduction in mitotic activity leads to decreased rate of cell expansion and elongation in the higher root regions. Heavy metal toxicity also reduces root cellular turgor leading to inhibition in cell enlargement.

Compacted soil

Compacted soil restricts the volume of soil that the root system can exploit due to the high mechanical resistance of the soil. Soil compaction can also affect plant growth as a result of water stress and anaerobiosis (smaller pores) imposed by the high bulk density. This slows down water (low amount) movement and restricts oxygen diffusion. Plants cope up with such stress by osmoregulation and/or production of hormones (like ethylene) to tolerate the stress.

Light Stress

This is discussed in the previous module.

Nutritional Disorder

This is discussed in the previous module.

Clearly, plants can **tolerate** or **avoid** environmental stress through various mechanisms of biochemical, physiological or morphological acclimation.

Hormonal Mediation of Stress Response

Many reactions of plants to environmental stress involve morphogenetic changes similar to those that are brought about by hormones. This indicates that stress causes changes in phytohormonal balance. Water stress, for instance, increases ABA and ethylene but reduces

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cytokinins. These, in turn, influence plant development (e.g., reduction of shoot growth, closure of stomata, early ripening in cereal grains, induction of dormancy in seeds and buds, etc.)

Endogenous ABA level is largely a function of plant water status. Thus, variations in ABA level can be a tool in screening drought-tolerant cultivars of plant species such as wheat and rice.

Correlation analysis of rice cultivars that are salt-sensitive, mildly tolerant, and salt-tolerant indicate a positive relationship between ethylene production and salt tolerance.

Studies have been made on the alleviation of stress by exogenous application of PGR. These include:

1. application of cytokinin and/or gibberellic acid to increase shoot dry weights and reduce epinastic growth of petioles in waterlogged tomato plants;
2. application of ABA or its analogues in leaves of plants under water stress to induce stomatal closure and reduce transpiration (also to tolerate low temperature stress);
3. seed treatment and/or incorporation of growth regulators in the medium to reduce the effect of mechanical impedance on barley and pea growing in compact soil; and,
4. use of synthetic auxin, 3,5-dihalogeno-4-hydroxybenzoic acid to promote the light-induced inhibition of root growth and improve root growth in saline or waterlogged medium (mixing this chemical in irrigation water can improve root growth in paddy culture).

Synthesis of new proteins like HSP and/or activaton/inhibition of enzymes are commonly observed in stressed tissues. This is due to changes in gene expression at the level of transcription and translation, probably signaled by hormones

Additional readings:

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Summary

- adaptation of crop species is reflected in its responses to the combined environmental, edaphic and biotic features of its growing environment
- adaptation reflects the heritable features of a plant that enable it to survive, reproduce, or both, in a given environment
- acclimation reflects the non-heritable modifications to the plant phenotype caused by exposure to environmental stresses that enable it to tolerate further and more severe stress.
- plants can tolerate or avoid environmental stress through various mechanisms of biochemical, physiological or morphological acclimation.

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Properties of Soils Affecting Nutrient Availability

Overview

The term "nutrient availability" refers to the availability of nutrients for plant growth. This is determined by the number of components in the soil that can be easily absorbed through the roots of plants while simultaneously serving as important nutrients for growing plants.

Learning Objectives

At the end of this topic, the participant should be able to:

1. understand the role of soil structure to the plant nutrition,
2. comprehend the different soil chemical properties and its importance in soil fertility, and
3. identify biological factors that impacts nutrient availability.

1.2: Properties of Soils Affecting Nutrient Availability

The increase of the global human population in the last decades has been accompanied by an increase in demand for horticultural crops which resulted from the rapid rise in horticultural crop production, including the production of ornamental plants and cut flowers. This phenomenon has been made possible by the increased adoption of free marketed policies and trade agreements across the globe. At the same time, the global horticultural expansion has

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been enabled by continuous product improvement. The global floricultural expansion has been beneficial to many emerging economies, although many challenges remain.

1.2.1. Mineralogical and Physical Properties

Soil Structure

Soil structure is the result of how the soil separates arrange themselves to form stable, large soil particles that determine unique physical properties, such as tilth, ease of manipulation, water infiltration and drainage rates, air exchange, etc. Particle arrangement and soil structural stability are enhanced by the content of soil organic matter that results in particle formation aggregated for ease of water infiltration and air exchange. An established soil structure can be altered by soil manipulation and compaction.

There has developed a terminology associated with particular soil structural types—such as blocky, granular, prismatic, blocky and granular—that has aided the identification of soil structural systems. A system for identifying soil structure has been developed using 5 soil designations that are associated with particular soil characteristics:

Table 1. Characteristics of soil designations.

Soil Designation	Soil Characteristics
Loose	Easy to work; neither sticky nor malleable
Sticky	Adheres to hands and implements
Plastic	Smooth and malleable, subject to smearing when cultivated
Friable	Breaks into crumbs under gentle pressure; not sticky; does not smear
Firm	Not friable even under high stress; clear; cloddy when cultivated

The following are types of soil structure and associated soil characteristics:

- Single grain: loose incoherent mass of individual particles as in sands.
- Amorphous: coherent mass showing no evidence of any distinct arrangement of soil particles.
- Angular blocky: faces of peds rectangular and flattened vertices sharply angular.
- Subangular blocky: faces of peds subangular, vertices mostly oblique or subrounded.
- Granular: spheroidal peds characterized by rounded vertices.
- Platy: horizontal planes more or less developed. Prismatic: vertical faces of peds are well defined and edges sharp.
- Columnar: vertical edges near top of column are not sharp (column may be flat or round-topped or irregular).

Soil mineralogy

The chemical composition of a soil mineral is not always constant, it has a distinct chemical formulation and crystal structure. Soil minerals have long been used to assess surficial

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material weathering rates, establish the age of soils and sediments, describe soil formation mechanisms, and reconstruct paleoenvironmental conditions during pedogenesis.
In soil science, the term clay refers to three different ideas.

1. Soils with a clay content of greater over 50%
2. Clay fraction of the soil (i.e. particles <0.002 mm in size)
3. Clay Mineral

Clays, like micas, produce flat hexagonal sheets. Clay minerals are common weathering and low-temperature hydrothermal alteration products (including feldspar weathering). Soils, fine-grained sedimentary rocks including shale, mudstone, and siltstone, and fine-grained metamorphic slate and phyllite all include clay minerals.

Clays are formed when rocks (silicate minerals) and water interact. There are two types of amorphous clay minerals which is composed of alumina (aluminum oxide, Al2O3), iron (Fe oxides) and silica (silicon oxide, SiO2) that are not crystalline and crystalline (structured) that has ordered atomic arrangement. Example of amorphous mineral is allophane with 10 nm meanwhile kaolinite is a crystalline clay mineral because of the ordered atomic arrangement.

Silicate minerals are silicate-group minerals that produce rocks. They are the most common and important class of rock-forming minerals, accounting for around 90% of the Earth's crust. A silicate mineral is a type of ionic compound in which the anions are mostly silicon and oxygen atoms.

There are many types of silicate mineral wherein it includes the phyllosillicates (kaolinite-serpentine, smectites, vermiculite, and micas) and tectosilicates (zeolites, quartz and feldspars.

- Kaolinite, Halloysite and Dickite – 7 Angstrom, 1:1 type
- Illite and Glauconite – 2:1 type (non- expanding)
- Beidellite, Nontronite and Montmorillonite – 10-18 Angstrom, 2:1 type (expanding)
- Vermiculite, Stevensite and Saponite – 10-15 Angstrom, 2: 1 type (less expanding)
- Chlorite – 14 Angstrom, 2:2 type

Soil water

As with any scientific system, there develop terms that define characteristics that relate to the topic. The following terms are used to define classes of plant available water, available water, and soil moisture status associated with crop availability.

1. Classes of plant-available water:
   - Saturation: Gravitational water, rapid drainage
   - Field capacity: Capillary water, slow drainage
   - Permanent wilting: Hydroscopic water, essentially no drainage
2. Available water between field capacity and permanent wilting:
   - Unavailable water: That below permanent wilting
   - Field capacity: The amount of water remaining in the soil after gravitational flow has stopped
   - Permanent wilting point: Amount of water remaining in the soil when plants begin to wilt—water is held so tightly that plant roots cannot absorb.

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3. Soil moisture status related to plant availability:
    - Saturation percentage: 1/10 atmosphere near saturation
    - Field capacity: 1/3 atmosphere, 1/2 saturation percentage
    - Permanent wilting point: 15 atmospheres, 1/4 saturation percentage

The water content of a saturated soil (i.e., its saturation percentage) is about twice the field capacity and about four times the permanent wilting point. The relationships among saturation percentage, field capacity, and permanent wilting point generally hold consistent for all soils, from clay loams to sandy soils. Consumptive water use is the sum of two factors: transpiration (loss of water as vapor from leaf surfaces) and evaporation (loss of water from the soil surface).

1.2.2. Chemical Properties

A huge amount of plant nutrients is bonded in complex molecules in the soil, making them unavailable to plants. Plants may utilize the smaller fraction since it is in simpler, more soluble forms. Chemical weathering and biological activities progressively convert complex molecules into simpler molecules. As a result, a soil's chemical fertility is influenced in part by how quickly plants can obtain nutrients in the form they require. Chemical and electro chemical properties such as pH, salinity, sodicity and cation exchange capacity affects the nutrient availability.

Cation Exchange Capacity

Ions are electrically charged particles created when electrons are removed or added from an atom or molecule. A cation is an ion with a positive electrical charge while an anion is an ion with a negative electrical charge. The number of exchangeable cations per unit weight of dry soil, or the overall negative charge in a soil, is referred to as cation exchange capacity. It's measured in centimoles of positive charge (cmol) per kilogram soil or micro equivalents (meq) per 100 g soil. The nutrient-supplying capacity of a soil is can be determined by its cation exchange capacity, which determines the supply of nutrients to the developing crop. Moreover, this depends also upon to type of clay minerals present and amount of clay and organic matter.

Soil pH

It is possible for the soil solution to be neutral, acidic, or alkaline. The soil pH is the measurement of how acidic or alkaline the soil is. On a logarithmic scale ranging from 0 to 14, the pH measures the quantity of positively charged hydrogen ions (H+) in the soil solution. It is acidic when a soil solution contains more H+ ions. The soil solution is alkaline when there are less H+ ions [i.e., more hydroxyl (OH) ions].

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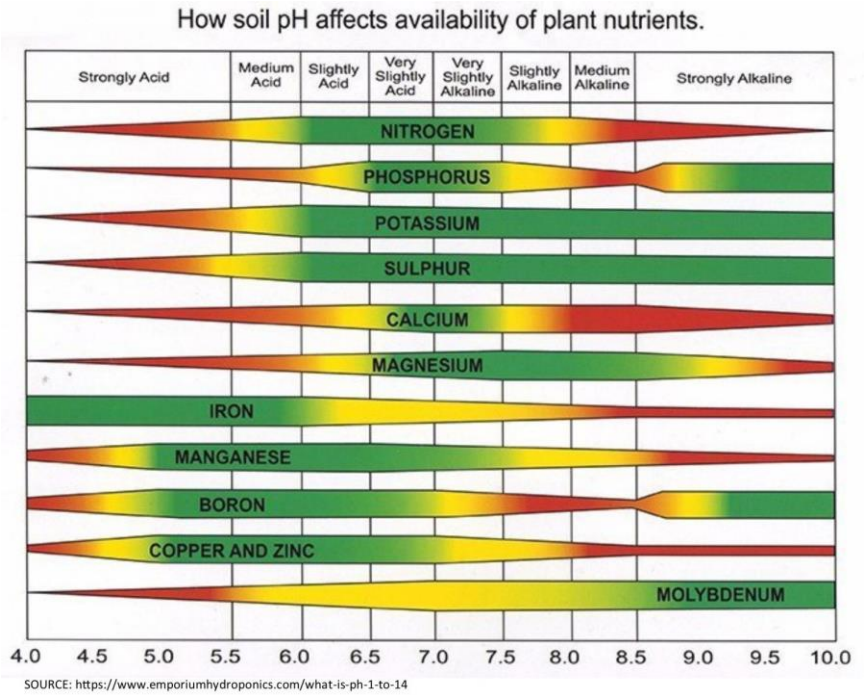


Figure 1. Effects of Soil pH on Nutrient Availability in the Soil.

Soil Salinity

A higher concentration of salts in the soil solution is known as soil salinity. As soil moisture is lost, mainly by evaporation, the concentration of soluble sodium, calcium, magnesium, and potassium salts in the soil solution rises. If the water table rises, these salts may already be present in the soil solution or they may be transported upward by capillary action from the ground water. Soluble salt concentrations can rise to the point where they obstruct plant growth. Saline soils are those that have a salt concentration in the plant root zone that is high enough to seriously impede plant growth.

Natural salinity is created by two natural processes that result in the accumulation of salts over time. The first is the weathering process, which breaks down rock and releases different types of salts, mostly sodium chloride, magnesium, and calcium, as well as sulphates and carbonates to a lesser amount. The most soluble salt is sodium chloride. The second is the wind and rain-borne deposition of sea salt.

Moreover, the Na^+ and Cl^- compete with nutrients like K^+, Ca2+, and NO3, salinity can create nutrition deficits or imbalances. N is one of the most growth-limiting nutrients in saline soils. High salinity and the Na concentration in the soil solution can impact plant potassium absorption. There's plenty of evidence that Na and the Na/Ca ratio influence K uptake and accumulation in plant cells and organs, and that salt tolerance is linked to K selectivity over Na uptake.

1.2.4. Biological Characteristics

Many living organisms are found in healthy soil, from large creatures, such as earthworms, to the smallest bacteria. Soil organisms help to decompose organic matter. The burrowing habit of the larger organisms incorporates the organic matter into the soil and also

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creates large pore spaces that aerate the soil and allow faster water infiltration. The smaller organisms, such as bacteria, actinomycetes, fungi, yeasts, algae and protozoa, further decompose the organic matter, which releases nutrients in a form that plants can use.

Soil Organic Matter

Organic matter is anything that is living or the remains of a living thing. However, in the context of soil composition, organic matter is a build-up in the soil of decayed plant and animal residues.

Organic soils, such as peats, contain from 20% to as much as 95% organic matter. Mineral soils contain anywhere from a trace to 15% or 20% organic matter. Organic matter is composed of about 57% organic carbon.

The benefits of organic matter in the soil include improving soil structure and increasing the nutrient and water holding capacity of the soil. Organic matter also provides a food supply for soil biology. Soils with low organic matter can have 'poor' structure, hold little water, and erode or leach nutrients easily. The exception is cracking clay soils where clay minerals have the main effect on structure. Soils with high organic matter levels have 'good' structure, good water-holding capacity, and reduced erosion and nutrient leaching.

Humus

A portion of soil organic matter exists as humic compounds. These compounds are defined as amorphous, colloidal, polydisperse substances and yellow to brown black in color. They are hydrophilic, acidic, and high in molecular weight, ranging from several hundreds to thousands of atomic units. Humus is the primary substance contributing to the cation exchange capacity (CEC) of a soil. It can also act as a chelate in the soil solution, thus affecting root ion absorption.

Adverse Effects of Organic Matter Additions

The frequent addition of organic materials, such as composts, animal manures, sewage sludge, organic waste products, etc., to a mineral soil, with time, have an adverse effect. With increasing organic matter content, a soil will have higher water-holding capacity, will remain cool with warming air temperatures and make the soil more difficult to cultivate when wet. Depending on the source of the organic material, elements may be added that would accumulate, and with time adversely affect the nutritional status of the soil and growing crop. For example, poultry manure contains substantial quantities of K that, with time, will adversely alter the cation ratio in the soil, eventually inducing either an Mg or Ca plant deficiency. Some organic waste products contain elements, particularly heavy metals, that can become toxic to plants or adversely affect the quality of produced product. It is advisable that an organic material before being added to a soil be assayed to determine its elemental content.

Crop Response to Fertilization

Overview

Plants require inorganic nutrients for normal growth, which has been known since the 18th century. Fertilizer became one of the requirements for maximizing the crop's productivity

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potential. Increasing agricultural production is one of the most important difficulties facing the world's efforts to achieve food security and poverty reduction.

Learning Objectives

At the end of the topic, the participant should be able to:

1. understand the effects of fertilizers to plants, and
2. know the concept of different nutrient interactions.

Discussion 1.3: Crop Response to Fertilization

1.3.1. Nutrient Response Curve

Models of crop production responses to applied nutrients are a relatively new addition to the approaches for recommending fertilizers. They have a role to play in combining nutritional data with data on other factors that influence yield and its response to nutrients added. It is common to see a systematic association between dietary concentrations or intake amounts of a nutrient and the degree of reaction. A model that takes these facts into account is needed to describe the nutrient-response relationship.

For thousands of years, mineral nutrients in various forms have been administered to crops. Science has tackled the problem of identifying important nutrients and suggesting the amounts, types, and timing required for optimal crop growth in the twenty-first century. More recent and advanced methods have sought to give recommendations for particular crops using tests such as soil-incubation tests, counts of tiller density, tissue tests, and cropping history a crop.

Crop simulation is the process of (1) testing on separate and preferably different data sets and, if necessary, recalibrating or reformulating (2) building a model based on existing knowledge of processes and (some) parameters. (4) industry problems will be applied with models (3) calibrating other parameters using a collection of field data and reformulating the model if the fit is poor

Simple regression-based models that are completely empirical are included in response-curve prediction models. This group of models differs from those below in that they are static, that is, they treat crop growth as a single step in which yield = potential is calculated using an integrated measure of the environment such as total rainfall.

1.3.2. Nutrient Interaction Effects

In order to realize the full potential, all plants require a sufficient supply of critical major, secondary, and micronutrients. No nutrient, however, should be considered secondary in importance because they all play a role in enhancing output or enhancing quality. Justus von Liebig demonstrated this principle with his Law of Minimum, which asserts that a single nutrient deficit is sufficient to reduce production.

When one nutrient influences the absorption and use of another, it is called a nutrient interaction. Interactions have been seen in the soil, at the root's surface, and within the plant. Other nutrient interactions have a less evident impact on crop health, such as boosting drought tolerance or susceptibility to insect damage.

There are numerous interactions that can have an impact on availability. When high quantities of a particular nutrient in the soil interfere with, and have a negative influence on, the

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uptake of other nutrients by a growing plant, these must be considered. Antagonistic nutrients are those that function in this manner.

High potassium (K) or ammonium-based fertilizer treatment rates for cool-season grasses reduces magnesium (Mg) intake, resulting in Mg deficient symptoms in grazing animals. Because calcium (Ca) and other cations compete for plant uptake, high rates of K fertilization may inhibit calcium (Ca) and other cation uptake.

In comparison to when either nutrient is utilized alone, positive interactions between nitrogen (N) and phosphorus (P) often result in improved crop yields and enhanced P uptake. When an ammonium-based fertilizer is used in conjunction with a P-based fertilizer, the P absorption and yields are generally increased.

On the other hand, P fertilizer can remove sulfate and molybdate from the surface of soil minerals, increasing their availability for plant uptake, due to their chemical similarities.

Although the interaction between P and zinc (Zn) has been well documented, the link between the two is frequently misunderstood. Some plants may respond to high P levels by reducing Zn uptake and yield, but this is not a universal reaction and relies on the crop and growing conditions.

In plant proteins, the correct ratio of nitrogen (N) to sulfur (S) falls within a rather narrow range. Single applications of S or N fertilizer can alter this ratio, resulting in lower growth and yield and even harvest quality.

Below are the effects of excess plant nutrients on imbalances of soil nutrients and on growth of plants.

Table 2. Nutrient Imbalances and its effects on yield quality.

Excess	Imbalance	Effects on Crops
Nitrogen	Copper and Sulfur	Softens Fruit and Poor Yield
Phosphorus	Iron and Zinc	Poor Yield and Fruit Quality
Potassium	Calcium and Magnesium	Poor Yield and Fruit Quality
Calcium and Magnesium	Potassium	Poor Yield and Fruit Quality

Soil Fertility Management

Module Overview

Soil fertility refers to a soil's ability to support plant growth and development by supplying required essential nutrients as well as desirable chemical, physical, and biological characteristics as a growing environment. The macronutrients nitrogen, phosphorus, and potassium, as well as sulfur, calcium, and magnesium, are all found in plants. Boron, chlorine, copper, iron, manganese, molybdenum, and zinc are all micronutrients.

A fertile soil includes sufficient soil organic matter, a pH range that encourages growth, and a thriving community of microbes that engage in nutrient cycling. Long-term productivity of managed ecosystems depends on practices that maintain or improve soil fertility.

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Learning Objectives

At the end of the topic, the participant should be able to:

1. identify nutrient deficiency symptoms in plants,
2. diagnose soil fertility issues and propose possible solutions, and
3. implement soil management techniques.

Discussion 1.4: Soil Fertility Management

1.4.1. Field Techniques for Deficiency Diagnosis

In crop fields, visual signs are an important diagnostic tool for determining nutrient deficiencies. However, visible nutrient shortage signs in field crops might be mistaken with other issues such as diseases or pest damage. It's critical to seek for specific signs that indicate vitamin deficiencies. Low nutrient levels in the soil are the most common cause of nutritional deficits in field crops. Consider nutritional issues in connection to all circumstances impacting plant growth, not just the levels of nutrients present in or supplied to the soil.

The presence of sufficient amounts of plant nutrients in the soil does not guarantee that they will be absorbed by the roots. It's possible that nutrients are present. Plants may be unable to absorb nutrients in forms that are not available to them, or other factors may prohibit them from doing so. The availability of nutrients can be influenced by unusually low or high soil pH levels.

Nutrient Mobility

Plant nutrients are divided into two categories according to its mobility in plants: mobile and immobile. Mobile nutrients are usually translocated to the growing sections when intake is limited, leaving a deficiency symptoms.to the older leaves. Immobile nutrients, on the other hand, cannot be translocated within the plant, resulting in deficits in the younger leaves.

- Mobile nutrients: Nitrogen, Potassium, Phosphorus and Magnesium
- Immobile nutrients: Sulfur, Calcium, Iron, Molybdenum, Manganese, Boron and Zinc

Deficiency Symptoms

- Nitrogen - Nitrogen is a mobile nutrient in plants, symptoms appear first on the oldest, lowest leaves and proceed up the plant if the shortfall is not corrected. Growth of lateral shoots is slowed and restricted. Plants show overall chlorosis (light greening of the entire plant) and yellowing of older leaves, which progresses to younger ones. Older leaves become necrotic and prematurely defoliate. Corn plants with spindly stalks or stems are caused by nitrogen shortage and are pale, yellowish-green in color.
- Phosphorus - The deficient plants are smaller and grow slower than phosphorus-abundant plants. Growth has been slowed. Some plants' elder leaves have a purplish color. The tips of the leaves are decaying and have a dark green color. Maturity is delayed, and the fruit and seed development is poor. When plants begin to close their canopy, signs of deficiency frequently subside.
- Potassium - Leaf edge chlorosis on new developed leaves can be followed by interveinal blistering and necrosis from the leaf edge to the midrib as potassium deficit progresses. Even if plants are supplied potassium, the chlorosis caused by potassium deprivation is

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irreversible. Potassium is required in substantial levels by most crops. Starting with the lowest leaves, the insufficiency manifests itself as a yellowing and necrosis of the leaf edges. Symptoms usually take a while to show after planting.

- Calcium - It is a component of plant cell walls that gives them structural strength. New foliage, buds, and roots with reduced growth are examples of deficiency symptoms. Younger leaves curl downward and brown at the edges and tips, a condition known as tip burn. They may also have unusually green foliage in some plants. The roots become stubby and short.
- Magnesium - It is a component of the chlorophyll molecule's structure. Early signs include yellowing of the leaf tips, which progresses interveinally toward the base and midrib, creating a herringbone pattern. Interveinal chlorosis is a condition in which older leaves become chlorotic between veins. Plant development slows, leaf size shrinks, and lower leaves are shed in severe deficit.
- Sulfur – The sulfur deficit results in a broad yellowing of the leaf, similar to nitrogen deficit, and can occur when the plant is rapidly accumulating biomass. Plant stunting and delayed maturity are additional signs. The youngest leaves may develop interveinal chlorosis.
- Iron - Its insufficiency is comparable to Magnesium insufficiency, although it affects young leaves and shoots rather than older leaves. On newly developing leaves and young shoots, there is a light green to yellow interveinal chlorosis. Shoots that die from the tip inwards are common. Newly emerging leaves may shrink in size and turn practically white, with necrotic patches, in severe cases.
- Manganese - It works as a nitrogen assimilation enzyme activator. A scattered interveinal chlorosis with weakly defined green regions around the veins can be seen in newly emerging leaves. Symptoms such as chlorosis and necrotic spots are typical. When there is a severe shortage, young leaves shrink and tip dieback can ensue.
- Zinc - Plant growth regulators, such as Auxin and Indole Acetic Acid, require zinc to activate (IAA). Younger leaves have chlorosis, bronzing, or mottling. Young leaves develop interveinal chlorosis, which is followed by restricted shoot growth, short internodes, and small, discolored leaves, giving the affected area a rosette appearance.
- Boron - It is taken as borate by plants from the soil. Plants become stunted and distorted as a result of this. As the main stem dies, side shoots known as "witches' broom" proliferate, ensuring that the lateral shoot's growth remains doormat. Loss of apical dominance is the term for this. New growth on blooming shrubs turns dark green, with cupped or puckered little brittle leaves with short internodes.
- Molybdenum - In most plants, molybdenum deficiency symptoms are linked to a build-up of nitrate in the damaged plant component. This is due to insufficient nitrate reductase activity. Reduced maize tasselling, pale leaves with interveinal and marginal chlorosis (yellowing) and necrosis (scald), and premature maize grain germination are all possible symptoms.
- Chlorine - Chloride is engaged in a number of important plant processes, including the chemical breakdown of water in the presence of sunshine, the activation of numerous enzymes, and the movement of cations (K, Ca, and Mg). Curling of leaflets, wilting of leaf tips, and chlorosis have all been reported in certain crops.

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- Copper - The earliest evident symptom of Cu deficiency may be chlorosis, wilting, and drooping of adult leaves. The chlorosis is interveinal, with color decreasing gradually as one moves away from the main veins. The lesser veins are usually less green than the main veins, although they may be strongly defined enough to give the chlorosis a mottled appearance. Necrosis can appear on chlorotic leaves in areas or patches that progress until the entire leaf is dead.

1.4.2. Organic and Inorganic Fertilizers

A fertilizer is a substance that provides essential nutrients to plants in order to boost their growth, output, or nutritional value. Fertilizers contain nutrients that all plants consume in big quantities.

Organic fertilizers, such as manure and plant waste, are by-products of everyday living. They deliver nutrients in a slow-release form that lasts longer in the soil. The nitrogen is usually in a complicated organic structure that is difficult to dissolve in water.

Inorganic fertilizers are usually far more concentrated than organic fertilizers, which are made from a variety of sources. The available nutrients are usually quite soluble, and if they aren't provided appropriately, they can quickly drain from the soil.

Organic Fertilizers

- Chicken Manure - The first stage in the effective operation of a manure handling and utilization system is to understand how much manure and plant nutrients are produced on a chicken farm. The digestibility of the ration, animal age, amount of feed wasted, amount of water lost, amount of bedding used, and number of times the chicken house is cleaned in a year all influence the nutrition content of chicken dung. Plants require all 13 necessary plant nutrients, which are all found in chicken dung. The nitrogen concentration of this manure is usually high.
- Cattle Manure - Using cattle manure to fertilize farms is a cost-effective and ecologically friendly way to boost crop yields. Cattle manure contains nutrients that can be used instead of commercial fertilizers. Cattle manure is a fantastic soil supplement for improving soil quality. Manure can boost crop yields by adding a lot of nutrients and organic matter to the mix. The benefits of the minerals and organic matter may not be apparent right away. It is usually high in nitrogen and potassium which are essential for plant growth and development.
- Vermicast - Vermicomposting is the process of turning organic waste into compost using earthworms. Vermiworms eat organic materials and subsequently excrete casts, which are dark, odorless, nutrient- and organically rich soil mud grains that are good soil conditioners. Vermicasts are a ready-to-use fertilizer that may be applied at a higher rate than compost because nutrients are released at rates that are preferable to growing plants. The nutrition content of vermicast varies depending on the earthworm meal used. At 75% to 80% moisture content, it normally contains 1.5% to 2.5% nitrogen, 1.25% to 2.25% P_2O_5, and 1% to 2% K_2O.
- Guano - The manure of seabirds and bats is referred to as guano. Because of its extraordinarily high level of nitrogen, phosphate, and potassium, guano is a highly effective fertilizer as manure. Guano was also used to make gunpowder and other explosive materials, but to a lesser extent. Guano is crucial to the environment because

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it helps to disperse nutrients. Bats' guano, which nourishes bacteria, fungus, invertebrates, and vertebrates, is often entirely reliant on bats for nutrition in cave environments. When bats are removed from a cave, animals that rely on their guano may become extinct. Guano also helps to shape caves by causing erosion due to its high acidity.

Inorganic Fertilizer

Nitrogen Fertilizer

- Urea [$CO(NH_2)_2$] contains 45–46% N. Nitrogen has excellent physical qualities, such as being less corrosive than other nitrogen fertilizers and having a lower tendency to cake than ammonium nitrate. Because of its high nitrogen content, it saves money on storage, transit, handling, and application. Urea is a liquid substance that can leach just as easily as nitrate. Initially, urea can enhance soil pH in the application zone due to the release of NH3, but over time, due to the nitrification of NH_4^+ to $NO3^-$, soil pH might fall below the original pH. Biuret, an impurity that can be phytotoxic, can be found in different levels in urea.
- Ammonium phosphates, with 11–13 % nitrogen and 48–62 % P_2O_5, provide both nitrogen and phosphorus. Diammonium phosphate (DAP) [$(NH4)_2HPO_4$] is another common source of both N and P, with 18–21% N and 46–53% P_2O_5. Both fertilizers are fully soluble in water.
- Ammonium sulfate [$(NH_4)_2\ SO_4$] has been utilized in the pineapple and sugarcane industries, as well as small farms, for many years. It has a nitrogen content of 21% and a sulfur content of 11%. If used on a regular basis over an extended length of time, it will reduce the pH of the soil.
- N (13%) and K_2O are two important elements found in potassium nitrate [KNO_3] (44 %). It is easy to apply since it is not hygroscopic (that is, it does not absorb moisture from the air), the NO_3^- is readily available, and it generates a minor elevation in soil pH.
- Calcium nitrate [$Ca(NO_3)_2$] contains 15% nitrogen and 34% calcium oxide. The NO_3^- is freely available, however it is a very hygroscopic substance. If not stored in moisture-proof packaging, it is prone to liquefaction.

Phosphorus Fertilizer

- Superphosphate [$Ca(H_2PO_4)_2$]. It has two types which is the ingle superphosphate (SSP) with 16–22% P_2O_5 and triple superphosphate (TSP or CSP) with 44–52% P_2O_5. Single superphosphate is created by processing rock phosphate with sulfuric acid, resulting in a S content of 11–12%. Triple superphosphate, on the other hand, is created by acidulating rock phosphate with phosphoric acid, resulting in a S content of only 1–1.5%. Both of these fertilizers contain phosphorus that is easily available to plants. These are pH-neutral fertilizers, meaning they have no influence on the pH of the soil.
- Rock phosphates [$Ca_{10}OHF(PO_4)_6$] contain 27–41% P2O5, which is slowly accessible to plants but lasts for many years in the soil. Apatite, which varies significantly in composition and solubility and is only available to plants in acid soils (<pH 6.0), is found in rock phosphates. With acid soils, rock phosphates should be

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broadcast and thoroughly mixed to provide maximum contact of the fertilizer with the soil, allowing for faster dissolution and increased P availability to plants. On certain long-term crops, such as pastures and perennials, rock phosphate can be highly beneficial.

- Ammonium Phosphate. Both nitrogen and phosphorus are provided by monoammonium phosphate, which contains 11–13% nitrogen and 48–62% P_2O_5. Another typical source of both N and P is diammonium phosphate (DAP) [$(NH_4)_2HPO_4$], which contains 18–21% N and 46–53% P_2O_5. Both fertilizers are completely water soluble.

Potassium Fertilizer

- Potassium nitrate [KNO_3] supplies 44 percent of the K_2O and 13 percent of the nitrogen in the body. It has a low hygroscopicity, is easily soluble, and raises the pH of the soil.
- Potassium chloride [KCl], also known as muriate of potash, is entirely water soluble and contains 60–63% K_2O. It's the most common potassium fertilizer on the market.
- Potassium-magnesium sulfate (K_2SO_4. $2MgSO_4$) contains 22 22 percent potassium, 11 percent magnesium, and 22 percent sulfur. It's a common ingredient in dry fertilizer formulas.
- Sulfate of potash is potassium sulfate [K_2SO_4], which comprises 50–53% K_2O and 17% S. It is used on crops that are susceptible to significant amounts of chloride (Cl–), such as potato and avocado. It is entirely soluble in water.

Calcium Fertilizer

- Calcium sulfate, gypsum [$CaSO_4.2H_2O$] is an amendment that provides calcium in a form that has little effect on soil pH, making it beneficial in soils with a pH that is suitable for plants. It has a calcium content of 23% and a sulfur content of 19%.
- Calcium nitrate [$Ca(NO_3)_2$] is a highly soluble compound that contains 15% nitrogen and around 20% calcium. It's useful when you need a lot of calcium and don't want to affect the pH of the soil too much.
- To alter soil pH and supply calcium, liming materials such as lime [CaCO3] and dolomite [$CaMg(CO_3)_2$] are utilized. Dolomite has around 22% calcium and 12% magnesium, while lime has about 38% calcium and 12% magnesium. The amount of Ca and Mg in the substance varies depending on the source.

Magnesium Fertilizer

- Magnesium oxide [MgO] is a 55% magnesium compound that raises soil pH. Because it is not easily soluble, it should almost always be absorbed into the soil.
- Magnesium sulfate, Epsom salts [$MgSO_4.7H_2O$] provides both Mg (9.8%) and S (13%) without affecting the pH of the soil. It's easily soluble and can be used as a foliar spray or in the soil.

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- Dolomite [$CaMg(CO_3)_2$] is a liming material that contains both Ca and Mg in varying proportions. One dolomite marketed in Hawaii contains around 22% calcium and 12% magnesium. Ground coral having 38 percent calcium and 0.6 percent magnesium is commonly sold as agricultural lime in Hawaii.

Sulfur Fertilizer

- Ammonium sulfate [$(NH_4)_2SO_4$] contains 24% sulfur in addition to nitrogen, and is used to provide sulfur to soil and plants. Because both ammonium and sulfate contribute to acidity, this is one of the most acidifying fertilizer components.

Micro Nutrient Fertilizer

- Iron chelates, iron EDTA [NaFeEDTA] is a foliar spray or soil application that includes 5–14% Fe.
- Iron (ferrous) sulfate [$FeSO_4.7H_2O$] contains around 19% Fe and can be applied to the soil or utilized as a foliar spray (1–2% $FeSO_4$).
- Zinc chelates, zinc EDTA [$Na_2ZnEDTA$], is a foliar spray that contains around 14% zinc. It can also be used on the ground.
- Zinc sulfate [$ZnSO_4.H_2O$] has a Zn content of about 35%. Zinc sulfate should be disseminated and absorbed into the soil or applied in a band because Zn is relatively immobile in soil. Foliage can also be treated with zinc sulfate.
- Copper chelates, copper EDTA [$Na_2CuEDTA$], is a soluble copper chelate that contains around 13% copper. It's usually applied as a foliar spray.
- Copper sulfate [$CuSO_4.5H_2O$] has a copper content of about 25% and can be applied to the soil or foliage.
- Sodium borate, often known as borax [$Na_2B_4O_7.10H_2O$], is a mineral that contains around 11% B and can be used in the soil or as a foliar spray. Because B can become hazardous at high concentrations, application rates vary depending on the crop and soil.

1.4.2. Nutrient Management Practices

The use of adequate fertilizers in a balanced manner is an important part of nutrient management. Fertilizers must be applied at the appropriate rate for optimal crop growth, taking into account crop requirements and agroclimatic factors. Negative externalities, on the other hand, should be reduced.

Soil Testing

To effectively manage soil nutrients, optimize productivity, and save expenses, soil testing is required. Soil tests are important for determining the state of soil fertility, although accuracy necessitates caution when sampling fields. The samples should represent a pretty homogenous area. A soil sample should be made up of 12-15 sub-samples collected from all around the field. It's critical that the sampling region is sufficiently uniform. Separate samples

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should be taken from areas with varying soil textures, slopes, or cropping or fertilization histories. The size of the field isn't as essential as the variability within it.

Soil Test Results Interpretation

The results of soil tests reveal the state of numerous chemical aspects of the soil. Nutrient levels, soil pH, buffer pH (some laboratories don't disclose this, but it's used to figure out how much lime you need), cation exchange capacity (CEC), and percent base saturation are among them. The amount of lime to apply will be determined by the pH of the soil and the crop(s) to be cultivated, that will be based on lab results. It also covers the nutrients in the soil that are deficient.

Lime Application

The pH of a soil is a good chemical indicator of its quality. Liming to adjust pH to the levels required by the crop to be grown can enhance the soil quality of acid soils. Carbonates, oxides, and hydroxides of Ca and/or Mg are commonly used as liming materials. It is almost always necessary for liming material to be a salt in order for it to be cost effective. Because a hydroxyl (OH^-) is not released to neutralize H^+ by creating water, a strong base and strong acid such as NaCl (sodium chloride) or $CaSO_4$ (gypsum) are ineffective in elevating pH levels (H_2O).

Application of Organic Fertilizer

Organic fertilizer is an important part of agricultural sustainability since it helps to improve soil fertility. It provides the advantages of being inexpensive, enhancing soil structure, texture, and aeration, boosting the soil's ability to retain water, and stimulating healthy root development. Minerals, animal sources, sewage sludge, and plants are all sources of organic fertilizer. Vegetables, animals, and residual materials all played a role in increasing soil organic matter content.

Inorganic Fertilizer Application

Soil provides the majority of the nutrients that plants require. Plant growth may be limited if any of these nutrients are in short supply. To meet plant demands, nutrients are recycled from plants to soil in normal settings. Agricultural crops, on the other hand, may demand more nutrients than wild vegetation. Only the correct amount of inorganic fertilizer will benefit the soil. To avoid overapplication of inorganic salts, it is highly recommended that inorganic fertilizer will be used based on soil testing or analysis.

Zero/Minimum Tillage

Soil disturbance is maintained lower using minimum tillage. Season after season, no soil disturbance is made in a zero tillage or no-till system. Increased organic matter, water retention, nitrogen cycling, microbial activity, soil fertility, and reduced soil erosion are all benefits of these methods. The smallest amount of soil disturbance ensures that the soil's microscopic organisms thrive and multiply. Their actions in the soil to break down materials are boosted. The nutrients acquired by previous crops are returned to the soil when the residues of those crops are allowed to decompose and mingle with the soil.

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III. FACTORS AFFECTING CROP PRODUCTION

To achieve optimum crop productivity, it is important that students learn and understand the different factors that affect crop production. In this module, factors affecting crop production such as environmental and genetic factors will be discussed.

At the end of the topic, the participants should be able to:

- Learn the different environmental factors that influence crop production.

A. ENVIRONMENTAL FACTORS

This includes abiotic factors (climate or above-ground factors and edaphic or soil factors) and biotic factors (pests and beneficial organisms).

A. Abiotic Factors

Climatic factors

Definition of terms:

Climate - the seasonal pattern of a particular place occurring from year to year.
- a composite of day-to-day weather conditions described in both averages and variability.

Weather - a momentary state of the atmosphere brought about by the combination of elements, e.g., temperature, pressure, moisture content, air movements, radiation, etc.
- day-to-day changes of the state or condition of the atmosphere

Macroclimate - the climatic environment one meter above the plant canopy

Microclimate - generally refers to the climatic environment one meter below the canopy in the case of tall plants or the climate within the leaf canopy for short (below one meter) plants.

The Climatic Elements:

1. Precipitation - is any form of water particles falling on the ground in liquid or solid form (rainfall, hail, snow, etc.)

Average rainfall in the Philippines = 2553 mm
- Luzon – 2724 mm;
- Visayas – 2391.7 mm;
- Mindanao – 2349.8 mm

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Role of water in plants

a. as a reactant in many biological reactions
b. enters into the structure of biological molecules
c. serve as medium of transport of nutrients and other substances
d. helps regulate plant temperature

Categories of plants based on need for moisture

a. xerophytes - desert plants
b. hydrophytes - aquatic plants
c. mesophytes - land plants; most economically important plants

Factors affecting amount and distribution of rainfall:

- Topography greatly influences the amount and distribution of rainfall
- Mountain ranges present barriers to clouds, causing them to rise to higher elevations and generally colder temperatures causing vapor to condense and water to fall on the windward sides as the clouds pass over, leaving the leeward side relatively dry. Example is Los Banos in Laguna and Sto. Tomas in Batangas. Sto. Tomas is drier than Los Banos due to the presence of Mt. Makiling.
- Air circulation patterns affect the seasonal distribution of precipitation.

Rain formation requires:

a. high relative humidity (RH)
b. sufficiently low temperature (below condensation point)
c. condensation nuclei
d. sufficiently low pressure

Drought – insufficiency of rainfall/moisture which seriously affect plant growth.

- Absolute drought – 29 consecutive days without rainfall of at least 0.25 mm.
- Partial drought – 15 consecutive days without rainfall of at least 0.25 mm.

2. Temperature - the degree of hotness or coldness of a body
   - every chemical, physiological and biological process in plants is influenced by temperature.

Three (3) cardinal temperatures:

a. minimum temperature – that temperature below which the velocity of the reaction becomes zero, due to the deactivation of enzymes.
b. optimum temperature – temperature where the velocity of the reaction is at maximum.
c. maximum temperature – that temperature above which the velocity of the reaction becomes zero, due to the desaturation of enzymes.

Temperature of the environment depends upon:

a. solar radiation – vertical rays are more energy efficient/unit area than oblique rays (in polar regions).
b. surrounding land masses or bodies of water.
c. altitude – for every 100-meter rise in elevation, there is a 0.6°C decrease in temperature.

In the Philippines: high elevation ➔ 13. 2 – 24.6°C
low elevation ➔ 23.3 – 31.5°C

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Classification of crops according to temperature requirement:

a. cool season crops – e.g., cole crops like cabbage, broccoli, cauliflower
b. warm season crops – e.g., rice, banana
c. tropical – e.g., coconut
d. sub-tropical – e.g., citrus

Effect of temperature on crops

- vernalization requirement of certain crops for flowering (e.g., celery seed exposed to 4.4-10°C for 10 days under imbibed condition)
- effect on crop maturation

3. Wind or air in horizontal motion

- normal wind speed in the Philippines = 7.2 km/h
- at 30 km/h = leaf tearing may already occur especially in banana and abaca

Effects of wind on plants

a. increase transpiration
b. destructive effects of strong winds, typhoon (e.g., crop lodging, grain shortening)
c. sterility due to loss of pollens
d. disease spore dispersal
e. reduced CO_2 levels especially in enclosed spaces
f. affects plant form

- Air circulation in the atmosphere results from the sun's radiation falling more directly on the tropical regions than on the polar regions, the warmer air rises and flow forward the poles, cools and sinks as cold polar air and then returns toward the equator as ground flow
- The interactions cause the establishment of regions, large and small, each with a different climate.

4. Solar radiation or light - energy given out by the sun through radiation

Three aspects important to plants:

a. light intensity
   - expressed in foot-candle or lux.
   - plants are generally spaced so that maximum leaf area is exposed to sunlight
   - some plants do not require high light intensity (shade-loving) because they have low light saturation point.
   - some plants require subdued light to survive. e.g., some ornamentals

b. duration or daylength – expressed in hours per day

c. wavelength – expressed in Angstrom or nanometers or identified by color
   - not all wavelengths of light are equally effective

in Photosynthesis - red and blue wavelength
in Photoperiodism - far red and red wavelength

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Effect of light on plants:

a. photoenergetic effect – direct effect on photosynthesis (intercepted radiation is important)
b. photocybernetic effect – effect on plant development (light quality is rather important than quantity of light)
c. photoperiodic effect (or response) – plant response as conditioned by daylength

Classification of plants according to light intensity requirements:

a. Heliophytes - sun loving
- light-saturated at about 5000-foot candles
- examples: banana, chrysanthemum, corn, cotton, cowpea, cucurbits, eggplant, papaya, peanut, sugarcane

b. Sciophytes - shade loving
- light-saturated at about 500 foot candles
- examples: ginger, African violet, ferns, Philodendron, coffee, begonia, black pepper

- Plants belonging to the intermediate group may be converted through acclimatization into either heliophytes or sciophytes.

Classification of plants according to photoperiodic response:

a. day neutral – will flower over a wide range of daylength
   Examples: banana, citrus, coconut, corn, tomato
b. short-day plant – requires a dark period exceeding some critical length to induce flowering
   Examples: coffee, kenaf, lima bean, rice, sesame, soybean, winged bean
   for cassava, sweet potato, taro and yambean, short-day condition is required for tuber formation
c. long-day plant – inhibited from flowering when the dark period exceeds some critical length
   Examples: aster, castor oil, onion, radish

5. Relative humidity - proportion/amount of moisture in the air

Average RH in the Philippines = 82%

Importance in Crop Production:
a. pest and disease incidences e.g., powdery wildew
b. postharvest behavior of commodities
   - very dry atmosphere - wilting
   - high humidity - not conducive to grain drying
   - enhance mold, aflatoxin buildup
c. supplying water requirement during crop production
   - low RH, high temperature - high evapotranpiration
   - high RH, high temperature - low evapotranspiration

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6. Gaseous Environment

Carbon dioxide
- hardly limiting in crop production except at high intensifies with no or little air circulation in dense plant population
- critical in enclosed environments like greenhouses

Air pollution
- becomes crucial as the world becomes more industrialized
- toxic substances: lead, sulfur dioxide, smog, carbon monoxide, hydrofluoric acid (HF)

7. Cloudiness
- cloud, including smog and fog, affect the amount of radiation received by plants. Most of the solar radiation is reflected by clouds.
- clouds are also believed to be responsible for the GREENHOUSE EFFECT:

- clouds, acting similarly as the glass covering of a greenhouse, stop the transfer of thermal radiant energy from the earth to the cold sky thereby slowing down the cooling process of the earth.
- on the global scale air and water molecules present on clouds trap long-wave radiation and reduce their outward flow to the space because the shorter wave lenghts reaching the earth's surface when re-radiated as longer waves cannot pass through the clouds as readily as the shorter wavelengths.

Classification of Philippine Climate (Corona Classification)

Type 1. Pronounced wet and dry season
Example: Ilocos, Occ. Mindoro, Antique, Negros Occ.
November – May = dry
June – October = wet

Type 2. No dry season with pronounced maximum rain period
Example: Most of Bicol, Samar, Leyte, Surigao, Agusan, Davao
November – January = maximum rain period

Type 3. No very pronounced maximum rain period with short dry deason lasting from one to three months
Example: Cagayan, N. Viscaya, Capiz, Cebu, Negros Oriental, Masbate, Mt. Province
February – April = dry season

Type 4. No pronounced maximum rain period and no dry season
Example: Isabela, Bohol, Cotabato, Lanao, Zamboanga, Bukidnon
Rainfall is distributed throughout the year

Climatic Stresses

1. Typhoon and weather variations

Typhoon - strong winds with speed greater than 121 kph.

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- tropical storm typhoon

Depression - usually originate from the Pacific Ocean; circulation (800 km diameter) with a low central pressure (-72 cm Hg) with air spirals towards the calm eye (30 km diameter)

2. Destruction of the ozone layer
   - reported to be 50% (Antartica)
   - Ozone – protective shield against the harmful UV rays
   - it is 6-30 miles above the earth

   Harmful effects:
   a. Depressed photosynthesis
   b. Reduced levels of seed protein, lipids and carbohydrates
   c. Deleterious effects on human (skin cancer, etc.)

3. Global Warming
   - increased concentration of carbon dioxide in the atmosphere
   - Carbon dioxide concentration might double to around 600 ppm in 30-75 years
   - Methane gas contributes to global warming
   - Results to high atmospheric temperature which affect the ocean level by a few mm per year
   - Ingress of saline water to a 15km distance inland
   - Ocean levels will rise by as much as 30 cm in year 2010 and up to 150 cm by 2050

4. “EL NIÑO”

What is El Niño?
- a periodic ocean – warming and atmospheric disturbance characterized by deficient rainfall or prolonged drought in some areas, while heavy rains, storms or harricanes occur in other areas of the globe
- for the past 30 years, the Philippines has been hit by some seven (7) El Niño episodes.
- The 1982-83 episode is rated as the most intense in the past century
- The 1997-98 episode, has comparable intensity – bringing immense damage to Philippine crops, water/electric supply aside from contributing to poisoning of sea foods (red tide)

Origin of El Niño
- From the Spanish word meaning “Boy child or Little child”
- Used to be considered as a local event along the coasts of Peru and

Ecuador - describing the appearance of warm ocean currents flowing the South and Central American coasts around Christmas time - believing that the temporary heavy harvest of fish was a gift of the Christ child.

El Niño Watch
- El Niño occurs in the Pacific Basin every 2 to 9 years. It usually starts during the Northern winter (December to February). Once established, it lasts until the first

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half of the following year, although at times, it stays longer. It exhibits phase-locking in annual cycles.

Climatic indicators of El Niño in the Philippines include

- delayed onset of the rainy season
- early termination of the rainy season
- weak monsoon activity
- weak tropical cyclones activity
- El-Niño is said to be triggered when the strong westward-blowing trade winds weaken and reverse direction.

Effects of El Niño

a. fish kill especially cold water fish
- tuna and milkfish catch declines

b. decrease in yield for most crops

c. human death

5. Acid rain

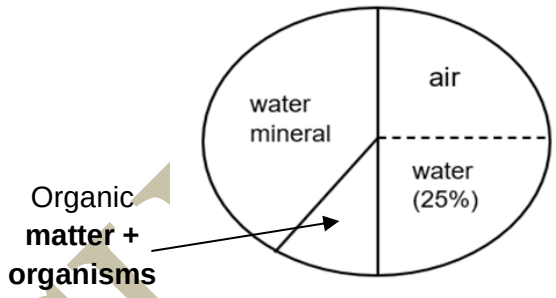
- Sulfur dioxide produces S which is released from natural sources and human activities
- Oxides of S and water will produce acid rain

6. Lahar

- Mt. Pinatubo eruption in the early 1990's
- lead to decreased in agricultural lands resulting to low production

Edaphic Factors - refers to the soil as a factor in crop production

A. What is soil?

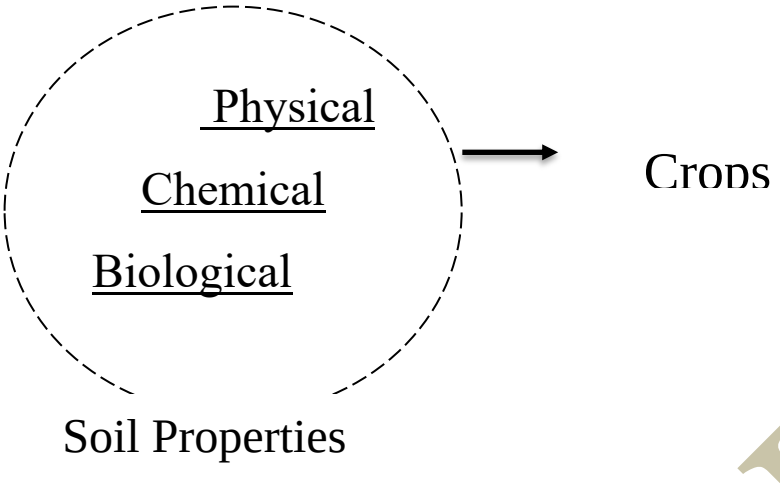


Soil as a three-phase system

- Solid – mineral + organic matter
- Liquid – soil solution
- Gas – various gases

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B. Soil Properties in relation to Crop Production



1. Physical Properties

1.1 Soil Texture
- the relative proportion of primary soil particles, i.e., sand, silt and clay in a particular soil

	Particle Size
Sand	> .02 µ
Silt	.02-.002 µ
Clay	< .002 µ

soil texture range:

Sand	Loam	Clay
<10% clay	~equal proportion of sand silt and clay	>4.5% clay

Importance of soil texture to crop production:

	Heavy soil	Light soil
Tillability	difficult	easy
Aeration	less	more
Water-holding	high	low capacity
Nutrient-holding capacity	high	low

Implications:
- the desirable soil texture for crops is that in between heavy and light, i.e., heavy soil, to allow for easy workability and sufficient water and nutrient holding capacity.

1.2 Soil Structure:
- the arrangement of soil particles into aggregates.

How soil structure is formed?
- ped - natural structure
- dod - structure formed with the application of external force, e.g., plowing

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- organic matter - binding agent

Types of soil structure
- blocky
- granular
- columnar
- no structure

Importance of soil structure to crops:
- to some extent, soil tillability is related to structure. There are soils that are relatively easier to break up because of their structure.
- soil structure influences the infiltration of water through the soil.
- soil structure influences soil aeration which is critical during seed germination and seedling emergence.

1.3 Soil Depth
- particularly, the top soil relative to the subsoil.

Physical Properties

- Top soil - from where crops obtain most of the nutrients. Also, where organic matter is concentrated.

- Sub soil - storage of nutrients and water, but often less fertile than top soil.

1.4 Bulk density — Properties related to
1.5 Soil porosity — degree of aeration and
1.6 Hydraulic conductivity — water holding capacity

1.7 Soil organic matter content
- composed of dead plant residues and wastes
  - for most mineral soils 5%
  - for most Philippine soils 2-4%

Importance:
- prevents loss of nutrients by forming complexes with nutrient elements
- facilties absorption and percolation of water into and through the soil
- increases water holding capacity
- source of nutrients
- improves penetration of roots
- Influences soil structure formation
- influences soil chemical properties may contain N, P, S, B, Zn
- determines the biotic composition

Humus – amorphous, collidal substance which is resistant to further decomposition
- improves soil structure
- increases CEC and water holding capacity of soil
- gives dark color to soil

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2. Chemical Properties

2.1 Soil pH or soil-reaction

- degree of acidity or alkalinity
- influences nutrient availability
- negative logarithm of H^+ activity
- 7.0 – neutral pH ($H^+ = OH^-$)
- decrease in soil pH acidity
- increase in soil pH alkalinity

pH Effects

- pH below 5.0 – Al, Fe & Mn become toxic
- Ca & Mo deficiency
- pH below 5.5 – Mo, Za, K & S deficiency
- pH 6-7 (neutral) – most nutrients are in available form
- pH above 7.5 – Al toxicity, salinity, Zn & Fe toxicity
- pH above 8.0 – formation of Ca phosphates
- pH above 8.5 – salinity, Zn & Fe deficiency

2.2. Cation Exchange Capacity

- ability of soil to absorb and release cations through the soil solution

how?

- soil particles can be colloidal (e.g., clay) and can contain excess (net) negative charges

how to measure?

- sum of exchangeable cations in a given wt. of soil expresses in me/100 g. soil

Nutrient Holding Capacity – CEC

Particular	**me/100g**
Humus	600
Good clay (Montmorillonite)	80 to 150
Not good clay (Kaolinite)	3 to 15
Sand	0
Good soil	20 above
Bad soil	5 and below

By M. Maeda and Y. Mastuo (Basic Knowledge of Soil)

3. Biological Properties

3.1 Macroscopic organisms
3.2 Microscopic organisms

a. Fungi
b. Actinomycetes and protozoa = decomposers (aerobic)
c. Bacteria (billion/g topsoil)
- Thiobacillus oxidizes S sulfate form
- Auxotrophic bacteria – oxidizes Mn & Fe to less available form

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- Nitrifying bacteria
- N fixing bacteria

d. Blue-green algae – Nitrogen fixation

3.3 Root – microoganism association found in the nuts of legumes – Nitrogen fixation

a. bacteria - rhizobia in nodules

b. fungi - mycorrhizal fungi
- converts phosphorous to biological forms

3.4 Organic matter and humus

OM 5% N,

Only 2% available to plant

Philippine soils

2.4% OM

high fertility	>3.5%
medium	2-3.5%
low	<2%

Soil Organisms

Fauna:

a. Macro -small mammals, insects (springtails, ants, beetles, grubs), millipedes, centipedes, sowbugs, mites, slugs, snails, earthworms, spiders

b. Micro - nematodes, protozoa

Flora:

a. Roots of higher plants

b. Algae – blue, blue green, diatoms

c. Fungi – mushroom, yeast, molds

d. Bacteria – aerobic/anaerobic
- autotrophic/heterotrophic

e. Actinomycetes

Benefits from Earthworms

- Burrowing – channels for drainage and aeration, entry of other animals, entry of water, nutrients, roots
- Mix the soil, “plows” the soil
- Incorporates crop residues
- Contribute to OM
- Humus enrichment
- Improves soil structure
- Control pests (e.g., leaf miner pupa, scub pathogen)
- Nutrient recycling

Topography
- whether the land is flat or sloping
- a major parameter in delineating lowlands and uplands

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- in the Philippines, the slope of the land is used as the major determinant. i.e., uplands - > 18% slope

- in crop production, topography is critical in:
  - irrigation and drainage
  - soil conservation
  - in flat lands, the problem can be in drainage.
  - in sloping lands, the problem can be in how to bring water up for irrigation.
  - poor drainage can result into soil fertility problems.
  - in sloping lands, soil erosion can be a major problem especially if crop production practices do not consider soil and water conservation strategies.
  - to a great extent, CEC is an indicator of soil fertility because most nutrients are taken up by plants in cation form.
  - the soil can also exchange anions
  - roots of plant also have their own CEC – the exchange therefore depends on the interaction of soil and root CEC.

B. Biotic Factors

- all living elements in the environment that can affect crop production
- includes: beneficial organisms pests

- Beneficial organisms
  - provide beneficial effects on crop production

- Pollinators
  - important role in the preservation of species and in biodiversity conservation

- Decomposers
  - an important part in the food chain which is related to energy flow in a crop production system.
  - a trophic level, usually consisting soil microorganisms (soil biotic factors)
  - specifically important in the maintenance of soil organic matter.

- Natural pest enemies
  - provide balance in a crop production system particularly in the control of pests
  - as bio-control agents against pests

- Pests
  - a collective term that includes insect pests, diseases, weeds, invertebrates and vertebrates
  - has always been a major limiting factor in crop production!
  - Damage can go as high as 100%.

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B. GENETIC FACTORS

1. Genotypes

The expressed physical characteristic of a crop (phenotype) is influenced by environment and genotype. Genotype is the genetic makeup of an individual and directly affects crop yield. There are specific genes that codes for better growth and yield in crops. To achieve optimum crop production, growers should take advantage of recommended varieties that are bred to produce high yield, resistance to pests and diseases, and tolerance to environmental stresses.

Highly recommended varieties/ cultivars

Rice

Rice (*Oryza sativa*) is the most important food crop grown and is the staple food for nearly half of the world's population. The Philippines is the eighth largest producer of rice and contributes to Asia's production which comprises ninety percent (90%) of total rice produced in the world. Rice area harvested in 2010 was 4.4 million hectares out of the 5.4 million hectares of arable land in the Philippines (Global Rice Science Partnership, 2013).

Condition	Varieties	Characteristics
Rainfed lowlands	NSIC Rc192 (Sahod Ulan 1)	• Average yield - 3.7 t/ha • Maximum yield - 5.5 t/ha • Maturity - 106 days after seeding • Height – 109 cm • Grain size – Medium • Milling recovery – 70.1%
	PSB Rc14 (Rio Grande)	• Average yield - 3.6 t/ha • Maximum yield – 6.1 t/ha • Maturity - 110 days after seeding • Height – 92 cm • Grain size – Long • Milling recovery – 65.78%
	PSB Rc68 (Sacobia)	• Average yield - 3.4 t/ha • Maximum yield – 4.4 t/ha • Maturity - 116 days after seeding • Height – 116 cm • Grain size – Long • Milling recovery – 63.5%
	NSIC Rc272 (Sahod Ulan 2)	• Average yield – 3 t/ha • Maximum yield – 6.4 t/ha • Maturity - 110 days after seeding • Grain size – Medium • Milling recovery – 66.6%

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Upland environments	PSB Rc80 (Pasig)	• Average yield – 5 t/ha • Maximum yield – 8.7 t/ha • Maturity - 112 days after seeding • Height – 92 cm • Grain size – Medium • Milling recovery – 68.6%
	PSB Rc9 (Apo)	• Average yield – 2.9 t/ha • Maturity - 119 days after seeding
	PSB Rc23 (Katihan 1)	
Areas prone to flooding	PSB Rc18 (Ala)	• Average yield – 5.1 t/ha • Maximum yield – 8.7 t/ha • Maturity - 123 days after seeding • Height – 102 cm • Grain size – Long • Milling recovery – 65.34%
	NSIC Rc194 (Submarino 1)	Can survive, grow, and develop even after 10-14 days of complete submergence • Average yield – 3.5 t/ha (normal), 2.5 t/ha (submerged) • Maturity – 112 DAS (normal), 125 DAS (submerged) • Height – 97 cm (normal), 93 cm (submerged) • Grain size - Long
	PSB Rc222	• Average yield – 6.1 t/ha (transplanted), 5.7 t/ha (direct seeded) • Maximum yield – 10 t/ha (transplanted), 7.9 t/ha (direct seeded) • Maturity – 114 days (transplanted), 106 days (direct seeded) • Height – 101 cm (transplanted), 98 cm (direct seeded)

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Tomato

Tomato (*Lycopersicum esculentum* Miller) is one of the most popular vegetable grown in many parts of the world. The fruit is commonly used as an ingredient in various food preparations and is considered as a profitable crop for off-season production.

Variety	Days to harvesting (DAT)	Productive period (months)	Yield (t/ha)	Fruit			Type	Planting season	Reaction to pests and diseases
				Color	Size	Shape			
Apollo for fresh market	35	2-3	30	Red orange	Medium large	oblong	Determinate	Dry season	TMV-MR bacterial wilt-MR
Magilas for fresh	30	1-2	30	Pink red	large	oblong	Determinate	Dry and wet season	Dumping off-R wilt-R TMV-R
Atlas (F1)	40	2-3	40	Red orange	Medium Large	oblong	Semi determinate	Dry and wet season	Bacterial-R wilt-R TMV-R
Atlas(rafted) (kamlong)	40	2-3	40	Red orange	Medium large	oblong	Semi determinate	Wet season	Bacterial wilt-R TMV-R
Cherry	50	4-5	40	Red orange	Medium	round	Indeterminate	Dry and wet season	Wilt R TMV -MR
TMV - Tomato Mosaic Virus. MR - Moderately resistant; R - resistant									

Coconut

One of the top agricultural exports of the country is coconut (*Cocos nucifera L.*). Also known as the “tree of life”, coconut comprises one-third of the country’s arable land (68 of 79 provinces). Previously considered as the world’s leading producer, the country has been overtaken by India and currently ranked as the second largest producer in the world.

VARIETY	CODE	ORIGIN	DISTINGUISHING FEATURES
TALL			
Baybay	BAY	Philippines	Copra per nut is quite heavy although nuts do not really look big; very thin husk; homogenous and produces high number of leaves per year; bunches with short peducle and nuts are trapped between leaf axils; stem quite robust and firm; tolerant to FDMT.

San Ramon	SNR	Philippines	Usually bears extra large nuts when young needing 3 to 3.5 nuts per kilo copra.
Tagnanan	TAG	Philippines	Produces nuts with good fruit corn position; 350 g copra per nut and very homogenous; good GCA with MRD
Laguna	LAG	Philippines	Produces a fairly large number of nuts; 190 g copra/nut; hybrid cross with CAT green dwarf good
Agta	AGA	Philippines	Husk of nut is streaked, making it look around and tough. Young nuts show black streaks
Macapuno	MAC	Philippines	A mutant form of the Laguna variety, nuts have soft and jelly-like meat. This character is lethal to the embryo which needs to be grown in artificial medium. Naturally occurring palms grown from normal nuts bear about 25%. MAC nuts may give 100% macapuno nuts.
DWARF			
Catigan	CAT	Philippines	Nuts are medium sized, round with prominent stigmatic tip; copra per nut is about 200 to210g; husk is thick; peduncle and bunch rachis are long; young nuts and petioles are green.
Coconino	CNO	Philippines	Nuts are very small, oblong and smooth; stem quite thin and high yielder of toddy. Copra per nut seldom exceeds 100g. Coconino is a green dwarf.
Kinabalan	KIN	Philippines	Green dwarf, nuts are extra

			large, robust stem and copra per nut sometimes exceeds 300g; homogeneous.
Pilipog	PIL	Philippines	Nuts are green and round; female flowers, tip of roots and base of shoots of newly germinated seedlings are pink; copra per nut is 100g.
Tacunan	TAC	Philippines	Another green dwarf; spikelets are very short; nuts are medium to large oblong with broad equatorial diameter and pointed stigmatic; and unopened spathes are flat on distal end.
Mangipod	MGP	Philippines	A green dwarf which retains hundred of ripe nuts on the crown due to drying of nuts on the tree. Highly self-pollinating and shows very strict dwarfism.
Tampakan	TMP	Philippines	Nuts are long and pointed. Shape of shell and cavity conforms exactly to the shape of inhole nut which has a very thin husk.

Peanut

Peanut (*Arachis hypogaea* L.), also called groundnut, is one of the field legumes grown in the country. Worldwide, peanut is considered as an important food crop, used as raw material for peanut oil (50%), confectionery (37%), and seed purposes (12%). On the other hand, the protein-rich vegetative part is utilized for feeding livestock.

Variety	Dry Pod Yield (t/ha)	Other characteristics
PSB Pn 1 (UPL Pn 10)	Dry season – 2.0 Wet season – 1.5	• Medium-seeded • Two-seeded • Moderately resistant to foliar diseases & bacterial wilt
NSIC Pn 02	Dry season – 1.87 Wet season – 1.34	• Resistant to peanut rust and Cercospora Leaf Spot (CLS) diseases • Moderately resistant to *Aspergillusflavus*
NSIC Pn 03	Dry season – 1.80	• Moderately resisntant to peanut rust

	Wet season – 1.86	and CLS • Mostly 3-seeded; • bunch type peanut with light brown seed coat
NSIC Pn 04	Dry season – 1.81 Wet season – 2.48	• Gives high yield and better bean quality • during dry season • Mostly 2 to 3 seeded; bunch type; with light brown seed coat • Moderately susceptible to CLS
NSIC Pn 05	Dry season – 1.86 Wet season – 1.77	• Moderately susceptible to CLS and peanut rust • Recommended as regional variety in • Central Visayas particularly in Bohol • Mostly 3 seeded; bunch type with salmon pink seed coat.
NSIC Pn 06 (Biyaya 14)	Dry season – 1.89 Wet season – 1.97	• Tolerant to leaf hopper • High yielder in wet season in light textured soil • Two seeded Spanish variety with plump • and oblong shape pods; shelling easy because of thin seed coat
NSIC Pn 07	Dry season – 1.91 Wet season – 1.86	• Moderately susceptible to CLS and peanut rust • Can be grown in dry and wet seasons; • gives better bean quality in dry season • Mostly 2 to 3 seeded; bunch type; peanut with pinkish seed coat
NSIC Pn 08	Dry season – 1.80 Wet season – 1.87	• Moderately susceptible to CLS and peanut rust • Can be grown in dry and wet seasons; • gives better bean quality in dry season • Mostly 3 seeded; bunch type; peanut with creamy or light tan seed coat
NSIC Pn 09	Dry season – 1.60 Wet season – 1.62	• Moderately susceptible to CLS and peanut rust • Can be grown in dry and wet seasons; gives better bean quality in dry season
NSIC Pn 10	Dry season – 1.54 Wet season – 1.66	• Moderately susceptible to CLS and • peanut rust • Can be grown in dry and wet seasons; • gives better bean quality in dry

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		season • Mostly 2 to 3 seeded; big sized beans; bunch type; peanut with pinkish seed coat
NSIC Pn 11 (Namnama -1)	Dry season – 1.63 Wet season – 1.60	• Resistant to CLS and peanut rust • Can be grown in the dry and wet seasons • Spreading type peanut making it a breakthrough for off-season (WS) production. • Ideal for both processing and boiling (table type) purposes
NSIC Pn 12 (Ilocos Pink)	Dry season – 2.3 Wet season – 1.93	• Two-seeded • Medium-seeded • Medium maturing

Lanzones

Lanzones (*Lansium domesticum* Corr.) is a seasonal fruit that is sold in bunches, resembling grapes but twice the size. The fruit is commonly grown in the southern part of the Philippines and is available late September to early November.

Variety	Characteristics
Langsat	Langsat has slender trees with upright branches, a scrappy habit and dark green foliage. The fruit spikes are long, carrying 15-25 ovoid thin-skinned fruits that exude latex until fully ripe. Included under this category are the Paete and Camiguin of the Philippines.
Duku	Duku is a spreading tree, with dense dome-shaped canopy of bright-green leaves, bearing shorter spikes, usually with few fruits. The fruits are normally larger and more rounded than the langsat fruit with a thick skin (up to 6 mm) and relatively free from latex. Duku fruits are generally sweet and aromatic. Due to thick skin, the shelf life of duku is generally sweet and aromatic. Due to the thick skin, the shelf life of duku is generally four to seven days longer than the langsat fruits.
Duku-langsat	Duku-langsat is intermediate between duku and the langsat. The best type under this group is the longkong.

Corn

Corn, maize, or scientifically known as *Zea mays* was first domesticated in Mexico and is considered as the most important crop in the United States and is a staple food in many places all over the world. Corn is a gramineous plant characterized by having a cane stalk as other grasses, although solid inside unlike other family members that have it void. Corn is mainly distinguished by its female inflorescence, called “corncob”, where the seeds or kernels are grouped along one axis. In addition, corn ear is covered by green bracts with papery texture, and ends in a kind of dark yellow plume, formed by the styles.

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Varieties	Characteristics
Ginintuan (IPB Var I)	This is a yellow flint open pollinated corn variety with moderate resistance to downy mildew and corn borer. It matures in about 110 days and has an average yield is 4.08 and 4.55 tons per hectare for wet season and dry season, respectively.
Philippine Super sweet No.1 (Asukar)	It is an open-pollinated yellow Super sweet variety with excellent taste. Fresh ears can be harvested 70-72 days after planting. It yields 7.2 tons per hectare (fresh ears) and grows to about 2.28 meters.
IPB 911	This yellow hybrid has an average yield of 7.2 tons per hectare. It matures in 105 days and is resistant to downy mildew and tolerant to corn borer.
IPB 913 (PSB Cn 92-94)	This hybrid has an average grain yield of 6.58 tons per hectare. It grows to an average of 2.0 meters and matures in 104 days during the wet season and 108 days during the dry season. It is resistant to downy mildew, moderately resistant to rust and stalkrot, and moderately tolerant to corn borer.
IPB 921 (PSB Cn 93-37)	This hybrid can adapt to all regions of the country and can be grown throughout the year. It yields 6.89 tons per hectare, matures in 104 days, and is resistant to downy mildew and tolerant to corn borer.
IPB 929 (PSB Cn 93-38)	This variety has a wide adaptability to different soil types. It yields 7.01 tons per hectare, matures in 106 days, and is resistant to downy mildew and tolerant to corn borer.
IPB 919 (PSB Cn 93-36)	This hybrid grows in any soil type and adapts to all regions of the country. It has an average yield of 6.35 tons per hectare, matures in 104 days, and is resistant to downy mildew and tolerant to corn borer.
IPB Var 7 (Dafrosa)	This is a yellow open-pollinated variety that matures 105 days after planting. It yields 5.57 tons per hectare and can be grown in any soil type in all regions of the country.
IPB 33161 (Nutrisweet)	This is a double endosperm mutant variety that has brittle and opaque genes which are responsible for the Super sweet taste of young kernels and high concentrations of two essential amino acids, namely lysine and trytophan. Its kernels at green stage are creamy yellow, sweet and very tender when cooked. It yields 5.74 tons per hectare and can be harvested 72-76 days after planting. It is resistant to rust and can be grown during the wet and dry seasons.

Tanco White (IPB Var 2)	This variety is high yielding, white flint (the hard grain form), open-pollinated (pollinated naturally without human intervention), and grows up to 2.25 meters and matures in 103 days. It is resistant to mildew and can produce 5.07 tons per hectare.
IPB Var 4	This variety is white, open-pollinated, and with a good husk cover, kernel depth, and broader leaves. It matures in 110 days and can yield up to 4.89 tons per hectare. It is moderately tolerant to corn borer.
Improved Makapuno	This is a better and glutinous variety of *Makapuno* with distint cone-shaped ears and grains filling up to the top. It matures after 68-70 days after planting. It produces 6.26 tons per hectare. It can be harvested early and thus, demands a higher price.
Lagkitan or Composite No. 2	This variety is open-pollinated, glutinous, and yields 6.5 tons per hectare. This variety is for table use with excellent eating quality. The plant grows up to 224 centimeters and matures after 68-70 days of planting.
DLU Pearl Sweet (PSB Cn 93-49)	This is an open-pollinated, glutinous type also for table use. Fresh ears can be harvested in 72 days during dry season and 70 days during the wet season. It yields up to 6.10 tons per hectare. It is well adapted to both dry wet seasons and can be grown to any type of soil-making it a very good cash crop.
SE, or sugar enhanced hybrids	It has the highest sugar content of any sweet corn varieties. They do not need to be isolated from other sweet corn categories when planted in the garden.
SH2, or super sweet hybrids,	They are very sweet and have a crisp texture. They must be planted separate from other corn varieties to prevent cross-pollination.
SU, or normal sugar hybrids,	have the highest yields. However, they are not quite as sweet or crisp as the SE or SH2 varieties. The flavor and texture is very good. SU varieties are considered the standard of quality among sweet corn varieties. No isolation is needed when planted with other sweet corn varieties.
Heirloom sweet corn	is known for its full flavor and substantial texture. Some heirloom varieties can be traced back more than 100 years. Heirloom sweet corn should be planted away from other corn varieties to prevent cross pollination.
LG Comp 1-91	Widely adapted across the agro-climatic regions of the country
USMARC 1191	Better than the check variety (IPB Var 1) in wet seasons in major corn growing regions
7PASO54	Widely adapted across the agro-climatic regions of the country

AH 193	Better adapted than the check variety in major corn growing regions in Luzon and Visayas
CPX 520A	Better adapted than the check variety in major corn growing regions in Luzon and Visayas
PAC 1358	Stable yielder despite stress conditions.
P-3008	Well adapted in major corn growing regions of Luzon, Visayas and Southern Mindanao
IES Cn4	The new variety can be grown in all regions of the country but seems to perform best in Isabela during the west season.
IPB 911	The new variety can be grown in all regions of the country but seems to perform best in Isabela and Bukidnon during the wet and dry season.
IPB 9204	Can be grown in all regions of the country but was observed to perform best in Isabela and Bukidnon during the wet season.
BS 9754	Can be grown in all regions of the country, but seems to perform best in Isabela, Cotabato, and Bukidnon, during the wet and dry season.
FE 817	Can be grown in all regions of the country, but seems to perform best in Isabela and Bukidnon, during the wet season.
FE 820	Can be grown in all regions of the country, but seems to perform best in Isabela and Bukidnon, during the wet season.
LG 3B	Performs better over a wide range of agro-climatic conditions
Cn 96-68 USM Var 12	Highly adapted in Northern Mindanao Area and Northern Luzon Area
Cn 96-69 Kristal	Performs well during dry and wet seasons in Northern Mindanao
Cn 96-70 BS 9881	Highly adapted in Northern Mindanao area and Northern Luzon area
Cn 96-71 CW 22	Can be planted after lowland rice; Performs well during both wet and dry season
Cn 96-72 CW 24	Can be planted after lowland rice; Performs well during both wet and dry season
Cn 96-73 CW 26	Performs best in Luzon during wet and dry seasons

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Cn 96-74 FE 8826	Highly adapted in Northern Mindanao area and Northern Luzon area
C-818	Performs well during dry and wet season in all locations but performs best in Northern Luzon and Northern Mindanao
C-838	Performs well during dry and wet season in all locations but performs best in Northern Luzon and Northern Mindanao
C-909	Performs well during wet and dry season in all locations; highly recommended for commercial production in Region II
MARIANA I	Performs well in whole of Philippines but is best grown in Northern Luzon and Northern Mindanao during dry season
MARIANA III	Performs well in whole of Luzon, Northern Mindanao and Bukidnon areas; grown during wet and dry seasons, but better in dry season
MARIANA	Performs well in whole of Luzon, Northern Mindanao and Bukidnon areas; grown best in dry season
C-939	Performs well in Luzon, Region II and Mindanao during wet and dry seasons
FE 325	Performs well in Luzon, Region II and Mindanao during wet and dry seasons
MARIANA II	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons.
BS 9889	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons; relatively tolerant to high density planting
BS 9890	Performs well in Luzon and Mindanao during dry season
C-848	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons.
IPB 9202	The variety performs well in Luzon, Visayas and Mindanao either during the wet season.
P-3013	Performs well in Luzon durign wet and dry seasons
P-3023	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons.
CW-48	Performs well in Luzon and Mindanao during dry season
CW-108	Performs well in Luzon, Visayas and Mindanao; best in Luzon during either wet or dry season
LG 5	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons.
CMU 9422	Performs well in Northern Luzon and Mindanao during wet season

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CMU 9424	Performs well in Luzon, Visayas and Mindanao during wet and dry seasons
P-30J32	Well adapted in Ilagan, Batac, La Granja, VisCa, USM, CMU and Ipil; Performs better in dry season than in wet season
P-3012	Performs best in IES, Ilagan, Isabela during either wet or dry season; good performance in Batac, Ilocos Norte, and Pangasinan during wet season
P-30B29	Highly adaptable in Mindanao particularly in Tupi, USM, CMU and Ipil; better in dry season than wet season
CW 208	Performs well in IES, Isabela during wet and dry season; good in Ilocos Norte in wet season; best in USM, North Cotabato during wet or dry season
P-30A10	Adapted in Ilagan, Batac, La Granja, USM, CMU and Banga. Good performance in Ipil and Ubay during dry season
CW 58	Performs best in Ilagan and USM during wet or dry season; good performance in CMU and VisCa.
DK 888	Location-specific recommendation in irrigated lowland in Visayas particularly in the provinces of Bohol, Iloilo, Leyte and Negros, dry and wet seasons, transplanting culture.
DK 999	Can be grown in Luzon, Visayas and Mindanao; best in Ilagan, La Granja, USM, CMU and Ipil during wet or dry season
DK - XPT 9674	Location-specific recommendation in irrigated lowland areas in Visayas particularly in the provinces of Bohol, Iloilo, Leyte and Negros, dry and wet seasons, transplanting culture.
AG 5324	Can only be grown in Mindanao during wet or dry season
AG 8314	Best adapted in Mindanao. Specific areas in Mindanao where very good performance was noted are CMU and USM during the wet and dry seasons.
C 7118	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasom; best in Ilagan, Batac and CMU.
CGX 04461	Can be grown in Luzon or Mindanao during wet or dry season.
CGX 04468	Can be grown in Luzon during wet or dry season; best in Ilagan and Batac.
CGX 06416	Can be grown in Luzon during wet or dry season. The new varities can be perform very well in Batac and Ilocos Norte.

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CPX 3600	Can be grown in Luzon, Visayas and Mindanao during wet or dry season
CPX 3601	Performs well in Visayas and Mindanao during the wet or dry seasons.
CPX 3627	Can be grown in Luzon, Visayas and Mindanao during the wet or dry seasons.
DK 9572	Can be grown in Luzon, Visayas and Mindanao during wet or dry season; best yield performance in the Ilagan in Luzon and USM in Mindanao.
P-30A38	Can be grown in Luzon and Mindanao during wet or dry season;best yield performance in the USM and CMU in Mindanao.
P-30A65	Can be grown in Luzon and Mindanao during wet or dry season;best yield performance in the USM and CMU in Mindanao.
TCT 1155	Can be grown in Luzon in either wet or dry season;best performance are in the Batac and Ilocos Norte.
CMU 9421	Can be grown in Mindanao in either wet or dry season.
CMU 9423	Can be grown in Mindanao in either wet or dry season.
CMU 9425	Can be grown in Mindanao in either wet or dry season.
AG 5325	Can be grown in Mindanao during wet or dry season.
AG 8314 B	Can be grown in Mindanao during wet or dry season.
AG 8374	Can be grown in Luzon particularly in major growing areas like Isabela during the wet and dry seasons.
C-7122	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.
C-7286	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.
C-8010	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.
P 30A55	Can be grown in Luzon and Visayas in either wet or dry season.
P 30D44	Can be grown in Luzon and Visayas in either wet or dry season.
BS 9901	Can be grown in Luzon in either wet or dry season.
NK 8820	Can be grown in Luzon in either wet or dry season.
C-9026	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.

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P-30R49	Can be grown in Mindanao during wet or dry season.
P-30M50	Can be grown in Visayas and Mindanao during wet or dry season
C-8619	Can be grown in Luzon in either wet or dry season.
C-8652	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.
IES GLUT 2	Can be grown in Luzon in either wet or dry season.
USMARC 9902W	Can be grown Luzon and Mindanao during wet or dry season.
USMARC 99149	Can be grown in Visayas and Mindanao during wet or dry season.
BS 9900	Can be grown in Luzon in either wet or dry season;particularly in Iligan,Echague,Batac and Pangasinan.
30P60	Performs well during both wet and dry season.
NK 8830	Can be grown in Luzon in either wet or dry season.
TSG 81	Can be grown in Luzon in either wet or dry season.
30W30	Can be grown in Visayas and Mindanao during wet or dry season.
NK 8840 (Super Jumbo)	Performs well during both wet and dry season.
NK 8850	Can be grown in Luzon and Mindanao in either wet or dry season.
DEKALB 9051	Can be grown in Luzon and Mindanao in either wet or dry season.
DEKALB 868 (PA 8201)	Can be grown in Mindanao during wet or dry season
30B80	Can be better grown in wet season than in dry season;it performs better during the wet season.
30P95	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons.
NK 8860	Can be grown in Luzon during wet or dry season.
Bio-seed 9888	Can be grown in Mindanao in either wet or dry season
Bio-seed 9909	Can be grown in Mindanao during wet or dry season
Bio-seed 9744	Can be grown in Visayas during wet or dry season
DK 3668w	Can be grown in Visayas and Mindanao during wet or dry season
DK 979	Can be grown in Luzon and Visayas in wet or dry season

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DK 9132	Can be grown in Luzon and Visayas in wet or dry season
CW 821	Best grown in Luzon in dry season, given that water supply is not limited
CW 811	Can be grown in Luzon in either wet or dry season provided that water supply is properly managed.
CW 801	Can be grown in Luzon in either wet or dry season
USMARC T X 010	Can be grown in Visayas during wet or dry season
USMARC T X 012	Can be grown in Visayas during wet or dry season
IES Cn 5	Can be grown in Luzon in either wet or dry season and fits well in lowland.
IES Glut 4	Performs well in both wet and dry season
Bio-seed 9744 Plus	Can be grown in Visayas and Mindanao during wet or dry season
Ghen 703	Can be grown in Luzon in either wet or dry season
Ghen 802w	Can be grown in Visayas and Mindanao during wet or dry season
NK 8870	Can be grown in Luzon and Visayas in wet or dry season
Los Baños White Gold	Can be grown in Luzon during wet or dry season.
IES Glut 6	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons
IES Glut 7	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons
TB 8108	Can be grown in Luzon and Visayas in wet or dry season
DK 878	Can be grown in Mindanao during wet or dry season
30M73	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons
30N34	Can be grown in Luzon in either wet or dry season
CW 831	Can be grown in Luzon in either wet or dry season
DK 838 YG	Can be grown in Luzon, Visayas and Mindanao during wet or dry season
30Y73	Can be grown in Luzon, Visayas and Mindanao during wet or dry season
30Y84	Can be grown in Luzon, Visayas and Mindanao during wet or dry season
30Y49	Can be grown in Mindanao during wet or dry season
30Y50	Can be grown in Visayas and Mindanao during wet or dry season

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30Y34	Can be grown in Luzon in either wet or dry season
30Y75	Can be grown in Luzon and Mindanao in either wet or dry season
30Y80	Can be grown in Luzon and Visayas in wet or dry season
30Y95	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons
DK 818-2 (RRC2)	Can be grown in Luzon, Visayas and Mindanao during wet or dry seasons
DK 868-1 (YG)	Can be grown in Mindanao during wet or dry season
DK 878-1 (YG)	Can be grown in Mindanao during wet or dry season
DK 909-1 (YG)	Can be grown in Luzon, Visayas and Mindanao in either wet or dry seasons
DK 9051-1 (YG)	Can be grown in Luzon, Visayas and Mindanao in either wet or dry seasons
DK 9161-1(YG)	Can be grown in Luzon in either wet or dry season
Nk 6204	Can be grown nationwide in either wet or dry seasons
Bio-seed 9899	Can be grown in Luzon and Mindanao in either wet or dry season
NK 6614	Can be grown in Luzon and Mindanao in either wet or dry season
Filipina 753	Can be grown in Mindanao during wet or dry season
NK 6624	Can be grown in Mindanao during wet or dry season
CW 847	Can be grown in Mindanao during wet or dry season
CW 848	Can be grown in Mindanao during wet or dry season
G 804	Can be grown in Visayas and Mindanao during wet or dry season
IPB Var 9	Can be grown in Mindanao during wet or dry season
USM Var 16	Can be grown in Luzon and Mindanao in either wet or dry season
USM Var 18	Can be grown in Luzon and Mindanao in either wet or dry season
IES 89-10	Can be grown in Luzon in either wet or dry season
IES 89-12	Can be grown in Luzon in either wet or dry season
TSG 80	Can be grown in Luzon or Mindanao during wet or dry season.
TSG 89	Can be grown in Luzon or Mindanao during wet or dry season.
Cw 18-69	Can be grown in Luzon or Mindanao during wet or dry season.

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CW 44-81	Can be grown in Luzon or Mindanao during wet or dry season.
NK 6745	Can be grown in Luzon or Mindanao during wet or dry season.
NK 32W	Can be grown in Mindanao during wet or dry season
30B42	Can be grown in Luzon in either wet or dry season
CW 850	Can be grown in Mindanao in either wet or dry season
CW 851	Can be grown in Luzon in either wet or dry season
CW 1441	Can be grown in low-elevation areas that are not water-logged; higher yield in dry season
TSG 398	Can be grown in low-elevation areas that are not water-logged; higher yield in wet season
TSG 388	Can be grown in low-elevation areas that are not water-logged in Mindanao; higher yield in dry season
TSG 366	Can be grown in low-elevation areas that are not water-logged in Mindanao
Bio-seed 9540	Can be grown in Luzon and Mindanao in wet or dry season
30W40	Can be grown in low-elevation areas in Negros Occidental, Leyte, Bohol and other areas in Visayas; performs well in wet or dry season but higher yield in wet season
USM Var 22	Can be grown in Luzon in either wet or dry season ; yield is higher in dry season
USM Var 20	Can be grown in Luzon and Visayas in wet or dry season
GSI 96	Can be grown in low-elevation areas but not water-logged areas in Mindanao
USM-DA Syn 0205 USM Var 24	Can be grown in low-elevation but not water-logged areas in Luzon during both wet or dry season
DK 818 RRC2/YG	Can be grown nationwide in either wet or dry seasons
DK 878YG	Can be grown in Mindanao in either wet or dry season
DK 878 RRC2	Can be grown in Mindanao in either wet or dry season
DK 878 RRC2/YG	Can be grown in Mindanao in either wet or dry season
DK 9070 RRC2	Can be grown nationwide in either wet or dry seasons
DK 9132 YG	Can be grown in Luzon in either wet or dry season
DK 9132 RRC2	Can be grown in Luzon in either wet or dry season

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DK 9132 RRC2/YG	Can be grown in Luzon in either wet or dry season
CW 1401	Can be grown in Luzon in either wet or dry season
CW 789	Can be grown in Luzon in either wet or dry season
NK 6654	Can be grown in Luzon in either wet or dry season ; yield is higher in wet season
NK 6645	Can be grown in Luzon, Visayas and Mindanao during wet or dry season.
NK 6226	Can be grown in Luzon or Mindanao during wet or dry season ; yield higher in dry season
NK 6626	Can be grown in Luzon or Mindanao during wet or dry season;higher yield in dry season.
Bio-seed 9541	Can be grown in Visayas and Mindanao during wet or dry season
Bio-seed 9542	Can be grown nationwide in either wet or dry seasons
PAC 105	Performs well during both wet and dry season
PAC 555	Can be grown in Luzon in either wet or dry season.
PAC 259	Can be grown in Luzon in either wet or dry season.
P 3482	Can be grown in Mindanao in either wet or dry season
Mt. Apo Gold 900y	Can be grown in Luzon in either wet or dry season.
IPB Var 13	Can be grown in Luzon or Visayas during wet or dry seasons
IPB Comp 3	Can be grown nationwide in either wet or dry seasons.It performs better during the dry season given adequate water supply particularly during critical growth stages.
Mt. Apo Cristal 800w	Can be grown in Luzon and Visayas during wet or dry seasons
IPB Var 8	Can be grown in Luzon and Mindanao during wet or dry season.
Var 6	Can be grown in Luzon and Visayas during wet or dry seasons
30G80	Can be grown nationwide in either wet or dry seasons
30T80	Can be grown nationwide in either wet or dry seasons
30T35	Can be grown in Luzon during wet or dry season.
DK 5885 RRC2	Can be grown nationwide in either wet or dry seasons
BC 9543	Can be grown in Luzon and Visayas during wet or dry seasons

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CW 1407	Can be grown in Mindanao except Bukidnon
30B48	Can be grown in Mindanao in either wet or dry season but it yields higher under higher irradiance.
NK 6297	Can be grown in Luzon and Mindanao in either wet or dry season but performs better during the dry season without severe water stress.
CW 159W	Can be grown in Visayas and Mindanao in either wet or dry season particularly in Cotabato Province.
NK 5012W	Can be grown in Visayas during wet or dry season but performs better during the dry season without severe water stress.
NK 5013W	Can be grown in Visayas during wet or dry season but performs better during the dry season without severe water stress.
X7B 445	Can be grown in Mindanao in either wet or dry season
Healer 101	Can be grown in Luzon during wet or dry season but yields are higher under high irradiance.
KK 168	Can be grown in Mindanao in either wet or dry season
S 6208	Can be grown nationwide in either wet or dry seasons
S 6218	Can be grown nationwide in either wet or dry seasons
P 3776	Can be grown in Luzon and Visayas during wet or dry seasons
TF 8129	Can be grown in Luzon and Mindanao in either wet or dry season but yields are higher under high light intensity.
PD 8230	Can be grown in Luzon and Mindanao in either wet or dry season but yields are higher under high irradiance.
BC 81163	Can be grown in Luzon and Mindanao in either wet or dry season
BC 42683	Can be grown in Luzon during wet or dry season particularly in Regions 1 and 2.
DANA 1107	Can be grown in Luzon or Visayas during wet or dry seasons
DANA 8255	Can be grown in Luzon or Visayas during wet or dry seasons
TSG 222	Can be grown in Luzon and Mindanao in either wet or dry season
TSG 361	Can be grown in low-elevation areas but not water-logged areas in Luzon and Mindanao.
CW 112w	Can be grown in low-elevation areas but not water-logged areas in Visayas.

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TSG 108w	Can be grown in low-elevation areas but not water-logged areas in Visayas and Mindanao
USM Var 30	Can be grown in low-elevation areas but not water-logged areas in Visayas.
BIO #1	Can be grown in Mindanao during wet or dry season particularly in Bukidnon and South Cotabato.
USM Var 28	Can be grown in low-elevation areas but not water-logged areas in Visayas.
MMSU Glut 1	Can be grown in Luzon during wet or dry season particularly in Ilocos and Quirino Provinces.
H 212 wx Glut	Can be grown in Luzon during wet or dry season particularly in Ilocos and Quirino Provinces.
CW 222	Can be grown in Luzon during wet or dry season
SAMPANN	Performs well during both wet and dry season
S 6219	Performs well during both wet and dry season
S 6259	Performs well during both wet and dry season
J757	Can be grown throughout the year provided water supply is adequate
J707	Can be grown throughout the year provided water supply is adequate
J505	Can be grown throughout the year provided water supply is adequate
IES 09-2w	Performs well during both wet and dry season
EG 501	This new hybrid is highly adapted to agro-climatic condition prevailing across Luzon and Visayas
S 6689	This new yellow hybrids is highly adapted to different agroclimatic conditions prevailing across Luzon, Visayas and Mindanao
S 6410	This new yellow hybrids is highly adapted to different agroclimatic conditions prevailing across Luzon, Visayas and Mindanao
PAC339	This variety is highly adapted to the agro-climatic conditions prevailing in major corn growing areas in Luzon and Visayas. This variety performs well during the wet and dry season
PAC 999 Super	This variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon, Visayas. This variety performs well during the wet and dry season
P 4097	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in corn growing areas in Luzon.
P 3668	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing in the major low, medium and high evaluation corn growing areas in Luzon

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X8B649	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in corn growing areas in Luzon and Visayas
CW 1630	This white hybrid performs well in low-elevation but not waterlogged areas in Luzon and Visayas. It can be grown throughout the year provided that water supply is adequate.
BIO 62556	This new hybrid is highly adapted to agro-climatic condition prevailing in Luzon particularly Regions 1 and 2
DK 6919	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in the major l;ow elevation corn growing areas in Luzon and Visayas. It can be grown both during wet and dry season but yields are higher under high irradiance
USM Var 32	This new white OPV is highly adapted to agro-climatic condition prevailing in Luzon
IPB Glut Q OB	This new open pollinated (white-kernel) glutinous variety is very well adapted to agro-climatic conditions prevailing across Luzon, and Mindanao. It performs better during the dry season given adequate water supply particularly during critical growth stages
IPB Glut Q LB	This new open pollinated (white-kernel) glutinous variety is very well adapted to agro-climatic conditions prevailing across Luzon, and Mindanao during wet and dry seasons. It performs better during the wet season.
DK 7878RRC2	This new yellow GMO hybrid is highly adapted to agro-climatic conditions prevailing in the major low elevation corn growing areas in Luzon and Mindanao. It can be grown both during wet and dry season but yields are higher under high irradiance.
P3645 YR	This new yellow GMO hybrid is highly adapted to agro-climatic conditions prevailing in the major low, medium and high elevation corn growing areas in Mindanao
DK 6818	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in the major low and high evaluation corn growing areas in Luzon, Visayas and Mindanao. It can be grown both during wet and dry season.
X5	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in low to medium areas in Visayas and Luzon
BIO 81163	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in Luzon, Visayas and Mindanao
BIO 91538	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in Luzon and Mindanao

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EG 701	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in Visayas
IES Cn 7y	This new open-pollinated yellow variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon Visayas and Mindanao. This variety performs well during wet and dry season.
IES Cn9	This new open-pollinated yellow variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon Visayas and Mindanao. This variety performs well during wet and dry season.
IES 10-04w	This new white open-pollinated variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon and Visayas. This variety performs well during the wet and dry season.
USM Var 34	This new white OPV is highly adapted to agro-climatic condition prevailing in Luzon
P 4097 YR	This new yellow GMO hybrid is highly adapted to agro-climatic conditions prevailing in the major low and medium evaluation corn growing areas in Luzon.
P 3990 YR	The new yellow GMO hybrid is highly adapted to agro-climatic conditions prevailing in the major low, medium and high evaluation corn growing areas in Luzon.
P 3774 YR	This new yellow GMO hybrid is highly adapted to agro-climatic conditions prevailing in the major low, medium and high evaluation corn growing in Luzon and Mindanao.
TEOSINTE 200	This new yellow hybrid variety is highly adaptable to agroclimatic condition prevailing in major corn areas of Luzon.
B1001236	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in Luzon
CSC Var 1	This new yellow open pollinated variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon. This variety performs well during the wet and dry season.
USM Var 40	This new white OPV is highly adapted to agro-climatic condition prevailing in Luzon and Mindanao
USM Var 38	This new white OPV is highly adapted to agro-climatic condition prevailing in Mindanao
USM Var 36	This new white OPV is highly adapted to agro-climatic condition prevailing in Luzon and Mindanao

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USM nch 33	This new yellow non-inbred hybrid variety classified under-pollinated variety performs well in low-elevation but not waterlogged areas in Luzon, Visayas & Mindanao. It can be grown throughout the year provided that water supply is adequate.
USM nch 35	This new yellow non-inbred hybrid variety classified under open-pollinated variety performs well in low-elevation but not waterlogged areas in Luzon, Visayas & Mindanao. It can be grown throughout the year provided that water supply is adequate.
IES Glut 10	This new white glutinous variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Mindanao. This variety performs well during the wet and dry season.
IES Glut 8	This new white glutinous variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Mindanao. This variety performs well during the wet and dry season.
IES Cn 11y	This new yellow open pollinated variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon and Mindanao. This variety performs well during the wet and dry season.
CW-RHB 701	This new yellow hybrid performs well in low-elevation but not waterlogged areas in Luzon. It can be grown throughout the year provided that water supply is adequate.
BH 102 G	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in Luzon.
EG 803w	This new hybrid is highly adapted to agro-climatic condition prevailing in Mindanao
SMQ 6349w	This new white corn hybrid variety performs well in low-elevation but not waterlogged areas in Visayas. It can be grown throughout the year provided that water supply is adequate.
USM nch 39	This new yellow non-inbred hybrid variety classified under open-pollinated variety performs well in low-elevation but not waterlogged areas in Luzon, Visayas & Mindanao. It can be grown throughout the year provided that water supply is adequate.
USM nch 37	This new yellow OPV is highly adapted to agro-climatic condition prevailing across Luzon, Visayas and Mindanao.
CVRC Cn 13	This new yellow open pollinated variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Mindanao only. This variety performs well during the wet and dry season.

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White Flint 12-06	This new white open pollinated variety is highly adaptable to the agro-climatic conditions prevailing in major corn growing areas in Luzon and Mindanao. This variety performs well during the wet and dry season.
B9540R	This new yellow glyphosate tolerant (RR) hybrid is very well adapted to agro-climatic conditions prevailing in Luzon. It performs well during the wet and dry season.
B9541R	This new yellow glyphosate tolerant (RR) hybrid is very well adapted to agro-climatic conditions prevailing in Luzon. It performs well during the wet and dry season even under moderately high population density due to its semi-erect foliage.
B9909R	This new yellow glyphosphate tolerant hybrid is highly adapted to agro-climatic conditions prevailing in the major low elevation corn growing areas in Mindanao.
B9909G	This new yellow stack trait hybrid is highly adapted to agro-climatic conditions prevailing in the major low elevation corn growing areas in Mindanao.
P3640WR	This 1st new white glyphosate tolerant (RR) hybrid variety is very well adapted to the agro-climatic conditions prevailing in low-elevation areas in Mindanao. It can be grown during the wet and dry season.
8899C	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in the major low elevation corn growing areas in Luzon and Visayas. It can be grown both during wet and dry season.
6777C	This new yellow hybrid is highly adapted to agro-climatic conditions prevailing in the major low elevation corn growing areas in Luzon, Visayas and Mindanao. It can be grown both during wet and dry season.
CW HR8	This new yellow hybrid performs well in low-elevation but not waterlogged areas in Luzon. It can be grown throughout the year provided that water supply is adequate.
CW-RHB 709	This new yellow hybrid performs well in low-elevation but not waterlogged areas in Luzon. It can be grown throughout the year provided that water supply is adequate.
EG 501	This new yellow hybrid is highly adapted to agro-climatic condition prevailing in major corn growing areas in Mindanao.

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USM Var 41	This new yellow OPV is highly adapted to agro-climatic condition prevailing in Visayas and Mindanao.
MMSU Glut 2	This new glutinous corn variety is highly adapted to agro-climatic conditions prevailing in Visayas.
DEKALB 6818S Genuity ®	This new GMO hybrid, DEKALB 6818S Genuity ® is highly adapted to agro-climatic conditions prevailing in major and potential corn growing areas in Luzon and Mindanao. It performs very well during the wet and dry season.
DEKALB 6919S Genuity ®	DEKALB 6919S Genuity ® is highly adapted to agro-climatic conditions prevailing in major and potential corn growing areas in Luzon and Mindanao. It performs very well during the wet and dry season.
DEKALB 7898S Genuity ®	DEKALB 7898S Genuity ® is highly adapted to agro-climatic conditions prevailing in major and potential corn growing areas in Luzon and Mindanao. It performs very well during the wet and dry season.
DK 6999	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon, Visayas and Mindanao.
B9545	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
B9540G	This new yellow stack trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
B9541G	This new yellow stack trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
P4124YR	This new yellow stack trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
P3774R	This new yellow glyphosate tolerant hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
P4097R	This new yellow glyphosate tolerant hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
NK6410BtGT	This new yellow trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.

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Agrisure NK8840 EZ Refuge	This new yellow trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
PR 8719	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon, Visayas and Mindanao.
PR 8531	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon, Visayas and Mindanao.
PR 9118	This new yellow conventional hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon, Visayas and Mindanao.
CVRC Cn 15	This new yellow OPV is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
MMSU Glut 3	This new white OPV glutinous variety is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
CVRC Glut No. 12	This new white OPV glutinous variety is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
DK 6999S	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
DEKALB 9898S	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
NK 306BtGt	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
NK 6505BtGt	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
P3774YHR	This new yellow biotech stacked-trait hybrid has an events of MON810 x TC1507 x NK603 also called intrasect, which are YieldGard corn borer protection plus Herculex plus Roundup Ready or tolerance to glyphosate. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.

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P4124YHR	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
B9545RR	This new yellow biotech-RR hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
J505	This new yellow biotech stack-trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
PAC 139	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Luzon, Visayas and Mindanao.
PAC 779	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
BC91528	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Visayas.
AH SUPREME 5150	This new biotech-stack trait hybrid is both glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
P3530YHR	This new yellow biotech-stack trait hybrid is highly adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
P3530R	This new yellow biotech hybrid has an event of NK603, Roundup Ready that is tolerant to glyphosate. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
PAC 777	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
K-509	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
K-511	This new conventional yellow hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Mindanao.
PAC 1510003	This new conventional white hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Visayas.

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PAC 1510066	This new conventional white hybrid is widely adapted to agro-climatic conditions prevailing across corn growing areas in Visayas.
IPB 1910	This new white open pollinated variety is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
PAC 339 VT2 PRO	This new yellow biotech stacked-trait hybrid with the events of MON89034 x NK603 is insect resistant and Roundup Ready. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
PAC 999 VT2 PRO	This new yellow biotech stacked-trait hybrid with the events of MON89034 x NK603 is insect resistant and Roundup Ready. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
B9545G	This new yellow biotech stacked-trait hybrid with the events of MON89034 and NK603, a glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
B9899G	This new yellow biotech stacked-trait hybrid with the events of MON89034 x NK603, a glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
H101G	This new yellow biotech stacked-trait hybrid with the events of MON89034 x NK603, a glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
H102G	This new yellow biotech stacked-trait hybrid with the events of MON89034 x NK603, a glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon.
DK8899S	This new yellow biotech stacked-trait hybrid with the events of MON89034 and NK603, a glyphosate tolerant and resistant to corn borer. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
SS6414BtGt	This new yellow biotech stacked-trait hybrid with the events of Bt11/GA21 Stack which is ACB resistant and Glyphosate tolerant. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in Luzon and Mindanao.
P4097YHR	This new yellow biotech stacked-trait hybrid has events of MON810 x TC1507 x NK603 also called intrasect, which are YieldGard corn borer protection plus Herculex plus Roundup Ready or tolerance to glyphosate. It is highly adapted to agro-climatic conditions prevailing across corn growing areas in

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	Luzon.
FARCO 58	Longer ear length than the check, high shelling recovery than the check, good husk cover.
MMSU BMIR Glut1	Longer ear length than the check, high shelling recovery, good husk cover.
CVRC Glut# 18-14	Good ear length, high shelling recovery than the check, good husk cover.
FARCO 88	Good husk cover, good ear fill, good ear length, high shelling recovery than the check.
K-508	Good husk cover, good ear fill, longer ear length than the check.
DEKALB®9118S VTDoublePro RIB Complete	Good ear length, good husk cover, excellent ear fill,high shelling recovery, can be planted in high density due to erect leaves and stand ability.
DEKALB®8719S VTDoublePro RIB Complete	Good ear length, good husk cover,high shelling recovery, can be planted in high density due to erect leaves and stand ability.
DEKALB®9919S VTDoublePro RIB Complete	Good ear length, good husk cover, excellent ear fill,high shelling recovery, can be planted in high density due to erect leaves and stand ability.

Eggplant

Eggplant (*Solanum melongena L.)* originated from India and is generally grown as a vegetable throughout the tropical, sub-tropical and warm temperate areas of the world. It is an important vegetable in countries like India, China, Japan and Philippines. It is termed as aubergine and brinjal (English,) talong (Tagalog), tarong (Ilocano) or bringhinas (Bisaya). Eggplants belong to the Solanaceae family, which includes tomatoes and peppers. In the Philippines, eggplant is considered one of the top vegetable crops in terms of volume of production (around 186,000 metric tons from January to June 2019). The size, shape, and color of eggplant varies from round to long and thin, and purple to white and green. East-West Seed offers a wide range of eggplant varieties to cater to the different preferences of consumers in various regions in the Philippines.

Varieties	**Characteristics**
Dingras Multiple Purple #1	95-105 days from planting to harvesting
E.G. Long Purple	90-100 days from planting to harvesting
Dumaguete Long Purple	95-105 days from planting to harvesting
UPL Eg-II	86 days of maturity, dark purple to blackish, broad to oval

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CA Cluster	55 days of maturity, purple
Mara	65 days of maturity, Purple, cylindrical, well adapted both for wet and dry season production, resistant to bacterial wilt
Mamburao	67 days of maturity, well adapted both for wet and dry season production, resistant to bacterial wilt
Tanglaw	55 days of maturity, well adapted both for wet and dry season production at Luzon, Visayas and Mindanao, resistant to bacterial wilt
Tanyag	60 days of maturity
Sikat	Purple, rounded, well adapted both for wet and dry season production at Luzon, Visayas and Mindanao, resistant to bacterial wilt
Sulit	Purple, rounded, well adapted both for wet and dry season production at Luzon, Visayas and Mindanao, resistant to bacterial wilt
Fortuner F1	Popular in most regions in the Philippines
Gwapito F1	Prefered in Nueva Ecija and some other parts of Northern Luzon.
Mistisa (A300)	white with purple stripes, grows to 20 centimeters long, has a long shelf life and acceptable quality and very prolific. It is tolerant to bacterial wilt and leafhopper and has resistance to shoot and fruit borer and phomopsis blight.
Century Round Allgrow	Plant height is 16 inches upright. Fruit is glossy with green calyx. Adaptable in lowland and in relatively cold areas.
Hybrid chhaya	Cluster bearing (2-3), resistant to bacterial wilt.
Hybrid Black Pride Allgrow	With long shelf life. Abundant fruiting.
Hybrid Eggplant Zam	High yielding, light green fruits.
Japanese type early Bird	A Japanese eggplant hybrid that bears numerous fruits. Has long productive life. Very attractive color and firm flesh. Ideal for tempura and frying.
Long purple ABC select	Moderately vigorous and uniform plants.
Long San Juan	Good tolerance against bacterial wilt. Glossy fruit.
Mayumi	High yielding (Pakbet type) moderately resistant to bacterial wilt.
Pepito	Fresh market, bacterial wilt resistant.
Spitfire 252	Fruits are glossy with attractive bright green calyx. A premium quality hybrid.

Pineapple

Pineapple (*Ananas comosus*), a tropical plant with edible multiple fruit consisting of coalesced berries, named for resemblance to the pine cone, is the most economically important plant in the *Bromeliaceae* family. It is originally from South America, most probably from the region between South Brazil and Paraguay.It is mainly grown for its fruits, either fresh or in processed

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forms. Pineapples may be cultivated from a crown cutting of the fruit, possibly flowering in 20-24 months and fruiting in the following six months.

In the Philippines, Pineapple is extensively cultivated in the Northern Mindanao, SOCCKSARGEN (Southern Cotabato, Cotabato Province, Sultan Kudarat, Sarangani, General Santos City), Bukidnon, Bicol and CALABARZON (Cavite, Laguna, Batangas, Rizal, Quezon) for domestic and foreign markets either fresh and for processing. The Philippines is the second biggest pineapple-exporting country in the world next to Thailand. In 2009, 57,687 ha were planted to the crop with a total production of 2,198,497 MT.

Varieties/Cultivars	Characteristics
Smooth Cayene (Hawaiian)	The most favored for canning and fresh fruit consumption. It matures in 18 months and yields 25,000 fruits per hectare. The fruit is cylindrical in shape (2.5–3 kg) and the flesh is pale yellow, soft and juicy with high sugar (from 13° to 19°Brix) and acid content. Well adapted to canning and processing. Leaves are spineless and cylindrical.
Queen (Formosa)	Queen variety is grown in some places for its fresh fruit. The small and very spiny plant gives a small fruit (0.5 to 1 kg), with a full yellow shell and small prominent eyes. The golden yellow pulp is crispy and sweet (14 to 18°Brix), with an excellent flavor and long shelf life.
Red Spanish (Native Philippine Red)	The Red Spanish (Native Philippine Red) is cultivated for its strong and delicate fiber in Western Visayas, particularly in the province of Aklan. The fruit is squarish in shape (1–2 kg) with pale yellow flesh and with pleasant aroma. Leaves are spiny.
Cabezona	It is the largest, measuring approximately 8-12 inches long when fully matured.

Soybean

Soybeans were a crucial crop in East Asia long before written records began. It originated inSsoutheastern Asia (including China, Japan, and Korea) that was domesticated more than 3,000 years ago for its edible seeds and young pods. Soybean (Glycine max (L.) Merrill) or soya bean is a legume crop belonging to the family Leguminosae or Fabaceae and sub-family Papilionaceae. The plant grows up to 1.5 meters tall, depending on variety. Erect stems are covered with thick brown hair. Leaves are compound, with 3 leaflets. They are alternate, trifoliate with ovate leaflets and short peduncle. The inconspicuous, stalkless white to purple flowers are borne singly or in small clusters in the axils (where leaf meets stem). The fruit is a broad, hairy, flattened legume or pod, around 10 cm (3 in) long, yellow to brown when fully mature and dried. The fruits are called pods measuring up to 7 centimeters long, containing 1 to 4 seeds which are colored yellow, black or green.

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Varieties	Characteristics
PSB Sy 1	Resistant to soybean rust and bacterial pustule
PSB Sy 3	Resistant to soybean rust and bacterial pustule
PSB Sy 8	Resistant to soybean rust and bacterial pustule
PSB Sy 6 (Tiwala)	Resistant to soybean rust and bacterial pustule
PSB Sy 7 (Tiwala 10)	Yellow seedcoat; has good seed storability; recommended for wet season cropping in So. Mindanao
NSIC Sy 01	Resistant to soybean rust and bacterial pustule
NSIC Sy 02	Resistant to soybean rust and bacterial pustule
NSIC Sy 05	Resistant to soybean rust and bacterial pustule
NSIC Sy 06	Resistant to soybean rust and bacterial pustule
NSIC Sy 07	Resistant to soybean rust and bacterial pustule
NSIC Sy 09	Resistant to soybean rust and bacterial pustule
Tiwala 14	proves to yield 14% to 70% more than the benchmark variety registered with the National Seed Industry Council (NSIC).
Tiwala 16	proves to yield 14% to 70% more than the benchmark variety registered with the National Seed Industry Council (NSIC).
Tiwala 18	proves to yield 14% to 70% more than the benchmark variety registered with the National Seed Industry Council (NSIC).
Tiwala 20	proves to yield 14% to 70% more than the benchmark variety registered with the National Seed Industry Council (NSIC).

Bittergourd

Bittergourd or ampalaya (Momordica charantia L.) belongs to the family Cucurbitaceae along with cucumber, squash, watermelon and muskmelon. Native to China and India, the fast-growing vine is grown throughout Asia. Ampalaya is a climbing vine that can grow as long as 5 m. It has tendrils, palmately veined leaves and cut nearly to the base into five or seven oblong-ovate, variously toothed lobes that form a heart-shaped base. It also has separate male and female flowers. The male flower has peduncle or stalk about 2-3 cm while the female flower has 3.5 cm long peduncle with small fruit. Flowering starts 45-55 days from sowing and lasts up to 6 months.

Varieties	Characteristics
Hybrid Socorro	Uniform, firm and glossy fruits. Good tolerance to diseases Green in color
La Trinidad	High tolerance to virus such as powdery mildew and the like Deep green in color
Sta. Lucia	A very prolific and uniform selection with thick flesh. Well adapted and vigorous. Good for year round production Bright shiny green in color
Trident 357	Very vigorous, large and uniform in size Dark green in color,
Bio choo Choo	Plant is vigorous with strong vines early maturing hybrid

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	Dark green in color,
Mestiza F1	light green, cylindrical in shape with an average length of 31 cm, weighing an average of 340 grams per fruit, fruits are very glossy, with smoother fruit surface, very high-yielding and attractive fruited Bitter gourd hybrid, has intermediate resistance (IR) to Cucurbit aphid-borne yellows virus (CABYV), a polerovirus also locally known as “namamarako”
Galactica F1	dark green in color and tapered in shape, shorter in length (average length of 28.0 cm) and weigh an average of 300 grams per fruit, fruits are glossy but has warty surface
Galaxy F1	dark green, cylindrical, longer and heavier fruits, has an average length of 33cm, 350 grams per fruit, fruits are moderately glossy with smoother fruit surface.
Sta Rita Strain	Has moderately bitter taste and is long in size, 45 DAP
Jade Star	Has bitter taste and is medium in size, 45 DAP
Native	Has bitter taste and is small in size, 50 DAP

Mango

Mango or *Mangifera indica* originated in India and has become the national fruit of the Philippines. It has evolved into different types and has long been part of the Philippine dessert, snack and even used in appetizers and main dishes.

Below are the different kinds of Philippine Mangoes:

1. Carabao Mango *(Manggang Kalabaw)*

The popular Philippine variety is the Carabao Mango, or known as champagne mango, sour and green when raw but orange-yellow and very sweet when ripe. Its flesh is very tender, soft and juicy with very little fiber or none at all. However, the carabao mango has evolved into different strains that are popular and exported worldwide.

2. Indian Mango

This mango came from India's Alphonso Mango. Best when eaten raw, this mango grows anywhere most particularly popular in the suburbs. The fruit is round in shape, with its thick flesh, slightly tart, firm and fibrous flesh are popular and mostly requested as street food.

3. Apple Mango

Similar to the characteristics of the Indian mango, the Apple Mango is much bigger and with a reddish tint on the top part of the of the fruit similar to the color of the red apple.

4. Horse Mango (Manggang Kabayo)

The shape of this mango is elongated and is usually large sized. It is also best eaten raw and has a tartly sweet taste. It has a firm flesh and large seed.

5. Pajo or *Pahutan* Mango

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This small mango is very sweet, juicy but very fibrous. It has a thick flesh and the way to eat it is to use your hands. It would be impossible to use any cutlery because of the fibers and the almost absence of flesh.

6. Evergreen Mango

This type retains in green color when ripe, and you can tell if it's overripe if you notice the yellow specks around the fruit. This mango variety is only found in Tangub City, Misamis Occidenatal, Philippines. It has a round shape and it's taste and texture is similar to the carabao mango. It is also very sweet with a tender juicy orange flesh.

Cacao

Cacao or *Theobroma cacao* literally translates as “food of the gods” in Greek. It is grown mainly for its seeds known as the cocoa beans which are used to make cocoa mass, cocoa powder and chocolate.

Cacao Varieties

1. Criollo is considered as the most prized, rare and expensive variety. It is native to Central and South America, and it is believed that it was the first cacao seed planted in the Philippines by Acapulco-Manila Galleon Trade in 1670. This variety is difficult to grow, as extremely susceptible to pests and diseases. Its beans are white to pale pink in colour and recornized as a superior quality, less bitter and more aromatic. It is considered as the “Prince of Cocoas” and an ingredient to premium chocolates.

2. Forastero, a native of the Amazon basin, is the most versatile variety and most commonly grown cocoa. It is significantly harder, disease resistant and high yielding. Its beans are purple-coloured and mainly used to give chocolate its full-bodied flavor. It has bitter taste, thus often blended with superior cocoas.

3. Trinitario, the hybrid of Criollo and Forastero, combines the best of the two varieties: the hardness and high yield of Forastero and the refined taste of Criollo. It is the predominant fine flavor cocoa and can be found in all the countries where Criollo cocoa was once grown including Southeast Asia and the Philippines. It is being used in about 10% of the world cacao supply.

Varieties from National Seeds Industry Council

Variety	Characteristics
BR 25 (CC-99-05)	Pod color- Reddish (red with green) when still young that turns yellow as it matures. Leaves are elliptical in shape with wavy leaf margins. Leaf length and width ratio is 11 cm is to 4.04 cm. First flowering starts at 16.12 months and fruiting follows at 17.70 months. Pod index is 23.1 pods/kg of dried beans. Pod length is 17.02 cm and has width of 7.07.
K 1	It has an elliptical leaf shape with smooth leaf margins. Leaf length is 31.21 cm with a mean width of 13.44 cm. It flowers at 23.20 months and bears fruit at 25.10 months Pod shape is Amelonado with a superficial ridge and a smooth surface. Rugosity also appears in some cases. Pod index is 19.20 pods/kg of

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	dried beans; its length is about 17.97 cm and has width of about 8.67 cm; pod wall thickness is around 1.80 cm. Its pod is red in color when still young and becomes yellow/orange when hits maturity. Bean color is violet and a hundred beans weigh about 182 grams. It is moderately resistant to known insect pests and diseases.
K 2	Leaf shape is elliptical with smooth leaf margins; leaf length is 32.73 cm with a mean width of 12.52 cm. It flowers at 21.10 months and bears fruit after three months or at 24.12 months. Pod shape is Amelonado and is in red color while still young and becomes yellow/orange when mature. It has pod wall thickness of around 1.40 cm with 34 beans per pod. Pod index is 25 pods/kg of dried beans. It is moderately resistant to known insect pests and diseases.
UIT 1(CC-99-02)	It has an elliptical leaf shape with wavy leaf margin. Leaf length and width ration is 22.36 cm to 8.13 cm. Pod shape is Cundeamor; pod length is 20.07 cm and width of 8.65 cm. Pod is yellow when old from the original color of green color when still young with wall thickness of 1.02 cm. Pod index is 21.69 pods/kg having 46 beans/pod; bean color is violet in color. Canopy diameter is 278 cm. It is moderately resistant to insect pests and diseases.
ICS 40	Leaf shape is elliptical with wavy leaf margin. Leaf length and width ratio is 29.95 cm is to 10.01 cm. Starts to flower at the age of 17.63 months and fruiting follows at 19.63 months. Pod shape is Cundeamor describe as a variety with elongated cylindrical fruit with ridges, a rugose surface, pronounced bottleneck and sharp point. Pod length and width ratio is 16.02 cm is to 9.45 cm. Pod color is green when young and turns yellow when mature with wall thickness of 1.35 cm. Pod index is 16.2 pods/kg with 44 beans per pod. Canopy diameter is 195 cm. It is moderately resistant to insect pests and diseases.
UF-18	Newly released variety by NSIC. It has big beans (approximately 80 dry beans/100 grams), has good yielding ability, has good pest resistance and with high butter content.

Sugarcane

Sugarcane, a plant that belongs to the genus Saccharum, a Latin word which means sugar. The genus has two species, namely *S. officinarum* (cultivated species) *and S. spontaneum* (wild species).

NSIC-Recommended varieties

Variety	Characteristics
NSIC 2002 Sc 01 or Phil 8013	Fast grower with moderately profuse flowering. At 8-10 months, it has yellowish to brownish green stalk while at maturity, it has purple stalk with drooping broad

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	leaves and semi-self-trashing. It is drought tolerant and the most widely planted variety in sugarcane producing areas. This variety has a potential yield (average yield across ecological test sites) of 103-ton cane/ha (TC/ha), 50 kg-ton/ton cane (LKg/TC), and 287 LKg/ha. It is resistant to downy mildew and smut, moderately resistant to leaf scorch, and susceptible to yellow spot.
NSIC 2002 Sc 02 or Phil 8477	Has greenish purple stalk with moderate wax, small to medium eyebuds, long and average dark green leaves, and erect to reclining stand. Commonly known as FVR variety, it is highly acceptable to the farmers as reflected by large area planted. This variety has a potential yield of 92 TC/ha, 2.32 LKg/TC, and 212 LKg/ha. It is resistant to downy mildew, smut, leaf scorch, and yellow spot.
NSIC 2002 Sc 03 or Phil 8583	It has yellow green to purple stalk, medium-sized (2.42 cm in diameter), waxy, and with bud grooves and round buds. It has erect leaves and self-trashing characteristics. This variety is also known as "Philsutech 1." Like the FVR variety, it is also highly acceptable to the farmers. Its potential yields are 106 TC/ha, 2.31 LKg/TC, and 251 LKg/ha. It is resistant to downy mildew, smut and lead scorch, but susceptible to yellow spot.
NSIC 2002 Sc 04 or Phil 8727	It is commonly known as "Philsutech 2, has purplish green stem, medium-sized reddish-purple cylindrical stalks, with corky cracks and patches in mature canes, few trichomes, oval eyebuds, green with purple base leaf sheaths, and is self-trashing. Its potential yields are 103 TC/ha, 1.82 LKg/TC, and 187 LKg/ha. It is resistant to downy mildew, smut, and yellow spot.
NSIC 2002 Sc 05 or Phil 8839	It is also known as Ming Ramos variety; has waxy, medium-sized purple stalk, with broad leaves, and is self-trashing. This variety has high sugar yield and acceptable to the farmers as reflected by the large area planted. It has a potential yield of 94 TC/ha, 2.11 LKg/TC, and 196 LKg/ha. It is resistant to downy mildew, leaf scorch, smut, and moderately resistant to yellow spot.
NSIC 2002 Sc 06 or Phil 8943	Fast germinator and fast starter. It has yellowish green cylindrical stalk with corky patches and cracks. The leaves are erect with semi-self-trashing characteristics. It has a potential yield of 116 TC/ha, 1.71 LKg/TC, and 201 LKg/ha. It is resistant to downy mildew, leaf scorch, smut but susceptible to yellow spot.
NSIC 2002 Sc)7 or Phil 91-1091	It is erect and fast grower; it has medium-sized (2.69 cm in diameter), cylindrical, and purple to greenish purple stalks. It is waxy, with long bud grooves. The

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	young leaves are erect with drooping tip while older leaves droop from middle to top. It has a potential yield of 105 TC/ha, 1.71 LKg/TC, and 173 LKg/ha. It is resistant to downy mildew, leaf scorch, smut but susceptible to yellow spot.
NSIC 2002 Sc 08 (PS 1) or VCM 84-947	It is erect, self-trashing and germinates well. It produces heavy tillers and sparse tassel. It shows tolerance to drought and good ratooning ability. It has a potential yield of 105 TC/ha, 1.71 LKg/TC, and 173 LKg/ha. It is resistant to downy mildew, leaf scorch, smut and moderately resistant to yellow spot.
NSIC 2002 Sc (PS 3) or VCM 84-524	It is an early maturing variety with good germination but slow early growth. It shows heavy tillering, erect torecumbent, has abundant trichomes that falls off at maturity, self-trashing, and reluctant to sparse tasselimg. It has a potential yield of 94 TC/ha, 212 LKg/TC, and 196 LKg/ha. It is resistant to downy mildew, leaf scorch, smut and yellow spot.
NSIC 2002 Sc 10 or VMC 84-549	It has a stalk diameter measuring 2.65 cm and semi-self-trashing. Its potential yields are 90 TC/ha, 187 LKg/TC and 163 LKg/ha. It is resistant to downy mildew, leaf scorch, smut and yellow spot.
NSIC 2002 Sc 11 or VMC 86-550	It has a stable and high LKh/TC. This highly self-trashing variety has a potential yield of 87 TC/ha, 1.93 Kg/TC and 162 LKg/ha. It is susceptible to downy mildew and smut but resistant to leaf scorch and yellow spot.
NSIC 2002 Sc 12 or VMC 87-95	It is heavy-tillering variety. It has a stalk diameter measuring 2.62 cm. the leaf is semi-self-trashing. This variety has a potential yield of 93 TC/ha, 1.7 LKg/TC and 144 LKg/ha. It is resistant to downy mildew, leaf scorch, yellow spot and moderately resistant to smut.
NSIC 2002 Sc 13 or VMC 87-599	It is known for having high sugar yield per hectare. Its stalk measures 244 cm in length and 2.69 cm in diameter. It has self-trashing leaf characteristics. This variety has a potential yield of 89 TC/ha, 1.81 LKg/TC, and 153 LKg/ha. It is resistant to downy mildew, leaf scorch, yellow spot and moderately resistant to smut.
NSIC 2002 Sc 14 (PS 2) or VMC 88-354	It is a good germinating, fast growing, heavy tillering and self-trashing variety. It is a good ratooner and sparse tasseler. This variety has a potential yield of 108 TC/ha, 1.73 LKg/TC and 184 LKg/ha. It is resistant to downy mildew, leaf scorch, yellow spot and smut.
PS 4 or VMC 95-152	It is characterized to be fair germinator, average grower, with moderate tillering and semi-self-trashing

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	growth habit. It thrives well in moist conditions and in loose or well-drained soils because of its slow to moderate growth. It is a good alternative variety for the early milling season in Davao, Bukidnon Sugar Milling Corporation (BuSCo), San Carlos, Ormoc, Balayan, Tarlac, Lopez, sagay, and Bogo areas. This variety has an average yield of 198 LKg/ha from 22 plant and 176 LKg/ha from 18 ratoons. It is resistant to downy mildew and rust and intermediate to leaf scorch and smut.
PS 5 or VMC 95-09	It is a good germinator with fair to heavy tillering, fast grower, and possesses good ratooning ability. It is characterized to be semi-self detrashing in growth habit, profuse tasseler and has high tonnage and drought tolerance. This variety is adapted in dry conditions of Type C and D weather patterns and in clayey soils. It is recommende in Bogo, Capiz, Balayan, Tarlac, Lopez and San Carlos. This variety has an average yield of 186 LKg/ha from plant and 168 LKg/ha from ratoon trials. It has intermediate resistance to downy mildew.

The Sugar Regulatory Administration (SRA) was able to develop superior varieties before the creation of NSIC. The following varieties are all high yielding and released for commercial planting.

Phil 8361	Phil 8360 has erect to recumbent and fast growth characteristics. Leaves are long, wide, and light green to green, reclining to lodging, erect with drooping tips, green leaf sheath with long inner and short outer auricle, no trichomes and highlyself-trashing. It has bobbin stalks that are greenish yellow when young and bright yellow at maturity with light wax and round protruding eyebuds with wings. It has a potential yield of 2.05 LKg/TC and 130.19 TC/ha. This variety is resistant to smut, downy mildew, and yellow spot, and moderately resistant to leaf scorch.
Phil 8835	Phil 8835 variety is erect and has medium growth habit with very sparse flowering characteristics. The leaves are narrow, erect with drooping tips, and semi-self-trashing. The leaf sheaths are yellowish green with short auricles and have few trichomes. The stalks are small in diameter (2.02 cm), waxy and purplish green. It has a potential yield of 2.29 LKg/TC and 114.24 TC/ha. It is resistant to downy mildew, smut and leaf

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	scorch and moderately resistant to yellow spot.
Phil 8839	Phil 8839 has erect to recumbent and medium growth habit. The leaves are semi-self-trashing, slight trichomes on leaf sheath, lamina on the first to second leaves are erect to the tip, and succeeding leaves are drooping at the center. Leaves are medium in diameter measuring 2.5-3.0 cm. Stalks are medium in diameter, waxy and purple. It has a potential yield of 2.24 Lkg/TC and 120.77 TC/ha. This variety is highly resistant to downy mildew, leaf scorch and yellow spot and resistant to smut.
Phil 8943	Phil 8943 is fast germinator, fast starter, with medium and erect to recumbent growth habit. The leaf sheath is yellowish green with trichomes. The first and second leaves are erect to tip and the succeeding leaves are medium, drooping at the center, and self-trashing. It has a potential yield of 1.91 LKg/Tc and 141.59 TC/ha. It is resistant to smut, downy mildew, and leaf scorch but susceptible to yellow spot.
Phil 90-0345	Phil 90-0345 is a fast-growing variety, has medium-sized stalks measuring 2.79 cm in diameter, waxy and cylindrical. The stalks are brownish green with long bud grooves. Matured canes have corky patches and aerial roots. Leaf sheath is green with very few trichomes. Leaves are broad that gradually droops at the center and non-self-trashing. It has a potential yield of 2.10 LKg/TC and 142.83 TC/ha. It is resistant to leaf scorch, smut and downy mildew and susceptible to yellow spot.
Phil 92-0051	Phil 92-0051 is fast grower and has erect growth habit. The leaves are narrow and erect with drooping tip. The leaf sheath is green without trichomes and semi-self-trashing. The stalks are medium in size (2.67 cm), waxy, yellowish green, and has bud grooves and round eyebuds with wings. It has a potential yield of 1.88 LKg/TC and 168.59 TC/ha. It is resistant to smut, downy mildew, leaf scorch, and yellow spot.
Phil 92-0577	Phil 92-0577 is fast grower and has erect to recumbent growth habit. It has large stalks measuring 3.03 cm in diameter, yellowish green and waxy with bud grooves. Eyebuds are round with wings. The leaves are broad, gradually drooping at the middle, with short inner auricle, moderate but fine trichomes, and semiiself-trashing. It has a potential yield of 179.13 TC/ha and 2.05 LKg/TC. It is resistant to smut, downy mildew, leaf scrotch and moderately resistant to yellow spot.
Phil 92-0751	Phil 92-0751 is fast grower and with erect to recumbent

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	growth habit. It has medium sized stlks of 2.62 cm in diameter, brownish purple, bobbin-shaped, waxy with bud grooves. Eyebuds are oval with wings. Young leaves are erect while older leaves droop. The leaves are medium with purplish green leaf sheath, short auricle, with few trichomes and self-trashing. It has a potential yield of 1.91 LKg/TC and 177.6 TC/ha. This variety is resistant to smut, downy mildew, leaf scorch, and yellow spot.
Phil 93-2349	Phil 93-2349 is fast grower and has erect growth habit. The stalks are cylindrical, brownish purple with corky patches. It has a potential yield of 2.28 LKg/TC and 127.67 TC/ha. This variety is resistant to smut, downy mildew, leaf scorch, and moderately resistant to yellow spot.
Phil 93-3155	Phil 93-3155 is fast grower and has erect growth habit with profuse flowering characteristics. The leaves are medium-sized that gradually droops with sshort inner auricle. The stalks measures 2.80 cm, cylindrical, greenish yellow with bud grooves. It has ovate eyebuds, moderate trichomes and self-trashing. It has a potential yield of 2.38 LKg/TC and 112.13 TC/ha. This variety is resistant to smut and downy mildew, and moderately resistant to leaf scorch.
Phil 93-3727	Phil 93-3727 is fast grower, has erect growth habit with sparse flowering characteristics. The leaves are medium-sized, drooping, with purplish green leaf sheath and short inner auricle. Stalks are medium-sized (2.88 cm), cylindrical, brownish yellow with no bud groove. It has oval eyebuds with moderate trichomes and is self-trashing. This variety has a potential yield of 2.33 LKg/TC and 112.62 TC/ha. It is highly resistant to smut, yellow spot and downy mildew, and moderately resistant to leaf scorch.
Phil 93-3849	Phil 93-3849 is fast grower with erect growth habit and sparse flowering characteristics. The leaves are narrow and erect with green leaf sheath and short inner auricle. The stalks are medium-sized, cylindrical, yellowish green with no bud groove. It has round eyebuds with wings, heavy trichomes and is self-trashing. It has a potential yield of 2.30 LKg/TC and 128.51 TC/ha. It is resistant to smut, downy mildew, yellow spot , and moderately resistant to leaf scorch.
Phil 94-0913	Phil 94-0913 has erect to recumbent growth habit with sparse flowering characteristics. The leaves are medium-sized with erect tips on younger leaves and drooping when mature. Leaf sheaths are green with purple base, few trichomes, and semi-self-trashing. Stalks are medium sized, barrel shaped, greenish

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	brown, waxy, with bud grooves and corky patches. It has a potential yield of 2.45 LKg/TC and 141.16 TC/ha. It is resistant to smut, downy mildew, and moderately resistant to yellow spot and leaf scorch.

HOT PEPPER

Hot Pepper (*Capsicum annuum L.*) is considered to be one of the popular high value crops in the country. It is a perennial plant with small tapering fruits. This crop is commonly planted during May to July and September to December, although offseason production is encouraged for higher profit.

Variety	Characteristics
Red Hot F1	It has superior yield and vigorous with good disease tolerance, it has dark green to bright red colored fruits which is market preferred.
Vulcan F1	High yielding variety, adapted for low-to-mid-elevation planting; its fruits are uniform and glossy.
Django F1	light green "sinigang" type, high yielding variety and for year-round planting. It is resistant to bacterial wilt.
Django Dos F1	This variety is high yielding and early-maturing. Its fruit is glossy, light green in color; has intermediate resistance to cucumber mosaic virus.
Super Heat F1	very strong plant vigor, intermediate resistance to bacterial wilt and adapted for growing in rainfed areas.
Hotshot	an open pollinated variety with moderately pungent and medium dark green to red color of fruit.

ONION

Onion (*Allium cepa L.*) is considered to be one of the most indispensable culinary ingredient in the world. Used as a popular spice in the country, onion is mainly grown in Central Luzon and Ilocos Region for local and export markets.

Bulb onions

Variety	Characteristics
Red Globe	bulbs medium-size, round to oval, purplish color skin; maturity is 100 days.
Excel	Bulbs small to medium size, flat, flesh white. Maturity is 90-100 days. Yields 58 tons/ha.
Granex	Medium-size bulbs with brownish yellow skin. Maturity is 90 days; yields 23 tons/ha.
Red Creole	Bulbs medium-size, flat to medium oblate, well-filled neck, skin medium thick and dull buff, red, good keeper with purplish skin. Maturity is 90-100 days; yields 18 tons/ha.
Crystal Wax	Bulbs medium-size, very flat and white. Maturity is 100 days; yields 10-20 tons/ha.

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Certified Creole	Maturity is 90-100 days; yields 10-20 tons/ha.

Green Onions

Varieties	Characteristics
Japanese bunching	long, slender leek-like silvery white stalks, light green color
Baguio green	Produces many divisions and flowers are delicate.
Boltsville bunching	long slender leaves

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2. Genetically Modified Organisms (GMO)

Learning Objectives:

At the end of this topic, the participant should be able to:

1. identify the genetically modified organism (GMO),
2. understand the economic importance of GMO, and
3. learn the different varieties of sweet potato.

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What does "GMO" mean?

Many individuals are aware with the popular acronym “GMO”, which stands for “genetically modified organism.” The more scientifically accurate term is “genetically engineered organism.”

This term refers to any organism (such as a plant or animal) whose genome (DNA) has been altered through the use of genetic engineering techniques, in which an undesired gene is removed, or a desired gene is isolated, copied, or synthesized, before it is prepared and then inserted into the host genome. The technology is often called “modern biotechnology” or “gene technology”, sometimes also “recombinant DNA technology” or “genetic engineering”. It allows selected individual genes to be transferred from one organism into another, also between nonrelated species. Foods produced from or using GM organisms are often referred to as GM foods.

In plants, the most common techniques for inserting a gene are:

1. *Agrobacterium*-mediated recombination (using a type of bacteria known for its ability to transfer DNA between itself and plants)
2. Biolistics, also known as microparticle bombardment (using a biolistic particle delivery system, also known as a “gene gun,” to insert DNA)
3. Electroporation (applying an electrical field to cells, in order to increase the permeability of their cell membranes, so that DNA can be introduced into the cell)

Once a cell has been transformed by using one of these methods, a new plant is grown from that cell via tissue culture. Then, tests are performed to make sure the plant contains the DNA that was added to the original cell.

There are a few different types of genetically engineered organisms. *Transgenic* organisms have been modified to contain gene(s) obtained from an unrelated or sexually incompatible species. *Cisgenic* organisms contain gene(s) from another organism of the same species, or from a sexually compatible species.

Objective for developing plants based on GM organisms:

To improve crop protection. The GM crops currently on the market are mainly aimed at an increased level of crop protection through the introduction of resistance against plant diseases caused by insects or viruses or through increased tolerance towards herbicides.

- Resistance against insects is achieved by incorporating into the food plant the gene for toxin production from the bacterium *Bacillus thuringiensis* (Bt). This toxin is currently used as a conventional insecticide in agriculture and is safe for human consumption. GM crops that inherently produce this toxin have been shown to require lower quantities of insecticides in specific situations, e.g. where pest pressure is high.
- Virus resistance is achieved through the introduction of a gene from certain viruses which cause disease in plants. Virus resistance makes plants less susceptible to diseases caused by such viruses, resulting in higher crop yields.
- Herbicide tolerance is achieved through the introduction of a gene from a bacterium conveying resistance to some herbicides. In situations where weed pressure is high, the use of such crops has resulted in a reduction in the quantity of the herbicides used.

Is the safety of GM foods assessed differently from conventional foods?

Generally consumers consider that conventional foods (that have an established record of safe consumption over the history) are safe. Whenever novel varieties of organisms for food use are developed using the traditional breeding methods that had existed before the introduction of gene technology, some of the characteristics of organisms may be altered, either in a positive or a negative way. National food authorities may be called upon to examine the safety of such conventional foods obtained from novel varieties of organisms, but this is not always the case.

In contrast, most national authorities consider that specific assessments are necessary for GM foods. Specific systems have been set up for the rigorous evaluation of GM organisms and GM foods relative to both human health and the environment. Similar evaluations are generally not performed for conventional foods. Hence there currently exists a significant difference in the evaluation process prior to marketing for these two groups of food.

The WHO Department of Food Safety and Zoonoses aims at assisting national authorities in the identification of foods that should be subject to risk assessment and to recommend appropriate approaches to safety assessment. Should national authorities decide to conduct safety assessment of GM organisms, WHO recommends the use of Codex Alimentarius guidelines.

How is safety assessment of GM food conducted?

The safety assessment of GM foods generally focuses on:

- direct health effects (toxicity)
- potential to provoke allergic reaction (allergenicity)
- specific components thought to have nutritional or toxic properties
- the stability of the inserted gene
- nutritional effects associated with genetic modification; and
- any unintended effects which could result from the gene insert

Issues and Concern for human health:

While theoretical discussions have covered a broad range of aspects, the three main issues debated are the potentials to provoke allergic reaction (allergenicity), gene transfer and outcrossing.

Allergenicity

As a matter of principle, the transfer of genes from commonly allergenic organisms to non-allergic organisms is discouraged unless it can be demonstrated that the protein product of the transferred gene is not allergenic. While foods developed using traditional breeding methods are not generally tested for allergenicity, protocols for the testing of GM foods have been evaluated by the Food and Agriculture Organization of the United Nations (FAO) and WHO. No allergic effects have been found relative to GM foods currently on the market.

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Gene transfer

Gene transfer from GM foods to cells of the body or to bacteria in the gastrointestinal tract would cause concern if the transferred genetic material adversely affects human health. This would be particularly relevant if antibiotic resistance genes, used as markers when creating GMOs, were to be transferred. Although the probability of transfer is low, the use of gene transfer technology that does not involve antibiotic resistance genes is encouraged.

Outcrossing

The migration of genes from GM plants into conventional crops or related species in the wild (referred to as “outcrossing”), as well as the mixing of crops derived from conventional seeds with GM crops, may have an indirect effect on food safety and food security. Cases have been reported where GM crops approved for animal feed or industrial use were detected at low levels in the products intended for human consumption. Several countries have adopted strategies to reduce mixing, including a clear separation of the fields within which GM crops and conventional crops are grow

How do we genetically alter sweet potatoes?

Fascinatingly, researchers discovered that sweet potatoes already contain DNA sequences from *Agrobacterium* within their own genome. Not only that, but sweet potato plants actively express some of these genes. In light of this discovery, sweet potatoes serve as the first known example of a naturally transgenic food crop

Like any cultivated crop, sweet potatoes have been genetically modified by humans over a very long period of time, through selective breeding, to produce improved varieties with desirable traits for flavor, texture, color, shape, pest and disease resistance, drought tolerance, and so on.

Different varieties are crossed with each other via cross-pollination. Next, the offspring of these new crosses are planted and grown in field trials, in which they are observed and tested for desirable traits. Additional tests (such as bake tests and fry tests) are conducted as well. Varieties which have potential for the market (or potential for crossing with other varieties) are kept for further research, with the ultimate goal of delivering impressive, new varieties that growers and producers will want to buy.

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Food, genetically modified (who.int)

Genetically Modified Cotton

Genetic factors influencing crop production involves the application of genetic engineering. Genetic engineering is a simple addition, deletion, or manipulation of a single trait in an organism to achieve the desired change. In the field of agriculture, GMCs, GM Crops, or biotech are plants utilized which involves modification of DNA using genetic engineering techniques (Shetty *et. al.*, 2017). Researchers in the 1980s developed more accurate and controllable genetic engineering technologies to produce plants with desirable traits. In 1982, an antibiotic-resistant tobacco plant, the first genetically engineered crop was produced.

In 1982, an antibiotic-resistant tobacco plant became the first GM crop. When tobacco plants were altered for herbicide resistance, the first field trials took place in 1986 in France and the United States. Plant Genetic Systems (Ghent, Belgium), founded by Marc Van Montagu and Jeff Schell in 1987, was the first company to genetically design insect resistant (tobacco) plants by integrating genes from *Bacillus thuringiensis* that produced insecticidal proteins (Bt).

Learning Objectives

This subtopic intends LEA reviewers to have key points to remember about GMO cotton under the Unit Chapter Genetically Modified Crops.

At the end of the topic, the participants should be able to:

1. define genetically modified cotton,
2. recall the historical trends of genetically modified cotton, and
3. identify recent developments in *Bt* corn

The following are the commercial genetically modified crops:

Cotton

The most prominent environmental influence of cotton production is the use of agrochemicals (especially pesticides) thus lead to the development and utilization of GMO cotton. Genetically modified cotton is either herbicide-tolerant, resistant to Lepidopteran pests, or both.

For more than 20 years cotton has been the most widely sown genetically modified (GM) crop in Mexico. This topic will discuss the background, history, and recent developments of Bt cotton.

* Characteristics- an insect-resistant genetically engineered cotton created using a gene from the soil bacterium Bacillus thuringiensis (Bt).Cotton plant cells produce crystal insecticidal proteins, known as Cryproteins, after the genes from B. thuringiensis are inserted. These insecticidal proteins are efficient against some of cotton's most dangerous caterpillar pests, such as tobacco budworm and bollworm larvae. Susceptible insects stop feeding within a few hours of their first bite and die within 2 or 3 days if they have ingested enough toxin.

The **three primary components** of the genetic package inserted into cotton DNA:

Protein gene referring to the cryprotein. The original Bt cotton types grown in the United States only had one Cry-protein gene, Cry1Ac.

Promoter referring a DNA segment that regulates the amount of Cryprotein generated and where it is produced in the plant.

A genetic marker that determines whether a gene was successfully inserted into the plant's DNA.

Limitations of GM cotton:

1. UV radiation, heat, a high leaf pH, or desiccation all quickly degrade the toxin.
2. Because the toxin has no contact impact, caterpillars must consume enough treated plant tissue to receive a deadly dose.
3. Older larvae are less harmful to Bt Cry-proteins.

*History- In 1995, cotton is one of the crops approved by the Environmental Protection Agency (EPA) in USA.

- In 1996, insect resistant GM cotton was grown commercially for the first time after six years of field trials.

- Known as Bt or Ingard® cotton, the cotton was developed by CSIRO, using a gene owned by Monsanto.

- The first transgenic cotton (BG I) carrying the lepidopteran pest resistance gene Cry1Ac was approved for commercial use in Australia and the United States in 1996, and GEAC approved it for commercial use in India on March 26, 2002.

- In Mexico, two Bt cotton types (DP 33B and DP 35B) outperformed the conventional variety (DP-5690) in terms of BW control, and the former required one fewer pesticide spray than the non-Bt cotton.

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- BG I, the first generation Bt cotton, was unable to protect the crop from BW population development in the later stages of the crop over the entire crop season. Bollgard II® is the second insect-resistant GM cotton certified in 2003, and it differs from Bt cotton in that it has two genes from the Bacillus thuringiensis (Bt) soil bacteria rather than one.

- WIDE STRIKE ™. Dow AgroSciences entered the GM cotton market in 2009, when the OGTR authorized WIDE STRIKE™, an insect resistant cultivar.

- In 2001, Roundup Ready® cotton and Roundup Ready/Bt cotton became commercially available in Australia for the first time.

- In February 2006, the Office of the Gene Technology Regulator (OGTR) approved the commercial release of Roundup Ready FLEX® cotton varieties.

- In 2006, another company entered the GM cottonmarket in Australia, when the OGTR approved Bayer CropScience's Liberty Link® cotton

List of Genetically Modified Cotton

- Ingard® or Bt cotton- inserted by a Monsanto-owned gene derived from the Bacillus thuringiensis (Bt) soil bacteria, and it allowed the plant to manufacture the Bt protein, which killed the cotton bollworm, heliothis, when it ate the leaves.
- Bollgard 1®- Rather than one, it has two genes from the Bacillus thuringiensis (Bt) soil bacteria. When the cotton plant's main caterpillar pest eats the plant, the genes generate proteins, and the plant dies.
- Bollgard 2®-contains two genes from the soil bacteria Bacillus thuringiensis (Bt), rather than one. The genes produce proteins in the leaves of the cotton plant and when cotton's major caterpillar pest eats the plant, it dies.
- Roundup Ready® cotton - achieved by conventional breeding of the two (2) GM varieties. It is the Monsanto trademark for its patented line of genetically modified crop seeds that are resistant to its glyphosate-based herbicide.
- Roundup Ready FLEX® cotton - Unlike Roundup Ready cotton, they have two copies of a glyphosate tolerance gene acquired from a soil bacteria instead than just one.
- Wide Strike ™ - with Cry1Ac and Cry1F proteins from Bacillus thuringiensis, as well as a vegetative insecticidal protein (Vip3A) from Bacillus thuringiensis (Bt) that provides superior season-long protection against a wide range of destructive lepidopteran pests, including the fall armyworm and the American bollworm, throughout the cotton plant.
- Liberty Link® cotton - genetically modified developed by Bayer Crop Science to withstand applications of glufosinate ammonium, a broad-spectrum herbicide marketed as Liberty®.
- Bollgard 3®- contains three different insecticidal proteins (cry1Ac, cry2Ab, and vip3A). This protein 'stack' contributes to resistance management as it more difficult for the pest to overcome all 3 toxins in unison.
    - Cry1Ac protoxin- crystal protein synthesized during sporulation by the gram-positive bacteria Bacillus thuringiensis (Bt).

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- Cry2Ab- a protein that promotes colloidosmotic lysis by binding to the midgut epithelial cells of lepidopteran (Manduca sexta) larvae.
- VIP 3a- a novel Bacillus thuringiensis vegetative insecticidal protein with a broad range of activities against lepidopterans.

*Worldwide Status and Recent Developments

According to the International Service for the Acquisition of Agri-biotech Applications (ISAAA), 160 million hectares of GM crops were produced globally in 2011, with GM cotton accounting for the third largest GM crop (15.4 percent of global GM area) after soybeans (47 percent) and corn (32%). In the United States, Brazil, Argentina, India, China, Pakistan, South Africa, Australia, Myanmar, Burkina Faso, Mexico, Colombia, and Costa Rica, genetically modified cotton was cultivated.

Herbicide resistance is the most common trait in GM crops, accounting for 59% of all GM crops, followed by combined or stacked traits of herbicide tolerance and insect resistance (26 percent), and insect resistance (15 percent).

Summary

- Genetically modified cotton is also a product of Biotechnology that aims to developed cultivars that can withstand pest attack. It can be insect resistant, herbicide resistant, or both.
- Cotton that has been genetically engineered (also known as genetically modified or GM) is grown on 25 million hectares around the world, especially in India, China, Pakistan, and the United States. South Africa, Burkina Faso, Sudan, Brazil, Argentina, Paraguay, Columbia, Mexico, Costa Rica, Burma, Australia, and Egypt are among the countries cultivating much lesser amounts of GM cotton.
- The commercial cultivation of cotton was commercially cultivated in 1996 due to very serious pest problem.
- The only GM crop now produced in India is insect-resistant Bt cotton. Monsanto launched it in India in 2002 under the brand name Bollgard, in a joint venture with Mahyco, an Indian seed company (CBAN, 2013).
- With regard to GM cotton, Better cotton Industry (BCI) has taken the position of being "technology neutral," meaning that it will neither encourage farmers to cultivate it nor seek to restrict their access to it. GM seeds are now used to cultivate nearly three-quarters of the world's cotton. BCI allows the use of GM cotton if it is legally available in the nation of use and there is an overall support package in place for farmers that includes training and access to a variety of farming options (BCI, 2013).
- List of transgenic cotton

Trade name	Protein/substance	company	Target pests
Bollgard I	Cry1 Ac	Monsanto	Tobacco budworm,Cotton bollworm, pink bollworm
Bollgard 2	Cry1 Ac and Cry 2 Ab	Monsanto	Tobacco

			budworm,Cotton bollworm, pink bollworm
Bollgard 3	cry1Ac, cry2Ab, and vip3A	Monsanto	Tobacco budworm,Cotton bollworm, pink bollworm
Widestrike	Cry 1 Ac and Cry 1Fa	Dow	Tobacco budworm,Cotton bollworm, pink bollworm
Roundup Ready® cotton	Breeding of 2 GM cotton varieties	Monsanto	Weeds
Roundup Ready FLEX® cotton	two copies of a glyphosate tolerance gene	Bayer	Weeds
Liberty Link	copies of a glyphosate tolerance gene	Bayer	Weeds

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Genetically Modified Crops (Bt Corn)

Overview

Bacillus thuringiensis is a species of bacteria that produced proteins that are toxics to certain insects. Because of this, it has been used as a safe microbial insecticide for over 50 years to control pest caterpillars. This is because specific pH levels, enzymes, and midgut receptors are required to activate and bind a given Cry toxin to midgut cells, which leads to pore formation in the insect's intestine and death (Federeci,2002). A "lock and key" analogy is useful

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to explain this specificity. If the midgut receptor is considered the “lock” and the Cry proteins is the “key” then insect death only occurs when the “lock and key” match.
There are a number of Cry toxins that are categorized by their spectrum of activity. For corn pest, primary Cry proteins are Cry1 and Cry2 for Lepidoptera and Cry3 proteins for Coleoptera (Schnepf et al. 1998). Corn can be genetically engineered to produce this specific Cry toxins. As such, aspect of Bt corn are similar to host plant resistance traits such as DIMBOA (2,4-dihydroxy-7-methoxy -1,4benzoxazin-3 one), which as high levels reduces damage by European corn borer (Klun et al. 1967). Seed providers often combine or stack traits for Lepidoptera and Coleoptera control into the same plant. Also, different types of Bt toxins targeted for the same insects are often combined into more effective plant protectants called pyramids. This multiple toxin approach is useful for managing insect resistance to Bt corn.

Learning Objectives:

This subtopic intends LEA reviewers to have key points to learn the Bt Corn under the Unit Chapter of Genetically Modified Crops
At the end of this module the participant should be able to:

1. understand what is Bt corn
2. know the mechanism of pest resistance in Bt corn, and
3. know the impact of Bt corn to human and in the environment.

Assessment Criteria:

- Nature and importance of Bt corn is identified based on the objectives of study.
- Mechanisms of action of Bt corn is determine and confirmed in accordance with the procedural works.
- Certain target pest is controlled and determined based on the efficacy of Bt works.
- Mode of Bt action is determined based on the data reported.
- Impact of Bt in the environment is assessed and evaluated based on the data collected.

Conditions

The LEA reviewer must be provided by the following:

- Module, flyer and hand out
- Review materials

Methodologies

- Modular
- Flexible learning

Assessment Methods

- Online written exam

What is Bt Corn?

Bt corn is a type of genetically modifies organisms, termed GMO. A GMO is a plant or animals that has been genetically modified through the addition of a small amount of genetic material from other organisms through molecular techniques. Currently, the GMOs on the

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market today have been given genetic traits to provide protection from pest, tolerance to pesticide, or improves its quality.

To transform a plant into a GMO plant, the gene that produces a genetic trait of interest its identified and separated from the rest of the genetic materials from a donor organism. Most organisms have thousands of gene, a single gene represents only a tiny fraction of the total genetic makeup of an organism.

A donor organism may be a bacterium, fungus or even another plant. In the case of Bt corn, the donor organisms is a naturally occurring soil bacterium, Bacillus thuringiensis, and the gene of interest produces a protein that kills Lepidoptera larvae, in particular, European corn borer. This protein is called the Bt delta endotoxin.

Bt Delta Endotoxin

The Bt Delta endotoxin was selected because it is highly effective at controlling Lepidoptera, larvae, caterpillars. It is during the larval stage when most of the damage by European corn borer occurs. The protein is very selective generally not harming the insects in others orders (such as beetles, flies, bees and wasps). For this reason, GMO that have the Bt gene are compatible with biological control programs because they harm insect predators and parasitoids, much less than broad-spectrum insecticides. The Bt endotoxin is considered safe for humans, other mammals, fish, birds and the environment because of its selectivity.

How Bt Corn does it works?

Mechanism of Action of Bt Cry Protein Insecticidal Activity

Bt is a Gram-positive, aerobic bacterium found in a variety of outdoor environments (Federici and Siegel, 2008). When nutrients or oxygen are insufficient for vegetative growth, the bacterium sporulates, producing a spore, and a parasporal body containing one or more insecticidal crystal proteins (Federici and Siegel, 2008). Three main types of insecticidal proteins are known to be produced by Bt microbes: Cry (for crystal) proteins, Cyt (for cytolytic) proteins, and toxins that are produced and secreted during vegetative growth. The diversity within and among these protein classes provides activity against a range of larval insects (van Frankenhuyzen and Nystrom, 2002; Crickmore et al., 2014). Cry proteins are active against lepidopteran insects and some variants are also active against dipteran insects, coleopteran insects, or nematodes. Cyt proteins are toxic to mosquito and black fly larvae and a few beetle species (Soberón et al., 2013). VIP proteins, which are produced and secreted during the vegetative growth stage, have activity against coleopteran and lepidopteran insect species. Although each of these components contributes to the overall insecticidal activity of microbial Bt products, the Cry proteins are considered the most important component of commercial Bt formulations. It is for this reason that the majority of commercially available GM crops with insecticidal activity have been developed to express one or more Bt Cry proteins, which are the focus of this review.

The mode of action of Cry proteins has been extensively studied and reviewed (WHO/IPCS, 1999; Betz et al., 2000; Siegel, 2001; Bravo et al., 2007; OECD, 2007; Federici and Siegel, 2008; Soberón et al., 2010). Unlike most chemical pesticides, which are contact insecticides, Cry proteins are effective only when ingested by the insect. The Cry proteins are approximately

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70–140 kDa (Höfte and Whiteley, 1989). Within the alkaline conditions in the Lepidopteran insect gastrointestinal (GI) tract, proteases cleave the protoxin (inactive) form into an active toxic/toxin form. The activated Cry molecules typically bind to specific receptors on mid-GI-tract epithelial cells of target larval insects and then oligomerize, forming pores consisting of four to six Cry molecules each in cellular membranes. These pores enable excess cations to enter the cell, causing osmotic imbalance. The affected midgut cells take in water, swell, and lyse, eventually resulting in death of the insect (Federici and Siegel, 2008; Soberón et al., 2010). Receptors to which Cry proteins bind in insects include cadherin-like glycoproteins, which appear to be a primary receptor for some Cry proteins, and glycosylphosphatidyl-inositol (GPI) membrane anchored receptors such as aminopeptidase N or alkaline phosphatase (Soberón et al., 2010). There is also compelling evidence for ABC transporters serving as receptors for Cry proteins in several lepidopteran species (Gahan et al., 2010; Baxter et al., 2011; Atsumi et al., 2012). Receptor binding is not sufficient to cause toxicity to the insect, as oligomerization is required for pore formation (Federici and Siegel, 2008; Soberón et al., 2010). The many permutations of species-specific insect GI-tract proteases, receptors, and tendencies of each Cry protein to oligomerize in a specific situation result in the biological specificity that provides targeted insecticidal activity and safety to NTOs of Bt sprays and GM crops that produce Bt proteins.

Mode of Action of Bt

Bt spores have to be ingested by the susceptible insect to cause mortality. The Cry toxin becomes active by proteoletic enzymes in the alkaline gut juice (pH 8-10). Most Cry toxins are actually pro-toxins of about 130 to 140 kDa, and after activation, they become 60-70 kDa (Bravo et al. 2007). The activated toxin passes through the peritrophic membrane and binds to specific receptors on apical microvillar brush order membrane of the epithelial cells of the midgut making pores through which the toxins penetrates to such cells that become swollen. The swelling continues until the cells lyse and separate from the basement membrane of the midgut epithelium. The alkaline gut juices then leak into the hemocoel causing the hemolymph pH rises that leads to paralysis and death of the insect (Soberon et al 2010). Broderick et al. (2006) mentioned that the naturally occurring bacteria in the gut (E. coli and Enterobacter) penetrate to the hemocoel through the disrupted epithelium caused by Bt toxins and multiply causing sepsis of the hemolymph and death of the insect. In the Bt-moderately sensitive insects, such as Spodoptera spp., the endospore has a considerable role in killing the insect by producing toxins during its vegetative growth in the hemolymph (Crickmore et al. 2014). The insect or any living organism that does not have the receptors in gut epithelial cells is not killed by *Bt* (Gill et al. 1992).

Most of the Bt technologies used in corn affect two groups of insects. In Tennessee, the ones we're most interested in controlling are the caterpillar pests. In the Midwest, where beetle pests are significant, different Bt traits are used to control corn rootworms. Bt corn grown in Tennessee contains toxins called Cry proteins. They may also produce a toxin called a Vip protein, and there are different kinds of Cry proteins such as Cry1, Cry2, Cry2Ab, etc. This class of Bt toxins has different spectrums of activity, and some are better at controlling corn earworms than others

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Impact of Bt corn to human and in the environment.

Characterization of Effects of *Bt* Corn Pollen.

The Cry1A proteins expressed in most commercial *Bt* corn hybrids are toxic to the monarch butterfly. Mortality, expressed as LD_{50}, was estimated at 3.3 ng of protein/ml diet, whereas growth inhibition (EC_{50}) was estimated to be 0.76 ng/ml. However, the expression of Cry1Ab endotoxin within pollen of various events varies considerably depending on the promoter gene involved. Expression is greatest in event 176 *Bt* corn (1.1–7.1 µg/gm pollen) (http://www.inspection.gc.ca/english/plaveg/pbo/dd9609e.shtml, http://www.epa.gov/pesticides/factsheets/fs006458t.htm), a line that is being phased out through 2003. This event exceeds, by nearly two orders of magnitude, protein expression in events Bt11 and Mon810 (0.09 µg/gm pollen) (http://www.epa.gov/scipoly/sap/2000/october/), which is near the current level of detection by immunoassay.

Bt proteins safe to human

Bt proteins, naturally occurring insecticides produced by the soil bacterium, *B. thuringiensis*, have been used to control crop pests since the 1920s (1), generally as microbial products. Many strains of *B. thuringiensis* exist that produce different Bt proteins varying in the insects they target, e.g., larvae of butterflies and moths, beetles, and mosquitoes. The insecticidal Bt proteins form crystalline protein bodies inside the bacterium, hence the name Cry proteins. Full-sized Cry proteins are inactive until eaten by target insect larva, and inside the midgut they are cleaved and become active. The smaller, active peptides bind to specialized receptors, creating holes in the gut membrane that cause contents to leak and kill the larvae. The precision of different Bt proteins for their targets resides in the specificity of their tight binding to companion receptors in the insect gut (2).

The possibility for allergenic effects of four maize Bt varieties was specifically investigated in potentially sensitive populations (11). Skin prick tests were performed with protein extracts from MON810, Bt11, T25, and Bt176 and from nontransgenic control samples in two sensitive groups: children with food and inhalant allergies and individuals with asthma-rhinitis. Immunoglobulin E immunoblot reactivity of sera from patients with food allergies was tested versus Bt maize and pure Cry1Ab protein. No individual reacted differently to transgenic and nontransgenic samples; none had detectable IgE antibodies against pure transgenic proteins.

A truncated version of the full-length 131-kDa Bt protein, containing only the insect-toxic fragment, is used to engineer some crops. For example, Mon810 maize contains a truncated cry1Ab gene that codes for a 91-kDa protein. The potential for mammalian toxicity of the truncated protein was assessed by administering purified, truncated Cry1Ab protein from E. coli to groups of ten male and female CD-1 mice at ≤4000 mg/kg body weight (12). These doses represented a 200–1000-fold excess over the exposure level predicted on the basis of human consumption of MON810 grain. Mice were observed up to 9 days after dosing; no treatment-

related effects on body weight, food consumption, survival, or gross pathology upon necropsy were observed for mice administered Cry1Ab truncated protein.

Despite extensive evaluations of Bt food safety, in June 2005 a Greenpeace press release, published in the New York Times and other international newspapers, stated, “There are strong warning signs that this GE Bt rice could cause allergenic reactions, as it did when tested on mice based on a study (13) and references therein”. However, in the Moreno-Fierros study (13) referred to in the press release, Cry1Ac was being tested as an oral adjuvant to boost vaccine titers. As such, the protein was used in large amounts and the stomach pH was raised to prevent degradation of Cry1Ac. It had been chosen as an adjuvant precisely because it is nontoxic to vertebrates (14).

The native Cry9c, a protein effective against lepidopteran insects, was engineered into a variety of corn called StarlinkTM. Researchers knew the Cry9C protein did not originate from an allergenic source and had no amino acid homology with known toxins or allergens in available protein databases. However, when StarlinkTM corn was created, the Cry9C protein had no history of human dietary exposure, and in addition it was not readily digestible and was stable at 90ºC (15), both hallmarks of certain allergens (see “Were Foods Made From Bt Corn Removed from the Market Because of Allergenicity Concerns?”); Cry9C also had biochemical characteristics that differentiated it from other previously reviewed Cry proteins (16). To determine with reasonable certainty that no harm would result from human exposure to this protein, it was necessary for the EPA to determine if proteins with these biochemical characteristics were likely to affect the safety of a food. Because it was slow to digest, it provided longer lasting protection against insect damage, but the altered digestibility characteristics in humans and its relative stability to heat caused regulators to delay approval of the crop for human consumption (although it was approved for animals) so that they could reexamine its potential as a human allergen (see “Were Foods Made From Bt Corn Removed from the Market Because of Allergenicity Concerns?”).

A positive aspect of safety regarding Bt corn is the lower levels of mycotoxins compared with non-Bt corn. Mycotoxins are toxic and carcinogenic chemicals produced as secondary metabolites of fungal colonization (17) that occur as a result of insects such as the corn earworm carrying the mycotoxincontaining fungi that infest the kernels following wounding. In some cases, the reduction of mycotoxins in Bt corn results in a positive economic impact on U.S. domestic and international markets. More importantly, in less-developed countries certain mycotoxins are significant contaminants of food and their reduction in Bt corn could improve human and animal health.

In 2002, APHIS announced the deregulation of a corn variety, Mon863, with increased rootworm (Diabrotica spp.) resistance. Food safety assessments by the company used 90- day mouse feeding trials to demonstrate safety (18); independent assessments also demonstrated the safety of Mon 863 (19, 20, 21). Mon 863 contains a variant Cry3Bb1 with seven amino acid

differences from wild-type Cry3Bb1 to enhance plant expression and insecticidal activity against corn rootworm (22).

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C. HUMAN FACTORS

Overview

Ecological conditions are not the only factors guiding farmer choices and decisions in a production system. Economic and social conditions play important roles in farmer choice decisions. It would be pointless to promote crops or techniques that do not correspond to the true interests of farmers or for which the needed resources are unavailable (Dufumier, 1994).

Efforts to create more favorable socioeconomic conditions are often more useful than the promotion of farmer adoption of technical options (Breman, 1995). Just as individual consumers have different preferences about products they consume, farmer characteristics, asset endowments, risk preferences, and intertemporal considerations affect their choices (Hanson et al. 2004).

Learning Objectives

At the end of the session, the reviewers should be able to:

1. identify the aspects of socio-ecological systems in agriculture;
2. identify and deliberate the approaches in farmers decision; and
3. categorize the risks and appropriate interventions adopted by farmers.

1. Preference of Farmers: Decision-Making in Agriculture

The decision-making process in agriculture is complex, as many factors and a multifaceted environment influences farmers in their decision-making approach. Bradford (2009) describes the farming environment as complex and uncertain due to aspects of various origin, such as being economically or environmentally successful and of uncertain outer influences, like weather or political frameworks.

Furthermore, farmers can be influenced by their surroundings and the information given by the community. Another essential influence in the decision-making of a farmer is lessons learned from the past and possible recovery processes. According to (Bradford, 2009) decision-makers also often rely on intuition or "gut feelings" in complex situations.
Personal factors like values, attitudes, and norms are taken into account in order to understand the context of societies. Wigley (1988) suggests that it would become evident that, unlike commercial farmers in developed countries, small-scale farmers in developing countries are influenced by factors more important than the market.

King and Burton (1989) also mention that understanding the behavior of farmers is particularly important in areas undergoing agrarian structural reform, or where such reform is proposed, because only through an adequate understanding of the interrelationship between land ownership values and behavior can the implications of such changes be fully understood. At the same time, a better understanding of motivation may lead to a more thorough explanation of farmer's economic behavior and to more realistic models of agricultural systems.

2. The Capability of Farmers (Economic)

Latest Trend

Nationwide, the nominal wage rate of agricultural workers in 2019 averaged PhP 331.10 per day. In addition, the poverty incidence in the country is highest in the Agriculture sector at 40.8%, the highest among the identified sectors (Philippine Statistics Authority, 2020).

According to the National Economic and Development Authority (2021), the country's poverty rate is between 15.5% and 17.5% in 2021, likely near 16.6% in 2018.

How profitable is it to farm?

The answer depends upon the choices a farmer about what crops to grow, where and what technologies to use, and many other short to long-term management decisions. Economists assume that farmers make choices to improve their utility or well-being. In particular, farmers tend to pursue activities that increase their income, reduce their financial and physical risk, reduce labor requirements, and are convenient or enjoyable (Stoorvogel et al. 2004).

Challenges Faced by our Farmers

According to the study conducted by Arida (2009) entitled *Problems in rice farming: a Filipino farmers' perspective* revealed that common problems encountered by rice farmers were: the high cost of inputs, low price of palay, lack of capital, labor problem, lack of postharvest facilities, pest and diseases, and irrigation system. Similarly, low palay price may imply that farmers who yield more encounters a problem with the unavailability of postharvest facilities or lack of drying pavements, thus forcing them to sell their produce fresh despite its low price.

In addition, according to the case study of Gonzales and Buted (2015), farmers borrow money as capital for a farming portion of it will be spent to buy food for the family. This is congruent with the findings of Palis et al. (2015) that according to the FGD they have conducted, Filipino rice farmers are trapped in the cycle of poverty since most of them have insufficient capital to commence farming.

A farmer's income and ability to obtain credit will also influence his/her choice of crops, farming systems, and willingness to invest in new crops, systems, or technologies (Knowler and Bradshaw 2007). A risk-averse farmer or one who is a credit or income-constrained (which often is the norm rather than the exception, particularly in developing countries) may be less likely to adopt new technologies, even if they are likely to reduce his susceptibility to risk or increase productivity or income over the long-run (Hanson et al. 2004).

However, having cash is not an assurance of a successful business most especially for small entrepreneurs like farmers but the need for financial planning is enhanced. Financial problems and small capital eventually follow poor production, sales, distribution, and inventory management (Gonzales and Buted, 2015).

The capability of Farmers (Social Knowledge)

Latest Trend

The Philippines might reach a critical shortage of farmers in just 15 years because they are already aging – 57 years old on average (IRIN 2013). This aging phenomenon of farmers, however, is not unique to the Philippines. It is a global phenomenon and is especially true among smallholder farmers in Asia (Rigg et al., 2019).

Diverted Interest

Most of the farmers today do not want their children to be rice farmers because of the physical, psychological, and financial difficulties they encountered. Our farmers view that their children would not have any future in farming. Instead, parent farmers dream that their children will have non-farm jobs, either in urban areas or abroad, to have a higher and more stable income and to get out of poverty (Palis, 2020).

Furthermore, the children's respect for parental authority even extends through their adulthood, suggesting that Filipino children often value, if not constantly, the advice and decisions of their parents. Filipino parents would have a great influence on the aspirations, career, and future of their children, and this would determine whether they become the next generation of farmers or not (Palis, 2020).

Social Capital of Farmers

Farmers' attitudes and desires are influenced by their society's culture. If it is customary in a certain community for farmers to scatter seed and plow it into the soil, people will grow up to believe that that is the only correct way of planting. Even if the benefits of other methods are explained to them, their strongly held attitudes may make it difficult for to them change.

Farmers' resource availability, education, and knowledge are especially important. Farmers may be risk-averse toward making changes in cropping decisions or adopting new agricultural practices or might have very conservative attitudes toward technology or lower or higher levels of concern for the natural environment (Musshoff and Hirschauer 2008, Serra et al. 2008).

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Challenges Faced by our Farmers

Lack of knowledge and information about the costs and benefits of adopting new technologies or conservation practices *or* lack of knowledge about how to implement such technologies or practices will also affect a farmer's propensity to adopt them (Chavas et al. 2010, Chavas and Kim 2010).

Even if farmers have a lot of information and can implement new technologies efficiently and at low cost, differences in intertemporal preferences or credit constraints may mean that farmers are unwilling to sacrifice current profits or income for long-term improvements in soil fertility, risk-reductions, or improved yields (Shively 2001, Sunding and Zilberman 2001, Coxhead and Shively 2002).

Well-directed social networks among the farmers can enhance the productivity of agriculture, which, in turn, will enhance the living standard of the agriculture-dependent population (Shaijumon, 2018).

Assessing Risks and Interventions

Risk is the chance of falling below a critical (i. e., minimum or subsistence) income level, plays a vital role in the farmers' production decisions related to choices and levels of inputs and outputs. Empirically, how farmers decide under risky conditions is best analyzed by taking into account their risk perceptions and risk attitudes or preferences (i.e., risk-averse, risk-taker, or risk-neutral) (Lucas and Pabuayon, 2011).

Farmers whose survival hinges on production may be more sensitive to income variability than to average income and often exhibit high aversion to risk (Belaid and Miller 1987). Prediction of behavior in risky situations depends on how much is known of the individual's willingness to take risks (Patindol 2001).

How farmers respond to risks is important in agricultural decisions. Policies to improve farmers' ability to manage risks should be considered. Enhancing farmer education can help minimize the effect of production risk on a farmer's welfare. Moreover, investment in farmer-training to build human capital will enable farmers to manage production risks better. It should not be discounted that the farmers' attitudes toward risk are major determinants of the rate of diffusion of new technologies among them and of the outcome of rural development programs. Moreover, farmer education should focus on adaptation strategies to climate change considering that farmers consider weather variability as a major source of risk in farming (Lucas and Pabuayon, 2011).

A greater role for government and research institutions in technology development programs—in communicating potential risks and in providing the farmers with accurate information on benefits and risks from new technologies as well as alternative farming options. If properly informed, farmers would be able to adjust and be willing to take risks to increase their productivity and profitability.

Farmer groups considered farming as relatively risky given the increasing price of fertilizer and environmental factors, which are beyond their control. Fertilizer cost was the most felt source of risk while weather condition was the most visible source of risk affecting their farming activities (Lucas and Pabuayon, 2011).

Summary

- ✓ Economic and social conditions play important roles in farmer choice decisions.
- ✓ Farmers can be influenced by their surroundings and the information given by the community**.**
- ✓ Personal factors like values, attitudes, and norms are taken into account in order to understand the context of societies.
- ✓ The nominal wage rate of agricultural workers in 2019 averaged php 331.10 per day.
- ✓ The poverty incidence in the country is highest in the Agriculture sector at 40.8%, the highest among the identified sectors.
- ✓ Common problems encountered by rice farmers were: the high cost of inputs, low price of palay, lack of capital, labor problem, lack of postharvest facilities, pest and diseases, and irrigation system.
- ✓ A farmer's income and ability to obtain credit will also influence his/her choice of crops, farming systems, and willingness to invest in new crops, systems, or technologies.
- ✓ The Philippines might reach a critical shortage of farmers in just 15 years because they are already aging – 57 years old on average.
- ✓ Most of the farmers today do not want their children to be rice farmers because of the physical, psychological, and financial difficulties they encountered.

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3. Cultural Practices (Eco-Social and Political Observations)

Overview

This module discusses some of the cultural practices in crop production as well as the various economic, social, and political determinants influencing the farmers' decision-making and implementation of management practices. Observable changes in their current practices are also discussed as new trends of farming are introduced.

Learning Outcomes

At the end of the topic, the participant should be able to:

1. identify the various cultural practices in crop production;
2. understand the impact of economic, social, and political factors influencing the farmers' decision; and,
3. know some changes in farmers' cultural practices in response to gained knowledge and technologies through extension activities.

Cultural Practices

The culture of society is the accepted way of doing thigs in that society. It is the way in which people live, their customs, traditions, methods of cultivation and so on (FAO, n.d.). Culture is not an accidental collection of customs and habits but has been evolved by the people to help them in their conduct of life. Each aspect of the culture of a society has a definite purpose and function and is, therefore, related to all other aspects of its culture (FAO, n.d.).

Cultural practices include all the crop production and management techniques which are utilized by the farmers to maximize their crop productivity and/or farm income. It includes decisions on crops/varieties to be grown, time and planting, tillage, pest and disease control, application of fertilizers and irrigation, and so on. It must be understood that farmers' practice greatly depends

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on the long-running culture and tradition of the locality. Yet, with proper diffusion of package of technologies, these practices may have gradually changed over time.

1. Planting Material Selection

In deciding what crop, variety, and material to cultivate, farmers cross check the crop suitability with their available farm resources and other existing conditions. Other than biotic (e.g., known pests and diseases in the area) and abiotic factors (e.g., climatic and edaphic factors), the selection of crop or planting material is determined by various socio-economic and political determinants.

Economic Factors

The crop/varietal selection is primarily determined by the available farm resources which includes the quality and quantity of land, planting materials, irrigation, mechanization, labor, facilities and capital for basic crop production and post-production operations. As such, farmers in less developed countries are constrained by limited access to recommended seeds and other farm inputs (www.projectreserve.com). The selection of crops and varieties is also highly dependent on its feasibility, profitability, and marketability.

Social Factors

Social determinants include the age, income, cooperative membership, household size, level of education, farm size, and number of contacts with extension agents (Onyeneke, 2017). The local diet of the farmer or society is also very important in this aspect (FAO, n.d.). It has also been shown in many studies that, less experienced farmers are less inclined to adopt modern technologies such as hybrid maize (Mphinyane and Terblanché, 2005).

Political Factors

The availability of financial incentives or crop and varietal subsidies, commodity pricing, international trade policy, other agricultural programs, and regulations set by the government are among the political determinants in selection of crop and plant varieties. The recommended use of certified high-quality seeds (NSIC-registered seeds), for example, helps the farmers decide the variety to cultivate (INQUIRER.NET). Rice tariff collections in the Philippines were used to finance the production-distribution of certified inbred seeds, crop diversification program, and expanded crop insurance on rice (GMA News Online, 2020)

2. Tillage

Tillage is the activity or process of preparing land for growing crops. The type of tillage is mostly associated with production cost. The capital for hired labor and/or tractor services or access to equipment for field cultivation are among the main deciding factors for the type of tillage practice.

Economic Factors

The lack of finances to afford mechanization have additional negative impacts on livelihoods of already poor farmers; while the presence of off-farm income (e.g., pensions, salary, other incomes) contributes positively to farm productivity through financing of the tractor rental for soil cultivation (Berg, 2013).

Reducing soil tillage is a great way to cut costs, labor and soil erosion, while promoting soil health and getting the same crop yields. In regions with low annual precipitation (less

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than 20 inches), farmers see the benefit of leaving crop residues on the soil surface to conserve soil moisture (UMN Extension, n.d.).

Social Factors

Tillage practices is also affected by social and gender issues. The adoption of 'conservation tillage', for example, depends on the culture and tradition of the locality. A typical example is the use of draft animals in some communities. The value attached to cattle in some communities cannot permit their use for traction (Lubwama, 1999).

In the long-time culture and tradition in many rural areas, the adoption of technology (e.g., conservation tillage) are mostly favored towards men than women. One reason is that men are more physiologically suited to operate such technology. Another reason is that when men teach women to use a new technology, they often find them "slow" to learn. Hence, more effort and strategy to increase the level of technology for women are needed (Lubwama, 1999).

3. Cultural Methods of Pest and Disease Management

Cultural practices are vital in reducing plant pests and disease losses. Farmers often utilize as many methods as possible to reduce the risk of crop failure (TAMU Extension, n.d.).

a) **Crop rotation. The rotation of unrelated crops** per season or year prevents populations of pathogens from building up to damaging numbers. One should not expect rotation to eliminate disease development, but it certainly aids in reducing damage from most diseases.

b) **Balanced fertilization. Fertilizer use** may have some bearing on development of certain diseases. Generally, unbalanced fertilization, particularly of nitrogen with other nutrients enhances foliage disease development and predisposes some plants to other diseases. Potash, on the other hand, helps reduce disease development when it is in balance with other elements.

c) **Deep burial of crop residues. This** helps control certain diseases by placing the organism contained in the residue at a depth where there is an oxygen deficiency. This reduces the population of the disease-causing organism and permits the crop to escape much of the damage.

d) **Planting on a raised bed.** This is helpful in preventing certain diseases such as Southern blight and some of the wilt diseases. This practice is advisable when growing leguminous crops such as peanuts and soybeans, and when growing vegetable crops in tight, poorly drained soils.

e) **Burning of crop residues.** This has been discouraged because of destruction of valuable organic matter and creation of an air pollution problem. The fact remains, however, that it is a highly effective means of eradicating some disease-causing organisms associated with crop residue.

f) **Time of Seeding.** Delayed planting may sometimes help the plant escape the chances of certain diseases (e.g., damping-off caused by *Pythium* sp.). Early spring planting of cotton may effectively help escape cotton root rot.

g) **Removal of alternate host plants.** Certain plants might serve as a host reservoir for virus diseases that attack cultivated crops. Hence, removal of these prior to planting will aid in preventing infection.

h) **Rouging.** The removal of diseased plants as they appear is often an effective method in helping reduce the spread of a destructive disease.

Economic Factors

Farmers make decisions concerning the allocation of resources between different activities according to the nature of the season and the circumstances of the individual household (Edwards, 1993). In most cases, smallholder farmers use as many of the above-mentioned control practices as possible to manage pest populations, while considering the costly chemical pesticides as the last resort.

Social Factors

The adoption of new technologies and knowledge (e.g., hybrid cultivars and its pest control) are constrained by the limited experience and knowledge of the farmer (Mphinyane and Terblanché, 2005). The farmers' perceptions of pests as well as on-farm pest levels also determines the farmers' cultural practice. For instance, some perceived stem borers to be serious pest of maize, and therefore, adopted planting of trap crops such as napier grass along the field borders (Berg, 2013).

Political Factors

In many developing countries, the government often impose restrictions on use of certain pesticides (especially those with high acute toxicity) and other unsustainable field sanitation practices such as burning. In these areas, the use of personal protective equipment or maintenance and calibration of pesticide use are not easily implemented. In response, governments prohibit the importation, distribution, sale, and purchase of highly hazardous pesticides (FAO, n.d.).

4. Fertilizer Application

Fertilizer application or fertilization is the supplemental application of plant nutrients to crop plants to augment the supply from natural sources. The practice of field fertilization has been found to be associated with three socio-economic factors in many developing countries. These factors include the farmers' tradition, land ownership status, age of farmers, and agricultural literacy.

Socio-Economic Factors

The traditional way means an approach which is based on non-mechanized fertilization. The approach has been established following closed senior farmers' fertilization and such agri-customs (habits, practice) existing in the society for a long time as they have no training and technological efficiency. The way has been established bearing a notion 'more fertilizer, more yield' (Rahman and Zhang, 2017).

Many of the farmers in developing countries cultivate their land on rental basis. In many cases, the land rental price is very high due to high competition among small or landless

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farmers; soso, they make their best effort to gain high yields within the limited tenure using high inputs such as fertilizers (Rahman and Zhang, 2017).

In some case, farmers at younger age (below 40 years) apply more excessive amounts of inorganic fertilizers than older farmers. Those who are landless, and no other source of income often use excessive amounts of fertilizers in their rental or own small lands for higher gain is inevitable. Coupled with no training and environmental consciousness that causes environmental losses, younger farmers have a strong impact on over-fertilization (Rahman and Zhang, 2017).

The farmers' level of literacy is also an important factor in fertilizer application. The years of experience, the level of education, awareness of organic farming have significant effect on the number of fertilizers utilized by farmers (Mohammadi et al., 2017).

Socio-Political Factors

In the Philippines, social and political factors have been observed to change the traditional practice of crop fertilization. The increasing awareness of consumers' preference for organically grown crops over non-organics have impacted towards favorable use of organic and bio-fertilizers. Politically, the recent change in agricultural paradigm and policies have shifted the farmers' practice from traditional fertilization to a balanced fertilization strategy (BFS). Balanced fertilization strategy is an optimum combination of organic and inorganic fertilizers (De La Rosa, n.d.).

5. Irrigation

Irrigation is the timely application of the proper amount of water to the plants. Enough water must be available to sustain crop production. The schedule and type of irrigation differs across areas as it depends on the farmers' capability, the selected crop, and method of determining the right time of irrigation.

Economic Factors

Farm irrigation is often associated with the availability of natural and infrastructural resources, and capital of farmers. For instance, in rice farming, those who are near water reservoirs have been practicing irrigated farming for a long time. In upland ecosystems, farmers rely on deep wells and underground water and invests on irrigation pumps and fuel. Whereas, in areas with inaccessible or limited irrigation systems, rainfed cultivation is the norm, given there is ample amount of rainfall in the area throughout the year.

In developed countries, the use of drip or trickle system is popular among farmers to efficiently utilize water resources. Scheduling is also determined by precision instruments such as tensiometers. In less developed countries, surface and manual irrigation is predominant, and time of irrigation is done on a regular basis or is determined by soil or plant observations (e.g., wilting point).

Social Factors

Culture and tradition have dictated the practice of irrigation in many areas for the longest period. However, the farmers' need and perception, the efficient diffusion of new technologies, and growing awareness towards sustainable agriculture can change the farmers' tradition. For example, more than 80,000 farmers, covering about 90,000 ha, have already adopted the 'Alternate Wetting and Drying' (AWD) technology in rice farming in the Philippines (Cabangon, 2013). This had paved way to the cultural practice of conserving water in irrigated rice ecosystems.

Political Factors

The main concern of farmers is the accessibility and affordability of irrigation systems. Here, responsible government agencies (e.g., NWRB, NIA, NEDA, LGUs, etc) play important roles by pursuing rehabilitation programs on communal irrigation systems, expanding research and development, formulating and reviewing water use policies,

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strengthening the participation of stakeholders, and training and capability building in planning and implementation of communal irrigation projects, among others (Dayrit, n.d.).

Summary

- Cultural practices vary across location, time, and the farmers' background; and these are dictated by several human factors including the social, economic, and political.
- Economic factors include the availability of farm resources, capital or income, subsidies, commodity prices, fuel prices, cost of land, labor laws, transport, and markets.
- Social factors include age of farmers, income, labor, cooperative membership, household size, level of education, farm size, land tenure, the type of crop production practiced, and the society's inclination to farming.
- Political factors include the political system (e.g., capitalism, commercialism, socialism, etc.) and government policies on land, seeds, fertilizers, irrigation, marketing, and trade.
- Cultural practices have changed over time due to the farmers' gained experience, extension activities done properly, and above all, the practicality of the introduced technology.

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IV. SUSTAINABLE AGRICULTURE

A. MAN AND CROPS IN AN ECOSYSTEM

Overview

Sustainable Agriculture (SA) can be described as a system of ecological farming practices, which is based on scientific innovations through which it is possible to produce healthy foods with respect for the land, air, water, and farmers' health and rights.

The basic principles are employed to satisfy the needs of humankind for healthy food, to improve the quality of environment, to maintain the natural resources base, to use non-renewable and on-farm resources in the most effective way, to implement the natural biological cycles and to support rural economic development as well as the quality of farmers' life.

Modern day world, and especially developing societies, faces numerous challenges; among which the progressive increase in the population and thus the increase of food

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consumption and the expansion of industrial production. These factors require shifting the progress of agriculture toward sustainability.

Agriculture is considered the biggest profession on earth; it provides the world with its need of daily nutrition, which may reach up to 7.3 billion tons of dairy products every year, and around 2.25 billion cups of coffee every day. Plus, agriculture occupies 40% of the Land on which we are living, and consumes 70% of water resources and 30% of green reserves around the globe.

Learning Objectives

At the end of the topic, the participant should be able to:

1. define Sustainable Agriculture,
2. understand Concepts of Sustainable Agriculture,
3. identify the components of Sustainable Agriculture, and
4. enumerate the sustainable Agriculture Practices

Let us first define ***Sustainable Agriculture;***

"Sustainable agriculture" was addressed by Congress in the 1990 "Farm Bill" [Food, Agriculture, Conservation, and Trade Act of 1990 (FACTA), Public Law 101-624, Title XVI, Subtitle A, Section 1603 (Government Printing Office, Washington, DC, 1990) NAL Call # KF1692.A31 1990]. Under that law, "the term sustainable agriculture means an integrated system of plant and animal production practices having a site-specific application that will, over the long term:

1. satisfy human food and fiber needs;
2. enhance environmental quality and the natural resource base upon which the agricultural economy depends;
3. make the most efficient use of nonrenewable resources and on-farm resources and integrate, where appropriate, natural biological cycles and controls;
4. sustain the economic viability of farm operations; and
5. enhance the quality of life for farmers and society as a whole."

Concepts of Sustainable Agriculture

Agricultural sustainability may be reach through three basic factors:

1. Taking into regard ensuring financial income, in other word the profitability, whereas agricultural activity shall be considered as an investment.
2. The creation of a new social order benefiting from these agricultural activities, as well as providing work, development, and training for targeted individuals.
3. The preservation of the environment and ensuring its diversity. Such factors could be fulfilled through exploiting each and every available resource, taking into account to protect and develop such resource in order to provide safe nutrition and agricultural produce over the long term.

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In fact such precautions are able in one hand to achieve sufficiency to agricultural societies that are engaged in these activities, as well as to non-agricultural societies and in the other hand to manage profitability for investors. Over the long term sustainable agriculture could preserve the surrounding environment and ensure the bio diversity. If agriculture would not conserve the environment and its diversity therefore there will not be any sustainability or continuity of agricultural investment, and thus there will not be provided any safe nutrition to consume anymore.



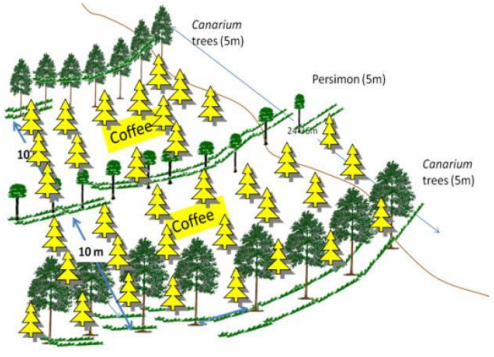
Concept of Sustainable Agriculture

Components of Sustainable Agriculture

The main components of both sustainable farming and conventional farming are exactly the same: soil management, crop management, water management, disease/pest management and waste management. It's the methods used that are often radically different. We'll discuss them in order, starting with soil management.

On a conventional farm, managing and maintaining soil fertility is as simple as running a soil test and applying the recommended doses of nitrogen, phosphorus, potassium and other nutrients to meet crop needs.

In sustainable agriculture, soil fertility is maintained and improved through a careful rotation of crops and generous amounts of compost and **green manure**, which are cover crops that are plowed back into the soil to enrich organic matter.



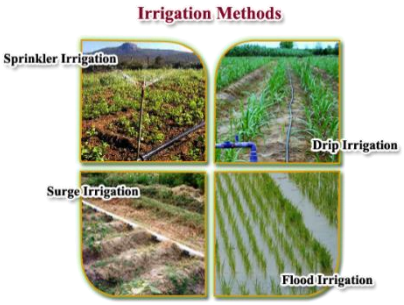


Monoculture is the term for agriculture that only produces one crop, year after year. The danger of monoculture is that it requires more and more chemical fertilizer to replenish lost nutrients, and more and stronger chemical pesticides and fungicides to kill off the bugs and diseases that evolve alongside the same crop year after year. Sustainable agriculture employs a broad crop diversity and careful rotation, so that nutrients are replenished naturally and no single pest or disease is allowed to get out of control.

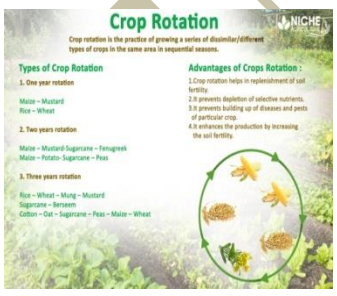


Unhealthy soils are easily eroded, and careless water management can allow chemical fertilizers, pesticides and fresh manure slurry to leach into rivers, streams and the drinking water supply.

Sustainable water management views water as a precious resource, efficiently watering crops using drip irrigation, which cuts down on erosion and evaporation. Efficient water use is hugely important in arid climates, where sustainable farmers plant drought-resistant crops and limit animal grazing.

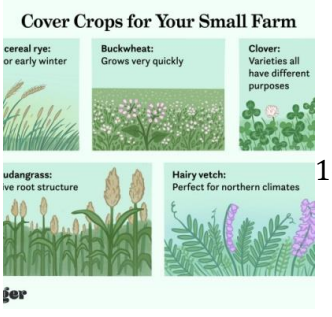


Sustainable Agriculture Practices



Rotating crops and embracing diversity. Planting a variety of crops can have many benefits, including healthier soil and improved pest control. Crop diversity practices include intercropping (growing a mix of crops in the same area) and complex multi-year crop rotations.

Planting cover crops. Cover crops, like clover or hairy vetch, are planted during off-season times when soils might otherwise be left bare. These crops protect and build soil health by preventing



erosion, replenishing soil nutrients, and keeping weeds in check, reducing the need for herbicides.

Reducing or eliminating tillage. Traditional plowing (tillage) prepares fields for planting and prevents weed problems, but can cause a lot of soil loss. No-till or reduced till methods, which involve inserting seeds directly into undisturbed soil, can reduce erosion and improve soil health.



Applying integrated pest management (IPM). A range of methods, including mechanical and biological controls, can be applied systematically to keep pest populations under control while minimizing use of chemical pesticides.

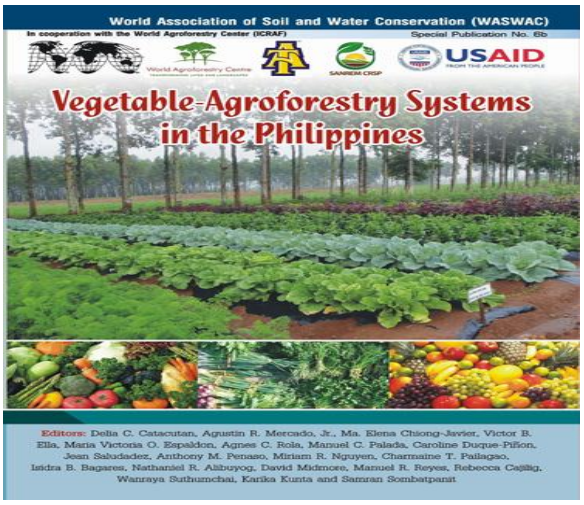




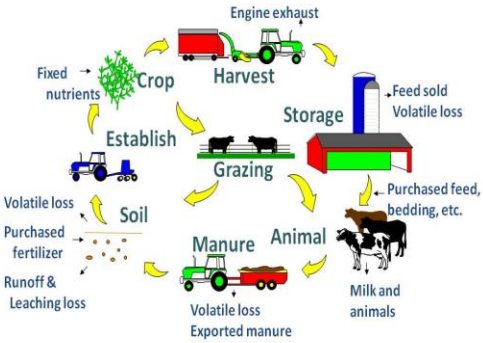
Integrating livestock and crops. Industrial agriculture tends to keep plant and animal production separate, with animals living far from the areas where their feed is produced, and crops growing far away from abundant manure fertilizers. A growing body of evidence shows that a smart integration of crop and animal production can be a recipe for more efficient, profitable farms.

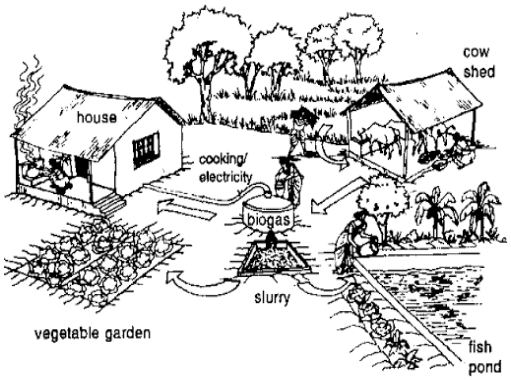
Adopting agroforestry practices. By mixing trees or shrubs into their operations, farmers can provide shade and shelter to protect plants, animals, and water resources, while also potentially offering additional income.





Managing whole systems and landscapes. Sustainable farms treat uncultivated or less intensively cultivated areas, such as riparian buffers or prairie strips, as integral to the farm—valued for their role in controlling erosion, reducing nutrient runoff, and supporting pollinators and other biodiversity.





Summary

Sustainable agriculture is a type of agriculture that focuses **on producing long-term crops and livestock while having minimal effects on the environment**. This type of agriculture tries to find a good balance between the need for food production and the preservation of the ecological system within the environment.

In the preservation the earth's natural resources, **sustainable agriculture** benefits the environment through the following factors maintain soil quality, reducing erosion, and preserving water.

Sustainable agriculture integrates **three main** goals – environmental health, economic profitability, and social equity. A variety of philosophies, policies and practices have contributed to these goals, but a few common themes and principles weave through most definitions of **sustainable agriculture**.

Sustainable agriculture practices

The following practices are considered to be under the sustainable agriculture

Rotating crops and embracing diversity. ...
Planting cover crops. ...
Reducing or eliminating tillage. ...
Applying integrated pest management (IPM). ...
Integrating livestock and crops. ...
Adopting agroforestry practices. ...
Managing whole systems and landscapes.

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1. Flow of Energy

Every ecosystem is complex. Many living and non-living components are involved in the two most important processes: the flow of energy and cycling of matter. Terrestrial producers are mostly plants-including some aquatic producers like algae and small organisms -. Consumers eat producers, other consumers, or both. Decomposers break down dead material, extracting energy and returning atoms back into the ecosystem to be used again. In terrestrial environments, decomposers recycle nutrients from dead plant or animal matter back to the soil while in aquatic environments, the matter is recycled back to the water.

In contrast to matter, energy flows and is not recycled. The initial input of energy comes from the sun in almost all ecosystems. Producers capture a fraction of the sun's energy and store it in molecules. Consumers use some of the energy they take in to keep their body processes running and the rest for growth and reproduction.

On average, only 10 percent of the energy contained in one level reaches the next higher level. The rest is lost to heat and maintaining body function. However, ecological efficiency varies

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greatly. It is generally highest in plants and lowest in animals that can maintain a stable internal body temperature like mammals or birds. The number of organisms at each level and the total biomass can be determined. There are usually fewer large animals than small animals. Biomass may be a better indicator of energy contained in the level than abundance.

Learning Objectives

At the end of the topic, the participant should be able to:

1. discuss the structure, functions, and types of ecosystem, and
2. explain how energy flow in the ecosystem

The regions of the earth's surface are remarkably diverse. In climate, topography and vegetation cover, deserts differ from mountain, which differ from foot hills and arctic tundra, which differ from flats and rolling plains. Oceanic provinces, inland seas, lakes, ponds and rivers differ in physical and chemical properties as well as in their arrays of organisms. With few exceptions, these systems run on energy from the sun.

An ecosystem is not a "place". It is a community of organisms functioning together and interacting with their physical environment through flow of energy.

A. Structure of Ecosystem

1. Trophic Levels

   Ecosystem has layered structure based on number of times energy are transferred from one organism to another, away from their initial energy input into the system. Thus, all organism that are the same number of transfer steps away from the energy input are said to be at the same **trophic level** (from troph, meaning nourishment).

   Energy flows in an ecosystem from one trophic level (feeding level) to the next. All organisms can also be classified as either autotrophs or heterotrophs

- Autotrophs
  - Producers: The first trophic level includes organisms that are called producers, or autotrophs, because they produce their own food. Auto- refers to "self," and troph- refers to "feeding," so autotrophs are self-feeders. Producers include plants, algae, and some bacteria that use photosynthesis to make their own food.
- Heterotrophs
  - Consumers: The second trophic level of organisms gets energy by eating other organisms. Consumers are further classified by the type of food eaten and by how many trophic levels of energy are involved.
  - Decomposers: Decomposers break down dead or decaying consumers and producers via chemical reactions and use the energy released from rearranging those molecules for growth and development. Decomposers can be fungi or bacteria.

Examples of Trophic Levels

Members of each level	Energy Source	Representative Organisms
Primary producer		

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Photosynthetic autotrophs Chemosynthetic autotrophs	Sunlight energy Oxidation of inorganic substance	Photosynthetic bacteria, plants Sulfur bacteria
Primary consumers herbivores	Primary producers	Grasshoppers, deer, periwinkles
Secondary consumers Primary carnivores	herbivores	Spiders, small squids
Tertiary consumers Secondary carnivores	Primary carnivores	Emperor penguins

2. **Food Chain and Food Web**

The general sequence of who eats whom is sometimes called a **food chain**. More typically, the same food resource is part of more than one chain, especially when that resource is at one of the low trophic levels.

Algae → fish → fisherman → shark

Yet this chain would be an oversimplification of feeding relationships, for it would exclude any number of alternatives. Most likely, crustaceans were also grazing on the algae. Small squids and assorted medium sized fishes might have been feeding on the crustaceans, some larger fishes were probably feeding on smaller ones. Sharks may have been moving in to feed on the large and medium sized fishes.

A food web is a network of may interlinked food chains, encompassing primary producers and an array of consumers and decomposers.

Flow of Energy in an Ecosystem

Energy flow is the flow of energy through living things within an ecosystem. All living organisms can be organized into producers and consumers, and those producers and consumers can further be organized into a food chain. Each of the levels within the food chain is a trophic level. Animals, including humans, get their energy by eating other organisms. We know that energy cannot be created or destroyed, so where does the source of energy in living systems begin, and how does it move through these systems?

- Not all trophic levels of a food chain receive the same amount of energy. Only about 10% of the Sun's energy absorbed by plants is transferred to the consumers at the next trophic level. Where does 90% of the energy go? Some of it is used to fuel body processes such as growth, repair, and reproduction. Most of it is transferred to the atmosphere as heat. We can represent the energy flow of an ecosystem in an energy pyramid.

- Energy moves through an ecosystem in a single direction. First it flows from the Sun to autotrophs, or producers. Then it flows from producers to consumers. Energy never flows backward from consumers to producers.
- When you draw a food chain, always begin with energy from the Sun to a producer. Don't forget that the arrows should point to the organism receiving the energy.

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- A small percentage of organisms use sulfur compounds from volcanic sea vents deep in the ocean as their source of energy. Instead of using Sunlight in photosynthesis, these organisms use chemicals to aid in chemosynthesis. They make their own food, just like in photosynthesis.

Primary Productivity

The rate at which the plants fix energy in organic compounds it is called the **primary productivity** of an ecosystem. The actual amount of energy stored in a given time interval depends on the balance between energy acquisition by photosynthesis and energy use by the producers (most often by aerobic respiration). It is also depends on how much of the stored energy is not utilized by the consumers. Thus,

1. The *gross primary productivity* is the total rate of photosynthesis in the ecosystem.
2. The *net primary productivity* is the rate of energy storage in plant tissues in excess of the rate of respiration during the measured time interval.
3. The *net community productivity* is the rate of energy storage in excess of heterotrophic consumption.

In a given ecosystem, net primary productivity is influence by such variables as the availability of sunlight and nutrients, temperature, rainfall, predation, duration of the growing season and the occurrence of fires. These variables influence not only the amount but also the distribution of net primary production. The more the favorable the environment the more growth is put into photosynthetic parts (shoots), hence the greater the productivity. Another variable affecting net primary productivity is the age of ecosystem.

Major Pathways of Energy Flow

Of all solar radiation reaching the earth, only about 1% or 2% become stored in organic matter. Then plant respiration reduces (often by half) the energy available for other members of the ecosystem. This tiny fraction becomes available in two forms,

- As living tissues of the producers
- As detritus (organic waste, dead tissues, and partially decayed organic matter)

All organism of the ecosystem lose energy to the surroundings (as low grade heat usually) during metabolic activity, hence there is a one way flow of energy out of ecosystem.
A **grazing food web** begins with consumption of living plant tissues by herbivores, which are in turn consumed by some array of carnivores. In **detrital food webs,** most of the net primary production is used by decomposers and detritivores (not by herbivores).These organism breakdown organic waste products or remains to simple inorganic substances that can be cycled back to the plants. The fraction of energy flows through grazing or detrital food webs varies from one ecosystem to the next (and often throughout the year in the same ecosystem).All ecosystems have both grazing and detrital food webs, and both are interconnected.

1. Energy flows into ecosystems from outside sources such as the sun.

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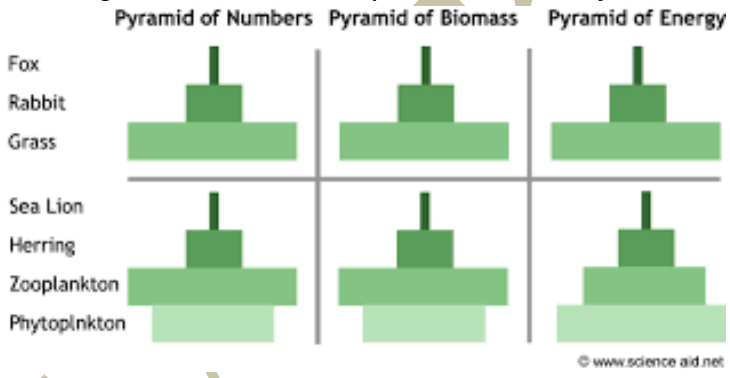
2. Energy flows through ecosystems by way of grazing food webs (based on the consumption of the living tissues of photo synthesizers) and detrital food webs (based on use of organic waste products and remains of producers and consumers).
3. Energy leaves ecosystems as a result of metabolic activities (aerobic respirations mostly) of each organisms. The energy is lost primarily as low grade heat.

Pyramid Representing the Trophic Structure

The trophic structure of an ecosystem is influence by the size of its component organisms, their different metabolic rates, and the loss of energy at each transfer. Often the trophic structure is represented as an ecological pyramid, with the producers forming a base for successive tiers of consumer above them.

Pyramid of Numbers

Sometimes the number of organisms in the ecosystem is depicted in pyramid form. Aside from monumental patience required to count all organisms, a pyramid of numbers does not completely define the trophic structure for an ecosystem. For one thing, it defines feeding relationships only at one moment and feeding relationships can change cyclically or permanently over time. For another thing, a pyramid of numbers does not take into account that the sizes of organisms being counted in each trophic level can vary.



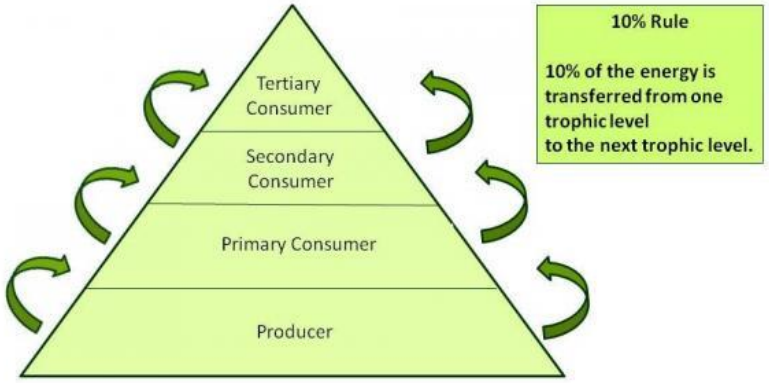
Pyramid of Biomass

Another approach is to weigh individuals in each trophic level instead of counting them. This would give us a pyramid of **biomass** (the total dry weight of all organisms at a given level. For most ecosystem on land, the pyramid of biomass have a large base of primary production, with ever smaller trophic levels perched on top. In contrast, in some aquatic ecosystems, the producers are tiny phytoplankton that row and reproduce rapidly. Here the pyramid of biomass can have small base, with the consumer biomass at any instant actually exceeding the producer biomass. The phytoplankton are consumed about as fast as they reproduce, it is just that the survivors (few as they maybe) are reproducing at a phenomenal rate.

Pyramid of Energy

When we wish to compare the functional roles of the trophic levels in an ecosystem, an energy pyramid is probably the most informative, for the pyramid shape is not skewed by overemphasis on variations in size and metabolic rates of individual. The energy pyramid reflects the laws of

thermodynamics (with energy losses being depicted at each transfer to another trophic level). The pyramid is always right side up with a large energy base at the bottom.
A pyramid of energy must be based on a determination of the actual amounts of energy that individual take in, how much they burn up during metabolism, how much remain in their waste products, and how much they store in their body tissues.



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2. Cycling of Nutrients

The nutrient cycle is a system where energy and matter are transferred between living organisms and non-living parts of the environment. This occurs as animals and plants consume nutrients found in the soil, and these nutrients are then released back into the environment via death and decomposition.
Nutrient cycling is strongly influenced by forest disturbances shaping nutrient availability and stoichiometry (ratios of available nutrients) immediately after disturbance and the species composition of the post disturbance landscape. While considerable work has focused on the effects of disturbances on nutrient cycling immediately post disturbance, less attention has been paid to the fundamental role of disturbances in nutrient cycling in the long term, as a component of the entire ecosystem

At the end of the session, the reviewers should be able to:

a. understand the basic concepts of nutrient cycling; and,
b. discuss the nutrient cycling in crop production

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Three Main Nutrient Cycles

The **nutrient cycle** is a system where energy and matter are transferred between living organisms and non-living parts of the environment. Animals and plants consume nutrients in order to live, and the nutrients are released back into the environment via death and decomposition. They are then available once again for *other* living things to consume and use for their own life functions.

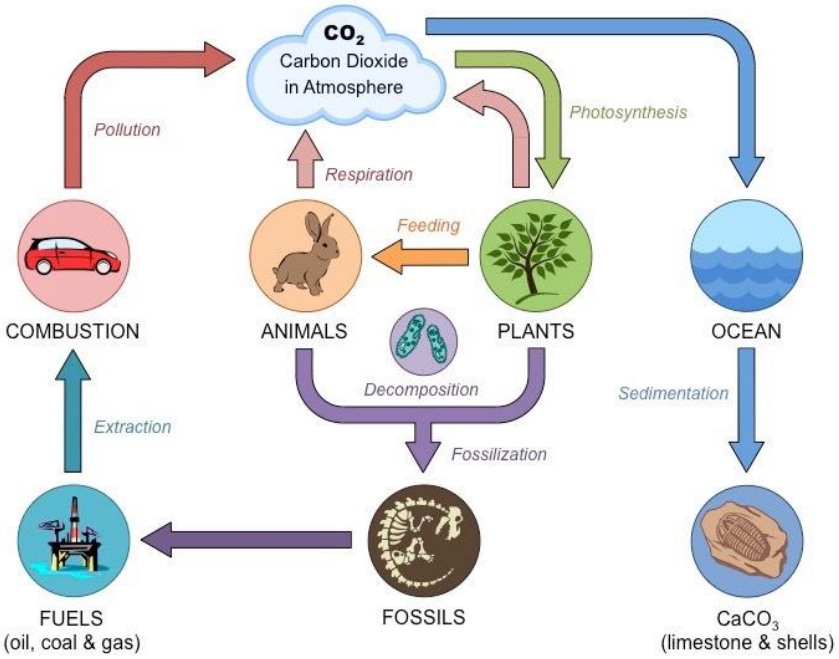
Living organisms need food to grow and to reproduce. Any food or element required by an organism to live, grow, or reproduce, is called a nutrient. Depending on the amount it is needed in, a nutrient can be classified as a macronutrient or a micronutrient . In nature, the nutrient elements and their compounds continuously move from the nonliving environment to the living organisms, and back to the nonliving environment. This cyclic movement of minerals from their reservoirs (air, water and soil), to the living components, and back to the reservoirs is called nutrient cycling or biogeochemical cycles.

Nutrients are chemicals required for plant and animal growth and other life processes. They are constantly recycled within Earth's biosphere. Nutrients spend different amounts of time in stores within the atmosphere, oceans, and land. Nutrients are stored for short periods of time in short term stores, such as living organisms and the atmosphere. Nutrients can also be incorporated into longer term stores, such as Earth's crust. Nutrient cycles describe the flow of nutrients in and out of stores as a result of biotic and abiotic processes. Without human interference, nutrient cycles are almost perfectly balanced. There are three main cycles that move nutrients through terrestrial and aquatic ecosystems:

a. the carbon cycle
b. the nitrogen cycle
c. the phosphorus cycle

The Carbon Cycle

How does the carbon cycle work?



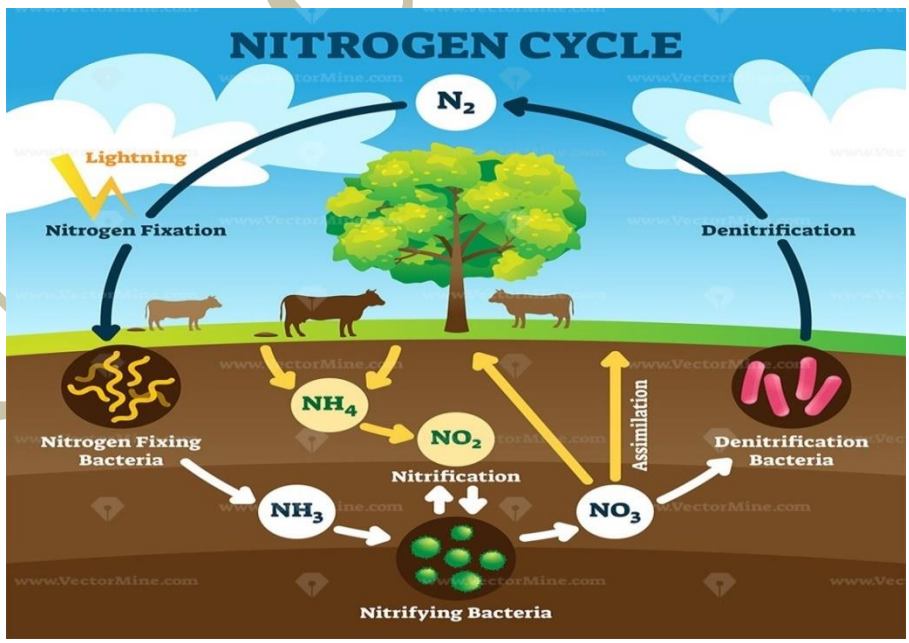
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Kindly include figure title and make your own figure and cite where it has been adaptedCarbon is an essential component of cells and life sustaining chemical reactions. Carbon is cycled through living and decaying organisms, the atmosphere, bodies of water, and soil and rock. Carbon moves between stores via six main processes:

a. Photosynthesis: Photosynthesis is a chemical reaction that converts solar energy and atmospheric carbon dioxide gas (CO2) into chemical energy.
b. Cellular respiration: During cellular respiration, plants and animals obtain energy by converting carbohydrates and oxygen (O2) into carbon dioxide and water.
c. Decomposition: Decomposers release carbon dioxide into the atmosphere through the decomposition of carbon-rich organic matter in soil.
d. Ocean processes: Dissolved carbon dioxide is stored in oceans. Marine organisms store carbon rich carbonate (CO32) in their shells, which eventually form sedimentary rock.
e. Volcanic eruptions
f. Forest fires

How do human activities affect the carbon cycle?
Human activities, such as fossil fuel combustion and land clearance, quickly introduce carbon into the atmosphere from longer term stores. These actions increase the levels of carbon dioxide, a greenhouse gas that contributes to global climate change.

The Nitrogen Cycle
How does the nitrogen cycle work?



Nitrogen is an important component of DNA and proteins. Most nitrogen is stored in the atmosphere, where it exists as nitrogen gas (N_2). It is also stored in bodies of water, living organisms, and decaying organic matter. Most organisms cannot use atmospheric nitrogen gas.

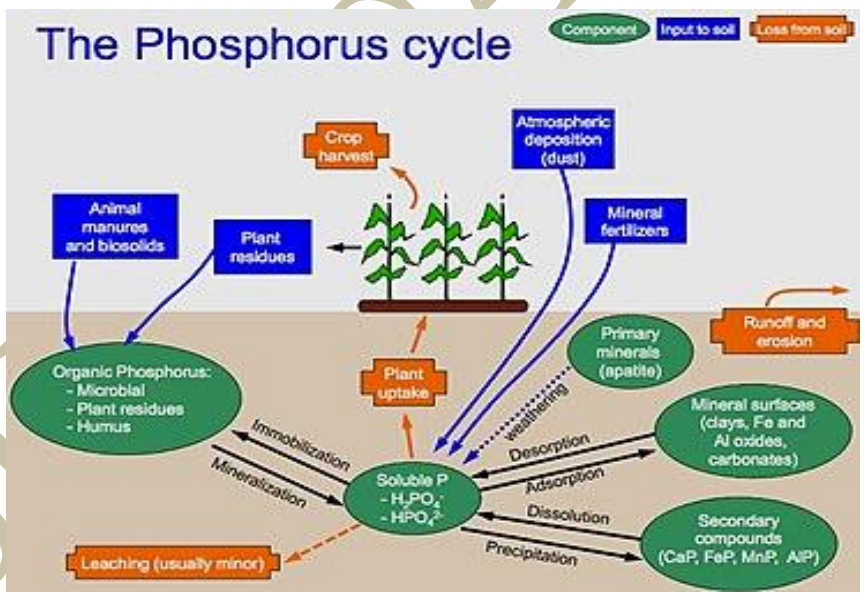
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The nitrogen cycle involves four processes, three of which make nitrogen available to plants and animals.

a. Nitrogen fixation: Nitrogen gas is converted into nitrate (NO_3^-) and ammonium (NH_4^+), compounds that are usable by plants. Nitrogen fixation occurs mainly through nitrogen fixing bacteria, and when lightning strikes in the atmosphere.
b. Nitrification: Ammonium is converted into nitrate and nitrite (NO_2^-) through the work of nitrifying bacteria.
c. Uptake: Useable forms of nitrogen are taken up by plant roots and incorporated into plant proteins. When herbivores and omnivores eat plants, they incorporate nitrogen into their own tissues.
d. Denitrification: Denitrifying bacteria convert nitrate back into atmospheric nitrogen.

How do human activities affect the nitrogen cycle?
Fossil fuel combustion and burning organic matter release nitrogen into the atmosphere, where it forms acid rain. Chemical fertilizers also contain nitrogen, which escapes into the atmosphere or leaches into lakes and streams. High levels of nitrogen cause eutrophication (too many nutrients) and increased algal growth in aquatic ecosystems, depriving aquatic organisms of sunlight and oxygen.

The Phosphorus Cycle
How does the phosphorus cycle work?



Phosphorus carries energy to cells. It is found in phosphate (PO_4^{3-}) rock and sediments on the ocean floor. Weathering through chemical or physical means breaks down rock, releasing phosphate into the soil from longer term stores. Organisms take up phosphorus. When they die, decomposers return phosphorus to the soil. Excess phosphorus settles on floors of lakes and oceans, eventually forming sedimentary rock. It remains trapped for millions of years until it is exposed through geologic uplift or mountain building.

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How do human activities affect the phosphorus cycle?
Commercial fertilizers and phosphate containing detergents enter waterways and contribute additional phosphate to the phosphorus cycle. Slash and burn forest clearance reduces phosphate levels, as phosphate in trees enters soil as ash. It leaches out of the soil and settles on lake and ocean bottoms, unavailable to organisms.

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B. PRODUCTION SYSTEMS AND CROP MANAGEMENT

Learning Objectives

At the end of the topic, the participant should be able to:

1. discuss the importance of land preparation, and
2. discuss the difference and characteristics of lowland and upland.

Land Preparation

- To provide a favorable soil environment for the germination and growth of a particular crop.
- It is done in accordance with the requirement of the crops, whether they grow under dryland or wetland system. The two systems of land preparation and water management (wetland and dryland) have contrasting features in terms of its physical, biological and chemical nature soils.

*land is prepared by the process of tillage.
JethroTull is considered as the father of modern tillage
Tillage
- a general term referring to manual or mechanical soil stirring actions necessary for the proper establishment and growth of crops
- is the manual or mechanized manipulation of the soil to provide a medium for proper crop establishment and growth.

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Types of tillage and the purposes of each.

A. Primary tillage involves the initial cutting of the soil where a crop has been grown and harvested, or when the ground is simply barren. This can be performed by using either an animal or tractor drawn or mounted implement.

B. Secondary tillage is the subsequent cultivation of the soil after primary tillage. This operation breaks the soil clods and incorporates materials thoroughly into the soil. The implement used is the harrow which is either comb-toothed or spike-toothed.

Two ways of preparing rice fields:

A. Wetland tillage is the usual way of preparing lowland fields for planting. The soil is tilled in a saturated or flooded condition.

B. Dryland tillage is the other method of preparing rice fields. This is for both lowland and upland fields where the soil is dry during land preparation. This is done to obtain a mellow, firm seedbed for sown seeds to germinate in, control weeds, and incorporate organic materials into the soil.

Types of land preparation:

- **Lowland/wetland preparation**- soaking, plowing, harrowing, leveling
- **Upland/dryland preparation**- plowing, harrowing/rotation, leveling

Tillage and planting equipment

- **plow-** animal-drawn moldboard, tractor-drawn moldboard, disc
- **harrow –**disc, comb-tooth, spike-tooth
- **Rotavator**
- **Furrower**- animal-drawn ('lithao'), tractor-drawn
- **Planting hole digger**

Characteristics of a well-prepared upland field:

- Soil is granular, friable, and compact enough to allow contact between seed and soil
- Sufficient moisture for germination and subsequent growth
- Level field to avoid water accumulation
- Free from weeds

Difference in terms of physical changes between upland and lowland land preparation:

- **Upland –** no flooding, no pudding, macro- and microspores are maintained, waters drains easily if there is no plowpan, tillage is easy especially at field capacity
- **Lowland –** flooding involved, pudding involved, macrospores are lost while microspores are maintained, plow pan is formed, tillage is hard when soil is dried

Difference in terms of chemical changes between upland and low land preparation:

- **Upland-** root zone is well-aerated, aerobic organisms are presents, no reduced zone, pH is stable, high N mineralization, P, K, Si, and Mo are less available, Cu and Zn are not affected, less generation of gases (i.e carbon dioxide, methane, nitrous oxide, hydrogen sulfide) and organic acids.
- **Lowland** – thin oxidized layers and reduced layer in root zone, aerobic and anaerobic organisms are present, increase in pH in acid soils and decreases in pH in alkaline

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soils, slow N mineralization, P, K, Si, and Mo are available, Cu and Zn are reduced, generation of gases and organic acid.

Advantages and disadvantages of wet and dryland tillage.

Wet tillage	Dryland tillage
Advantages	
✓ Improves weed control	✓ Requires less water
✓ Facilitates incorporation of organic materials and soil amendments	✓ Helps control golden snail
✓ Provides good soil condition for crop establishment	✓ Soil aeration
Disadvantages	
✓ Needs high water requirement	✓ High power requirement
✓ Soil structure is destroyed	✓ Lack of suitable equipment
✓ Unfavorable for subsequent establishment of rice-based crops	

Types of Tillage Operations:

- **Primary tillage –** the most aggressive tillage operations; undertaken when the soil is wet enough to allow the field to be plowed and strong enough to give reasonable levels of traction; after the crop harvest or at the beginning of the next wet season; includes the plowing operation which is opening of the compacted soil with the help of different plows
  - type of tillage that inverts, cuts or shatters the soil to a depth of 15-36 cm, leaving the soil rough.

- **Secondary tillage-** any working completed after primary tillage; shallower and less aggressive; includes the operations performing after plowing such as leveling, discing, and harrowing
  - tillage operation that follow primary tillage with the purpose of preparing the final seed bed for planting, seedling establishment and weed control

- **Seedbed preparation-**for germination of seeds; carried out by using hand tools or implements like harrow, rollers plank, and rider

- **Inter tillage-** carried out in the standing crops; after sowing or planting and prior to the harvesting of crop plants; includes gap filling, thinning, weeding, mulching, top dressing of fertilizes, hoeing, etc.

Form of Tillage Operations

Conventional tillage – combined primary and secondary tillage operations normally done in preparing a seed bed for a specific crops in a given geographical locations.
- results in a residue-free soil surface at planting time.

Conservation tillage–any tillage system that reduces loss of soil or water compared to clean tillage.

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- it is a basic tillage strategy that entails practices in which some crop residue remains on the soil surface after the operation. It maintains residue from previous crop on soil surface. The chief goals of conversation tillage are: 1) to reduce soil erosion and 2) conserve moisture.

The common types of conversion tillage include:

a. **No tillage or zero tillage.** A system of cropping whereby a crop is seeded directly into a seeded not tilled since the harvest of the previous crop.

b. **Mulch tillage-** a system that leave crop residue to serve as mulch, as in stubble-mulch tillage that aims to conserve moisture and to protect the soil from wind and water erosion by leaving crop residues on the soil surface.

- soil tillage which employed plant residues or other materials to cover the ground surface

c. **Strip tillage-** it is also called strip-till or zone tillage. It entails the disturbance of narrow strips in the soil where seeding is done. The interrow zone remains undisturbed and covered with crop residue.

d. **Minimum tillage-** it involves considerable soil disturbance but to lesser extent than conversational tillage. Some crop residue is left on the soil surface. Minimum tillage is also called reduced tillage.

- the amount of tillage required to create proper soil conditions for seed germination and seedling establishment

e. **Ridge tillage-** a small band of soil on the ridge is tilled. The soil from the top of the ridge is mixed with crop residue between ridges. The debris reduces soil erosion and increase water retention.

Tillage systems – refers to the nature and sequence of tillage operations in preparing a seed bed for planting

The advantages of using animals include:

- ✓ Cheap to maintain
- ✓ Multi-purpose use for meat and milk,
- ✓ Self-replacement and
- ✓ Manure production

Disadvantage of using animals include:

- ✓ Animals have limited daily working hours
- ✓ Animals need feeding and protection against pests and disease.
- ✓ Slow
- ✓ High man/power ratio
- ✓ Limited range of working conditions
- ✓ Animals require training

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CROP SCIENCE 2

UNIT I
INTRODUCTION
ISSUES IN CROP PRODUCTION AND DEVELOPMENT

I. Introduction

Increasing globalization affects agricultural production and trade and has consequences for the sustainability of both conventional and organic agriculture. During the last decades, agricultural production and yields have been increasing along with global fertilizer and pesticide consumption. This development has been especially pronounced in industrialized countries and some developing countries, wherein food security has increased, greater varieties of food have been offered and diets have changed towards a greater share of meat and dairy products (Knudsen et al., 2006).

However, this development has led to a growing disparity among agricultural systems and population, where especially developing countries have seen very few improvements in food security and production. Many rural households in developing countries lack the ecological resources or financial means to shift into intensive modern agricultural practices as well as being integrated into the global markets. At the same time, agricultural development has contributed to environmental problems such as global warming, reductions in biodiversity and soil degradation.

Furthermore, pollution of surface and groundwater with nitrates and pesticides remains a problem of most industrialized countries and will presumably become a growing problem of developing countries. The growing global trade with agricultural products and the access to pesticides and fertilizers have changed agricultural systems. Easier transportation and communication have enabled farms to buy their inputs and sell their products further away and in larger quantities and given rise to regions with specialized livestock production and virtual monocultures.

II. Learning Objective

At the end of the session, the participant should be able to identify the global trends in agriculture and food systems highlighting the issues in crop production and development.

III. Learning Content

A. World Agriculture: Trends and Impacts

Agriculture and food systems have changed very much over the last 50 years. Agricultural development has seen a rapid advance of agricultural technology in industrialized countries with the green revolution in the 1960s being counteracted by an increasing public awareness of environmental protection and sustainable development that evolved in the 1980's (FAO, 2000).

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In the 1990s, an increasing globalization occurred that has continued into the 21st century. The current wave of globalization was made possible by technological breakthroughs in transportation and communication technologies and affordable fuel in tandem with various efforts to liberalize international trade and investment flowers (FAO, 2003).

B. Agricultural Intensification

The agricultural revolution and globalization have had an enormous impact on agriculture and food systems in the developed countries.

- Development in industry, biotechnology, transport and communication have affected agriculture in different ways.
    - Industrial developments have provided the means for motorization and large-scale mechanization, mineral fertilization, treatment of pests and diseases (pesticides, veterinary drugs etc.) and the conservation and processing of vegetable and animal products in developed countries.
    - Developments in biotechnology supplied through selection, high yielding plant varieties and animal breeds have been adapted to the new means of production (FAO, 2000).
        - The latest biotechnological developments are the genetically modified crops, grown primarily in USA, Canada and Argentina.

The increase in fertilizer use and the use of improved varieties have been among the important factors for the increased food production.

- A third of the increase in world cereal production in the 1970s and 1980s has been attributed to increased fertilizer use (FAO, 2003).
- World fertilizer consumption grew rapidly in the 1960s, 1970s and 1980s.
- The global uses of pesticides have increased considerably during the second part of the 20th century.
    - Some of the problems with diseases and insects have increased with the increased use of nitrogen fertilizers due to a higher susceptibility of the crop to attack at higher nitrogen input (Olesen et al., 2003).
    - Future pesticide consumption is likely to grow more rapidly in developing countries than in developed ones (FAO, 2003).
    - A great part of the antibiotics produced today are used as treatments against infectious diseases or as growth promoters in animal production, especially for pigs and poultry.
- Agricultural systems have changed with the introduction of mineral fertilizer, pesticides etc.
    - With the use of mineral fertilizer, cash crop production no longer relies on soil fertility building or use of manure.
    - With the introduction of mechanization, agriculture has also been freed from the need to produce forage for draught animals.
    - The use of agricultural chemicals and GMO crops has partly released agricultural holdings from former crop rotation systems used to control weeds, insects and

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diseases.

C. Environmental Impacts

Human activities and the provision of foods for the growing world population put increasing demands on the natural resources of the earth. In some areas of the world, agricultural land use increases at the expense of forests and other natural terrestrial ecosystems.

Four major indicators of environmental sustainability:

a) Loss of land resources by soil erosion and soil degradation.
b) Loss of biodiversity involves a reduction in the number of living species on the earth and thus a loss of genetic resources.
c) Global warming
d) Use of fertilizer in high amounts per hectares and the large amounts of manure

D. Socio-Economic Impacts

Developments in agriculture and food systems such as industrialization and globalization have had socio-economic impacts all over the world, both for the millions who are engaged in farming and for the urban populations.

- The present phase of globalization, characterized chiefly by the proliferation of wireless communications, satellite television and the Internet, may be seen as the final outcome of a process that began in the mid-19th century with the first network technologies; the railroads and the telegraph.
- Social, economic and political life has become increasingly dominated by the rise and spread of technologies of mass production and mass transport that are highly intensive in the use of energy, minerals and capital (The Ecologist, 1993).

E. Challenges in Agricultural Production

1. Climate change

According to the most recent assessment report of the Intergovernmental Panel on Climate Change (IPCC), published in 2014, levels of anthropogenic emissions of greenhouse gases (GHGs) are now at their highest in history (Porter et al., 2014).

Agricultural production and its effect on land use are major sources of these emissions. Charting environmentally sustainable pathways for agricultural development has a central role to play, therefore, in mitigating climate change. The impacts of climate change are expected to be most adverse in low- and middle-income countries, where millions of people depend on agriculture and are vulnerable to food insecurity.

The impact of climate change is very comprehensive but its far reaching effects are now clearly visible on agricultural sector, on which relies the food production and economy of the

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world. As climate change and agriculture have inextricable links, abrupt changes in climatic conditions at such a rapid pace has threatened the food security at global scale.

As per Food and Agriculture Organization (FAO) data published in 2016, if the current situation of GHG emissions and climate change continue then by the year 2100 there will be decline in the production of major cereal crops (20–45% in maize yields, 5–50% in wheat and 20–30% in rice). Hence if the trends continue, in extremely near future crop losses may increase at an unprecedented rate which will substantially contribute to reduced production, spiked food prices, and it will become difficult to cope up with rising needs of growing population.

a. ***Flooding incidence***- Climate change has also resulted in increased and sporadic incidences of floods in last few years. Report by European Academies' Science Advisory Council (EASAC) suggests that extreme events including floods have increased by 50% in last 10 years and are now occurring at rate of four times higher than in comparison to 20 years back. These floods have resulted in washout of topsoil and nutrients from the soil, resulting in low productivity for several years to come, unless and until corrective and pro-active remediation strategies are not worked upon.

   Rise in sea levels or prevalence of heavy rainfall could result in decline of agricultural lands in coastal regions. This has also resulted in salinity of soils in coastal regions leading to stresses in crops such as decreased respiration, photosynthesis, and transpiration, ultimately jeopardizing food availability and security in such regions.

b. ***Drought*** - Extreme drought conditions, frequently occurring due to climate change, exacerbate the productivity of crops by causing nutrient immobilization and salt accumulation in soils making them dry, unhealthy, saline and finally infertile. Such barren lands become non-arable with course of time and are eventually abandoned by farmers leading to economic losses and social issues. Projected climate variations are not limited to only increase in drought conditions.

c. ***Soil degradation*** - Climate change is resulting into an exceedingly high rate of land degradation causing enhanced desertification and nutrient deficient soils. The menace of land degradation is increasing by the day and has been characterized as a major global threat. Land degradation is supposed to influence lives of 1.5 billion people and 15 billion tons of fertile soil is lost every year due to anthropogenic activities and climate change.

d. ***Water shortages*** - If global temperatures continue to rise, rainfall will increasingly become a beast of extremes: long dry spells here, dangerous floods there – and in some places, intense water shortages.

e. ***Increased pests and diseases*** - Climate change is one factor driving the spread of pests and diseases, along with increasing global trade. Climate change can affect the population size, survival rate and geographical distribution of pests, and the intensity, development, and geographical distribution of diseases.

However, when temperatures and precipitation levels get too high, this can slow the growth and reproduction of some pest species and destroy them by washing their eggs and larvae off the host plant, he explains.

This would explain why many pests are moving away from the tropics towards more temperate areas. Pests like warmer temperatures – but up to a point. If it is too hot or too cold, populations grow more slowly. Since temperate regions are not currently at the optimal temperature for pests, populations are expected to grow more quickly in these areas as they warm up.

Crop diseases are following a similar pattern, particularly when it comes to pathogens like fungi.

2. Natural resource degradation

a. ***Deforestation*** - Agriculture is estimated to be the proximate driver for around 80 percent of deforestation worldwide.

   Deforestation is the conversion of forested areas to non-forest land use such as arable land, urban use, logged area or wasteland. According to FAO, deforestation is the conversion of forest to another land use or the long-term reduction of tree canopy cover below the 10% threshold. Deforestation can result from deliberate removal of forest cover for agriculture or urban development, or it can be an unintentional consequence of uncontrolled grazing (which can prevent the natural regeneration of young trees). The combined effect of grazing and fires can be a major cause of deforestation in dry areas. Deforestation implies the long-term (>10 years) or permanent loss of forest cover.

b. ***Loss of biodiversity*** - Temperate grasslands, savannahs, and shrub-lands have experienced heavy conversion to agriculture, more so than other grassland types including tropical and subtropical grasslands, savannahs, and woodlands. In many places the introduction of non-native species has negatively affected grassland ecosystems leading to a decrease in biodiversity.

c. ***Soil erosion*** - Half of the topsoil on the planet has been lost in the last 150 years. In addition to erosion, soil quality is affected by other aspects of agriculture. These impacts include compaction, loss of soil structure, nutrient degradation, and soil salinity. These are very real and at times severe issues.

   The effects of soil erosion go beyond the loss of fertile land. It has led to increased pollution and sedimentation in streams and rivers, clogging these waterways and causing declines in fish and other species. And degraded lands are also often less able to hold onto water, which can worsen flooding. Sustainable land use can help to reduce the impacts of agriculture and livestock, preventing soil degradation and erosion and the loss of valuable land to desertification.

The health of soil is a primary concern to farmers and the global community whose livelihoods depend on professionally managed agriculture that starts with the dirt beneath our feet.

d. ***Flood and water quality*** - Agriculture is both a major cause and casualty of water scarcity. Farming accounts for almost 70% of all water withdrawals, and up to 95% in some developing countries. We will have to use our natural resources more wisely as time goes on and when it comes to water there is no exception. For example, the choice of crop greatly impacts the amount of water that is needed.

e. Cultivation resulting to depletion of nutrients in various ecosystems

Changes in the climate and atmosphere can have serious effects on plants, including changes in the availability of certain nutrients. In a world of continual global climate change, it is important to understand the strategies that plants have evolved to allow them to cope with some of these obstacles.

Nutrient depletion in soils adversely affects soil quality and reduces crop yield and consequently poses a potential threat to global food security and agricultural sustainability.

f. Loss of biodiversity resulting to increased incidence of pest and diseases and loss of natural predators

Food security is threatened by an alarming increase in the number of outbreaks of transboundary pests and diseases of plants and animals. These pests and diseases jeopardize food security and have broad economic, social, and environmental impacts.

IV. Summary

- Increasing globalization and production in agriculture has primarily benefited the industrialized countries and certain developing countries that are integrated into the global markets.
- However, the development hides a growing disparity among agricultural systems and population, where especially developing countries have seen very few improvements in food security and production.
- Many rural households in developing countries lack the ecological resources or financial means to shift into intensive modern agricultural practices as well as being integrated into the global markets.
- Intensive agriculture especially in industrialized countries has contributed to environmental problems: pollution of surface and groundwater with nitrates and pesticides, global warming, reductions in biodiversity and soil degradation, and virtual monocultures and specialized livestock productions.

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UNIT II

A. CROP PRODUCTION AND AGRICULTURE AND TRADE

I. Learning Objective

At the end of the session, the participant should be able to understand the concept of crop production and agriculture and trade.

II. Learning Content

A. What is crop production?

Crop production is the science, art, and business of producing crops. It includes but not limited to the application of the knowledge in crop science to produced basic needs and gain monetary returns from produced crops to individual growers at the minimum environmental hazard thereby sustaining the environment.

Crop production is a complex operation which is dependent to three factors: the crop, environment and the management or the human factor. The synergy of these factors will lead to a successful crop production endeavor. It covers the whole range of operations from planting to harvesting including the primary aspects of crop processing.

Crop production as a science

As a science, crop production utilizes theories derived from various scientific disciplines especially in allied fields such as botany, chemistry and physics. Crop production aims at increasing productivity of plant parts with economic value this can be done though the management of the morphological and physiological responses at a given environmental conditions. Theories used in the management are inevitably product of scientific methods. Thus, a formal training in agricultural science is of advantage but does not mean the sole requirement in performing an efficient crop production management.

Crop production as a business

One of the major purposes of crop production is on provision of food and other basic need to the humanity it engaged into a careful planning, organized works and activities to produce said products. It is aimed at higher productivity to increase profitability. As a business the farmers are the managers, they made decisions so that this economic activity will prosper. They harmonized the factors of crop production to provide an environment that enhance productivity thereby increasing profits.

Crop production as an art

The art in crop production is continuously evolving. In primitive culture it is manifested in day-to-day activities of the crop producers in the farm this includes but not limited to selection of a specific crop and/or variety to produce, crop improvement, land preparation, cultivation, pest and disease management, harvesting and post-harvest technologies. The way they conceptualize,

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invent, and apply these ideas to their respective farm is a form of art. Decisions were made based on their experiences, strong interaction with nature and exchange of ideas with fellow farmers.

In modern agriculture these artistic values are carried on but at improved levels. A crop producer now decides on day-to-day farm activities using several techniques shared through various information channels. The quad media (print, broadcast, internet, and social media) in this era plays an important role in uplifting crop productivity.

The crop production sectors enter through different era of technological development. From the primitive agriculture through slash and burn, it enters to the mechanical era, chemical era in now in the age of biotechnology and information technology. This evolution is a reflection that each day an invention is made as manifestation of its artistic values.

1. Global scenario of crop production

Most of the races depend on cereals as their staple. Thus, it is one of the largely produced and most traded commodity by quantity. In 2018, one third of the total crop produced worldwide is cereals, followed by sugar crops (24 %) and vegetables (12 %). Oil crops, fruit, and roots and tubers each accounted for 9 to 11 percent of the total production.

According to FAO (2020), only four individual crops accounted for half the global production of primary crops in 2018. This includes sugarcane (21 % of the total, with 1.9 billion tons), maize (13 %, with 1.1 billion tons), rice (9 %, with 0.8 billion tons) and wheat (8 %, with 0.7 billion tons).

In terms of oil crops, there was a significant increase in total world production between year 2000 and 2017. The total vegetable oil production is actually doubled in 2017 compared to 2000. Palm oil had the largest increase it overtook soybean oil as the main vegetable oil produced in 2006. The other main vegetable oils are rapeseed and sunflower oil.

China, USA, India and Brazil were considered as the top producers in the world. Sugarcane is largely produced in Brazil, maize and soybean in the United States of America while rice, wheat and potatoes are largely produced in China. India ranks second to brazil in terms of sugarcane production and next to China in terms of rice, wheat and potato production.

2. The Philippine crop production

The Philippine crop agriculture covers the area planted with permanent crops and area planted with temporary crops. In 2019, crop production area covers around 13.298 million hectares, 50% of it is harvested with corn (2.5M ha) and palay (4.6 M ha). In terms of volume in production, sugarcane, rice, coconut, banana and corn were the top 5 produced crop in the Philippines in the year 2019 (PSA, 2020).

Based on the data of Philippine Statistics Authority (2020), rice was the largely produced crop in 2019 in terms of area harvested. Total land area devoted for rice production is around 4,651,500 hectares. It is followed by coconut (3,651,900 ha) and then corn (2,516,700 ha).

Luzon is the largest producer of palay wherein 60.7% of the country's production is coming from this island. On the other hand, Mindanao is also considered as the largest producer of corn and coconut in the country. In 2021, 51.4% of the total corn production and 59.5 % of the country's coconut production were coming from Mindanao. Around 72.3% of sugarcane was produced in Visayas. Moreover, large amount of our major crop export - banana and pineapple were also produced in Mindanao.

Crop industry has been one of the Filipinos sources employments. In 2016, growing of perennial crops employed the greatest number of workers with 62,362 or 43.3 % of the total. This was followed by growing of non-perennial crops and animal production with 33,339 workers (23.1%) and 20,635 workers (14.3%), respectively.

B. What is agriculture and trade?

Agriculture and trade involve the production and marketing of products that are produced through farming industries. It creates a greater access on the variety of agricultural goods, commodities are produced organically, genetically engineered, and some are grown conventional. A product should be competent in terms of its quality such as the appearance, taste, and smells, etc. Trades is the ways of bringing fresh fruits and vegetables to neighboring countries, trading is also a way of supplying products in a country where such commodities are not available or not growing.

Agricultural trade is basically important in alleviating poverty and hunger in the world. It creates thousand jobs for the people, trade helps in a number of ways by allowing countries to take advantage of their radically different factor endowments, with land-abundant countries providing exports and land-poor countries taking advantage of much more efficiently produced imports. Trade liberalization can help by increasing production efficiency in agriculture, allowing improvements in dietary diversity and increasing access to food. Allowing trade substantially reduces the volatility of food prices by diversifying sources of supply.

1. The Philippine crop trade

In 2019, the country's earnings from agricultural exports accounted to 9.4% of the total export. The top crop exports in the said year were fresh banana, coconut oil, and pineapple and products which share is accounted to 52.5% of the total value of the agricultural exports of the country. Japan and China were the major destination of our exported bananas while Netherlands and USA were the destination of our exported coconut oil. Pineapples were mostly shipped to China and USA (PSA, 2020). On the other hand, major crop and crop products accounted to 28.9% of the total value of agricultural imports. Wheat (including spelt) and meslin and soybean oil/cake meal from USA and rice from Vietnam are among the major imported commodity.

2. The General Agreements on Tariffs and Trades (GATT)

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- It was signed on October 30, 1947 by 23 countries with legal basis in minimizing barriers to international trade by reducing quotas, tariffs, and subsidies while maintaining the created guidelines of GATT.
- On January 1, 1948, GATT was take into effect leading to the creation of the world Trade Organization (WTO). On January 1, 1995, 125 nations were signatories to this agreement covering about 90% of global trade.
- The council for trade in goods is responsible for GATT and consists of representatives from all WTO countries.
- The council has 10 committees that addresses market access, agriculture, subsidies, and anti-dumping measures.
- GATT was created to form rules and regulations that protect on the importation of goods to member countries.
- GATT is responsible in the imposition of tariffs to the member countries.
- The first meeting was held in Geneva, Switzerland included the 23 countries.
- The second meeting was held in Annecy, France in April 1949, and tariffs was the main topic.
- The third meeting was held in Torquay, England in September 1950 and 38 countries were then involved, tax were reduced as much as 25%.
- For the first time in 1956 Japan became involved in the GATT with 25 other countries in Geneva, Switzerland, topics were again on the tax reduction.
- GATT held eight meeting from April 1947-December 1993.
- In 1964, GATT began to work on the reduction of monopoly pricing policies which attacking the countries that may resulted to global issues that includes addressing agricultural disputes and working to protect intellectual policy.

4. The ASEAN Free Trade Area (AFTA)

- On August 08, 1967, the ASEAN organization was established in Bangkok with five original member countries, these are Indonesia, Singapore, Malaysia, Philippines, and Thailand.
- Trade among member countries was insignificant when ASEAN was established.
- AFTA is a trade union agreement by the Association of Southeast Asian Nation (ASEAN) that supports manufacturing of products in all ASEAN countries. It is an agreement that covers free trading, and imposition of minimal tariffs to the member countries.
- The ASEAN Free Trade Area was established in 1992 with 10 member countries.
- After the establishment of AFTA, a total of trade among ASEAN countries grown from US$ 44.2 billion in 1993 to US$ 95.2 billion in 2000.
- The Common Effective Preferential Tariff (CEPT) Agreement for AFTA requires tariff uniform and low tariff rates on the products traded within the member countries.

AFTA Objective

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1. To uplift and increase the competitiveness of production of the ASEAN regions for the world market through elimination and imposition of uniform tariffs and non-tariff barriers among ASEAN members. It allows product to be market to all the member countries that provides better quality products at lower costs. It drives economic growth and increase innovations in the country.
2. Helping both export and investment in the long run
3. Support manufacturers import raw materials with low tariff.
4. Provides one stop shop for all member countries by providing the products from each member countries to each other.

III. Summary

- Crop production is an art, science and business.
- Crops can be classified based on their production and practices and their utilization.
- The world crop production is dominated by China, USA, India and Brazil.
- Cereals are largely produced and most traded crop commodity worldwide.
- ASEAN was established to creates agreements between member countries to export their products to each other by imposing uniform tariffs. This was done to eliminate the monopoly pricing that may result to high cost that will hinder underdeveloped countries to imports products for their people. Agricultural trading helps eliminate food hunger and helps people in the production of more foods and creating more jobs.

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UNIT II
B2. FOOD UTILIZATION

I. Introduction

World population is consistently increasing from the past decades and can be seen to be still progressing in the coming years. As of June 2021, the total number of people currently living in the world is estimated to be around 7.9 billion and is expected to increase to about 9.7 billion on the year 2050. The increasing rate of population growth encompasses higher demand of food supply. However, the stability in terms of availability and accessibility of quantity and quality foods are beset with various problems on the production level, natural phenomena, wastage of food at the table and storage as well as the lack of efficient techniques of utilizing harvest which have affected the amount and distribution of food supply over the population. Every year around the globe, it is estimated that 1.3 billion tons of food are being wasted from which 30% are cereals, 45% are fruits and vegetables, 20% are oilseeds and pulses and 45% are roots and tubers. This percentage of food wastage accounts 1/3 of all the produced food for human consumption from which some are due to lack of efficient utilization (FAO UN, 2021). Proper food utilization is therefore an essential method in order to minimize food wastage, permit equal access and increase available food for the population. Philippines has a level of hunger that is moderate with a score of 19.0 which ranks 69th out of the 107 countries in terms of global hunger index year 2020 (GHI, 2020).

II. Learning Objectives

At the end of the session, the participant should be able to:

1. Understand food utilization and its significance;
2. Identify food products derived from crop utilization; and
3. Identify new techniques of crop utilization for food.

III. Learning Content

A. Definition of Food Utilization

Food utilization is one component for the attainment of food stability interrelated with food availability and food accessibility. Food availability is attained when there are sufficient supply of quantity and quality food either from domestic production or import. On the other

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hand, food accessibility is when an individual or groups of people are able to acquire adequate amounts of food.

Food utilization is the proper and wise use of harvest according to purpose. In its process, it employs proper methods of processing as a major way of utilization, proper storage techniques and the knowledge on nutrition.

Food utilization aims to address how much food the people eat, what and how they eat. It assures that food products are safe, properly preserved or improved on their nutritional quality and with longer shelf-life for the consumer's convenience and choice to meet their dietary needs (Wigboldus *et al*, 2011; USAID, 2011).

B. Sources and Considerations of Food Utilization with their Product Examples

Crops are grown mainly for human consumption. The produce may either be consumed as raw food or cooked or in their processed form. They are also utilized in the making of animal feeds and other non-food products. Groups of crops for food utilization include cereals, vegetables, legumes or pulses, roots and tubers, spices and condiments, fruits and those of plantation crops.

The knowledge on the nature and characteristics of the crops are among the essential considerations in the proper determination of applicable type/s of utilization that fits on a specific produce or harvest. Harvest could come from the different parts of a plant such as the root, stem, leaf, flower, fruit and seed. The table below shows the numerous ways and by-products of crops utilized amongst smallholder farmers and processing industries.

Table 1. Groups, Types or Examples of Crops and Their General Utilizations

Groups of Crop	Types/ Examples	General Utilizations
1. Cereals	barley, corn, millet, oat, rice, rye	• Prominently utilized as food for meal (staple) and/or snack • Flour for bread and other bakery products specially wheat and oat • Fermented into alcoholic beverages (e.g., rice wine) and nutritive drinks • Processed into syrup - e.g., from corn, sorghum • Porridge • Pasta • Cakes and biscuits • Noodles • Cooking oil - e.g., corn oil • Non-dairy probiotic drinks
2. Vegetables	leafy vegetable, crucifers, solanaceous crops, cucurbits, tree vegetable and other vegetables	• The immature or mature portions like roots, stems, leaves, flowers, fruits and seeds are eaten as fresh

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		salad, riped or cooked with staples, • Pickles • Sauces • Cakes and flakes • Noodles • Chips • Wine • Dehydrated food products - e.g., okra fruit • Beverages (some mixed with fruits) • Cooking oil • Confectionaries - e.g., from cucumber seed kernels • Food powders as food additives – e.g., tomato powder
3. Legumes or Pulses	soybean, pigeon pea, cowpea, mungbean,	• Eaten with staples either cooked directly or sprouted seeds are cooked • Cooking oil - e.g., from soybean, groundnut • Canned products - e.g., brine, sugar, purees • Noodles • Fermented milk - e.g., yoghurt from soybean • Sweeteners • Bean soups • Flours for cake • Pudding - e.g., from cowpea • Cheese - e.g., soybean cheese or tofu • sauce • Butter - e.g., peanut butter
4. Roots and Tubers	sweet potato, potato, yacon, radish, yambean, tugi, gabi, cassava, arrowroot, carrot)	• Eaten as raw (specially yacon) but mostly cooked • Starch • Wine • Juices – e.g. ,carrot juice • Cakes • Flour – e.g., from cassava, taro • Chips – e.g., potato chips
5. Spices and Condiments	*Spices* are added in minimal quantities during the cooking	• Used to color foods

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	process. e.g., black pepper, ginger, onion, garlic *Condiments* on the other hand, are flavoring substances having a sharp taste and are usually added to food after cooking. e.g., tamarind fruit pulp, vanilla, cinnamon, turmeric	• Food powders - e.g., atsuete for curry, tamarind pulp powder for sour soups • For special flavorings - e.g., onion, powders of garlic, pepper, ginger
6. Fruits and Plantation crops	Tree fruits - e.g., mango, lanzones Nut fruits - e.g., cashew, pili Small fruits - e.g., strawberry, grapes Oil crops – e.g., palms	• Normally eaten as raw • Processed into dried fruit, fruit juice, canned fruit, frozen fruit, jam, syrup, jellies, marmalades, pickles and alcoholic beverages • Some fruits have seeds that are edible and made into roasted seed like cashew • Unripe fruits are used as vegetables - e.g., papaya, young jackfruit cooked in coconut milk, young marang fruit for curry • Cooking oil - e.g., palm oil • Beverages or non-alcoholic drinks - e.g., coffee, tea, cocoa • Fermented beverages – e.g., tuba from coconut, banana wine, bignay wine, dragonfruit wine, strawberry wine • Candies • Butter – e.g., pilinut butter

Obtained from: Bautista 1994, Igra 2017, Notes on Agricultural Science.

C. Factors Affecting Utilization of Crops for Food (Modified from Swidiq *et al*, 2012)

1. Availability of resources and needed materials
2. Quality of produce
3. Price of processed product
4. Labor costs
5. Capital investment in processing
6. Limited information on the extent of utilization
7. Socio-economic constraints – There must be alternative systems on the limiting factors on the adoption and utilization by the farmers for consistency.
8. Culture of the people – This factor is a very important consideration especially for the processing factories as the taste of the people varies from culture to culture.

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UNIT III. SITE CHARACTERIZATION FOR CROP PRODUCTION

A. BIOPHYSICAL AND ECONOMIC DETERMINANTS

I. Introduction

Location and site tremendously influenced the crops grown, management practices, cropping patterns, and inputs needed for economic yields. A crop performs best and is least costly to produce if grown under the most favorable environmental conditions. In crop production, therefore, the grower must consider the site and the specific environmental factors that influence plant growth and development to attain optimum yield and maximum profit (Escasinas, 2015).

It is essential to identify and observe the site's physical, biological and socio-economic conditions before deciding what crop production or business to establish in a particular area. A planning activity before actual production is necessary to determine and decide what crop and enterprise to develop or set up in a given area or location.

This module deals with the *whys* and *hows* of site characterization and evaluation. The skills acquired in this module may help answer whether the land intended for a certain agri-venture could best support the crops to be grown (Teka and Haftu, 2012). These will be carried out by matching site conditions with the crop requirements. These requirements include soil physical properties, water holding capacity, depth, type of soil, acidity, biological composition, topography, drainage, soil texture, cation exchange capacity, state of erosion, climate, rainfall, temperature, wind, natural vegetation, the situation related to market and transportation facilities and farm size (Penzhorn, 2012; Sulieman et al., 2015). Thus, it is vital to evaluate, identify and characterize the soil and other factors affecting crop production in a given agro-ecological unit under a defined management system (Burrough, 1996). In addition, setting up new production enterprises in an area where certain crops have become dominant would be advantageous because of the proven adaptation of the commodity and the experience of the farmers in the area.

II. Learning Objectives

At the end of the session, the participant should be able to:
1. discuss the biophysical and economic factors to be considered in site characterization and evaluation; and
2. develop awareness in site characterization and crop-environment suitability.

III. Learning Content

Location refers to the geographical position or place where the crop enterprise is located in relation to a city, town, highway, or any point of reference. It is a geographical position, given certain physical (climate and soil), biological and socio-economic factors at the macro-level. On the other hand, site refers to the actual place where any given crop is produced. It is the localized area with its microenvironment.

A. Methods for Site Characterization

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1. Setting goals and objectives. This involves a long-term vision for the quality of life (human and social resources), the future of the farm (environmental and natural resources), and how the farm enterprise will provide income/profit (economic and financial resources). Thus, the agripreneur has to identify the factors that influence successful crop production, has to assess site suitability or crop suitability, and analyze the possible site-crop enterprise relationship. This will be followed with the formulation of the farm plan – steps, measures, requirements to implement the plan, and the sketch of the farm to show present and future scenarios.

2. Characterize the biophysical environment. The agripreneur has to do soil sampling and soil analysis to assess the physical properties of the proposed farm. Other information needed includes secondary climate data from a nearly agrometeorological station. Likewise, there must be an actual observation and identification of existing vegetation/plant species in the site. The land features such as slope and elevation must also be measured. Thus, an actual survey and mapping of the field is a must.

3. Characterize the socio-economic environment. In farm planning, the agripreneur has to engage in actual observation and gathering of information on current prices of farm inputs, including the prices of farm products sold in the market. Moreover, one has to conduct an interview of farmers and their households about the type of farming system employed as well as the conduct of interview with key informants about

B. Factors to Consider in Site Selection

1. Bio-physical factors

a. Climatic Factors

- Rainfall Pattern. With upland farms, usually rainfed, planting crops should be in timing with good rain. A planting calendar based on rainfall patterns should be adopted to fully utilize photosynthesis. The climate also influences cropping patterns as this depends on the rainfall intensity and distribution as it affects soil workability, crop growth and development, and maturity.
- Temperature. Plants have specific temperature requirements to grow vigorously at different stages of growth. In the choice of crops, the growing season has to be considered, which are also influenced by altitude, altitude, and slope of the place. Extremes of temperature adversely affect physiological processes and inhibit plant growth. Different crops and even varieties of a crop require specific temperature ranges for best performance. Temperature requirements can also vary with growth stage. Not only the temperature but also daily changes in temperature affect the plant response. In tomato and potato, night temperature is important for normal growth and yield formation.
- Typhoon/Drought Occurrence. In areas where there are 'typhoon belts', the crops to be planted are creeping species (sweet potato, etc) and the annuals so that these could be harvested with less effect from strong winds or the planting of drought tolerant crops.
- Wind Velocity. Still air is beneficial to crops as this aids the pollination process. But wind velocity also cause soil erosion and therefore planting wind breaks are necessary.

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- Light/ Solar radiation – Light intensity, duration and quality affects dry matter accumulation as this is used for photosynthesis. Plants have higher dry matter accumulation under optimum sunlight. All aspects of plant growth are affected by light directly or indirectly. The aspects of light important in crop production are light intensity and daylength or photoperiod. The favorable light intensity range varies with the kind of plant. Certain plants perform better in full sun than under shade, while other plants perform better under shade than in full sun. Day length controls the flowering of short-day plants and long day plants but not that of day neutral plants. It also affects bulbing, tuber formation and stem elongation.

b. Soil Factors

- Soil texture refers to the relative proportion of clay, silt and sand. Soil structure depends on the kind and amount of clay in the soil. Clay particles work with organic matter to hold water and nutrients, binds soil particles together; holds more water and nutrients than sandy soils; has higher cation exchange capacity than sandy soils. It should be noted that medium textured soils like loam and silt loam are best soils for diversified upland crops. Paddy rice prefers poorly drained clayey soils.
- Soil depth also greatly affects crop production. The effective soil depth refers to the thickness of the soil to layers coarser than loamy fine sand, sand, gravels, stones, rocks or any impermeable layer or to depth to water table. This also refers to the thickness of the soil to layers coarser than loamy fine sand, sand, gravels, stones, rocks or any impermeable layer or to depth to water table. Some research results showed that deeper soils are more productive than shallow soils. Crops can endure a longer drought when they thrive on soils having a higher water holding capacity. The thicker the effective soil depth, the more the volume of soil can be extracted of plant nutrients and water. The ideal soil for annuals crops must be relatively shallow depths but for perennial crops, greater depths should be advantageous.
- Soil drainage is the removal of water from the soil through surface run-off and by flow through the soil to underground spaces. It should be noted that well-drained soil promotes favorable conditions for plant growth and microbial activity. Drainage is important for crops that are sensitive to water logging and is affected by slope, permeability, vegetation and land use, and amount of rainfall.

  Available water is the water content of a soil held between field capacity and permanent wilting point expressed in percent. The field capacity represents the maximum amount of water a soil can hold against normal drainage. In case of inability of soil moisture, the plant will experience permanent wilting point and will start to wilt.

  Soil with poor drainage has limited agricultural use because of wetness. These soils are not suitable for upland crops unless adequately drained but they are best for paddy rice.
    - Medium textured soils hold the most water in available form
    - Coarse textured soils hold little water because of their low water content at field capacity
    - Fine clayey soils supply limited available water because they return large amount of water at permanent wilting point.

- Inherent Soil Fertility. Soil fertility refers to the status of the soil in terms of its ability to provide the plants adequate amount of nutrients in balance proportion and on readily available

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form necessary for normal plant growth. It is related to pH, organic matter, CEC, available P, extractable K and base saturation

a. Soil pH affects plant growth on its nutritional needs. It influences the rate of plant nutrient released by weathering, the solubility of all materials in the soil and the amount of nutrient ions kept on the CEC sites.
b. Organic matter content affects soils fertility in terms of CEC and nutrient supply. The higher the organic matter content, the higher is the CEC supply of available N, P and S.
c. Available P. It is the second most deficient nutrient in most soils; is derived from mineralization of organic matter. P availability is related to soil pH. A pH range from 6.5 – 7.5 is usually best for P availability.
d. Exchangeable P. It is essentially needed in large amount by plants which come mostly from mineral solids. K is most available at pH 6 – 7.5.but in leached soils, available K declines.
e. Percentage base saturation. This refers to the percentage of CEC occupied by basic cations such as Ca, Mg, K and Na measured at pH 7 or 8. Cations like H and Al are excluded because they are acid produced.

c. Topography

Physical features of the land defined by the contour, elevation, and depressions. This is the natural shape and slopes of the land. The terrain or contour of the land affects cropping patterns. The lowlands are utilized for producing most likely annuals (cereals, etc.) whereas in the uplands with high altitudes, vegetables are raised.

d. Biological Factors

- Cropping patterns. These are temporal and spatial arrangement of crops in the field which are based on type of climate and soil, availability of farm inputs and technical knowhow of the farmer. Therefore, one cannot deviate much unless the community as a whole, changes the pattern. The risk from pest damage is the field is greater if planting is not synchronized with that of the community.
- Varietal selection. The variety to be planted in the farm depends on its adaptability, the preference or purpose for planting it, its resistance to pests and diseases, maturity indices, high yielding and good eating quality. The information on the yields of best farms in the area must be obtained and used for upgrading operations.
- Insects & diseases. The prevalence of insects and disease must be considered, thus planting synchronously is favorable and the variety should be resistant to pests and diseases. Cavite, Laguna, Batangas and nearby provinces should not be planted with papaya due to prevalence of ringspot virus.
- Weeds. There must be thorough land preparation before planting and there must be maximum utilization of the land to prevent growth of weeds (increase plant density).
- Animals. Astray animals in the farm affect production. However, draft animals are beneficial to farmers. There must be compatibility of the animal component to the crop components.

2. Socio-Economic Factors

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Economic factors - this refers to the cost of land, local taxes, labor supply situation and level of skills, facilities (electricity, communication, equipment, accessibility), and the local market situation and preferences of the populace. The supply and demand situation should be known so that operations can be adjusted for harvesting to coincide with slack periods of supply when are high.

- Off-farm income sources. Farm management involves capital and the capital of a farmer might not be enough hence, he has to secure off-farm income to augment his finances.
- Labor market. There is shortage of labor during "peak months" and labor is expensive and if labor is available, it is expensive and demand higher price. Therefore, in the farm, operations should be done on staggered basis such that labor is available at off-seasons
- Market & Credit. In deciding what to raise in the farm, the farmer has to the consider the demand (quantity and quality) of the products at high price based on customer preference; the marketing outlets and the ROI. Supply and demand situations should be known so that operations can be adjusted for harvesting to coincide with the slack periods of supply when prices are high. Also, the farmer has to develop storage and selling strategies to avoid risk. On the other hand, credits are supplied by banks or middlemen. Credits help farmers a lot but this may also tie down their produce to lower prices. On the other hand, capital for inputs and services are readily available.
- Transport facilities. These are necessary for marketing of farm products as well as transport of supplies and inputs. Insufficient transport facilities in the farm results to slow turnover of outputs to cash, and may even result to wastage of produce.
- Channels of market information. Media and technicians inform farmers of the recent product demand and prices of commodities in the market. Farmers have to raise goods of preference to consumers that command high prices.
- Landholding, inheritance pattern and land tenure status. Available land resources and farm distribution affect crops grown and the farming system used. For land owners, they can plant on extensive scale whereas others on small scale based on farm sizes. Land tenure status influences production as there are agreements and disagreements between landlord and tenants. With fragmented holding, fields may be of different soil types due to complex management practices.

C. Examples of bio-physical features of land and crop adaptation (Lantican, 2001)

1. Submerged conditions during most of the crop's life cycle e.g., rice, taro water chestnut (apulid) and *Sesbania rostrata* for green manuring
    - Most crops are sensitive to waterlogging
    - Jute and sorghum can tolerate periodic waterlogged conditions
    - Corn cannot tolerate waterlogging and sensitive to even short period (36 hrs) of waterlogging
    - Rambutan, papaya and durian are very sensitive to under-drainage or water-logged conditions
    - Sorjan – a system of raising upland crops - (vegetables & peanuts) in raised beds alongside submerged lowland rice

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2. Humid condition and where rainfall is more uniformly distributed which is suitable to all perennial crops such as mangosteen (Mindanao) and abaca (Mindanao, Eastern Visayas and Bicol Region). Mango produces quality and tasty fruits in areas where there is a dry spell.
3. Cool climate or mild climate in the highlands of Mt. Province, Mt. Kanlaon (Negros Island) and in Bukidnon-Lanao plateau are suitable for vegetable species like white potato, cabbage, peas, carrots, head lettuce, asparagus, etc. The suitable fruit species are litchi (lychee), longgan, ponkan mandarin orange, strawberry, grape and macademia nut.

4. Coastal areas are suited for the following crops: coconut, cashew, tamarind and talisay (*Terminalia catappa*) which is a source of dye for clothing materials and nipa palm (*Nympha fruticans*) can thrive well in saline conditions. The limit of elevation for coconut is 600m above sea level.

5. Partial shade is good for ginger, black pepper, vanilla, lanzones, rambutan, mangosteen, cacao, coffee, banana, arrowroot, pachouli and many other crops can be grown under coconut trees, ipil-ipil and madre de cacao (*G. sepium*).

6. Low pH 4.3 – 5.0 as in Bukidnon is suitable for rubber, pineapple, cassava, sweet potato and *Stylosanthes humilis*,a forage legumes that nodulate at pH 4.0-4.5. Acid tolerant crop varieties of corn have been developed by CIMMYT, IPB-UPLB and CMU; mungbean, peanut and cowpea in IPB-UPLB and rice in IRRI and CIAT-Colombia.

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B. SOCIO-CULTURAL

I. Introduction

Socio-cultural factors influence the production system of the farmers. The sociocultural environment involves the religion, norms and customs, and social institutions which strongly influence the productivity of the farming system. Socio-cultural factors contribute to individual differences and have a basis in society and culture. Religion plays an important role in agriculture because it contains a culture's ideas about the relationship between people and nature. Norms are the acceptable standards of behavior of a society, while customs are the ways people actually behave. Every society has established institutions for governing interpersonal relationships. Of

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particular interest are the traditional institutions of governing land, stratifying the populations, and exchanging labor.

II. Learning Objectives

At the end of the session, the participant should be able to:

1.) Identify the socio-cultural factors that influence the production system of the farmers;
2.) Understand how these factors affects farm productivity; and
3.) Determine strategies base on identified socio-cultural factors to address issues and problems concerning agricultural productivity.

III. Learning Content

Socio-cultural Factors Affecting Farming System

1. Organization and Leadership

People who hold recognized positions of authority are known as formal leaders. They are usually easy to identify once the pattern of leadership in the society is understood. Some inherit their position; others are elected, and others are appointed by someone in higher authority. Leadership may be shared by several people or be held by a single person. In most social communities there are religious leaders, such as priests, as well as secular leaders, such as elected councilors and village heads. Formal and non-formal leaders help members in understanding the current pattern of resource distribution and these leaders would mobilize the members to form cooperatives and avail of and have access to support and technical assistance.

2. Ethnicity and idiosyncrasies

Values, outcomes and traditions are inherent to all people and this affects farming system. A large body of research has demonstrated that household-level motivations, cultural and social values, and socialization have a primary influence on farm structure, management, and adaptation (Gasson and Errington, 1993; Lobley and Potter, 2004; Salamon, 1992; Bennett, 1982). Social and cultural factors are influenced by farmer race and ethnicity. Each ethnic group has unique historical and cultural legacies that influence their goals, motivations, values, access to land, and resources, which, in turn, influence the way each group structures their farms and envision the future.

3. Aspirations and attitudes

Aspirations help farmers to strive more for certain definite purpose – child's education or alleviate income. Farmers' attitude also limit production but if they are taking risk as opportunities then they have to diversity their farms to have continuous cash flow, much food, wider host range of pests and diseases and if one crop fails, there are still others to be harvested.

It is considered as a process of bringing desirable change in the behavior of the farmers to adopt innovations relating to agriculture in such a way that they are clear and convinced of their utility (Khan, 2005).

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4. Intellectual and technical perspective

Those who engaged in farming are the middle age group and are of low educational attainment. They lack technical know-how but with wider farming experiences. The knowledge and decision making they have affected the productivity of his farm. His preference and perception with regards to the crops to grow or enterprise to establish or technology to accept or reject partly determines the nature of the production system. This influence by the farmers Intellectual and technical perspective.

Merleau-Ponty (2002) highlights another approach of decision-making, that of individual perception, which is one of the major influencing factors when it comes to personal decision-making. Perception according to Bradford (2009) is the way in which people approach different impacts and accept different information. The principles of perception can be categorized in two different ways: intuition and decision bias. Intuition can either be described as an experience-based decision, where patterns of behavior have been collected in someone's long-term memory and can be applied to situations in various combinations (Simon, 1987) or like Gigerenzer (2007) sees intuition as a guess, hunch or hypothesis which is triggered by the unconscious. Intuition has also been described as a direct and immediate action which happens without conscious thoughts or judgement (Fishbein 1987; Greer 1988)

5. Tradition, customs and beliefs

Indigenous knowledge encompasses spiritual relationships, relationships with the natural environment and the use of natural resources, relationships between people, and is reflected in language, social organizations, values, institutions and laws. These knowledge systems are usually embedded in naturalistic epistemologies and belief systems, which differ radically from those of scientific systems (IUCN, 1997).

Religion and agriculture have functioned as natural bedfellows throughout recorded history. This is more evident in the manner in which religious traditions across the world have habitually shaped agricultural practice and development. Whatever traditions, customs and beliefs of the people influence the type of crops or animals to raised. They have to follow on what they "believe" in terms of farming.

Researches have shown that the cultural factors affect the acceptance of diffusion of new technologies. It was discovered that the adoption of information technology products was not fully understood, partially because of the lack of fit between technology and culture (Kalliny and Hausman, 2007). Cultural factors are leading constraints to the effectiveness of extension work. In some cultures, access to certain products may not be possible in local markets and, therefore, adoption of such products may be difficult and slow. Language according to Macionis and Plummer (2005), was the key to the world of culture and a system of symbols that allows members of a society to communicate with one another. They posit that these symbols take the form of spoken and written words, which were culturally variable and composed of the various alphabets and ideograms used around the world. Although language differences can impede the communication of improved technology unless they are taken into account, it was the major means

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of cultural reproduction which was the process through which one generation passes culture to the next (Macionis and Plummer, 2005).

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UNIT III. SITE CHARACTERIZATION FOR CROP PRODUCTION
C. POLITICAL DETERMINANTS

I. Introduction

Agricultural systems are bound by political determinants. These determinants have profound effects on how a country's agricultural system operate. It is crucial to understand these determinants if we aim to achieve a sustainable state of our agricultural industry. These determinants range from government policies on production and trade, government policies on support to the farming industry and government institutions involved in the formulation and implementation of such policies.

II. Learning Objectives

At the end of the session, the participant should be able to:

1. know the different government agencies under the Department of Agriculture;
2. understand how the Philippine government has endeavored to adapt policy changes to address the pressing needs of food production;
3. identify some policy responses in relation to the Covid-19 pandemic;
4. familiarize with some Philippine laws relevant to agriculture; and
5. identify and explain how these different political factors influence the farmer/grower/producer in the characterization and evaluation of a particular site for crop production.

III. Learning Content

The Philippines in Agricultural Context

The Philippines is a country with a land area of approximately 300,000 km^2. Relative to its mid-size land area is a population of 107 million. The country is ranked 13th among the world's most populous countries. Agriculture is an important sector for the Philippines, accounting for a quarter of total employment and 9% of GDP. Farms tend to be small sized with the average

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landholding at just 1.3 hectare. Moreover, its farming population has an average age of 57 years old, raising concerns on the future of the agriculture industry.

The Philippines Support to Agriculture

According to the Organization for Economic Cooperation and Development (OECD), an intergovernmental organization founded in 1961 to stimulate economic progress and world trade, the Philippines level of support to farmers, measured as a share of gross farm receipts (%PSE), averaged 27% in 2017-19. This is higher than the OECD average and one of the highest among all emerging economies.

The dominant form of support to Philippine producers is Market Price Support (MPS). MPS also reflects existing trade barriers. These barriers are in the form of tariffs and Tariff Rate Quotas (TRQs).

The main beneficiaries of price support policies in the Philippines are the rice producers. Aside from rice producers, price support policies are also provided for sugarcane, pork and poultry. This support comes in the form of high import tariffs. In this way, local producers are provided with a legal level of safety net in preventing swift downward price swings due to oversupply.

Domestic Policy Developments in 2019-2020

During pre-pandemic months, there were important institutional changes related to domestic market regulations and rice trade. These changes were introduced by the Rice Tariffication and Liberalization Law (RA 11203). One of the notable changes introduced by RA 11203 is the transfer of food safety regulatory function from the National Food Authority (NFA) to the Bureau of Plant Industry (BPI). These functions included issuing of permits, licenses, or registering trade and importation of rice.

Trade Policy Developments

On January 1, 1995, the Philippines joined the World Trade Organization. During the country's WTO membership, it has been applying quantitative restrictions (QRs) on rice imports.

In March 2019, the country replaced quantitative restrictions (QRs) on rice imports with a tariff system. In this tariff system (RA 11203), a single tariff of 35% is applied without quota for ASEAN countries. For the non- ASEAN countries, a tariff rate quota (TRQ) is established.

Policy Responses During the Covid- 19 Pandemic

At the course of the COVID- 19 pandemic, the Philippine government issued provisions to continue agricultural production. These issuances include the following:

1) Farming and fishing activities shall continue.
2) Provided health safety protocols are followed, all farmers, farm workers, fisher folks and agribusiness personnel shall be exempted from home quarantine.

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3) Agricultural supply stores and outlet nationwide must be allowed to operate under a skeletal force.
4) Essential farm personnel (including veterinarians, farm hands, farm and fisher folks) that works at production areas, bearing proper documentation, are also requested passage at quarantine checkpoints.
5) Backyard vegetable gardens are promoted by the Department of Agriculture as source of available and healthy food during the quarantine period.
6) LGUs have been distributing seed packs in order to provide planting materials for backyard and containerized gardening in the urban areas.

Philippine Laws Relevant to Agriculture

Code No.	Title
R.A. 10601	Agricultural and Fisheries Mechanization Act (AFMech Law)
R.A. 11511	An Act Amending Republic R.A. No. 10068 or the Organic Agriculture Act of 2010
R.A. 10068	Organic Agriculture Act of 2010 and its Implementing Rules and Regulations (IRR)
R.A. 8435	Agriculture and Fisheries Modernization Act (AFMA)
R.A. 11203	Rice Tariffication and Liberalization Law

1. R.A 10601 - Agricultural and Fisheries Mechanization Act (AFMech Law)

AFMech Law mandates the formulation of a comprehensive national policy on cost-effective and environmentally-safe agriculture and fisheries to achieve food security and increase farmers' income.

2. R.A. 11511 – An act amending R.A. 10068 or the Organic Agriculture Act of 2010

Under this newly enacted law, a participatory guarantee system will be established. This is a locally-focused quality assurance system developed and practiced by people engaged in organic agriculture. The National Organic Agriculture Program-National Program Coordinating Office will also be created to oversee the implementation of the National Organic Agriculture Program. The law will likewise restructure, strengthen, and empower the Bureau of Agriculture and Fisheries Standards in providing assistance to farmers and fisherfolk.

3. R.A. 10068 – Organic Agriculture Act of 2010

This act declares that the policy of the State shall be to promote, propagate, develop further and implement the practice of organic agriculture in the Philippines in order to enrich the fertility of the soil, increase farm productivity, reduce pollution and destruction of the environment and prevent the depletion of natural resources. The act provides for the establishment of a comprehensive organic agricultural program and of a National Organic Agricultural Board which shall carry out the policy and program. Composition, organization and powers and functions of the Board are provided for in the text. The Board shall constantly devise and implement ways and means of producing organic fertilizers and other farm inputs and of helping to alleviate the problems of industrial waste and community garbage disposal. The Bureau of Agriculture and Fisheries Product Standards is the authority responsible for granting official accreditation to organic certifying bodies and entities, and for the registration of organic food and organic input producers. The Act further provides for: labelling requirements; retailing of organic produce; research, development and extension (RDE); creation of organic agriculture RDE networks; incentives; penalties; etc.

4. R.A. 8435 - Agriculture and Fisheries Modernization Act (AFMA)

Enacted on December 22, 1997, AFMA is a policy instrument defining measures to modernize Philippine agriculture for the country to compete in the global market.

5. R.A. 11203 - Rice Tariffication ad Liberalization Law

RA 1120 replaced quantitative restrictions on rice imports as follows:
- 35% tariff for ASEAN imports
- 50% for non-ASEAN imports.

The RLL has also removed the regulatory functions of the National Food Authority (NFA) over rice importation and streamlined the process of acquiring import permits. Consequently, big businesses with massive capital have managed to dominate the industry and ship in massive amounts of cheap imported rice.

Agricultural policies in the Philippines are designed and implemented by a complex system of institutions. The government could strengthen institutional coordination between the Department of Agriculture and other relevant departments and institutions that develop and implement programs supporting agriculture; strengthen transparency and accountability of publicly-funded programs; accelerate efforts to build a solid policy-relevant statistical system; and integrate monitoring and evaluation mechanisms into the policy process.

How and in what way then can these political policies/factors influence the site characterization and evaluation for crop production?

The Philippine government has focused most of its market price support on rice, sugarcane, pork and poultry thereby engaging in the production of these four major commodities makes it more enticing to farmers. This may however affect the diversity of production in the country. Take for example corn production, which is second to rice, a staple food and may have the potential for

production in marginal areas, and yet has no attached agency focused on the improvement of its production.

The value of agriculture as vital economy in the country could not be more emphasized during the COVID-19 pandemic. Marching orders were implemented for continued production, marketing and other operations related to agriculture. Also, the value of diversifying production was emphasized during the pandemic wherein each household was encouraged to establish backyard gardens especially among the 4Ps beneficiaries.

Organic agriculture seems to be at the better position now more than ever with the recognition of the participatory guarantee system. This does not only allow cheaper certification but a more rapid, wider and personal certification system of produce as organic.

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UNIT IV. PRODUCTION PRACTICES
A. LAND PREPARATION FOR ANNUAL AND PERENNIAL CROPS (PRACTICES FOR SOIL EROSION CONTROL)

I. Introduction

Do you want to help our soils get healthy? If yes, then this module is for you! Observe along the highways when you are traveling to other places. Look at the crops in the mountain top and at the bottom. How's your observation? It usually appears that the crops below have a good stand while on top are poor growth. What does this mean? This implies that the layers of a soil profile had been exposed tremendously to agents of erosion.

Fortunately, the government agencies and non-government organizations have looked and stepped toward the community to seek appropriate actions. They aimed to conserve soil while increasing income from crop and animal farming. To sustain the area from the present generation to the next, practices for soil erosion must be realized.

Let's go and be part of heroic deeds for Mother Nature!

II. Learning Objectives

At the end of the session, the participant should be able to:
1. Review the layers and contents in a soil profile;

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2. Discuss the soil erosion in the Philippines;
3. Assess the various soil and water conservation practices;
4. Develop a simplified chart for contour farming; and
5. Categorize the plants according to its uses for contour farming establishment.

III. Learning Content

A. A review on soil profile

When we think of soil, it is a medium for plant growth. However, it will realize us that there is a development and formation in soil over some time. It can be noted that this development can be traced back to weathering of rocks and leading to its first manifestation of soil formation the so-called, soil horizon C <Parent Material>. In short, the layers of soil are laid out in a soil profile as influenced by factors of soil formation. Among these are parent materials, climate, biota, topography, and time.

If we look at the soils from our mountains, topsoil from the hill is deposited and accumulated on the flat land which becomes a thick A horizon. Generally, summits are well-drained and have strong horizon development, whereas shoulder and back slopes are influenced by erosion and have shallower topsoils and less infiltration. Exposure to excessive rainfall influences and aggravates these occurrences.

Our biota consists of elements that are needed for plants to grow. These are sunlight, air, nutrients, and water, in like manner, the soil is composed of inorganic (45%), organic (5%), water (25%), and air (25%). An improvement of soil for crop production is vital because nutrients are mostly found in topsoil. If this topsoil will be lost then the consequence is a reduction of crop yield.

B. Soil erosion in the Philippines

It appears that soil erosion is the most widespread among the process of soil degradation and it leads to being a threat to food security (Asio *et al*., 2009). The Philippines is distinct to its mountainous and rugged topographical positions of about 60 percent and has more than 18 percent slope. Forest cover has been reduced due to deforestation and land-use change. This is also aggravated by occurrences of 15-20 typhoons per year wherein soils are lost when it is exposed to excessive rainfall since our country has a wet tropical climate with an annual rainfall ranging from about 1,000 mm in the south to more than 4,000 mm in mountainous areas particularly along the eastern coast of the archipelago.

The islands of Luzon, Visayas, and Mindanao have greatly varied but severely eroded soils with 33, 21, and 46 percent, by NAP (2004). An alarming graphical data is presented below that there are regions in the country that are beyond 80, which means an average soil loss rate in t^{-1} ha^{-1} per year. Thus, this phenomenon must be addressed and taken into holistic consideration especially those farmers who migrated from lowland to upland with steep slopes. Appropriate farming strategies and the right attitude of upland farmers are keys to a sustainable and ecological approach to farming.

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C. Existing Practices in the Philippines for Soil Erosion Control

1. Cropping Systems and Agroforestry for Perennial Crops

Namuco *et al* (2010) presented the systems of a culture for fruit and plantation crops. These crops can be grown together with livestock and in combination with crops for smallholders whereas the corporate farms are practicing monocropping. The hilly lands must be conserved and maintain through agroforestry because of the inevitable increase of upland dwellers tilling their lands.

a. *Sod system*- it consists of grasses, shrubs, and other important vegetations that are to be grown with regular trimming and mowing to control their growth especially during the time of fruiting for easy execution in harvesting and caring of fruits. Adopting this will have increased organic matter accumulation and reduction of soil erosion.

b. *Mulch system*- it is a cheap utilization of on-farm waste such as rice straw, bamboo leaves, etc. for the mulching orchard. This will lead towards a reduction of water run-off during the rainy season but an increase of organic matter is evident. Careful monitoring should be carried out to avoid burning the entire orchard during dry months.

c. *Clean culture*- it is the opposite among all cropping systems. It views that the crops such as pineapple are grown with regular tilling, weeding, and leaving soil bare at all times. Occurrences of pests and depletion of organic materials could be at threat when continuously happening.

d. *Covercropping*- it is similar to sod system, however, it composes of leguminous cover crops (kudzu, calapogonium, centrosema, etc) to be grown and distinctly grows as creeping, viny or bushy. Practically, the kudzu (*Pueraria phasioloides*) and calapogonium (*Calapogonium mucunoides*) are recommended to be planted while the trees are still young. Centrosema (*Centrosema pubescens*), on one hand, is an ideal one even if the trees have attained their full size.

e. *Coconut-based multiple cropping systems*- crops may be planted depending on the growth phases of coconut. Ecological benefits include reduction of soil erosion with total productivity and income could be attained thru efficient use of land, light, and labor under coconut trees.

> *Phase I*- refers to the first six years from planting of coconut: sun-loving plants (corn, watermelon, banana, vegetables)
>
> *Phase II*- refers to 7th to 20th year: Shade-tolerant plants (cacao, coffee, lanzones, gabi)
>
> *Phase III*- refers to 26th year and beyond: in a combination of shade and sun-loving plants

f. *Agroforestry*- it drives pressure to reforest the marginal lands due to population increase in upland areas. This is a system wherein the forest trees coexist with crops and farm animals. These

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are domesticated by farmers as stewards to meet their needs including food, fuelwood, and cash income.

Here are some places as to agroforestry practices is concerned.
1. Hanumoo farm in Mindoro Oriental
2. Naalad farmers in Cebu
3. Tausugs in Sulu
4. Mauyong by Ifugaos in Cordillera
5. Tagbanuas of Palawan
6. Ikalahans of Nueva Vizcaya

Furthermore, the agroforestry system has its shoots called LandCare. It is the land degradation at the community level, which emerges the landcare system seeing it for long term health of the land. International Rice Research Institute (IRRI) had its first selection at Claveria in Misamis Oriental (Mindanao), to test the upland rice under acidic soils. Since then, it was handed over by International Centre for Research in Agroforestry (ICRAF) for upscaling the agroforestry-based farming system and soil conservation.

One of the key features of Landcare is the multisectoral approach between and among community, LGUs, and NGOs. There are appropriate technologies under this system:
a. natural vegetative strips (NVS)
b. nursery establishment for agroforestry development
c. timber trees with NVS

Types of barriers to control soil erosion:
1. *Natural barrier*- leaving a strip of the existing vegetation
2. *Planted barriers*- the planting of living wall forming hedgerows
3. *Constructed barriers*- these are rockwall, pole walls, bamboo walls

Let us focus on plants in making the best choice for the farm as per the recommendation of ICRAF.

COMMON NAME/SCIENTIFIC NAME	**ADVANTAGE**	**DISADVANTAGE**
Natural Vegetative Filter Strips (NVS)		
Cogon (*Imperata cylindrica)*	> no establishment cost > source of animal fodder	> less biomass for mulch > source of additional weed in alleyway
Forage Grass		
Napier (*Pennisetum purpureum*)	> good animal fodder > planting material is widely available	> high labor for cutting and fertilizer requirement
Pasture Legume		

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Forage peanut (*Arachis pentoi*)	> high biomass	
Forage Legume		
Flemengia (*Flemengia macrophylla*)	> source of organic fertilizer > shade tolerant	> planting materials are not widely available > some labor requirement
Leguminous Trees		
Ipil-ipil (*Leucaena leucocephala)*	> excellent source of organic fertilizer as green mulch > adapted to alkaline and slightly acid soils	> susceptible to psyllid > poor drought tolerance
Cash Perennials		
Banana	> Fruits are readily saleable > Grows better along contour	> may compete with food crops
Timber Trees		
Mahogany	> source of fuel and building material	> slow growing tree
Other species		
Sunflower	> less competitive with food crops	> can spread over alleys during fallow periods

2. Sloping Agricultural Land Technology (SALT 1)

The original system's underlying objectives were to:

- control soil erosion;
- help restore soil structure and fertility;
- be efficient in food crop production;
- be applicable to at least 50 percent of the country's hillside farms;
- be easily duplicated by upland farmers using local resources and without the need for external loans;
- be culturally acceptable;
- be fully productive in as short a time as possible;
- require minimal labor;
- be economically feasible and ecologically sound.

Two (2) packages of technology have been modified from the system:

a. *Row Crop SALT system*- non-permanent crops or annual crops are planted in the alleys such as corn and beans.

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b. *Permanent Crop SALT system*- all the alleys can be planted with bananas, coffee, fruit trees, or other permanent cash crops.



Source: Adang (2019) Sloping Agricultural Land Technology (SALT): A sustainable farming options for the uplands

Other type of SALT systems:

- Simple Agro Livestock Technology (SALT 2)
- Sustainable Agro Forest Land Technology (SALT 3)
- Small Agro Fruit Livelihood Technology (SALT 4)

3. Vegetable-based upland technology

Dano *et al* (2004) had a report that appropriate use of technologies in combating soil erosion could be led to an increase in income under the trees environment. Vegetables as intercropped with trees particularly fast-growing species (*Gmelina arboria, Eucalyptus spp*.) could be possible and generates income. They concluded that this system is promising the incorporation of tree-vegetable production than plant trees alone.

This study was conducted from 1998 to the present (2004) in a research farm located at the foot slopes of Mt. Kitanglad in Lantapan, Bukidnon Province, Southern Philippines.

D. Contour Farming Establishment

Have you ever seen the rain splashing on the ground and, soil detaches particles subsequently transports from hilltop downhill? This will lead to soil illness and plant-crop destruction. It does not solve the problem without the help of individuals in a community to conserve soil and water for long-term sustainability. Contour farming is a technique in controlling the flow of water and thereby prevents soil from getting lost. Farm practices that aid water and soil conservation such as crop rotation, relay cropping, cultivation along the lines, incorporation of organic matter into the soil, tree wood and wood products, mulching and cover crops. A well-established contour farm should have these important activities:

1. Stabilizing gullies- one way to stabilize the horizontal development of gully on the farm is to construct gully dams. This is to collect sediment, form a barrier to pond water and reduce the

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erosive power of the water streaming off. Planting trees, shrubs, and grasses could be of great help for controlling soil erosion in gullies.

2. Locating the contour with an A-Frame- an instrument to be used in contour establishment is A-Frame which locates contour lines and works like the carpenters level.

Here are some tips in making the A-Frame following UDP (2002):
a. A 1-meter stick should be nailed in 2 bamboo sticks (2 meters each) in shaping A-frame.
b. A stone should be tied in a string and hang 10 to 20 cm below the crossbar where legs of the A-frame are joined.
c. Put the frame instrument on level ground for the calibration test. Mark with A and B where its legs are on the ground considering that a mid-point of a crossbar is in position. Move the frame from one place to another such as leg of A transfers to B, leg of B transfers to A, and so on.

3. Distancing between contour lines- contour is first located at the middle of the slope followed by a second contour at a vertical interval of 1 to 2 meters. This can be done at the eye height of the farmer.

Factors to consider in distancing between contour lines:
a. contour lines vary on each slope
b. the steeper the slope the closer the lines and vice versa
c. lines more than 15 meters apart is not recommended due to barrier breakdown
d. upon plowing, make the initial plow line adjacent to the upper contour barrier then to the bottom. Plow the remaining part of the field is repeated.

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UNIT IV. PRODUCTION PRACTICES
B. PLANTING MATERIALS SELECTION AND PREPARATION

I. Introduction

Aside from yield quantity and quality as basic considerations, varieties to be grown should be selected based on market requirements, farmer preference, and adaptability to the locality. Other considerations may include soil type and nutrient levels, water availability, prevailing temperatures and humidity, and pest and disease history. For efficient chemical, water and other input utilization, planting materials may be selected based on their nutrient and water use efficiencies, and pests and disease resistance.

Selection of planting materials considers soil and site suitability/compatibility. For efficient chemical, water and other input utilization, planting materials may be selected based on their nutrient and water use efficiencies, and pests and diseases resistance. Crop varieties/species known to be toxic for human consumption are not grown. Source of planting material, the necessary seed treatments and related documents. The seed and planting materials should be of high quality. Sourcing and/or procurement of seed and planting materials from the Accredited Seed Growers or Plant Nursery Operators are encouraged.

II. Learning Objectives

At the end of the session, the participant should be able to:

1. identify the characteristics of good planting material; and
2. know the methods and preparation of good planting material.

III. Learning Content

1. Sexual Propagation

Sexual propagation is propagation by seed or spores. The plants will become fertilized and produce the seed that will grow into a new with both of the characteristics of the parent plants. It is the easiest and cheapest way of growing plants.

Factors to consider when selecting seeds

- Seeds should be viable
- Seeds should be clean (No spots, cracks, holes, growths)
- Seeds should be taken from mature, healthy fruits
- Do not use seeds from diseased plants
- Do not use from plants that did not perform properly
- Dry seeds properly before storing (air dry)

Characteristics of good quality seed

- Higher genetically purity
- Higher physical purity for certification
- Possession of good shape, size, color, etc., according to specifications of variety

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- Higher physical soundness and weight
- Higher germination (90 to 35 % depending on the crop)

Seed or planting materials of a crop or a plant is a very important component of crop production as good seed interacts with the good environment around it and determines the health of the crop or plant. If the environment is conducive and the seed is poor, production is likely to be less.

There are several advantages of sexual propagation:

- It may be quicker and more economical than asexual propagation.
- It may result in new cultivars and vigorous hybrids.
- For some plants, it may be the only means of propagation.
- It provides a way to avoid transmission of particular diseases, such as viruses.
- It maintains genetic variation, which increases the potential for plants to adapt to environmental pressures.

2. Asexual Propagation

Asexual propagation is a type of plant reproduction using roots, stems, or leaves of a parent plant. Various asexual propagation methods exist such as growth from tubers, bulbs, rhizomes, and stolons, taking stem cuttings, leaf cuttings, cane cuttings, grafting, division, layering, and tissue culture.

A. Separation and Division

Separation and division are the easiest and quickest ways to propagate many plants. Separation uses naturally occurring vegetative structures such as bulbs and corms. Individual bulbs or corms are separated from a clump. Most of the spring- and summer flowering bulbs are propagated this way. Division involves digging up the plant or removing it from its container and cutting (dividing) the plant into separate pieces. Division uses specialized vegetative structures such as rhizomes and tubers. Indoor plants that can be propagated by division include ferns, snake plant, prayer plant, and African violet. Outdoor plants that can be divided include many perennials such as daylily, hosta, iris, liriope, and verbena. Separation and division are normally done in the fall or early spring.

B. Layering

Stems still attached to their parent plant may form roots where they come in contact with a rooting medium. This method of vegetative propagation is generally successful because water stress is minimized, and carbohydrate and mineral nutrient levels are high. The development of roots on a stem while the stem is still attached to the parent plant is called **layering**. A layer is a rooted stem after it has been removed from the parent plant. Some plants propagate naturally by layering, but sometimes plant propagators assist the process. Layering is enhanced by wounding one side of the stem where the roots are to form or by bending it very sharply. The rooting medium should always provide aeration and a constant supply of moisture. Some common forms of layering are as follows. However, keep in mind these protocols are sometimes modified.

Tip layering. Dig a hole 3 to 4 inches deep in the rooting medium. Insert the tip of a current season's shoot and cover it with soil. The tip grows downward first, then bends sharply and grows upward. Roots form at the bend. The recurved tip becomes a new plant. Remove the tip layer and plant it in late fall or early spring. Examples of plants propagated this way include purple and black raspberries, trailing blackberries, and dewberries. The aforementioned plants also do this naturally.

Simple layering. Bend the stem to the ground. Cover part of it with soil, leaving the remaining 6 to 12 inches above the soil. Bend the tip into a vertical position and stake in place. The sharp bend often induces rooting but wounding the lower side of the bent branch may help. Simple layering can be done on most plants that have low-growing branches. Examples include azalea, forsythia, boxwood, honeysuckle, rhododendron, and wax myrtle.

Compound layering. Bend the stem to the rooting medium as for simple layering, but alternately cover and expose stem sections. Wound the lower side of the stem sections to be covered. This method works well for plants producing vinelike growth such as heart-leaf philodendron, pothos, and grape.

Mound layering. Cut the plant back to 1 inch above the ground in the dormant season. Dormant buds produce new shoots in the spring. Mound soil over the new shoots as they grow . Roots develop at the bases of the young shoots. Remove the layers in the dormant season. Mound layering works well on apple rootstocks, cotoneaster, daphne, quince, and spirea.

Air layering. Air layering can be used to propagate large, overgrown houseplants (such as rubber plants or dieffenbachia that have lost most of their lower leaves) as well as some woody ornamentals such as camellias. The process varies depending on whether the plant is a monocot or dicot. For monocots, such as *Dracaena fragrans* ‘Massangeana’ (corn plant), make an upward cut about one-third of the way through the stem. This is normally done on a stem about 1 foot from the tip. The cut is held open with a toothpick or wooden matchstick. Dust the wound with rooting hormone and surround with damp, unmilled sphagnum moss. Wrap the moss with plastic and hold in place with twist ties or electrician's tape. Aluminum foil can also be used; it does not require twist ties or tape to hold it in place.

The process for dicots is similar except a 1-inch ring of bark is removed from the stem. Scrape the newly bared ring to remove the cambial tissue in order to prevent callus from forming. Wrap and cover using the same procedure as that described for monocots.

After the rooting medium is filled with roots, sever the stem below the medium. The new plant requires some pampering after planting until the root system becomes more developed.

Natural forms of layering. Sometimes layering happens without the help of a propagator. Runners and offsets are specialized structures that facilitate propagation by layering. A runner produces new shoots where it touches the growing medium. Plants that produce stolons or runners are propagated by severing the new plants from their parent stems. Plantlets at the tips of runners may be rooted while still attached to the parent or detached and placed in a rooting medium. Examples include strawberry and spider plant.

Plants with a rosetted stem often reproduce by forming new shoots, called offsets (offshoots), at their base or in the leaf axil. Sever the new shoots from the parent plant after they have developed their own root system. Unrooted offsets of some species may be removed and placed in a rooting medium. Some of these must be cut off, whereas others may simply be lifted from the parent stem. Examples include date palm, haworthia, bromeliads, and many cacti.

C. Budding and Grafting

Budding and grafting are methods of asexual propagation that join parts of two or more different plants together so they unite and grow as one plant. These techniques are used to propagate cultivars that do not root well from cuttings or to alter some aspect of the plant (for example, to create weeping or dwarf forms).

Most fruit and nut trees are propagated by budding or grafting.

The scion consists of a short stem piece with one or more buds and is the part of the graft that develops into the top of the grafted plant. The rootstock (also called stock or understock) provides the new plant's root system and sometimes the lower part of the stem. Seedling rootstocks are used commonly.

Six conditions must be met for bud grafting to be successful:

1. The scion and rootstock must be compatible (capable of uniting).
2. The scion and stock plant must be at the proper physiological stage of growth.
3. The cambial region of the scion and the stock must be in close contact - touching if possible.
4. Proper polarity must be maintained (the buds on the scion must be pointed upward).
5. Immediately after grafting, all cut surfaces must be protected from drying out.
6. Proper care must be given to the graft after grafting.

Once the bud has healed and broken to form a shoot, the top of the stock plant is cut back (removed) to the bud union.

Not all plants can be grafted. The two plants should be closely related taxonomically. Grafting between clones and seedlings of the same species is usually successful.

Budding

Budding, or bud grafting, is the union of one bud, with or without a small piece of bark, from one plant (scion) onto a stem of a rootstock. It is especially useful when **scion** material is limited. Budding is usually done during the growing season when the bark is slipping (soft and easy to peel back from the cambium), June through August in the Southeast U.S., but it can also be done in late winter or early spring.

T-budding—This is the most commonly used budding technique. When the bark is slipping, make a vertical cut (same axis as the rootstock) through the bark of the rootstock, avoiding any buds on the stock. Make an intersecting, horizontal cut at the top of the vertical cut

(forming a “T” shape), and loosen the bark by twisting the knife slightly at the intersection. Remove a shield shaped piece of the scion, including a bud, some bark, and a thin section of wood. Push the shield under the loosened stock bark at the "T". Wrap the union, leaving the bud exposed. Many fruit trees are propagated by this method.

Chip budding—This method can be used when the bark is not slipping. Although all the basics in handling budwood and stock are the same for chip budding and T-budding, the cuts made in chip budding differ radically. The first cut on both stock and scion is made at a 45° to 60° downward angle to a depth of about ⅛-inch. After making this cut on a smooth part of the rootstock, start the second cut about ¾-inch higher and draw the knife down to meet the first cut. (The exact spacing between the cuts varies with species and the size of the buds.) Then remove the chip. Cuts on both the scion (to remove the bud) and the rootstock (to insert the bud) should be exactly the same size. Although the exact location is not essential, the bud is usually positioned one-third of the way down from the beginning of the cut. If the bud shield is significantly narrower than the rootstock cut, line up one side exactly. Wrapping is extremely important in chip budding. If all exposed edges of the cut are not covered, the bud dries out before it can “take.” Chip budding has become more popular over the past five years because of the availability of thin (2 mil) polyethylene tape as a wrapping material. This tape is wrapped to overlap all of the injury, including the bud, and forms a miniature plastic greenhouse over the healing graft.

Grafting

Grafting is the union of the stems of two plants to grow as one. There are several kinds of grafting; which method to use depends on the age and type of plants involved.

Whip and tongue grafting—This method is often used for stems ¼-inch to ½-inch in diameter. The scion and rootstock are usually of the same diameter. This graft provides excellent cambial contact and heals quickly. Make a sloping cut 1-inch to 2½-inches long at the top of the rootstock and a matching cut on the bottom of the scion. On the cut surface, slice downward into the stock and up into the scion, so the pieces will interlock when brought together. Fit the pieces together such that the cambium matches on at least one side, and then wrap and tie or wax the graft union.

Cleft grafting—Cleft grafting is one of oldest and most widely used methods of grafting. The two most common circumstances when one would use this technique are (1) when creating a weeping ornamental tree and (2) when upgrading an older cultivar of a fruit tree to a newer, improved one while maintaining the existing root system. Cleft grafting is best done in early spring when the buds on the stock plant are beginning to swell but before active growth has begun. The scions, however, must be fully dormant. Collect scion wood ⅜ to ⅝-inch in diameter during the dormant season and refrigerate until needed.

Cut the limbs to be reworked at a right angle to the main axis of the branch. Use a heavy knife to split the branch, making a 2-inch vertical cut through the center of the stub to be grafted. Be careful not to tear the bark. Keep this cut wedged apart. Prepare two scion pieces 3 to 5 inches long; cut the lower end of each scion piece into a wedge. Insert the scions at the outer edges of the cut in the stock. Tilt the top of the scion slightly outward and the bottom slightly inward to be sure

the cambial layers of the scion and stock touch. Remove the wedge propping the slit open, and cover all cut surfaces with grafting wax. If both scions grow, one can be removed later.

Some other types of grafting include splice, side, stub, side veneer, and side tongue. Check reference books for additional information. Each method has advantages for special situations.

D. Micropropagation

Involves the application of tissue culture techniques to propagate plants from very small plant parts (parts of leaves, stems, shoot tips, root tips, single cells, and pollen grains). The small plant part is grown (cultured) in a test tube, petri dish, or other sterilized container with a culture medium and precise environmental conditions. The container and growing medium must be sterilized. Because plants often harbor bacterial and fungal spores, the plant tissue must also be disinfected.

Micropropagation is a rapidly growing part of the plant propagation industry. It is not practical for most home gardeners because of the very specific requirements of the culture media and the constant efforts that must be made to avoid possible contamination from disease organisms. For nurseries, special care must be taken in transporting micropropagated plants from the lab to the store because they are not acclimated to outdoor growing conditions.

V. Summary

Selection & preparation of planting materials is the most important consideration when you develop an orchard/ farm because it is an investment, what you intent to plant is also what you will harvest in the near future. A quality planting material plus good agricultural management and practices will result in quality harvest.

Farming is a form of business because at the start you will invest an area, a suitable area for a desired crop to plant. Planting a recommended, quality planting materials will be the next consideration and so with the technology that is necessary to be employed for a quality produce.

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UNIT IV. PRODUCTION PRACTICES
C. PLANTING METHODS

I. Introduction

Managing crop production involves a number of practices to be employed. Along with these, the choice of planting method and system of planting to be used is just among of important factors to be considered prior to planting in the field. Planting is considered crucial in achieving superior plant growth and optimizing yield. In this lesson, you will learn some of the popular planting methods used by farmers or growers, including the advantages and disadvantages of each method. Further, you will be introduced to the different planting systems adopted by farmers in growing various crops, calculate the population density and planting materials needed in a given area.

II. Learning Objectives

At the end of the session, the participant should be able to:

1. Familiarize the different methods of planting for agronomic and horticultural crops;
2. Discuss the advantages and disadvantages of each planting method;
3. Estimate the planting materials required in a given area; and
4. Calculate the population density for different planting systems

III. Learning Content

Along with other considerations, selection of suitable planting method to be adopted is vital for successful farming venture. In general, choosing which method (direct or transplanting) to use depends on a number of factors such as the species to be grown, growing condition or season and location, grower's familiarity and skills, financial capability, yield and return on investment.

Planting Methods

a. Direct method or seeding – this method involves sowing of seeds directly in the soil (in the field) eliminating the need for nursery. Most upland crops such as corn, upland rice and agronomic legumes are classical examples of directly seeded crops.

1. Broadcasting – spreading of seeds evenly in area without distinct arrangement and spacing (no rows and furrows) either manually or mechanically.
2. Drilling – sowing of seeds along a row or furrow at uniform depth either manually or mechanically. Seeds are covered thinly with soil.
3. Dibbling – also known as hill method in which holes are made on the soil at a specified distance using a stick or machine implement where seeds are sown in the hole and covered thinly with soil.

b. Transplanting – this method makes use of a nursery where seeds are germinated and seedlings are raised prior to planting them in the field. Most vegetables, fruit trees and lowland rice varieties are usually transplanted.

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Transplanting vs. direct seeding

Methods of planting	Advantages	Disadvantages
Direct method or direct seeding	• Most economical (less cost and labor required) • Crop generally matures early compared to transplanted • Less disturbance and not subjected to stress	• Higher cost associate with higher seed requirement • Not satisfactory seed germination and seedling survival • Uneven stand of vegetative growth • Greater weed competition at young stage • Plants are prone to lodging due to less root anchorage
Transplanting	• Low seed requirement • Save cost on expensive hybrid seed • Better germination and higher seedling survival • Greater control over growing conditions at seedling stage • Improve land-use efficiency • Assure complete stand and crop uniformity	• Cost is high due to labor and materials used for raising seedlings, and labor for transplanting • Risk of transplanting shock if procedures are not observed • Extend the growing season

Planting Methods for Agronomic Crops

- *Direct seeding*. This is done either by hill, drill or broadcast method. Seeding is usually done after the application of basal fertilizer (organic or inorganic).

  For upland crops such as peanut, mungbean and soybean or even upland rice, seeds of these species are usually sown either by hill or dibbling and drill method in the prepared rows or furrows. One week after seeding, extra seedlings are thinned leaving only the allowable number of plants per hill or linear meter. e.g., 15 plants per linear meter of soybean and mungbean and 2-3 plants per hill for peanut.

  For lowland rice, pre-germinated seeds are usually used and broadcasted (most common) evenly over the properly prepared and leveled field. A hectare area usually requires 100 to 150 kg of rice seeds using this method. Irrigation water is introduced to field after 5-7 days from planting followed by the application of pre-emergence herbicide to control weeds.

- *Transplanting*. This is the commonly used alternative planting method for lowland rice. This method produces seedling by wet/dry seedbed or modified dapog method. At planting, a transplanting marker made of lumber passed through over a well-prepared and leveled field

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in one direction and then crossed in another direction to form a square. Seedlings are then transplanted in the provided marker at a predetermined distance.

For wetbed method, the pre-germinated seeds are sown in a well-prepared wet seedbed. A 50kg seed is required in a 1-hectare field to be sown in a 400 m^2 seedbed. Seedlings produced by this method can be transplanted in the field at 25-30 days from sowing.

The dry seedbed method is common in rainfed areas where the pre-germinated seeds are sown in dry seedbed (1 to 1.5 m wide bed). A 500 m^2 seedbed is usually enough to accommodate 50kg seeds intended for a 1 hectare field. Seedlings produced with this method are ready for transplanting similar to wetbed method.

For dapog method, the pre-germinated seeds are sown in a concrete pavement periodically irrigated with fresh water. This method will require a seedbed size of 40 m^2 to accommodate 50-60 kg seeds for a 1-hectare planting field. Seedlings are usually transplanted earlier at 10-14 days from sowing.

Planting Methods for Vegetable Crops

- Crops that are usually transplanted. Species under this group are usually having smaller-sized seeds and slow-growing at the onset of germination. Hence, transplants are usually established first in the nursery prior to planting in the field. e.g. cabbage, pepper, broccoli, cauliflower, celery, tomato, lettuce, eggplant

- Crops that are usually direct seeded. This group has medium to large-sized seeds which facilitates direct sowing in the prepared plots or furrows in the field. Crops are characterized as fast-growing after germination; hence, they can quickly establish in the field. However, if seeds are expensive particularly the hybrid ones, growers usually produced transplant to ensure higher percentage of germination and survival. e.g. beans, kangkong, sweet corn, melons, bitter gourd, cucumber

- Crops that should be directly seeded. This group is never transplanted because the tip of their taproots may be damaged in the process, resulting in deformed or forked roots which are undesirable. e.g. carrot, radish, turnips, beets

Factors to be considered whether to transplant or direct seeded

- Cost and availability of the seeds. Direct seeding always requires 3-4 times more seed than transplanting. When the cost of seed is high, as in hybrid seed, transplanting may be recommended instead of direct seeding.
- Quality of land preparation. Direct seeding of small-seeded crops is impractical when the field is not thoroughly pulverized during land preparation. Large soil clumps make it difficult to control depth of seedling, resulting in poor emergence. As a rule, small seeded crops such as lettuce and celery must be a serious problem in direct seeded, show growing crops, such as onion and celery

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- Root-regeneration ability of the crops. Some crops such as legumes do not easily regenerate roots, hence do not easily recover from transplanting shock. The opposite can be said of solanaceous and cruciferous crops. Cucurbits are intermediate in rooting behavior and can be transplanted if the procedure is done early enough at the cotyledonary leaf stage.
- Multiple cropping. When vegetables are grown after another crop, it is advisable to start the seedlings even before the previous crop is harvested. This allows planting of the seedling immediately after harvesting the previous crop, reducing the period when conditions are not ideal. Hence, they are usually direct seeded, unlike pepper and celery, which grow very slowly in the initial growth stage. And when condition is favorable, it is more practical to transplant, to allow more intensive management of the seedlings in a small area of the seedbed than in the open field. The result is as good start for the crop, which is often translated into a higher yield and better product quality.

Seedling Production

- Soil medium preparation. Mix compost and aged rice hull at the ratio of 2:1, or equal part of compost, garden soil and sieved sand or rice hull charcoal. The mixture should be sterilized by placing it in a vat or metal container and put over the fire for at least one hour with regular misting and mixing. The mixture should be moistened enough and allowed to cool off before using.
- Seedbox method. Prepare a seedbox 50 cm long, 30 cm wide and 10 cm deep. Place the sterilized mixture into the seedbox. Make rows of about 5 cm apart and about 1 cm deep. Then seeds will be drilled thinly in the rows of about 1 cm deep and cover them lightly with fine soil
- Seedbed method. Prepare a lot of about 1 meter wide and of any desired length depending on the quality of seeds for germination. Then add to the soil a 2:1 mixture of rice hull charcoal and compost. The seedbed will be sterilized by burning over it in a 5-cm layer rice straw or any dried leaves. Make rows 5 cm apart and 1 cm deep and drill the seeds into the rows, cover lightly the seeds with soil and then water the seedbed early in the morning
- Modified seedbox method or Cellular method. Prepare small pots made of used paper, banana leaves (lokong), coconut fronds or bamboo (bagakay). The pots will be filled with the same kind of sterilized soil used in the seedbox or seedbed methods. 2-3 seeds will be sown in each small pot, 7-10 days after germination, thin the seedlings to only a healthy per pot

Care of seedlings

- Pricking. This is done by transferring the seedlings 7-10 days after germination to another plot or individual container or seedling tray or seedbed following a planting distance of 5 x 5cm. This operation is usually employed if seedbox or seedbed method is used. The newly pricked seedlings should be irrigated and protected from direct sunlight for 1-2 days. To give the seedlings a head start, a starter solution should be applied at the rate of 1 tablespoon of ammonium sulfate or urea per gallon of water. This should be followed by the application of fresh water to avoid potential salt injury to young leaves.

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- Hardening. The is the process of gradually exposing the seedlings to sunlight 7-10 days after pricking with concomitant reduction of watering and allowing the seedlings to exhibit temporary wilting. In some cases, reduced fertilization with nitrogenous fertilizer is also practiced.

Recommended seeding rate and spacing of some important crops

Crop	Seeding rate (per ha)	Spacing (cm)
Rice	40-60 kg	20 x 20 or 25 x 25 at 3-5 plants/hill
Corn	15-20 kg	50 x 75 at 2 plants/hill
Sorghum	8-10 kg	50 furrow at 10-15 plants/linear meter
Mungbean	18 kg	50 furrow at 15-18 plants/linear meter
Peanut	90-125 kg (shelled)	20 x 60 at 3 plants/hill
Soybean	30-60 kg	60 furrow at 15-18 plants/linear meter
Squash	200 g	50-100 x 100
Watermelon	695 g	100 x 150
Tomato	80 g	50-75 x 50
Eggplant	75 g	50-75 x 50
Sweet pepper	200 g	50 x 50
Cabbage	150 g	75 x 40-50
Pechay	700 g	15-20 x 15-20
Radish	3.25 kg	30 x 10
Stringbean	8 kg	100 x 25
Okra	2.87 kg	75 x 30-50
Pipino	1.20 kg	75 x 30-50

Source: adapted from Pava and Abellanosa, 2003.

Systems of Planting for Perennial Crops

There are factors to consider in selecting the system of planting to be used. These are the slope of land, crops to be grown, purpose of utilizing spaces in between rows, convenience, and cultural practices. The commonly adopted planting systems by fruit and plantation crop growers include the following.

- Square system
  This may be the simplest among the planting systems hence it is commonly used. In this system, planting is done at each corner of the square and the distances between all plants either between hills or rows are equal. Since plants are distanced equally farm management practices such as spraying, pruning, weeding, harvesting, among others can be done with ease. Possible cash crops as intercrops can be planted in between spaces conveniently.

$$\text{No. of plants} = \frac{\text{Area } (m^2)}{(\text{Planting distance})^2}$$

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- Quincunx system
  This system of planting makes use of the square with an additional plant at the center. This maximizes area utilization during the early growth and fruiting stages of the plantation. The planting of filler plants provides an additional income to the grower in the early life of the orchard. This filler plant can be gradually removed when main crops become larger to reduce overcrowding.

$$\text{No. of plants} = \frac{\text{Area } (m^2)}{(\text{Planting distance})^2} + \left[\left(\frac{\text{Length}}{S} - 1 \right) \left(\frac{\text{Width}}{S} - 1 \right) \right]$$

- Triangular system
  In this system, plants are planted at equidistant to one another from every direction to form an equilateral triangle. This system has no special advantage over the square system except providing more open space for the trees and for intercrops. In addition, it can accommodate more number of plants per unit area.

$$\text{No. of plants} = \frac{\text{Area } (m^2)}{(\text{Planting distance})^2 \times 0.866}$$

V. Summary

The planting methods and systems layout are just as crucial as with other cultural management practices for crop production. Proper selection of this method can have implication on the performance of the crop in the field. Better growth and optimum yield can be achieved if suitable planting method is adopted and other factors are considered and managed. For instance, attention to details is required during land preparation direct seeding will be used. Additionally, one must ensure that every seed sown must be at the same depth below the soil surface uniform germination and seedling growth.

Prior to planting, the lay outing is a crucial operation in a way that each plant should be planted at an ideal spacing for proper plant growth and development and for accommodating the optimum number of plants per unit area in order to maximize yields. The selection of planting system should aim at allowing easy application of other production practices in addition to maximizing land utilization and aesthetic look of the farm.

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UNIT IV. PRODUCTION PRACTICES
D. WATER SUPPLY ANG MANAGEMENT
(IRRIGATION PRACTICES)

I. Introduction

Irrigation makes a major influence on agricultural production by making a whole range of crops viable in an otherwise unreliable climate and helping insure against drought. However, irrigation does not automatically guarantee favorable return of investment. It is a high-cost exercise, using water from increasingly limited supplies, and raises environmental concerns in the community. Many of the pressures facing some farmers have been caused by a lack of understanding in the past of the best practices necessary for irrigation management.

Crop production depends on the successful implementation of the agricultural and water management technologies. This is vital as world population continue to grow, food production should also continue in a sustainable way. The implementation of technologies is also important to minimize environmental degradation resulting from agricultural activities. Agricultural and natural resources engineers are applying scientific principles for the optimal use of natural resources in agricultural production. Water is the scarcest resource. The importance of the judicious use of water in agricultural sector for sustaining agricultural growth and the retardation of environmental degradation needs no elaboration. Judicious use of water for crop production requires knowledge of water conveyance and application methods, their designing, strategic management of water resources, land and watershed management, etc. Increasing efficiency in conveyance and pumping systems are also of great concern. Irrigation management strategy practiced in normal soils may not be appropriate in problematic soils such as saline soils. (M.H. Ali 2010)

II. Leaning Objectives

At the end of the session, the participant should be able to:

1. Define what is irrigation; and
2. Know the management practices involved in irrigation for optimum crop production.

III. Learning Content

A. Definition of Irrigation

Water is considered as one of the most important requirements of agricultural productivity. The application of water to soil is essential for plant growth and it serves the following functions:

i. It supplies moisture to the soil necessary for seeds germination and other biological processes during plant growth.
ii. It cools the soil and the surroundings thus making the environment more favorable for plant growth.
iii. It helps washes out or dilutes salts in the soil.

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iv. It helps in tillage operations by softening clods.
v. It enables fertilizers application.
vi. It decreases the adverse effects of frost on crops.
vii. It ensures crop success against short-duration droughts

Irrigation – the artificial application of water to soil for enhancing the naturally available moisture in the root-zone soil for optimum crop productivity

To keep the optimum amount of moisture in soil, irrigation supplies water to the land where water is scarce and drainage withdraws water from the land where water is in excess. The purpose of providing irrigation and drainage is to maintain moisture in the root-zone soil within the range required for maximum agricultural productivity.

B. Types of Irrigation Systems

- Center pivot irrigation
- Drip irrigation
- Lateral move irrigation
- Localized irrigation
- Manual irrigation
- Sprinkler irrigation
- Sub-irrigation
- Surface irrigation

Center pivot irrigation – an automated sprinkler system which has a number of metal frames involving a rotating pipe or boom that holds and supplies water to a circular area of an agricultural field through sprinkler heads or nozzles (CDC, 2016)

Drip irrigation – is an irrigation system in which water is applied directly to the root zone of plants by means of applicators such as orifices, emitters, porous tubing and perforated pipe operated under low pressure. These applicators can be placed on or below the surface of the ground or can be suspended from supports.

Lateral move irrigation – is a system which tends to be less expensive but requires more labor than other irrigation system. Through this, water is distributed through a series of pipes , each with a wheel and a set of sprinklers, which are rotated either by hand or with a purpose-built mechanism. The sprinklers move certain distance across the field and then need to have the water hose reconnected for the next distances (CDC, 2016).

Localized irrigation – a method of applying water around each plant or a group of plants results in wetting only a small area of the soil surface and sometimes a root zone part.

Manual irrigation – is a method which is probably the most basic type of irrigation method. It is very used in many horticultural enterprises such as nurseries and fruit trees that cannot use automated fixed irrigation system. Water is distributed across land through manual labor and watering cans. This system is very labor intensive.

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Sprinkler irrigation – water is distributed by overhead high-pressure sprinklers or guns from a central location in the field or from sprinklers on moving platforms (CDC, 2016)

Sub-irrigation – this system is the most effective in areas with high water tables. The water is distributed across the land by raiding the water table, through a system of pumping stations, canals, gates and ditches.

Surface irrigation – in this system, the water is deliver to crops using gravity-fed, overland flow of water and there is, no mechanical pump involved. It adapts easily to flat topography and can function without outlet drainage facilities. It is also widely utilized therefore a well-known system which can be operated without any high-tech applications (Taghvaeian, 2017).

C. Irrigation Methods

These irrigation methods are further subdivided as indicated below:

(a) Flooding:

- From field ditches - Where water is applied from field ditches without any levees to guide its flow, or otherwise restrict its movement, the method is designated ordinary flooding. Flooding from field ditches is well adapted to some lands that have such irregular surfaces that the other flooding methods are impossible.

- Border - The border method is suitable to soils of wide variation in texture. It is important, however, to study the physical soil properties in advance of preparing land for border irrigation. Rather impervious subsoils overlain by compact loams permit long border strips, whereas open soils having porous, gravelly subsoils necessitate short narrow strips.

- Check - The check-flooding method consists of running comparatively large streams into relatively level plots surrounded by levees. The check method of irrigation for grain and forage crops is advantageous in localities where large irrigation streams arc available and also on projects which depend on direct flow from widely fluctuating streams.

- Basin - The basin method of flooding is essentially the check method especially adapted to irrigation of orchards. From the supply ditch the water is conveyed to the basin, either by flowing through one basin and into another, or preferably by small ditches constructed so that the water may be turned directly from a ditch into each basin

(b) Sub-Irrigation:

- Controlled by internal supply ditches.
- Uncontrolled by excess application of water to higher lands.

(c) Sprinkling - Sprinkler irrigation is a method of applying irrigation water which is similar to natural rainfall. Water is distributed through a system of pipes usually by pumping. It is then sprayed into the air through sprinklers so that it breaks up into small water drops which fall to the

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ground. The pump supply system, sprinklers and operating conditions must be designed to enable a uniform application of water

Types of Sprinklers Systems

- Sprinkler –
- Riser pipe
- Lateral distribution pipe
- Main line pipe
- Fixed pumping plant

D. Selection of Irrigation Method

The following are the factors that will influence the decision on selecting what irrigation method is appropriate to apply:

- Funds available to invest in irrigation hardware.
- Operating costs of labour, fuel and maintenance.
- Ability to apply smaller quantities of water at more frequent intervals.
- Availability of labour, and the possibility of automation (even in part).
- Skills of the labour force and management.
- The area irrigated and the scale of operations.
- The value of the crop, its sensitivity to irrigation management at critical growth stages, and the probability of rainfall during the irrigation season.
- Personal preference and experience.
- Additional purposes, such as fertilizer application or frost control.

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UNIT IV. PRODUCTION PRACTICES
E. PEST MANAGEMENT

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I. Introduction

In agricultural production, pests are organisms that diminish the value of resources in which man is interested as they interfere with the production and utilization of crops and livestock used for food and fiber. The term "pest" applies to all noxious and damaging organisms including insects, mites, nematodes, plant pathogens, weeds and vertebrates (Fernadez-Cornejo & Jans, 1999). In economic perspective, agricultural pest is an animal or plant whose population density exceed some unacceptable threshold level, resulting in economic damage (Fernadez-Cornejo & Jans, 1999). All of the given definition based on agricultural and economic perspective are true. But as much as the definition must be known, managing them is of utmost importance. Knowledge of the pest and its management it is a great tool to protect crops from agricultural losses.

Pest management involves a set of techniques to reduce pest populations or their detrimental effects. The underlying philosophy in pest management is that, "pest should be managed, not eradicated" and are "inevitable components of agricultural system" (Fernadez-Cornejo & Jans, 1999).

Pest management techniques can be broadly classified into chemical, cultural, and biological. It will be further discussed as the we dive in into the session.

II. Learning Outcomes

At the end of the session, the participant should be able to:

1. understand the concept of pest and pest management; and
2. explain the concept, challenges, advantages, and disadvantages of Integrated Pest Management.

III. Learning Content

A. Pest Management Techniques

A.1. Cultural Control

Cultural control is the deliberate alteration of the production system by targeting the pest itself through agronomic practices to avoid or reduce pest injuries to crops. Crop rotation, intercropping, sanitation, trap crops and pest resistant crop plants are few examples of cultural control.

A.1.1 Tactics to prevent, reduce or delay pest colonization of the crop

- **Site selection** involves locating the crop field in such a manner that pests cannot easily get their way back.
- **Intercropping** - a practice that involves growing two or more crops in proximity to produce a greater yield on a given piece of land by making use of resources that would otherwise not be utilized by a single crop. Intercropping may concentrate the pest in a smaller, more manageable area so that it can be controlled by appropriate tactic.

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Based on spatial arrangement, intercropping can be divided into the following four categories

(i) *Row intercropping*: Growing two or more crops at the same time with at least one crop planted in rows.

(ii) *Strip intercropping:* In this method two or more crops grow together in strips wide enough to allow separate production of crops using mechanical implements but close enough for the crops to interact.

(iii) *Mixed intercropping***:** Growing two or more crops at the same time with no distinct row arrangement.

(iv) *Relay intercropping***:** Growing of two or more crops simultaneously during part of the life cycle of each crop. In this method, a second crop is planted after the first crop has reached its reproductive stage of growth, but before it is ready for harvest.

Based on growth pattern and compatibility, inter-cropping may also be divided into the following groups:

(i) *Parallel cropping*: Under this cropping two crops are selected which have different growth habits and have no competition between each other. This technique allows the crops to exhibit their full yield potential.

(ii) *Companion cropping*: In companion cropping the yield of one crop is not affected by the other. In other words, the yield of both crops is equal to their pure crops. This technique reduces the risk of total crop failure.

(iii) *Multistory cropping***:** Growing plants of different heights in the same field at the same time is termed as multistory cropping. It is mostly practiced in orchards and plantation crops for maximum use of solar energy even under high planting density

- **Trap crops** are grown as a control measure to lure pests away from the cash crop. Pests are either prevented from reaching the crop or concentrated in certain parts of the field away from the main crop. The principle of trap cropping relies on pest preference for certain plant species, cultivars or a certain stage of crop development.
- **Altering the time of planting** - Plantation and harvest dates of some crops can be rearranged to reduce or to avoid potential pest damage. Early planting ensures that seedlings have reached a non-susceptible or tolerant stage when the pest appears. Planting needs to be done only after the emergence or immigration of the pests leaves the pests without hosts. Early harvest date may prevent pests from reaching damaging population densities or overwintering stage by harvest

A.1.2 Tactics to reduce survival of pests by creating adverse biotic and abiotic conditions

- **Sanitation** or source reduction involves eliminating food, water, shelter, or other necessities that are important for pest survival. In crop production, sanitation includes practices to remove weeds that harbor pest insects or rodents, to eliminate weed plants before they produce seed, to destroy diseased plant material or crop residues, and to keep field borders or surrounding areas free of pests and pest breeding sites
- **Crop rotation** interrupts the normal life cycle of pests by placing them in a non-host habitat. It is highly effective to prevent different weeds, soilborne plant pathogens and root-living arthropods to thrive.

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- **Plant and row spacing** – It is important in preventing plant pathogens that usually require a certain moisture threshold to germinate and grow. In contrast, by increasing plant density, it is possible to 'dilute' the damage caused by pests to individual plants.
- **Tillage and destruction of breeding area -** Tillage (soil-turning and residue-burying practices) and seed bed preparation can reduce the number of soil-living pest organisms.

A.1.3 Tactic to reduce injury caused by pests to crop plants

Plants that are resistant to pest attack are less preferred by pests. Resistant crops adversely affect the pests' normal development and survival. Resistant varieties can tolerate the damage without an economic loss in yield and/or quality. Constitutive plant resistance is easy to use, cheap and compatible with other pest management tactics. Induced resistance to herbivores and pathogens reduces plant exposure to autotoxic environment of secondary compounds.

A.2. Physical and Mechanical Control

Physical and mechanical controls either kill insects and small rodents, or make the environment unsuitable for them by attacking, or setting up barriers. These methods are used for crop production and household pest management.

- **Barriers** - Row covers are useful to keep insects away from plants. These are knitted tenuously with plastic or polyester fiber so that plants can still absorb sunshine and moisture from the air.
- **Traps** - Devices like fly paper or sticky boards, covered with sticky and poisonous substance, are used to attract insects. Insects that are attracted by those traps land upon the surface and get glued.
- **Fire -** Farmers consider fire to destroy insect breeding grounds. Fire burns the soil-top and kills insects present there. However, firing may kill beneficiary insects as well.
- **Temperature** - Different thermal conditions can be used to kill insects or to prevent their infestation. Cold storage prolongs the shelf life of agro-products, and retards the development of pests. Heat treatment is also effective in killing insect larvae in certain types of products. For example, mangoes are submersed in hot water baths (at 115 °F for 68 min) in order to kill the eggs and larvae of fruit flies prior to export.
- **Radiation -** Gamma radiation kills all stages of pests in storage conditions. This is a common method employed to kill insects or larvae during export or import of large quantities of grains, fruits and vegetables.
- **Ultrasonic vibrations -** Moths are often sensitive to bats' ultrasonic signals. Imitation of the bat's echolocation system helps in driving away the lepidopterous insect pests from the area.

A.3 Biological Control

Biological alternatives can be used as a replacement of chemical pesticides to leave the ecosystem undisturbed. Biological alternative options can be broadly classified as: (a) Biological Control, (b) Biopesticides, (c) Semiochemicals, and (d) Transgenic Organisms.

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Biological control (biocontrol) is the use of natural enemies to reduce the damage caused by pests. Natural enemies are predators, parasitoids, insects or other arthropod species. Biopesticides, also known as biological control, are based on pathogenic microorganisms or natural products which usually kill pests. The term biopesticide may also be used, more widely, to describe the application of biological agents, pathogens, predators, or parasitoids. In addition, botanicals, semiochemicals and transgenic plants sometimes be described as biopesticides. According to the US Environmental Protection Agency (EPA), biopesticides are certain type of pesticides derived from natural materials such as animals, plants, bacteria and certain minerals.

There are three different approaches of biological control.

- **Importation** involves the enforcement of the natural enemies of a pest to a new locale where the pest does not inhabit naturally. The process involves determination of pest-origin and consequently collection of appropriate natural enemies associated with the pest. Selected natural enemies are then passed through rigorous assessments, testing and quarantine processes to ensure their appropriate use. Finally, the selected natural enemies are mass-produced and distributed.
- **Augmentation** involves the supplemental release of natural enemies to boost up the natural inhabitant population. In addition, the cropping system may also be modified to favor or augment these natural enemies called as habitat manipulations. During the critical time of the season, a small number of natural enemies to pest is released, which is called inoculative release.
- **Conservation -** In conservation method, biological control action is taken to enhance the effectiveness of existing natural enemies to pests in the ecosystem. As natural enemies are already adapted to the habitat and target pest, their conservation becomes simple and cost-effective. An example of such cost-effectiveness is growing nectar-producing crop plants in the borders of rice fields. These provide nectar to support parasitoids and predators of planthopper pests and thus effectively reduce pest densities by 10–100 folds. This also diminishes the necessity to spray insecticides by 70%, and consequently boosts overall crop yield by 5%.

A.3.1 Biopesticides

Biopesticides are certain types of pesticides derived from natural materials such as animals, plants, bacteria, and certain minerals. For example, canola oil and baking soda have pesticidal applications and are considered as biopesticides.

Biopesticides fall into three major classes:

- **Microbial pesticides,** which consist of a microorganism (e.g., bacterium, fungus, virus or protozoan) as active ingredient, are used to control different types of pests. Each active ingredient is specific to its target pest. For instance, some fungi are capable of controlling certain weeds, while certain fungi are specific to kill insects. The most widely used microbial pesticides are subspecies and strains of Bacillus thuringiensis (Bt).
- **Plant-Incorporated-Protectants (PIPs)** are pesticidal substances, which are produced from the genetic materials of plants. These materials are produced by applying genetic engineering. For example, gene from the Bt pesticidal protein can be introduced into the genetic material of plant. The plant then produces substances that destroy the pests.

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- **Biochemical pesticides or herbal pesticides** are naturally occurring substances that control (or monitor) pests and microbial diseases. Biochemical pesticides are often called botanical pesticides when they are derived from plant extracts. Biochemical pesticides include substances like insect sex pheromones and various scented plant extracts. Pyrethrin, extracted from chrysanthemum plants, is a highly effective botanical insecticide. Pyrethrin acts rapidly on insects causing immediate knock down.

A.4. Chemical Control

The most common method of pest control is the use of pesticides—chemicals that either kill pests or inhibit their development. Pesticides are often classified according to the pest they are intended to control. For example, insecticides are used to control insects; herbicides to control plants; etc. Pesticides also include chemosterilants and growth regulators, which are used to interfere with the normal reproduction or development of the pest.

The effectiveness of a pesticide in controlling a pest depends on:
• Using the right pesticide for the target pest (Selection)
• Applying the pesticide at the right time (Pest Scouting/ Monitoring)
• Applying the pesticide in the right amount (Mixing/Application)
• Applying the pesticide to the right place (Application)

If any of these factors is wrong, the effectiveness of the pesticide application will be reduced, control of the pest will be poor, costs will be increased, and additional (probably also incorrect) applications will be applied.

Use of Pesticide in Philippine Setting

The use of pesticides in Philippine agriculture continues to increase despite the adoption and promotion of the Integrated Pest Management (IPM) Program. Insecticides constitute approximately 55%, fungicides 22% and herbicide 16% of the pesticides used in the country for rice, corn, vegetables and plantation crops. The extensive use of pesticides and the risks they pose to human health and the environment are now the focus of national concern and interest. Environmental problems to be addressed as a consequence of pesticide use are: (a) pollution of ground and surface water: (b) increased resistance of pest populations to pesticides; and (c) impact on non-target organisms; i.e., reduction of beneficial insects, soil micro and macro-organisms; biomagnification, and pesticide residue transmittal within the food web (Davis, 1993).

Hence, an increased social pressure prevails pesticide use and creates pressure to replace them gradually with biopesticides, which are safe to humans and non-target organisms. Biopesticides possess an extensive variety of microbial pesticides, biochemicals derived from microorganisms and other natural sources. Biopesticides also include processes involving genetic modification of plants. Biopesticides have been proven effective to control target pest without altering ecological balance, however, there are different barriers exist to make biopesticides more effective. To overcome these barriers, many agricultural societies have adapted and implemented other alternative contextures, such as cultural tactics, physical,

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mechanical and biological tactics, and behavioral controls using semiochemicals and transgenic crops (Hossain, Rahman & Khan, 2017).

B. Integrated Pest Management (IPM)

IPM is an approach which means "the careful consideration of all available pest control techniques and subsequent integration of appropriate measures that discourage the development of pest populations and keep pesticides and other interventions to levels that are economically justified and reduce or minimize risks to human health and the environment". IPM emphasizes the growth of a healthy crop with the least possible disruption to agro-ecosystems and encourages natural pest control mechanisms.

As mentioned above, IPM uses all suitable techniques or strategies to keep pests below levels that cause unacceptable crop loss. These different strategies are usually placed under the main headings of mechanical, cultural, biological, and chemical methods of pest management.

B.1. IPM Concepts and Principles

Concepts

- IPM is farmer-based
- IPM is location specific (even down to the field level or crop growth stage),
- IPM is a combination of all suitable techniques.
- IPM must be considered as an integral part of crop production together with all agronomic techniques
- IPM considers the economics of pest management.
- IPM optimizes pesticide use, eliminates unnecessary use of pesticide, and promotes safe handling and application for the protection of health and the environment – as little as possible, as much as necessary.

General Principles of Integrated Pest Management

1. **Prevention and Suppression**

   The prevention and/or suppression of harmful organisms should be achieved or supported among other options especially by crop rotation, use of adequate cultivation techniques, use of resistant/tolerant cultivars and standard/ certified seed and planting material, use of balanced fertilization, liming and irrigation/drainage practices, preventing the spreading of harmful organisms by hygiene measures, and protection and enhancement of important beneficial organisms, e.g., by adequate plant protection measures or the utilization of ecological infrastructures inside and outside production sites.

2. **Monitoring**

   Harmful organisms must be monitored by adequate methods and tools, where available. Such adequate tools should include observations in the field as well as scientifically sound warning, forecasting and early diagnosis systems, where feasible, as well as the use of advice from professionally qualified advisors.

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3. **Decision-making**

Based on the results of the monitoring the professional user has to decide whether and when to apply plant protection measures.

Robust and scientifically sound threshold values are essential components for decision-making. For harmful organism's threshold levels defined for the region, specific areas, crops and particular climatic conditions must be taken into account before treatments, where feasible.

4. **Non-chemical methods**

Sustainable biological, physical and other non-chemical methods must be preferred to chemical methods if they provide satisfactory pest control.

5. **Pesticide selection**

The pesticides applied shall be as specific as possible for the target and shall have the least side effects on human health, non-target organisms and the environment.

6. **Reduced pesticide use**

The professional user should keep the use of pesticides and other forms of intervention to levels that are necessary, e.g., by reduced doses, reduced application frequency or partial applications, considering that the level of risk in vegetation is acceptable and they do not increase the risk for development of resistance in populations of harmful organisms.

7. **Anti-resistance strategies**

Where the risk of resistance against a plant protection measure is known and where the level of harmful organisms requires repeated application of pesticides to the crops, available anti-resistance strategies should be applied to maintain the effectiveness of the products. This may include the use of multiple pesticides with different modes of action.

8. **Evaluation**

Based on the records on the use of pesticides and on the monitoring of harmful organisms the professional user should check the success of the applied plant protection measures.

B.2. Advantages and Disadvantages of IPM

Advantages:

a) Lower cost intervention
   - Traditionally, the use of the pesticides to control the pest invasion would account to lots of cost.
   - Also, these pesticides need to be imported as well
   - The application of IPM would lessen the financial burden.
   - Moreover, different techniques involved in IPM are more sustainable with long lasting benefits

b) Benefits to the environment

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- Use of the pesticides are often linked degradation of the environment causing some more additional problems
- IPM is an eco-friendly approach and the effects on the environment is always considered before the application of any interventions
- Less use of pesticides won't affect the fertility of soil

c) Minimizes residue hazards of pesticides
   - It is obvious that in an IPM schedule the use of pesticides will be considerably reduced, hence the pesticide residue hazards will also get automatically minimized.

d) Anti-Resistance
   - The IPM model in itself is the anti-resistant mode for pest control.
   - It discourages the use of chemicals and thus creates less cases of anti-resistance.
   - Pesticides are used only when the other alternatives are not satisfying.

e) Useful and best intervention for the general public
   - Assurance of safe, reliable and low-cost pest control
   - The pest control will not affect the crops
   - Moreover, it is safe and affordable for the general public as well

Disadvantages or Limitations:

a) More involvement in the technicalities of the method
   - IPM needs to be planned
   - IPM demands more attention and dedication
   - Requires expertise of various field
   - All those involved in the IPM needs to be educated and trained which often requires much time.

b) Time and energy consuming
   - Application of IPM takes time.
   - Much time is needed in planning itself.
   - As IPM strategies differs from region to region, a separate plan is required for each region.
   - The expected results of intervention may take long time to be achieved.

Challenges in IPM:

- Financial constraints in initiation of the IPM
- Lack of involvement of the experts
- Inadequate research on pests
- Lack of supportive government and policies
- In developing countries where traditional farming is still practiced, farmers are unknown to the concept of IPM as well as IPM seems difficult to practice.

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UNIT IV. PRODUCTION PRACTICES
F. CROPPING SYSTEMS

I. Introduction

Cropping system is the most important component of a farming system. It indicates the yearly sequence and spatial arrangement of crops and fallow in an area. It is also based on the soil,

climate and the availability of water to be used as available resources. The cropping system has different types like monocropping which is practice of growing a single crop year after year on the same land, in the absence of rotation through other crops or growing multiple crops on the same land.

Every individual is aware of the bitter truth of the overpopulation problem which is increasing day by day at an alarming rate in today's world. Due to this rise in population the farmland is going down gradually. 92 million farm households are under marginal farm category which possesses < 1 ha as operational holding and out of this, 70 % of the farmers are having an area of below 0.5 ha land. By 2050, more than 95 % of holdings are estimated to be under the small and marginal category (Agarwal, 1995). In this context, it has become an essential part for the farming community to use their land holdings up to the maximum. Under such circumstances, a very effective method of farming can help the farmers to earn benefits from the same piece of land.

II. Learning Objectives

At the end of the session, the participant should be able to:
1. Identify and explain different types of cropping systems; and
2. Understand the advantages and disadvantages of cropping systems.

III. Learning Content

A. Monocropping

Monocropping is the practice of growing one crop on the land. This practice was recognized as a very economical way to provide farmers with a way to earn money, grow large amounts of a staple crop, like soy, corn, or wheat, and sell these crops off to any company willing to use it for food or fuel.

The problem with this is that the destruction that is done to the environment when monocropping is being engaged is far greater than the benefits it brings. The depletion of the nutrients in the soil, the use of pesticides on the crops, and not the entire crop is used for food. This leaves land that is even more reliant on the pesticides to fight off predators which in return puts more chemicals into the system. Finally, the entire reason we grow food is to eat it. This type of farming does not provide the diversity needed in our diets or to our ecosystem.

This is a modern cropping system in which different but specific crops are grown on the same piece of land in a logical or scientific sequence. The component crops are selected on the basis of complementary or supplementary relationship; deep soil feeders (e.g. yams) should follow shallow soil feeders (e.g. maize. The system is chemical intensive, and problematic because farmers practice a lot of deforestation and shorter fallowing.

B. Multistorey System

In a multistorey cropping system two or more crops of different heights are grown simultaneously on a piece of land in a certain period. A multi storey cropping system accommodates crops of different heights, canopy patterns and root system to maximize use of sunlight, nutrients, sustainable land use and maintain an ecological balance. In this system of cropping, the possibility of more efficient use of resources like sunlight, water, soil and nutrients is leading to increased biological diversity, more crops per unit area and sustainability of production.

Key of successful of a multistorey cropping system is the crop with diverse feature (growth habit, root depth and crop duration) such as coriander, radish, beetroot, cluster bean, cowpea etc. a useful in making multi-tier cropping systems.

Growing more than one crop at a time in the same field, farmers minimize soil erosion, which is a serious drawback of the monoculture cropping system. The practice of the multi storey cropping system is highly successful in plantation crops such as coconut, cashew and coffee for enhancing sustainable productivity and realizing higher income per unit area.

Advantages of multistorey cropping ystem

1. It gives maximum production from small plots. This can help farmers cope with land shortages along with income per unit area increasingly substantially.
2. Minimizes the risk of crop yield loss.
3. Improves physical properties and health status of the soil.
4. Including legumes in the cropping pattern helps maintain soil fertility by nitrogen fixation in the soil.
5. Different types of crops can be produced thereby providing a balanced diet for the family.
6. Weeds are suppressed due to high density planting.
7. Saves the crop from climatic aberrations like high rainfall, soil erosion, landslides etc.
8. Maintain an ecological balance.
9. Provides suitable micro-climate conditions that benefit the winter crops.
10. Efficient use of resources available.
11. Helps in maintaining ecological balance.

Components of a multistorey cropping System

A model with multiple cropping system, requires systematic planning with regard to selection of crops, planting, manuring and other management practices. The following points are to be considered while raising a garden with multi-storey cropping system.

1. Base crop: It is the main component of the system. The crop which is selected as the base crop should have tall growing, widely spaced and perennial nature. The wider spacing provides more scope for growing other crops.

2. Other crop: Locally adapted crop suits the best for growing in the interspaces of the base crop. The root system of these crops should be of varying depth so that they can draw nutrients from different layers of the soil to avoid competition among the crops. Secondly, the crop should be of different heights so that proper sunlight can be received by all the crops uniformly without any

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hindrances. Moreover, the canopy should be also of different size and structure to avail sufficient wind velocity. The crop in the inter spaces should be capable of growing under partial shade or shade to some extent because of the shade by the base crop.

Coconut Based Multistorey Cropping System

According to a study conducted in Central Plantation Crops Research Institute, Indian Council of Agricultural Research, Kasargod, area under each coconut tree spaced at 7.5 m × 7.5m is 56.25 m2. The maximum concentration of roots found to reach 12.57m2, which comprises 22.2% of the total area which means a provision of about 77.8 % is facilitated for growing other crops. The crop canopy coverage and solar energy utilization of single coconut plantation indicates that around 45–50 % of the sunlight reaches the ground surface without any hindrance which can be again an added advantage for the other crops (Nimbolkar et al., 2016).

Thus, it provides opportunities for growing short and medium duration inter-crops like banana-turmeric-ginger-pineapple-vegetables-papaya, sorghum-legumes, sunflower, tapioca, sweet potato, guava etc., following crop rotation in the initial 5 years of plantation (CPCRI, Kasargod, 2014-15). After 20-25 years, since 40 % sunlight will be provided in the inter row spaces, therefore shade loving crops can be taken up as other crop in this system like Coconut + Black pepper + Cocoa + Pineapple/ Turmeric / Ginger (Mathew et. al., 1993).

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STATE UNIVERSITIES AND COLLEGES – ASSOCIATION OF COLLEGES OF AGRICULTURE IN THE PHILIPPINES (SUC-ACAP)

SELF-ASSESSMENT QUESTIONS

Crop Science

SUC-ACAP-LEA

2021

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PREFACE

Republic Act 8435 or the Agriculture and Fisheries Modernization Act of 1997, simply known as AFMA, was enacted on December 22, 1997. It is a policy instrument defining measures to modernize Philippine agriculture for the country to compete in the global market. A basic human resource development framework in modernizing agriculture under the Act is to enhance the professional competency of agriculturists by upgrading the standards of the agriculture and fisheries education that will ensure to similarly upgrade the quality and sustainability of, and promote global competitiveness, at all levels.

One strategy for professionalizing the agriculture sector is the Licensure Examination for Agriculturists (LEA). First implemented in 2003, LEA serves as a standardized evaluation tool to identify graduates of Agriculture and Agriculture-related degree programs who are academically and technically fit to be admitted in the agricultural profession. The three-day examination consists of multiple-choice type questions in the following component subjects: Crop Science, Soil Science, Crop Protection, Animal Science, Agricultural Economics and Marketing, and Agricultural Extension and Communication.

The COVID-19 pandemic has been a major concern of all higher education institutions. When review sessions for licensure board exams are already supposed to be grinding, SUCs are confronted by the reality that face-to-face review is undeniably not possible. Under the circumstance, universities face new pressures to provide an engaging, motivating, and effective teaching and learning experiences for both reviewees and reviewers.

Following this, the SUC-ACAP, with its goal of advancing the agriculture professional competency among their member-institutions, and the College of Agriculture and Food Science, UP Los Baños embarked on a joint initiative to enhance the capability of the SUC faculty particularly those engaged in the conduct of review classes in preparation for the Licensure Exam for Agriculture (LEA). To achieve this, the joint initiative conducted three online lectures, namely, 1) Innovative Teaching Methods and Tools Designed for Online Review Classes by no less than the Chancellor of the UP Open University, Dr. Melinda dP. Bandelaria. This online course aimed at introducing the faculty participants to the particularities of conducting review classes online with the use of innovative methods and tools; 2) Test Construction: the Multiple-Choice Type Test from a seasoned UP faculty, Dr. Blanda R. Sumayao, which gave the faculty-participants the opportunity to review the principles and rules for constructing multiple-choice type tests; and 3) Design and Development of Review Materials for LEA, again with Dr. Bandelaria with the objective of providing the faculty-participants the necessary knowledge and skills in designing and developing review materials and planning for the delivery of online review sessions.

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The Project initiative also gave the faculty-participants the opportunity to review their course syllabus viz-a-viz CMO #14 S. 2008 or the CHED-prescribed outline which enabled them to design and develop this harmonized syllabus for each of the six core subject matter area. The review materials are thus a product of many minds and talents, that is, of faculty-participants from the different SUCs. I would like to acknowledge all the writers/contributors who willingly presented themselves to the initiative and for strictly following deadlines in submitting drafts; to all University Presidents and the Deans of the Colleges of Agriculture who have been highly committed to the objective of this Project initiative and for the support provided to the faculty-participants; to all Cluster and Sub-Cluster Coordinators for the constant reminders and follow-ups; and to the SUC-ACAP Secretariat for the untiring administrative support.

And lastly, we are grateful to the College of Agriculture and Food Science, UP Los Baños for being true to its commitment of being a public service University, providing scholarly and technical assistance to its sister and brother SUCs. Specifically, we thank the faculty of the CAFS who served as Subject Area Coordinators/Specialists who unselfishly guided our faculty participants throughout their journey in the project, as well as the Technical Assistants and Administrative Staff who have lend full support to attaining the objectives of the joint initiative.

As I have always reiterated in many occasions, let us be reminded that student preparation for the licensure is just one of the factors that would determine performance. It is still incumbent upon us to ensure that we will be able to provide the necessary prerequisite services and institutional support in order for students to ultimately pass successfully and help advance the students' professional careers post licensure exam.

The preparation for publication of these review materials was supported by the SUC-ACAP.

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December 2021

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FOREWORD

Unlike in *Lord of the Rings* and *Harry Potter* films, there are no wizards, Balrogs, Voldemort, precious rings, or elderly wands, on the journey to becoming a professional agriculturist. But there are narrow bridges and mazes that you must travel and cross over. One of these challenges is the Professional Licensure Exam for Agriculture.

In the current climate where globalization, changing natural conditions, production, and social composition; migratory movements, among others, have significantly impacted most agricultural landscapes in the developing world, more than ever, the Licensure Exam for Agriculture has become all the more relevant as it prepares graduates of Agriculture to embrace the tasks ahead of them with utmost professionalism by way of recognizing their expert knowledge and practice. The Licensure Examination for Agriculture was established to recognize the expert knowledge and practice of aspiring professional agriculturists.

This Review Material produced by the faculty participants of the joint initiative of the UPLB College of Agriculture and Food Science (UPLB-CAFS)- State Universities and Colleges-Association of Colleges of Agriculture in the Philippines (SUC-ACAP) is a comprehensive resource that covers all the six core subject areas in the field of agriculture namely, Agricultural Economics, Agricultural Extension, Animal Science, Crop Protection, Crop Science, and Soil Science. I commend the faculty of the CAFS who unselfishly devoted their time in coaching and mentoring the faculty participants as well as in providing direction during the development of the review materials. The materials hope to serve as a valuable review tool to students of SUC-ACAP member institutions.

I wish you the best in your journey and beginning your career as a professional agriculturist.



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CROP SCIENCE 1

Instruction: Choose the letter of the correct answer.

UNIT I INTRODUCTION

A. AGRICULTURAL DEVELOPMENT

1. The International Labor Organization reported that there are 58% total employment services of the country. How many percent of this is employed in the agricultural sector?
   A. 50%
   B. 48%
   C. 33%
   D. 23%

2. Which if the following is the typical water requirement of rice in Asia?
   A. 500-1000 mm
   B. 1000- 1100 mm
   C. 1100-1200 mm
   D. 1300−1500 mm

3. Based on the world's record of 2020, the volume of milled rice produced by the Philippines is:
   A. 14.7 tons
   B. 12.93 tons
   C. 11.93 tons
   D. 12.93 tons

4. Which type of farming system rely on rainfall as the main source of water for irrigation?
   A. irrigated lowland farming
   B. upland farming
   C. rainfed farming
   D. irrigated upland farming

5. This method allows the crops to be grown in soil-less condition which uses mineral-enriched water solutions.
   A. aquaponics
   B. hydroponics
   C. aeroponics
   D. nursery

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B. THE PHILIPPINE AGRICULTURE

1. It is the Philippine national tree that produces pretty yellow flowers and a source of hardwood for many uses.
   A. Mahogany
   B. acacia
   C. narra
   D. ipil-ipil

2. The staple food of the Filipinos.
   A. banana
   B. corn
   C. rice
   D. sweet potato

3. The *hacenderos* and *sacadas* class structures in Philippine agriculture started during this era.
   A. Japanes colonialism
   B. American colonialism
   C. Spanish colonialism
   D. Green revolution

4. A type of crop-based farming system where there is a complete integration of animal husbandry and crop production components. In this system, animals provide manure for crops, and crop residues are used to feed animals.
   A. lowland rice-based system
   B. smallholder scale mixed farming system
   C. integrated smallholder farming system
   D. agroforestry system

5. The rice granary of the Philippines.
   A. Isabela
   B. Nueva Vizcaya
   C. Nueva Ecija
   D. Laguna

6. An act which promotes the conservation and protection of wildlife species and their habitats to promote ecological balance and enhance biological diversity and regulate the collection and trade of wildlife.
   A. RA 8435
   B. RA 10068
   C. RA 9147
   D. RA 7586

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7. The National Integrated Protected Areas System (NIPAS) Act of the Philippines.
   A. RA 8435
   B. RA 10068
   C. RA 9147
   D. RA 7586

8. The Philippines' biggest agricultural export.
   A. banana
   B. coconut
   C. rice
   D. corn

9. The second top agricultural export crop of the country.
   A. coconut
   B. rice
   C. corn
   D. banana

10. These are ecosystems that are modified and managed by man to produce outputs that are useful to him.
   A. agroecosystem
   B. farming system
   C. natural system
   D. artificial system

11. These are things that take longer than a person's lifespan to be replaced. They can take millions of years to form.
   A. natural resources
   B. renewable resources
   C. non-renewable resources
   D. none of the above

12. A variety of rice developed by IRRI in 1968 and also called the "miracle rice."
   A. IR5
   B. IR6
   C. IR7
   D. IR8

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C. MAJOR CROP STATISTICS OF THE PHILIPPINES

1. A major grass in Central Luzon.
   A. mungbean
   B. sugarcane
   C. millet
   D. sorghum

2. A major fruit in ARMM.
   A. mangosteen
   B. marang
   C. durian
   D. strawberry

3. Major field crop and vegetable in Southern Mindanao.
   A. corn and potato
   B. rice and pechay
   C. banana and rose
   D. corn and cabbage

4. A major fruit in Western and Central Visayas.
   A. onion
   B. mango
   C. lanzones
   D. pineapple

5. The center of origin of rice (*Oryza sativa*) is:
   A. India
   B. Philippines
   C. Africa
   D. Mexico

6. Which of these crops can be classified both as an agronomic and horticultural crop based on their uses under Philippine condition?
   A. rice
   B. okra
   C. sorghum
   D. mungbean

7. Next to rice and corn, the top 3 crops of the Philippines include:
   A. coconut, banana, abaca
   B. coconut, sugarcane, banana
   C. coconut, wheat, sugarcane
   D. coconut, wheat, sugarcane

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8. Crops that could be propagated by crown suckers and slip.
    A. abaca
    B. pineapple
    C. banana
    D. gabi

9. Industrial crops classified as beverage.
    A. coconut
    B. papaya
    C. cacao
    D. vanilla

10. The enlarged fleshy underground part of the following crops are generally classified as stems, except.
    A. cassava
    B. potato
    C. ginger
    D. gabi

UNIT 2. PHYSIOLOGICAL PROCESSES AFFECTING CROP PRODUCTION

A. RESPIRATION

1. Which of the following factors affect respiration?
    A. age of the plant
    B. temperature
    C. level of oxygen
    D. carbon dioxide concentration
    E. all of the above

2. Which of the following statements is TRUE about respiration?
    A. Active parts of a plant respire more than the dormant parts.
    B. Stored orthodox seeds respire less than recalcitrant seeds.
    C. Leaf buds respire less than developing leaves.
    D. Young tissue respires more than old tissue.
    E. None of the above

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B. MINERAL NUTRITION

1. Which of the following elements is the central component of a chlorophyll molecule?
   A. C
   B. N
   C. H
   D. Mg
   E. S

2. Which of the following is a mobile nutrient within plants?
   A. P
   B. Mo
   C. Ca
   D. Fe
   E. O

3. Which of the following is considered a beneficial element?
   A. Al
   B. Zn
   C. Si
   D. Mn
   E. Cl

C. GROWTH AND DEVELOPMENT & HORMONES AND GROWTH REGULATION

1. The process by which cells become specialized into different types with different functions.
   A. Development
   B. Growth
   C. Differentiation
   D. Morphogenesis

2. Plant hormones can be defined or described as
   A. organic chemical substances that control physiological processes in plants
   B. enzymes that catalyzes growth and development
   C. organic compounds that is effective if applied in low concentrations
   D. All of the above
   E. A and C only

3. Which herbicides belong to the auxin-type
   A. glyphosate
   B. diquat
   C. paraquat
   D. 2,4-D
   E. Propanil

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4. The ratio of the leaf surface to the ground area occupied by a crop.
   A. CGR
   B. NAR
   C. LAI
   D. SLA
   E. LWR

5. Plant hormone that was first isolated from fungus that causes bakanae disease in rice
   A. Auxin
   B. Cytokinin
   C. Gibberellins
   D. Ethylene
   E. Abscisic acid

6. Spraying this hormone would develop parthenocarpy in fruits and induce growth of Thompson grapes.
   A. Salicylic acid
   B. Cytokinin
   C. Gibberellin
   D. Ethylene
   E. Abscisic acid

7. Zeatin is a type of ____.
   A. Auxin
   B. Cytokinin
   C. Gibberellin
   D. Ethylene
   E. Abscisic acid

8. A hormone responsible for closing of stomates in drought-stressed plants.
   A. Auxin
   B. Cytokinin
   C. Gibberellin
   D. Ethylene
   E. Abscisic acid

9. Hormones that have roles in apical dominance
   A. Auxin and Gibberellins
   B. Auxin and Abscisic acid
   C. Cytokinin and Ethylene
   D. Cytokinin and Auxin
   E. Abscisic acid and ethylene

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10. Hormone used in hybrid rice seed production to produce male sterile flowers
    A. Auxin
    B. Cytokinin
    C. Gibberellins
    D. Abscisic acid
    E. Ethylene

11. Expansion of a leaf or increase in leaf area is an example of:
    A. Development
    B. Growth
    C. Differentiation
    D. Morphogenesis

12. A fruit ripening hormone
    A. Auxin
    B. Cytokinin
    C. Gibberellins
    D. Abscisic acid
    E. Ethylene

13. Which of the following can describe the swelling of a piece of wood when placed under water?
    A. Development
    B. Growth
    C. Differentiation
    D. Morphogenesis
    E. None of the above

14. The increase in volume of dead seeds during germination testing is a sign of:
    A. Development
    B. Growth
    C. Differentiation
    D. Imbibition of water
    E. Dedifferentiation

15. These hormones are involved in the resistance of plants to pathogens because of the production of pathogenesis-related proteins.
    A. Brassinosteroids
    B. Auxin
    C. Methyl jasmonates
    D. Cytokinin
    E. None of the above

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16. The allometry of top growth to root growth, usually expressed as the shoot – root (S–R) ratio, is an internal growth mechanism in plant under:
    A. Circadian rhythm
    B. Correlation
    C. Endogenous rhythm
    D. Annual rhythm

17. Agronomists generally define growth as:
    A. Increase in dry matter
    B. Process of cell division and elongation
    C. Irreversible increase in size
    D. Series of qualitative changes

18. The transition from vegetative to reproductive stage is:
    A. Development
    B. Growth
    C. Differentiation
    D. Morphogenesis

19. A net gain of assimilate per unit leaf area and time.
    A. CGR
    B. NAR
    C. LAI
    D. SLA
    E. LWR

20. The changes that a bean plant undergoes from seed germination to maturation and senescence is termed as ____.
    A. Development
    B. Growth
    C. Differentiation
    D. Morphogenesis

D. PLANT MOVEMENT

1. Pea tendril coiling around a support is an example of:
    A. Thermotaxis
    B. Thigmonasty
    C. Thigmotaxis
    D. Thigmotropic

2. Curving of plant toward light is called
    A. Thigmotropism
    B. Gravitropism
    C. Phototropism
    D. Thismonasty

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3. Sleep movement are mediated by turgor changes in specialized motor cells located in
   A. pulvinus
   B. stem
   C. leaf blade
   D. roots

4. Roots perceived gravity by a mechanism that involved sedimentation of
   A. chloroplast
   B. statoliths
   C. ribosomes
   D. mitochondria

5. Which of the following is not true for nastic movements?
   A. Depend of the direction of the stimulus
   B. Are independent of the direction of the stimulus
   C. Genetically predetermined
   D. Driven by growth or turgor changes

6. The rapid movement of leaflets in Mimosa when touched is an example of what type of movement?
   A. growth movement
   B. movement with locomotion
   C. turgor movement

7. It refers to non-directional responses to stimuli.
   A. Tropic movement
   B. Tactic movement
   C. Nastic movement

8. It refers to a movement due to unequal growth on two sides of plant organs like stem, roots, tendrils, buds.
   A. growth movement
   B. movement with locomotion
   C. turgor movement

9. It refers to a nightly process wherein the leaves and the petals of flowers close at night and open during the day.
   A. nyctinastic
   B. chemonastic
   C. seismonastic

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10. It refers to a movement of young stem in zigzag style due to alternate changes in development of opposite side of the apex.
    A. epinasty
    B. nutation
    C. hyponasty

E. CROP ADAPTATION

1. The capacity of plants to adapt to their environment enables them to survive under wide range of changing environmental conditions.
    A. Physiological ecology
    B. Ecosystem
    C. Ecology
    D. Crop adaptation

2. Two principles that govern plant response to environment
    A. Vegetation and flora
    B. Saturation and limiting factor
    C. Tolerance and inhibition
    D. Flora and fauna

3. Plant response gradually changes with continuous exposure to the environmental factor (e.g., development of drought resistance)
    A. Direct
    B. Triggered
    C. Conditioning effects
    D. Homeostasis

4. Internal conditions are constantly maintained in spite of changes in the environment (e.g., internal concentration of growth hormones)
    A. Direct
    B. Triggered
    C. Conditioning effects
    D. Homeostasis

5. Factors affecting ecological plant distribution except:
    A. Water
    B. Temperature
    C. Topography
    D. Atmosphere

6. Refers to the plant's fitness for the unfavorable environment.
    A. Resistance
    B. Water stress
    C. Tolerance
    D. Genetics

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7. Too much UV-B radiation may result in the following except:
   A. Inhibition of root elongation
   B. Reduced leaf area
   C. Chlorosis
   D. Decreased seed viability

8. Restricts the volume of soil that the root system can exploited due to high mechanical resistance of the soil
   A. Compacted soil
   B. Heavy metals
   C. Environmental pollutants
   D. Flooding

9. Plants with similar genetic make-up that exhibit morphological and physiological differences due to varied natural environments.
   A. Ecophenes
   B. Ecotypes
   C. Resistance
   D. Genetics

10. Plants that can thrive in relatively acidic environments are known as _____.
   A. Calciphytes
   B. Acidophiles
   C. Acidopytes
   D. Halophytes

11. The plant's capability to maintain tissue activity through accumulating high Na^+ in older leaves, rather than in younger leaves, where leaf senescence appears first
   A. Tissue tolerance
   B. Drought acclimation
   C. Drought stress
   D. Vesicular Transport

12. The following are the effect of heavy metal on the physiological processes of plants, except for:
   A. Inhibited cell division in the root meristematic zone, thus resulting in diminished growth
   B. Reduces root cellular turgor leading to inhibition in cell enlargement
   C. Adversely influences the cell division machinery that is responsible for the cell cycle regulation in plants
   D. None of the above

13. Which of the following traits help plants to overcome water loss?
   A. Higher osmotic pressure in plants
   B. Higher aboveground plant growth under drought
   C. Decreased number of stomata and sunken stomata in the epidermal cells
   D. All the above

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14. Ability of plants to continue normal physiological activities in very dry conditions for prolonged periods of time
    A. Drought avoidance
    B. Drought tolerance
    C. Tissue tolerance
    D. Root plasmolysis

15. Which of the following bacteria has been added to wheat varieties that improved water relations and improved cell wall elasticity indicating that there is combined higher water status plus increased elastic adjustment for enhanced drought tolerance?
    A. Azospirillumbrasilense Sp245
    B. Azospirillumbrasilense Sp254
    C. Azotobacterbeijerinckii Sp245
    D. Azotobacterbeijerinckii Sp254

16. Which of the following cellular macromolecules can be oxidized and damaged by reactive oxygen species (ROS)?
    A. protein
    B. membrane lipid
    C. nucleic acids
    D. All the above

17. Halophytes can tolerate high levels of salts in their cytoplasm because
    A. Halophytes that accumulate large amounts of Na^+ ions (i.e., includers) pump Na^+ into the cytoplasm by specific transporters in order to avert accumulation of ions in the vacuoles
    B. They prevent intracellular Na^+ ion accumulation (i.e., excluders) through maintaining Na^+ in their shoots and leaves
    C. They regulate of ion transport from root to shoot to cope with adverse effects of salinity
    D. The vacuoles of the halophytes remodel the composition of their membrane lipids to avoid Na+ flow back into the mitochondria

18. The temperature identified as damaging for key reproductive processes in different crop species
    A. 21-25
    B. 26-29
    C. 30-40
    D. None of the above

19. Ultra-morphological alterations revealed in various plant parts under various heavy metal toxicities
    A. severe plasmolysis
    B. deficiency/absence of chromatin materials
    C. disruption of nuclear and plasma membranes
    D. All of the above

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20. Types of halophytes that have specific salt excretory structure originating from the epidermis
    A. Glycophytes
    B. Recretohalophytes
    C. Aero-halines
    D. Aqua-halines

UNIT III. FACTORS AFFECTING CROP PRODUCTION

A. PROPERTIES OF SOILS AFFECTING NUTRIENT AVAILABILITY

1. Which of the following is TRUE?
    A. High organic matter increases water holding capacity of soil.
    B. Organic matter makes up 50% in typical soil.
    C. Humus contributes to increased salinity.
    D. High sodium concentration leads to acidity of soil.

2. Which form of N is ideal for plant uptake?
    A. N2
    B. NO3-
    C. NH3+
    D. N2O

B. Crop Response to Fertilization

1. Which of the following pH range is ideal for crop growth?
    A. 4.5-7.0
    B. 4.5-6.0
    C. 5.5.-7.5
    D. 5.5-8.0

2. High amount of nutrient in the soil can lead to
    A. mineralization in the soil
    B. unavailability of other nutrients
    C. loss of other nutrients in the soil
    D. none of the above

C. Soil Fertility Management

1. Which of the following leads to nutrient loss in the soil?
    A. evaporation
    B. leaching
    C. surface run-off
    D. composting

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2. Which of the following affects the availability of nutrients for root uptake?
   A. soil depth
   B. soil tilth
   C. soil pH
   D. soil texture

D. Genetically Modified Cotton

1. The GM cotton is the _____ largest GM crop?

   Answer: 3rd

2. What is the most common trait in GM crops?

   Answer: herbicide resistant

3. How many of all GM crops are herbicide resistant?

   Answer: 59

4. True or false? Bollgard II is the second insect-resistant GM cotton certified in 2003, and it differs from Bt cotton in that it has two genes from the Bacillus thuringiensis (Bt) soil bacteria rather than one.

   Answer: True

5. Fill in the blank. In 2001, _____ cotton and _____/Bt cotton became commercially available in Australia for the first time.

   Answer: Round up Ready

6. The first transgenic cotton (BG I) carrying Cry1Ac was approved for commercial use in Australia and the _____ in 1996.

   Answer: USA

7. The Office of the Gene Technology Regulator (OGTR) approved the commercial release of _____cotton varieties in February 2006.

   Answer: Roundup Ready FLEX

8. The ____is the third protein in Bollgard 3.

   Answer: VIP3a

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9. The private agency that developed Liberty Link® cotton.

   Answer: Bayer Crop Science

10. GM cotton was 1st commercially cultivated in _____.

    Answer: Mexico

E. Genetically Modified Bt Corn

1. What donor organisms are found on Bt corn?
   A. Soil fungus
   B. Soil bacterium
   C. Soil mycelia
   D. Soil virus

2. The target insect killed by the proteins found in Bt corn belong to:
   A. Thysanoptera
   B. Coleoptera
   C. Diptera
   D. Lepidoptera

3. Will insects develop resistance to the toxins produced in Bt corn?
   A. It is impossible for insects to develop resistance to Bt corn.
   B. It is unlikely that insects will develop resistance to Bt corn.
   C. Under certain conditions insects may develop resistance to Bt corn.
   D. It is almost certain that insects will develop resistance to Bt corn.
   E. Insects are already resistant to the toxins produced in Bt corn.

4. Does Bt corn or Bt cotton only kill specific pests that damage the crop?
   A. The Bt toxin kills all insects.
   B. The Bt toxin kills European corn borer and other flying insects.
   C. The Bt toxin kills the European corn borer and its close relatives.
   D. The Bt toxin only kills the insects for which it is targeted.
   E. The Bt toxin repels but doesn't kill insects.

5. When did crops become resistant to herbicides?
   A. Crops have always been resistant to some herbicides.
   B. After the introduction of Bt corn in 1997.
   C. After the introduction of Roundup-ready soybeans in 1996.
   D. Crops are not resistant to herbicides.

6. Can genes escape from genetically modified crops and jump to other plants?
   A. Yes, and often do.
   B. Only to some crops, but those crops aren't genetically modified.
   C. Only during rare climatic conditions.
   D. No, genes cannot move from species to species without human intervention.

SAP-IEA

7. If we make plants that survive in regions where they normally wouldn't survive, such as very cold regions, could this cause unexpected ecological changes?
    A. No, the crops are only suited for cultivated fields.
    B. It is possible that a crop might invade the surrounding ecosystem.
    C. It is certain that the crop would move from the fields into the surrounding ecosystem.

8. How long does it take to develop a new genetically modified crop?
    A. Twenty years.
    B. Ten years.
    C. Five years.
    D. One year.
    E. Six months.

9. Can scientists predict with certainty where an inserted gene will go on a plant chromosome?
    A. With modern genetic techniques, scientists can insert genes precisely.
    B. Genes are inserted on the proper chromosome, but there is no control on where it goes on the chromosome.
    C. Scientists have a general idea of where the gene will go and what it will do to the plant.
    D. It's just a shot in the dark.

10. Bt corn is safe for humans because?
    A. It was slow to digest, it provided longer lasting protection against insect damage.
    B. Lower mycotoxins compared to non Bt
    C. Their mode of action requires a basic pH and some specific receptors and enzymes.
    D. all of the above

UNIT IV. SUSTAINABLE AGRICULTURE

A. Man and Crops in an Ecosystem

1. This concept refers to the ability to meet the demands of the present without interfering with the capacity to meet the needs of future generations.
    A. sustainability
    B. net productivity
    C. biodiversity
    D. agroforestry

2. It refers to the complex system of interactions between the living and non living organisms..
   A. habitat
   B. agroecosystem
   C. ecosystem
   D. community

Flow of Energy

1. It refers to the rate at which the plant fixes energy in organic compounds.
   A. Primary productivity
   B. Net productivity
   C. Gross productivity
   D. Community productivity

2. A geographic area where plants, animals and other organisms, as well as the landscape work together to form a bubble of life.
   A. Structure
   B. Place
   C. Ecosystem
   D. Community

3. An arrangement of organisms of an ecological community according to the area of predation in which it uses the next usually lower member as food source.
   A. Food chain
   B. Food sequence
   C. Food source
   D. Food producer

4. How many percent of the energy is transferred from one trophic level to the next trophic level?
   A. 5%
   B. 10%
   C. 15%
   D. 20%

5. These are organisms which produce their own food?
   A. Heterotrophs
   B. Autotrophs
   C. Consumer
   D. Decomposers

6. The second trophic level of organisms that gets energy by eating other organisms.
   A. Consumers
   B. Decomposers
   C. Producers
   D. Representative

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7. Refers to the total rate of photosynthesis in the ecosystem.
   A. Net primary productivity
   B. Net community productivity
   C. Gross community productivity
   D. Gross primary productivity

8. It is the rate of energy storage in plant tissues in excess of the rate of respiration during the measured time interval.
   A. Net primary productivity
   B. Net community productivity
   C. Gross community productivity
   D. Gross primary productivity

9. How many percent of energy in the ecosystem is transferred to the atmosphere as heat, to fuel body processes such as growth, repair and reproduction.
   A. 80%
   B. 90%
   C. 70%
   D. 60%

10. How does energy move through the ecosystem?
   A. Single direction
   B. Two directions
   C. Backward direction
   D. Sideward direction

Cycling of Nutrients

1. These are chemicals required for plant and animal growth and other life processes.
   A. Nutrients
   B. Substance
   C. Amendments
   D. Ingredients

2. It is an essential component of cells and life sustaining chemical reaction. It is recycled through living and decaying organisms in the atmosphere, bodies of water, soil and rock.
   A. Nitrogen
   B. Phosphorus
   C. Carbon
   D. Hydrogen

3. A system where energy and matter are transferred between living organisms and non- living parts of the environment.
   A. Nitrogen cycle
   B. Nutrient cycle
   C. Carbon cycle
   D. Phosphorus cycle

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4. What do you call a chemical reaction that converts solar energy and atmospheric carbon dioxide gas into chemical energy?
   A. Photosynthesis
   B. Respiration
   C. Decomposition
   D. Absorption

5. Which of the following is not included in the processes of nitrogen cycle?
   A. Nitrogen fixation
   B. Nitrification
   C. Denitrification
   D. Incorporation

6. The following are the main processes in the carbon cycle, which one is NOT?
   A. Photosynthesis
   B. Conversion
   C. Cellular respiration
   D. Decomposition

7. In this process the ammonium is converted into nitrate and nitrite through the work of nitrifying bacteria.
   A. Denitrification
   B. Nitrification
   C. Uptake
   D. Nitrogen fixation

8. In this process the nitrogen gas is converted into nitrate and ammonium compounds that are usable by plants.
   A. Denitrification
   B. Nitrification
   C. Uptake
   D. Uptake

9. What is being formed when fossil fuel combustion and burning organic matter releases nitrogen into the atmosphere?
   A. Acid rain
   B. Carbon dioxide
   C. Smoke
   D. Ammonium

10. It is one of the most important components of DNA and proteins.
    A. Nitrogen
    B. Carbon
    C. Phosphorus
    D. Potassium

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CROP SCIENCE 2

UNIT I. SITE CHARACTERIZATION FOR CROP PRODUCTION

A. BIOPHYSICAL AND SOCIO-ECONOMIC DETERMINANTS

1. This factor in site selection pertains to the environmental condition of the site.
   A. ecological
   B. economical
   C. sociological
   D. operational

2. The existing plants/vegetation in the site is an aspect under:
   A. ecological
   B. economical
   C. sociological
   D. operational

3. The following are economic factors of site selection, EXCEPT:
   A. cost of land
   B. local taxes
   C. labor cost
   D. relative humidity

4. Which among the following is an example of a biological factor?
   A. presence of insect pests
   B. income of the farmers
   C. amount of rainfall
   D. accessibility to the road

5. In a high elevation area with low temperature and high relative humidity, what crops should not be grown there?
   A. cabbage and cauliflower
   B. cereal crops
   C. irish potato and carrots
   D. grapes and strawberries

6. Physical features of the land defined by the contour, elevation, and depressions. This is the natural shape and slopes of the land. Typical land features and contour forms are flat, ridge, valley, hill, slide.
   A. topography
   B. soil
   C. land
   D. abiotic factor

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B. POLITICAL DETERMINANTS

1. The government agency that assists in planting tree species is _____.
   A. BFAR
   B. DA
   C. PCARRD
   D. DENR

2. The corporation that helps farmers who are engaged in sugarcane farming is _____.
   A. IRRI
   B. BPI
   C. SRA
   D. DA

3. For farmers to have the new rice varieties, help is available from _____.
   A. DA
   B. PHILRICE
   C. BPI
   D. IPB, College, Laguna

4. For recommendations on soil fertilization/amendments, farmers need help from _____.
   A. BSWM
   B. BAR
   C. FPA
   D. BPI

5. The food regulatory function of NFA has been transferred to _____.
   A. Agricultural Training Institute
   B. Fertilizer and Pesticide Authority
   C. Phil Rice Research Institute
   D. Bureau of Plant Industry

6. In response to the Covid -19 pandemic, the Dept of Agriculture is promoting _____.
   A. *plantitas* production
   B. swine raising
   C. rice production
   D. backyard vegetable gardens

7. These are among the crops which have gov't price support policies:
   A. potatoes and coconuts
   B. corn and vegetables
   C. rice and corn
   D. rice and sugarcane

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8. Agricultural and Fisheries Mechanization Act (AFMech Law) is
    A. R. A. 8435
    B. R. A. 11203
    C. R. A. 10068
    D. R. A. 10601

9. The organic Agriculture Act of 2010 is _____.
    A. R. A. 11203
    B. R. A. 8435
    C. R. A. 10601
    D. R. A. 10068

10. Training of farmers for agricultural production practices, etc. is done by _____.
    A. BPI
    B. UP at Los Baños
    C. BAR
    D. ATI

UNIT II. PRODUCTION PRACTICES

A. LAND PREPARATION FOR ANNUAL AND PERENNIAL CROPS (PRACTICES FOR SOIL EROSION CONTROL)

1. Which horizon does refer to the first sign of soil formation wherein bedrocks are unconsolidated?
    A. Horizon A
    B. Horizon B
    C. Horizon C
    D. Horizon D

2. This cover crop can be grown even if the trees have attained their full size and canopy.
    A. Kudzu
    B. Calopogonium
    C. Bluternnate
    D. Centrosema

3. What do you call a tool instrument that is used for locating contour lines?
    A. Plow board
    B. A-Frame
    C. Digging bar
    D. Tooth harrow

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4. Which barrier to control soil erosion on-farm does not belong to the group?
    A. rock wall
    B. Pole wall
    C. bamboo wall
    D. river wall

5. The plant root concentration is mostly found in _____.
    A. Horizon A
    B. Horizon B
    C. Horizon C
    D. Horizon D

6. This system can raise the income of farmers, yet it can be a threat to diversity and soil deterioration.
    A. Sod system
    B. Clean culture
    C. Mulch system
    D. Cover cropping

7. Coconut is one of the top exporter crops in the country. To augment farmers' income, there shall be cash crops to be grown under the trees. Which plant could not tolerate sunlight exposure?
    A. corn
    B. watermelon
    C. banana
    D. gabi

8. Which type of erosion could be potentially controlled and stabilized on-farm with high water streaming?
    A. rill erosion
    B. sheet erosion
    C. ravine erosion
    D. gully erosion

9. Which of the following barriers can be grown at 0-2000 masl and it is shade tolerant which serves as organic fertilizer?
    A. Sunflower
    B. Cogon
    C. Ipil-ipil
    D. Flemengia

10. Which among statements does NOT describe a soil profile?
    A. It has different layers with distinct characteristics called soil horizons.
    B. The soil horizon R represents bedrock that has consolidated rocks.
    C. Topsoil from uphill is usually brought by water with erosive power downhill.
    D. The soil profile constitutes layers: A, B, K, and D.

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11. Soil erosion is prevalent in the mountainous areas in the country. What do you think is an element that can greatly affect and aggravate this phenomenon?
    A. wind
    B. light
    C. water
    D. gas

12. Upland dwellers are continuously doing activities against the environment such as slash and burn, cutting of trees, etc. What drives them to do so?
    A. Politics in urban areas are corrupt. They do not want to be part of their lies and promises.
    B. There is no money in urban areas. They will eventually get hungry and in poverty.
    C. Their siblings were attending school. They planned to become professionals and keep away from farming and wanted to migrate to urban-populated areas to make a living.
    D. There are a lot of trees to cut for wood and fuel. The soil has more nutrients and is good for the cultivation of crops.

13. In calibrating an A-frame to locate contour lines, the crossbar should be_____.
    A. at the left mark of the crossbar.
    B. at the right mark of the crossbar.
    C. at the midpoint mark of the crossbar.
    D. at any marks of the crossbar.

14. Which of the following practices has led to high erosion risk?
    A. Plowing downhill
    B. b. Plowing across the slope without hedgerows
    C. a and b
    D. Plowing along contour lines with vegetative covers

15. *Imperata cylindrica* is used to control erosion. Which type of barrier does this plant belong to?
    A. Alley crops barrier
    B. Constructed barrier
    C. Natural barrier
    D. Planted barrier

B. PLANTING MATERIALS SELECTION AND PREPARATION

1. Method of plant propagation using seed.
    A. grafting
    B. asexual
    C. sexual
    D. micropropagation

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2. Method of asexual propagation under aseptic condition.
   A. grafting
   B. cutting
   C. tissue culture
   D. budding

3. One of the oldest and most widely used methods of grafting.
   A. T – budding
   B. side grafting
   C. cleft grafting
   D. cuttings

4. Cuttings from runners are usually used to propagate ___.
   A. citrus
   B. black pepper
   C. ornamental plants
   D. durian

5. Stems still attached to their parent plant may form roots where they come in contact with a rooting medium __.
   A. Embryo culture
   B. layering
   C. grafting
   D. cutting

6. Characteristic of a good quality seed.
   A. dormant
   B. immature
   C. viable
   D. none

7. The easiest and quickest ways to propagate many plants.
   A. Grafting
   B. cutting
   C. separation & division
   D. tissue culture

C. PLANTING METHODS

1. It is a seeding method where seeds are deposited in a row or furrow at a uniform depth either manually or mechanically.
   A. foliar
   B. broadcast
   C. transplant
   D. drill

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2. Calculate the number of guava seedlings needed to establish a 0.09 hectare (could be 1 ha) (30m x 30m) field using a quincunx system of planting with spacing of 120cm between rows and 120cm between hills.
   A. 1201
   B. 1102
   C. 1020
   D. 1212

3. Calculate the number of abaca seedlings needed to establish a 1.15 hectare field using a triangular system of planting with spacing of 8.5m between rows and 8.5m between hills.
   A. 183.81
   B. 183.79
   C. 183.82
   D. 183.80

D. WATER SUPPLY AND MANAGEMENT (IRRIGATION PRACTICES)

1. Water is essential for plant growth for the following reasons, except:
   A. it enables the application of fertilizer
   B. it cools the soil and its surroundings
   C. it decreases the adverse effects of drought on crops
   D. it helps washes out or dilutes salts in the soil

2. What is the purpose of irrigation and drainage?
   A. Maintain soil moisture for maximum productivity
   B. To provide water for wilting crops
   C. Maintain soil composition which is essential for planting
   D. Limit the number of pests for crops

3. Which of the following is not an irrigation method?
   A. Sub-irrigation
   B. Watering
   C. Basin Flooding
   D. Sprinkling

4. It is the man-made application of water to soil to increase the moisture of the root zone soil naturally accessible for maximum crop yield.
   A. Irrigation
   B. Tilling
   C. Harrowing
   D. Mowing

5. Which of the following factors of planning irrigation considers the difference between average and extreme weather conditions which in turn determine the reliability of supply and the ability of the system to keep water up to the crop?

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A. Water Source
B. Water Quality
C. Water Characteristics
D. Water Quantity

6. These are the factors that will influence the decision on selecting what irrigation method is appropriate to apply, except:
   A. Operating costs of labour, fuel and maintenance.
   B. Ability to apply larger quantities of water at more frequent intervals.
   C. Personal preference and experience.
   D. Additional purposes, such as fertilizer application or frost control.

7. This method consists of running comparatively large streams into relatively level plots surrounded by levees.
   A. Ditch Flooding
   B. Check Flooding
   C. Border Flooding
   D. Basin Flooding

8. It is a method of applying irrigation water which is similar to natural rainfall.
   A. Rainfall Method
   B. Sub-irrigation Method
   C. Sprinkling Method
   D. Flooding Method

9. The following are sources of water supply for irrigation, except:
   A. regulated watercourses
   B. groundwater
   C. unregulated watercourses
   D. seawater

10. Soil characteristics may be different between different locations within a field, and change with depth.
    A. Variability of Soil Characteristics
    B. Physical Characteristics
    C. Topography
    D. Chemical and Biological Characteristics

11. It is an automated sprinkler system which has a number of metal frames involving a rotating pipe.
    1. Lateral Move Irrigation
    2. Drip Irrigation
    3. Center Pivot Irrigation
    4. Automatic Rotating Irrigation

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12. It strongly influences how uniform water can be applied through different irrigation methods to provide optimal soil water conditions for crop growth and marketable yields.
    A. Irrigation planning decision
    B. Water management decision
    C. Irrigation management decision
    D. Water irrigation decisions

13. The following are factors on which the most appropriate irrigation method for an area depends upon, except:
    A. management skills
    B. physical site condition
    C. crops to be grown
    D. irrigation plot

E. It is the application of water to an area typically levelled to zero slope and surrounded by dikes or check banks to prevent runoff.
    A. Enclosed irrigation
    B. Furrow irrigation
    C. Border irrigation
    D. Basin irrigation

15. The water is distributed across the land by raiding the water table, through a system of pumping stations, canals, gates and ditches.
    A. Manual irrigation
    B. Sub-irrigation
    C. Sprinkler irrigation
    D. Surface irrigation

E. PEST MANAGEMENT

1. Which is NOT a cultural pest management technique?
    A. trap
    B. crop rotation
    C. sanitation
    D. trap crop

2. It is the ability to endure pest attack without suffering from significant loss of yield.
    A. Resistance
    B. Economic threshold
    C. tolerance
    D. economic damage

3. It is a biological control technique that enhances the effectiveness of existing natural enemies to pests in the ecosystem.

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A. Augmentation
B. Conservation
C. Inoculation
D. Importation

4. Choose the letter of the correct answer.
   I. Growing two or more crops simultaneously during part of the life cycle of each crop is mixed intercropping.
   II. Growing two or more crops at the same time with no distinct row arrangement is known as relay intercropping.
      A. Both statements are false
      B. Both statements are true
      C. I is true, II is false.
      D. I is false, II is true.

5. A deliberate alteration of the production system by targeting the pest itself through agronomic practices to avoid or reduce pest injuries to crops.
   A. Mechanical Control
   B. Cultural Control
   C. Chemical Control
   D. Biological Control

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ANSWER KEY FOR CROP SCIENCE 1

Unit I. INTRODUCTION

A. Agricultural Development

1. D
2. D
3. C
4. C
5. B

B. The Philippine Agriculture

1. C
2. C
3. C
4. C
5. C
6. C
7. D
8. B
9. D
10. B
11. C
12. D

C. Major Crop Statistics of the Philippines

1. B
2. C
3. A
4. B
5. A
6. D
7. B
8. B
9. C
10. A

UNIT II. PHYSIOLOGICAL PROCESSES AFFECTING CROP PRODUCTION

A. Respiration

1. E
2. C

B. Mineral Nutrition

1. E
2. A
3. C

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C. Growth and Development

1. C
2. E
3. D
4. C
5. C
6. C
7. B
8. E
9. B
10. C
11. B
12. E
13. E
14. D
15. E
16. B
17. A
18. A
19. B
20. A

D. Plant Movement

1. B
2. C
3. A
4. B
5. A
6. C
7. C
8. A
9. A
10. B

E. Crop Adaptation

1. D
2. B
3. C
4. D
5. C
6. A
7. A
8. A
9. A
10. B
11. A
12. D

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13. C
14. B
15. A
16. D
17. C
18. C
19. D
20. B

UNIT III. FACTORS AFFECTING CROP PRODUCTION

A. Properties of Soils Affecting Nutrient Availability

1. A
2. B

B. Crop Response to Fertilization

1. C
2. B

C. Soil Fertility Management

1. B
2. C

D. Genetically Modified Bt Corn

1. B
2. D
3. D
4. C
5. A
6. A
7. B
8. B
9. D
10. A

Unit IV. SUSTAINABLE AGRICULTURE

E. Man and Crops in an Ecosystem

1. A
2. C

Flow of Energy

1. A
2. C
3. A
4. B

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5. B
6. A
7. D
8. A
9. B
10. A

Cycling of Nutrients

1. A
2. C
3. B
4. A
5. D
6. B
7. B
8. C
9. A
10. A

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ANSWER KEY FOR CROP SCIENCE 2

UNIT I. SITE CHARACTERIZATION FOR CROP PRODUCTION

A. BIOPHYSICAL AND ECONOMIC DETERMINANTS

1. A
2. A
3. D
4. A
5. B
6. A

B. POLITICAL DETERMINANTS

1. D
2. C
3. B
4. A
5. D
6. D
7. D
8. D
9. D
10. D

UNIT II. PRODUCTION PRACTICES

A. LAND PREPARATION FOR ANNUAL AND PERENNIAL CROPS (PRACTICES FOR SOIL EROSION CONTROL)

1. C
2. D
3. B
4. D
5. B
6. B
7. D
8. D
9. D
10. D
11. C
12. D
13. C
14. C
15. C

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B. PLANTING MATERIALS SELECTION AND PREPARATION

1. C
2. C
3. C
4. B
5. B
6. C
7. D

C. PLANTING METHODS

1. D
2. A
3. D

D. WATER SUPPLY ANG MANAGEMENT (IRRIGATION PRACTICES)

1. C
2. A
3. B
4. A
5. D
6. B
7. B
8. C
9. D
10. A
11. C
12. B
13. D
14. D
15. B

E. PEST MANAGEMENT

1. A
2. C
3. B
4. A
5. B

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